

IWCF

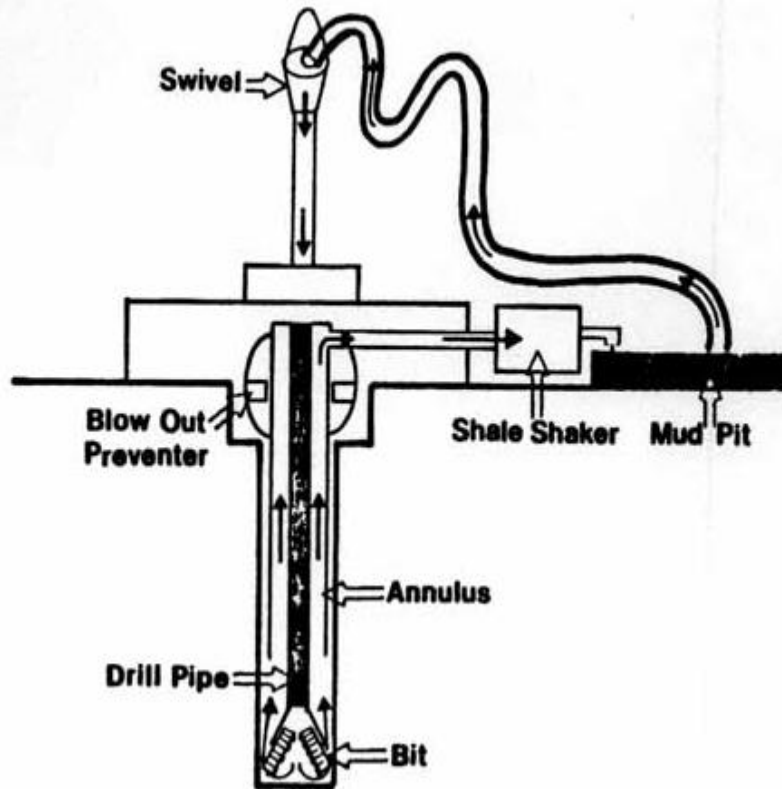


Well Intervention Pressure Control Course

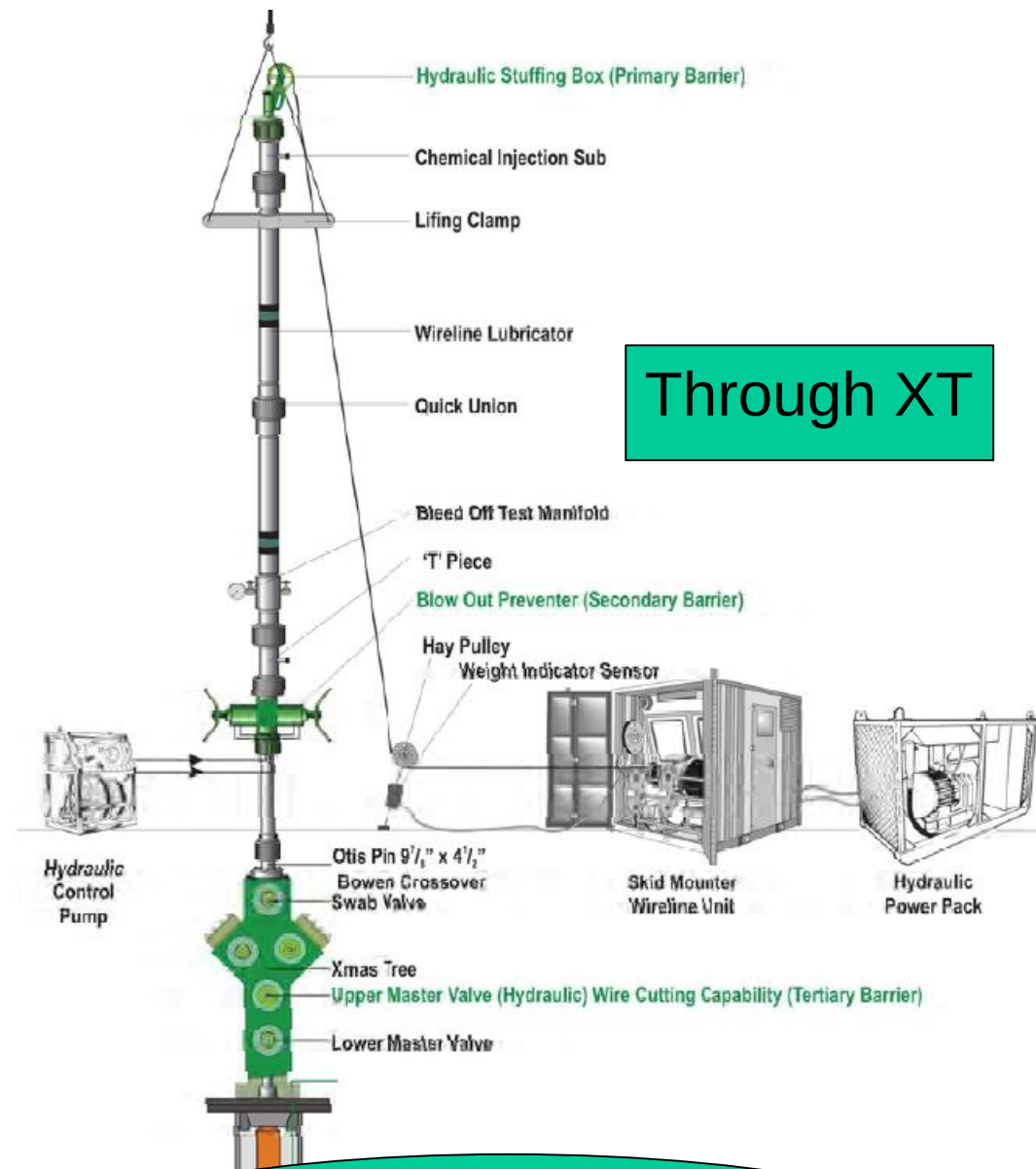
Course Content

1. Completion Operation
2. Completion Equipment
3. Wireline
4. Coiled Tubing

BOP used



Work Over
(Well Control) – 2 Barriers



Through XT

Well Intervention
(Pressure Control) – 3 Barriers

Well Control Event/ Incident

Impact of a Well Control Incident:

- Human Impact
- Environmental Impact
- Asset impact
- Over regulations
- Business disruptions
- Local and Public Image Impact

Attending the IWCF Well Control course will help to:

- Deepen theoretical and practical understanding of how to manage well site operations and maintain well control at all times.
- Maintain trust of stack holders
- Avoid over regulations
- Raising the competence
- Act as an induction for new recruited staff and giving them the information they need to work in the different positions.

- ✓ **Who shuts in the well and is responsible for safe working if a problem occurs during an intervention operation?**
 - The operator of the intervention equipment
- ✓ **For smooth and safe-running of a job;**
 - Hold a pre-job safety meeting with all personnel involved
 - Make sure that all personnel know what to do if a problem should arise
 - Only use tested, inspected and well-maintained equipment
- ✓ **If a well-control incident occurs;**
 - The well-team, well foreman (operator's representative) and foreman hold a pre-job meeting to allocate roles and responsibilities for the shut-in and for controlling the well if an incident should occur.
- ✓ **For good kick-off (toolbox) meeting;**
 - Get everyone involved to attend the meeting and go through plans, ask for feedback and comments, modify the plan if necessary and make sure that everyone understands in properly
- ✓ **If a problem has occurred with a well and it has been shut in;**
 - State the problem on paper, give possible causes, and hold a meeting with all parties involved. State the action to be taken (draw up a plan) if any, and then ask the onshore support team to comment on your conclusion or intended action
- ✓ **What are the reasons for holding meeting prior to any operation?**
 - Define rules and responsibilities
 - Discuss well control incident detail
 - Well control actions and the forward plan are fully discussed by all involved in the operation

Causes of Unplanned Well Inflow

- Wells inflow when the reservoir pressure exceeds the wellbore (bottomhole) pressure.
- Some of the most common causes of kicks include:
 - Not keeping the hole full (Loss of Hydrostatic Pressure)
 - Lost circulation
 - Swabbing
 - Underbalanced pressures

➤ Loss of Hydrostatic Pressure

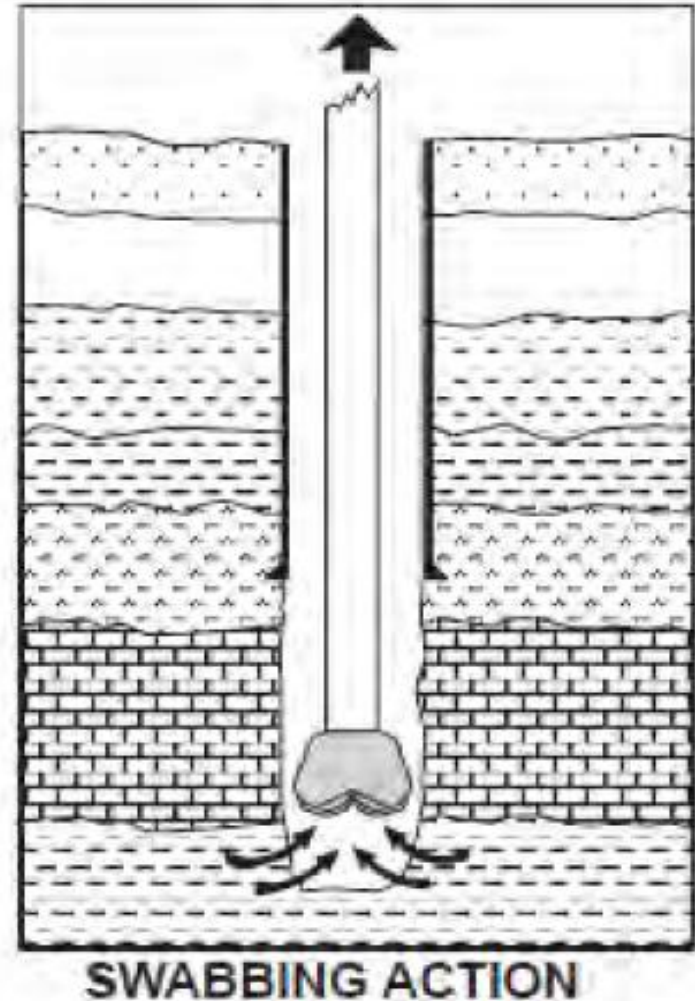
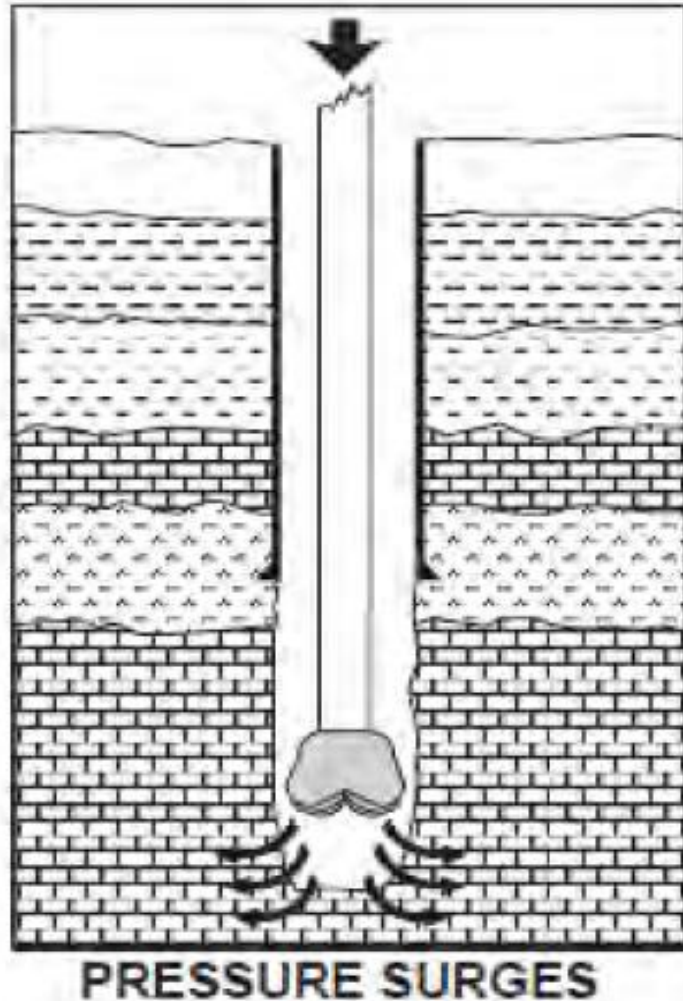
- Hydrostatic pressure is the number one defense against kicks.
- When tripping out of the hole, a volume (of steel) is being removed from the well.
- As the steel is removed, the fluid level & hydrostatic pressure in the well drops.
- If the hydrostatic pressure drops below the reservoir pressure, the exposed zone will flow. Therefore it is extremely important to fill the hole with fluid while tripping out of the hole.
- The volume of fluid needed to fill the well should be equivalent to the volume of steel (tubing) pulled out of the well.

➤ Losses

- Loss of circulation leads to a drop of both the fluid level and hydrostatic pressure in a well
- Losses can occur in any formation zone.
- Losses can occur in one zone while another zone is producing
- Three main causes of lost circulation are:
 1. Excessive pressure overbalance
 - when wellbore pressures exceed the fracture pressure
 2. Excessive surge pressure
 - increase in bottomhole pressure caused by the downward movement of the work string
 - minimized by monitoring trip speed and return fluid volumes while running in the well
 3. Poor formation integrity
 - use of a bridging agent to plug off the troublesome formation

➤ Swabbing

- Swabbing is caused by the upward movement of pipe in a well and results in a decrease in bottomhole pressure.



- **A live production well has a lot of sand and scale in the casing below the perforations. There are several hundred feet of open perforations and zones are at a much lower pressure than the others are. A work string is to be run through the completion to clean out the well to TD. Which of the following statement is true?**
 - a. The Thief zone (S) may stop proper circulation back to surface.
 - b. Reverse circulation is best when there is a thief zone.
 - c. A high pump rate will be required to overcome any losses into the thief zone (S)
 - d. It may be necessary to use a fluid containing LCM (lost circulation material). Such as sized salt particles, to stop the losses into the thief zone
 - e. The circulation system will have to permit pumping down the work string/completion annulus and the completion/casing annulus at the same time to have enough flow rates to overcome any losses into the thief zone

Completion Operations

Outlines

- I. Overview
- II. Introduction to Well Control
- III. Introduction to Barriers
- IV. Risk Management
- V. Circulating System
- VI. Testing
- VII. Influx Characteristics & Behavior
- VIII. Shut In Procedures
- IX. Well Control Methods
- X. Contingency Planning

- **Hydrostatic Pressure:**

HP is the pressure developed by column of fluid at given TVD.

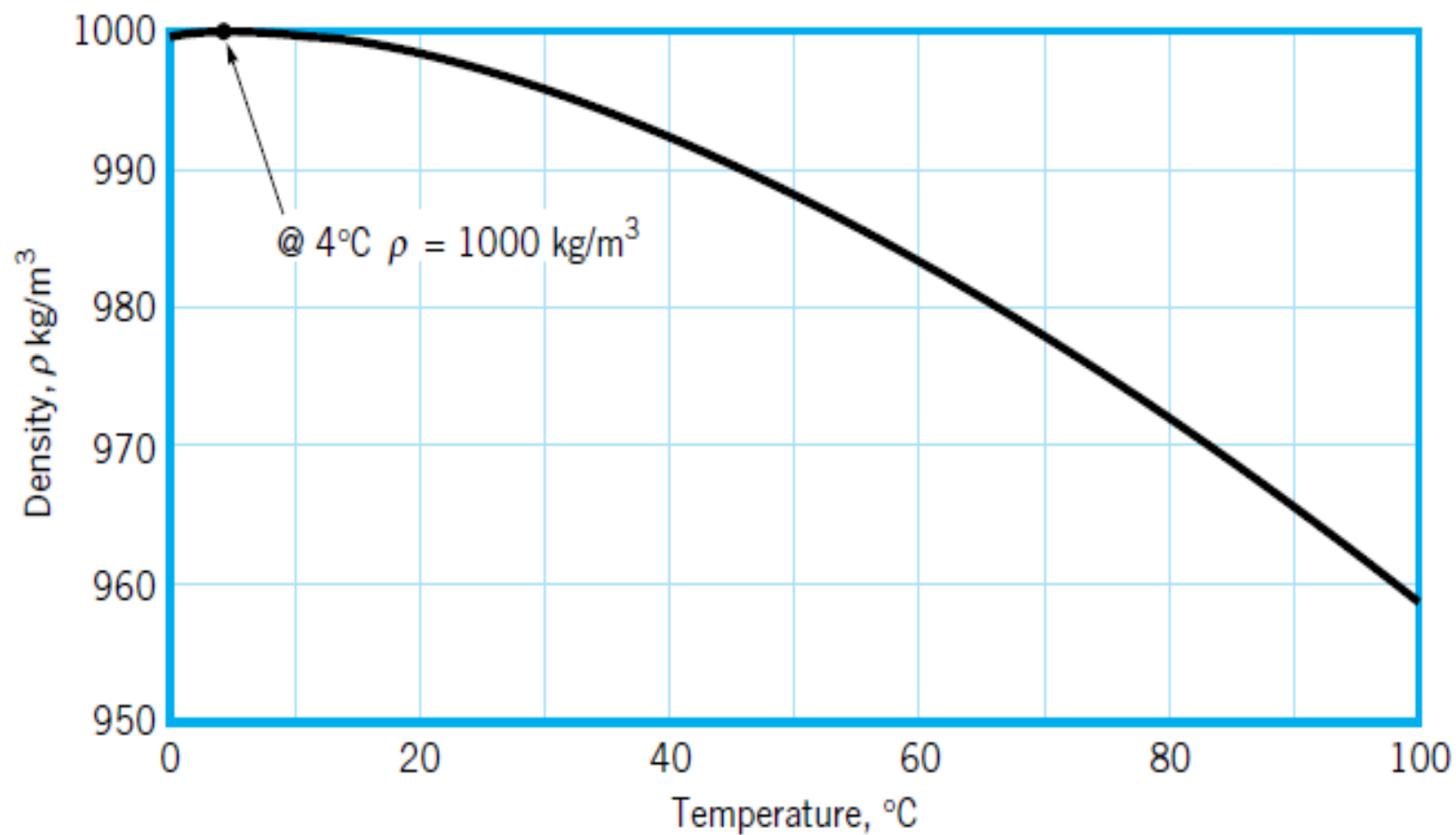
“Hydro” means fluid & “static” means stationary. HP is pressure created by stationary column of fluid.

Hydrostatic Pressure, **psi** = Density, **ppg** X 0.052 X True Vertical Depth, **ft**

Pressure Gradient, **psi/ft** = Fluid Density, **ppg** X 0.052

Pressure Gradient, **psi/ft** = Specific Gravity x 0.433

High bottom hole temperatures could affect the hydrostatic pressure gradients resulting in???



Density of water as a function of temperature.

- **Formation Pressure:**

- ✓ The formation pressure is defined as the pressure at which a fluid or gas exists in the pores of a permeable rock. This is also called pore pressure, OR, The pressure within the pore spaces of the formation rock.

- ✓ Formation pressures are normally classified into three groups:

- 1. **Normal Pressure**

- If the fluid in the pores is subject to hydrostatic pressure only.
Normal pressure is between 0.433 and 0.465 psi/ft.

- 2. **Subnormal Formation Pressures**

- Subnormal pressures are below 0.433 psi/ft.
 - If pore fluids are free to move, or escape, the grains lose some of their support and move closer together. This process is called compaction.

- Formation Pressure:

- ✓ Formation pressures are normally classified into three groups:

- 3. **Abnormal Formation Pressure**

- If the resulting formation pressures are above 0.465 psi/ft they are said to be abnormal.
 - The formation pressure is greater than normal formation pressure, this can be caused by:
 1. Compaction of the formation containing the fluid.
 2. Folding, faulting,...etc.

- Fracture Pressure

- ✓ Definition: The maximum amount of pressure a formation can withstand before the formation breaks down as a fracture is initiated, followed by loss of fluid, also defined as **Formation breakdown pressure**.

PRESSURE BASIC & BASIS

1. Fundamentals of Fluids & Pressure

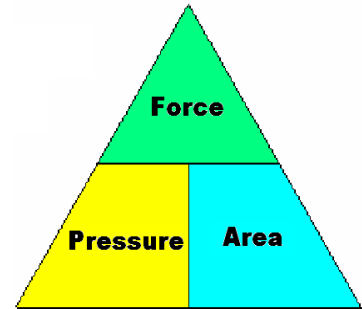
Pressure is defined as Force per unit area exerted by fluid i.e.

$$\text{Pressure} = \text{Force} \div \text{Area}$$

Using Δ concept, the formula can be changed:

$$\text{Force} = \text{Pressure} \times \text{Area}$$

$$\text{Area} = \text{Force} \div \text{Pressure}$$



2. Basic laws of physics concerning fluids & gas

- Liquids are not compressible
- Gases are compressible
- Liquids & gases are both FLUIDS
- Pressure in a fluid is transmitted equally in all directions.

Define

- a.** Fluid gradient **b.** Specific gravity **c.** Hydrostatic pressure

PRESSURE BASIC & BASIS

3. Fresh Water

Fresh water weighs the same per unit volume anywhere in the world that is why it's called "*Universal Fluid*" & used as *STANDARD* for comparison with other fluid.

Fresh Water Facts:

- Density of 8.33 lbs/gal
- PG of 0.433 psi/ft
- SG of 1
- API 10°

4. Specific Gravity (SG)

The ratio of the density of a substance compared to density of another substance which is used as the standard (*Fresh Water*).

PRESSURE BASIC & BASIS

5. API Gravity

API gravity is another value used to express relative wt of fluids & was introduced by American Petroleum Institute to standardized the wt of fluids at base temperature of 60° F. Water as standard fluid had been assigned the value of 10° API gravity.

To convert API gravity to SG, the following formula is used.

$$\text{SG} = 141.5 \div (131.5 + \text{API}^\circ)$$

6. Gas Correction Factor

Refer to “Gas Correction Factor” table. To find the GCF of 0.6 gravity at 7500 ft simply look down at 7500 ft column & move horizontal to 0.6 gravity column & find the CGF.

Well Depth	Correction Factors			
	0.6 Gravity	0.7 Gravity	0.8 Gravity	0.9 Gravity
3,000	1.064	1.075	1.087	1.098
3,500	1.075	1.089	1.102	1.115
4,000	1.087	1.102	1.117	1.133
4,500	1.098	1.115	1.133	1.151
5,000	1.110	1.129	1.149	1.169
5,500	1.121	1.143	1.165	1.187
6,000	1.133	1.157	1.181	1.206
6,500	1.145	1.171	1.197	1.224
7,000	1.157	1.185	1.214	1.244
7,500	1.169	1.204	1.232	1.264
8,000	1.181	1.214	1.248	1.282
8,500	1.193	1.239	1.266	1.304
9,000	1.206	1.244	1.282	1.324
9,500	1.218	1.259	1.302	1.345
10,000	1.232	1.275	1.320	1.366
10,500	1.244	1.289	1.338	1.388
11,000	1.257	1.306	1.357	1.410
11,500	1.270	1.322	1.376	1.433
12,000	1.282	1.338	1.395	1.455
12,500	1.297	1.354	1.415	1.477
13,000	1.311	1.371	1.434	1.500
13,500	1.324	1.388	1.455	1.523
14,000	1.338	1.405	1.475	1.548
14,500	1.352	1.422	1.495	1.573
15,000	1.366	1.438	1.515	1.596

PRESSURE BASIC & BASIS

7. Case Study

What is the differential pressure in psi exists between the tubing & annulus at the SSD assembly located at 8960 ft MD (8200 ft TVD) in the tubing?

Available Information:

- **The tubing/casing annulus is filled with 9.2 ppg brine full.**
- **The well is S/I at surface with CITHP of 600 psi.**
- **There is a gas cap of 0.6 SG gas down to 4000 ft fluid level**
- **32° API oil from fluid level' to 12000 ft.**

9. Volume

Find annular volume above packer in bbl
of the following well.

3½", 9.2 ppf Tubing:

9⅝", 47 ppf Casing:

9750 ft MD (9120 ft TVD) Packer:

Annulus Fluid Capacity between 9-5/8" x 3-1/2" Tubing

Casing	ppf	ft³/Lin.ft	Lin.ft/ft³	Bbl/Lin.ft	Lin.ft/Bbl
9⅝"	29.30	0.3607	2.7723	0.0642	15.5763
9⅝"	36.00	0.3468	2.8835	0.0618	16.1812
9⅝"	43.50	0.3308	3.0229	0.0589	16.9779
9⅝"	47.00	0.3237	3.0892	0.0577	17.3310
9⅝"	53.50	0.3100	3.2258	0.0552	18.1159

$$= 0.0577 \times 9750$$

$$= 562.6 \text{ bbls}$$

PRESSURE BASIC & BASIS

10. Well Pressure

Well Data: Well Depth: 9150 ft MD (7900 ft TVD)

Formation PG: 0.570 psi/ft

Gas PG: 0.08 psi/ft

1. Find Well BHP:

2. Find Maximum Surface Pressure:

3. Working Pressure for Wellhead Equipment:

4. Find the Kill Fluid Density:

WHP	Equipment Rating, psi
0 - 1500	3 K
1501 - 3500	5 K
3501 - 7500	10 K
7501 - 12500	15 K

Test Pressure = 1.5 working pressure

PRESSURE BASIC & BASIS

11. Pumping

Given Data:	Casing Capacity:	0.04049 bbl/ft
	Tubing Displ. Cap:	0.00829 bbl/ft
	Tubing Capacity :	0.01190 bbl/ft
	Pump Displacement:	0.0899 bb/stroke
	Tubing Shoe @:	9000 ft MD (7800 ft TVD)

1. How many strokes to displace Tubing String?

2. How many strokes to displace the Entire Wellbore?

PRESSURE BASIC & BASIS

12. Pumping Time

Given Data:	Tubing Depth:	8750 ft MD (8125 ft TVD)
	Tubing Capacity:	0.00387 bbl/ft
	Annular Capacity:	0.00970 bbl/ft
	Pump Rate:	1.25 bpm

- 1. Calculate the time to pump bottoms up (Forward Circulation).**
- 2. Calculate the time for complete circulation.**

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• Barrier Theory

- ✓ **Barrier Definition:** any device, fluid or substance that prevents the flow of well bore fluids.
 - It may be Mechanical or fluid.
 - Located down hole (mechanical plugs, columns of fluid), or topside (valves, plugs).
- ✓ **Well barriers:** are envelopes of one or more dependent well barrier elements to prevent fluids or gases from flowing unintentionally from a formation, into another formation or back to surface.
- ✓ **Well Barrier Element:** is defined as a component part of a well designed to prevent fluids or gases from flowing unintentionally from a formation, into another formation or to escape at surface.
- ✓ **Independent barriers:** means each barrier shall be able to be closed and contain the well bore fluids by its own control system irrespective of any power failure.
- ✓ **Overbalance:** Greater pressure than formation pressure
- ✓ **Underbalance:** Less pressure than formation pressure
- ✓ **Inflow Test:** Bleed pressure from above & monitor build-up
- ✓ **Positive Plug:** Hold pressure from up & down

- Barrier Theory

- **Minimum Barrier Requirements**

- A. A minimum of two independent and tested barriers shall be available at all times to prevent flow to the environment.
- B. As far as possible the barriers should be tested in the direction of flow of the well.
- C. The barriers must be capable of being operated independently of each other.
- D. If one barrier fails it must be possible to put compensatory measures into effect immediately. During this phase, efforts to re-establish the barrier and return the well to double barrier protection, are the only type of work permitted.
- E. In principle, tested and approved barriers shall not have a leak rate greater than 0, before any work is conducted.
- F. The owner of the well may accept deviation of the test from this requirement in certain circumstances.

- A Xmas tree is to be changed out on a production well. A deep-set positive plug has been set in the tailpipe below the packer. A second plug will be set in the tubing hanger before the tree can be removed. What is the correct first action to take before setting the second plug?
 - a. Inflow test the deep-set plug
 - b. Pressure up on the tubing to test the deep-set plug.
 - c. Pressure up on the annulus to test the deep-set plug from below.
 - d. If the wellhead pressure is not rising, the second plug can be run immediately.
- A newly completed well has a sliding sleeve (SSD) just above the packer. The sleeve has just been closed with the tubing full of diesel and the annulus full of brine. The tubing is open to the perforation. How should the SSD be tested to ensure it is closed?
 - a. Flow the well
 - b. Pressure up the annulus
 - c. Bleed down the annulus
 - d. Bleed down the tubing

- **Barrier Envelope:**

- ✓ A "Well Barrier Envelope" is defined as a combination of one or more Well Barrier Elements that together constitute a method of containment of fluids within a well that prevents uncontrolled flow of fluids into another formation, or, to escape at surface.
- ✓ “Envelope” is a concept that is utilized in considering a barrier as part of a total system that involves pressure or fluids.
- ✓ The barrier itself (e.g. a wireline BOP) will not be of much help if the complete system is not intact (Xmas tree, completion string, extension tubing).
- ✓ Wells in which the oil has to be lifted or pumped out (i.e. artificial lift pumps, rod pumps) need the same control barriers as wells in which the oil flows out of its own accord.

Example of Barrier Envelope

PACKER/TUBING ENVELOPE

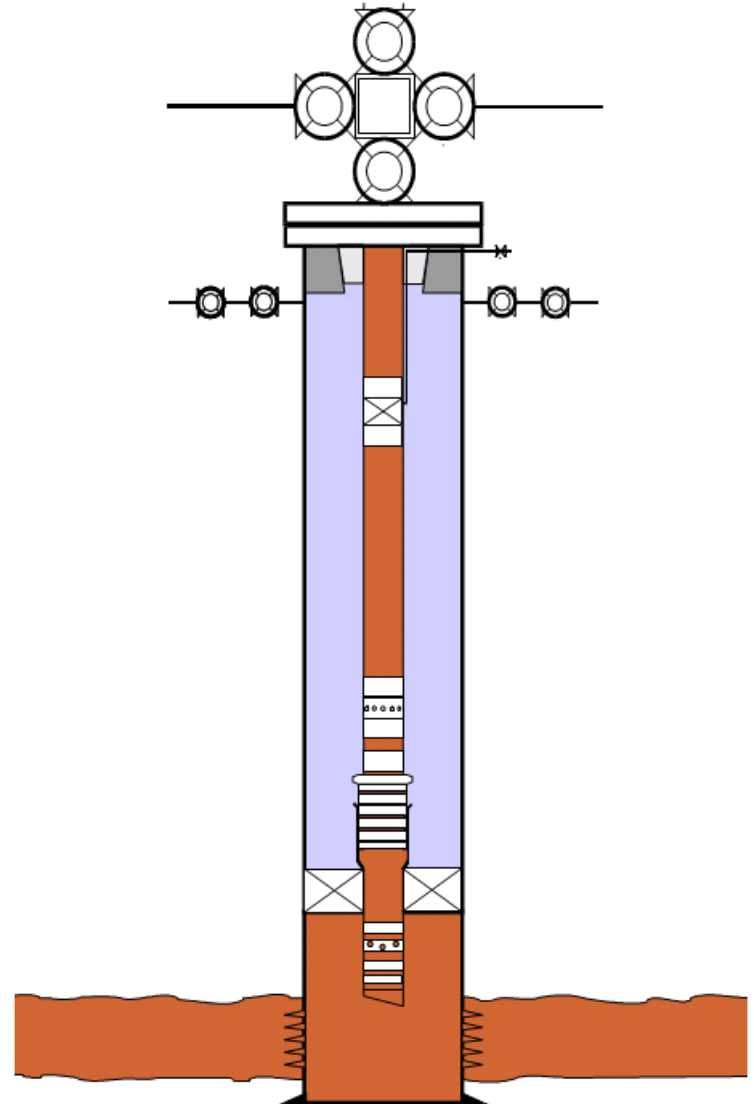
Barrier elements include:

- Casing below Packer
- Packer
- Tubing
- Tubing Hanger
- Xmas Tree

PRODUCTION CASING ENVELOPE

Barrier elements include:

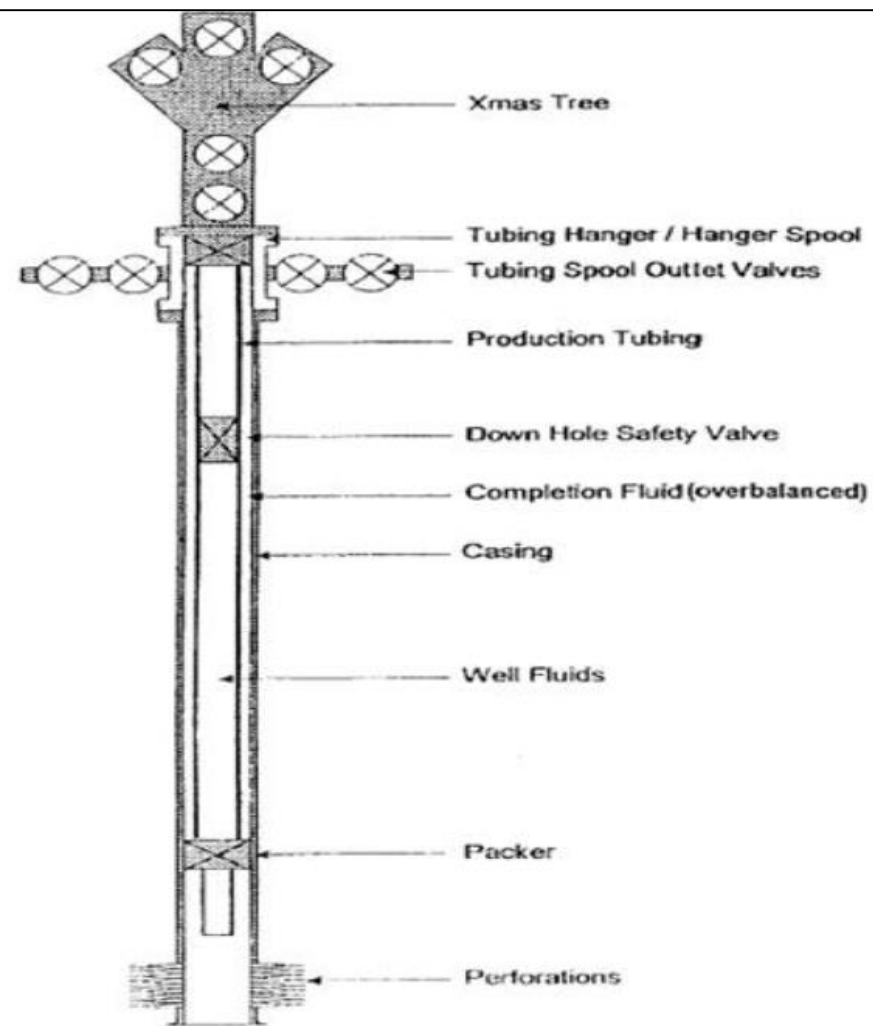
- Production Casing
- Side Outlets Valves
- Tubing Hanger
- Packer



A. Pressure has been observed in the completion annulus. If the DHSV is closed and the wellhead pressure bled off, the annulus pressure falls with it. A leak at which two barrier elements could cause this?

B. There is a leak at the flange between the tubing hanger spool and the Xmas tree. What can be closed to stop this leak?

C. There is a leak at the packer. What initially prevents the wellbore fluids from reaching the production casing?



- A Xmas Tree
- B Tubing Hanger/Hanger Spool
- C Tubing Spool Outlet Valves
- D Production Tubing
- E DHSV
- F Overbalanced Completion Fluid
- G Well Fluid
- H Production Casing
- I Packer

- **Barrier Classification:**

- **Primary Pressure Control (Used during normal operations)**

- ✓ Primary pressure control is the system, which provides the first line of defense from an uncontrolled well flow, e.g. mud column.
- ✓ This is the first object that prevents flow from a source.
- ✓ Its function to prevent unintentional flow to the environment or other formations.
- ✓ It differs according to well servicing intervention method.

- **Secondary Pressure Control (support of normal operations as contingency)**

- ✓ Secondary pressure control is the system, which provides the second line of defense, in the event that primary well control cannot be properly maintained.
- ✓ Secondary barriers are brought into action if primary barrier should fail.
- ✓ The secondary barrier is to prevent unintentional flow if the primary barrier fails. This is generally provided by the BOP system.

- **Tertiary Pressure Control (Used in emergencies)**

- ✓ It is not always available but may be an additional third and final line of defense in the event that secondary well control cannot be properly maintained.

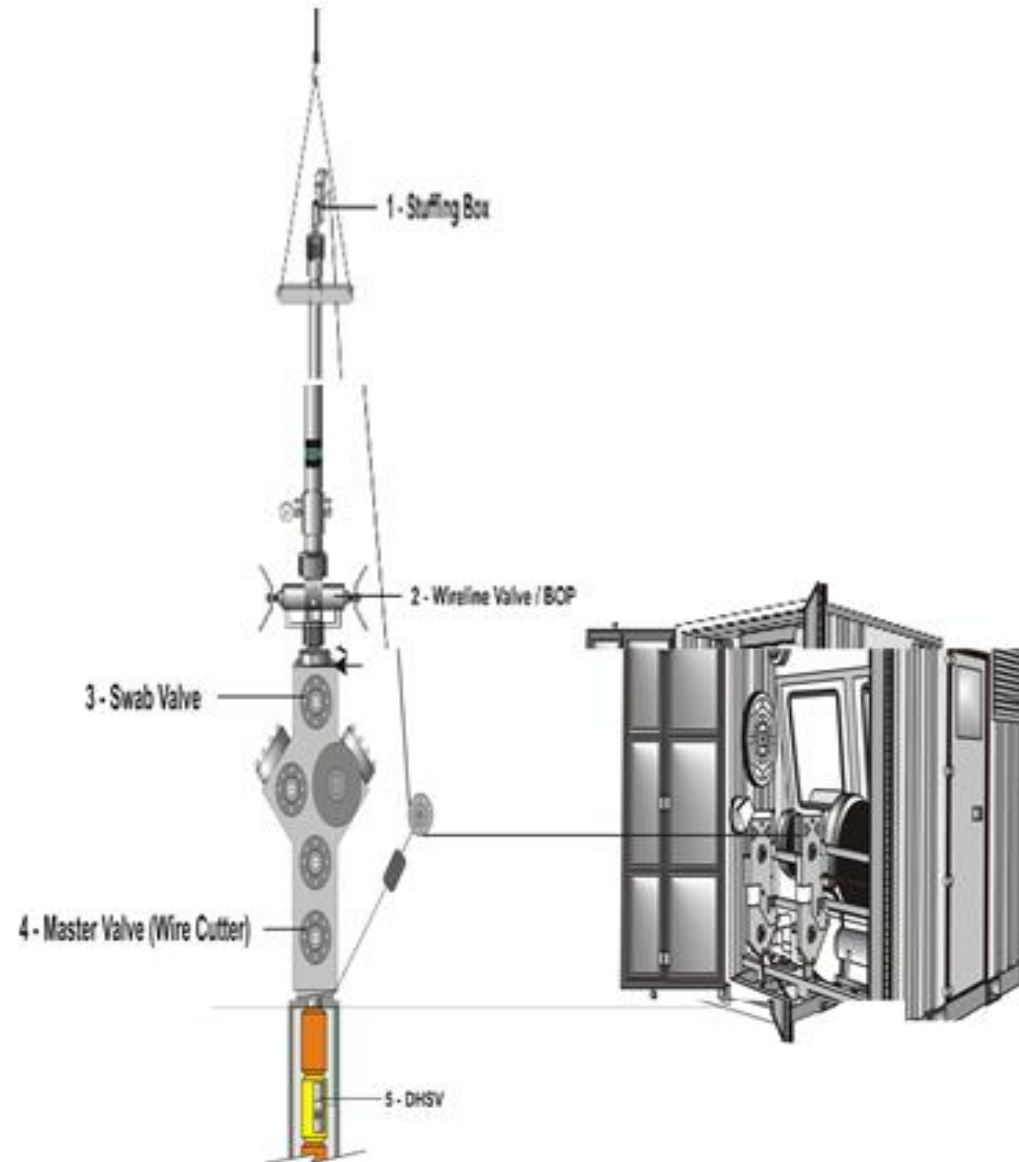
- **Barriers**

- **Sequence of Barrier Operation**

- ✓ The primary barrier is the first line of defense and on live Wells is usually in continuous operation.
- ✓ If there is a failure or potential failure of the primary barrier, the secondary barrier is brought into operation.
- ✓ The tertiary barrier is the last line of defense and it usually cut the Wireline or pipe, and is the last resort.
- ✓ In live well interventions, it is not generally necessary to provide kill facilities unless there is higher risk due to extreme high pressure or the presence of high concentrations of H₂S.

- **Barriers**

- **Sequence of Barrier Operation (Slick Line)**



• Barriers

➤ Sequence of Barrier Operation (Slick Line)

A. Primary

- Stuffing box and lubricator system.
- Check valve if WL breaks and is ejected from the lubricator.
- X/T valves when installing or removing tools from, the lubricator.

B. Secondary

- WL BOP rams/valve which can close and seal around the wire.
- Xmas tree Swab valve, if the wire is broken and ejected.

C. Tertiary

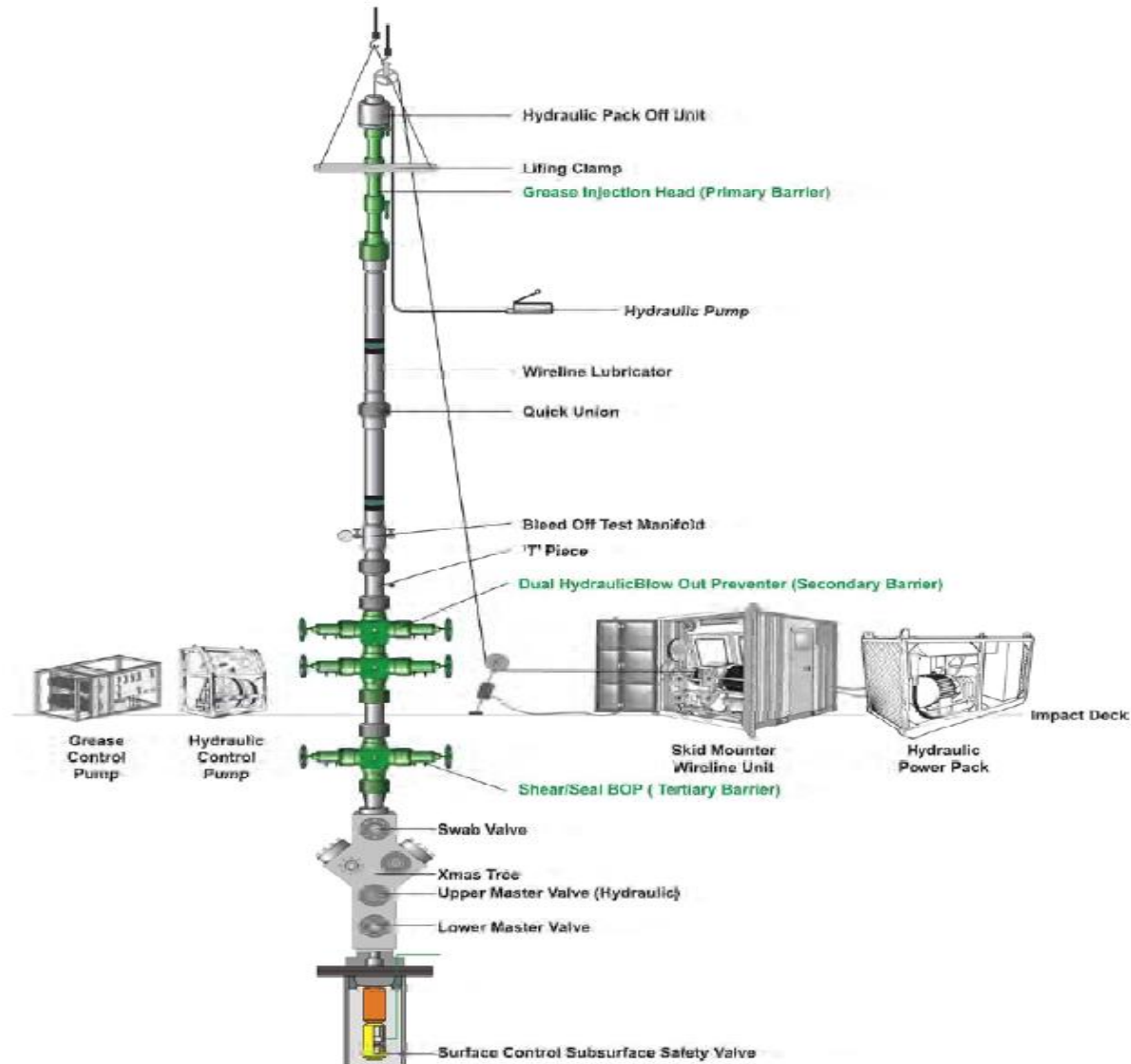
- Wireline cutting valve (Usually UMW designed to cut wire)
- BOP/Shear Seal Valve installed directly on top of the Xmas Tree.

✓ The valve used to cut wireline should be UMW for two reasons:

- If the lower master is used and damaged, it requires the well to be plugged before repair.
- If the swab is used and damaged the well cannot be used for production as there is no longer double barrier protection from the production fluid.

- **Barriers**

- **Sequence of Barrier Operation (Braided Line)**



Typical Braided Line Rig up

- **Barriers**

- **Sequence of Barrier Operation (Braided Line)**

- A. Primary**

- Grease head seal and lubricator system.
- Check valve if the wire breaks and is ejected from the lubricator.
- Xmas tree valves when installing into, or removing tools from, the riser.

- B. Secondary**

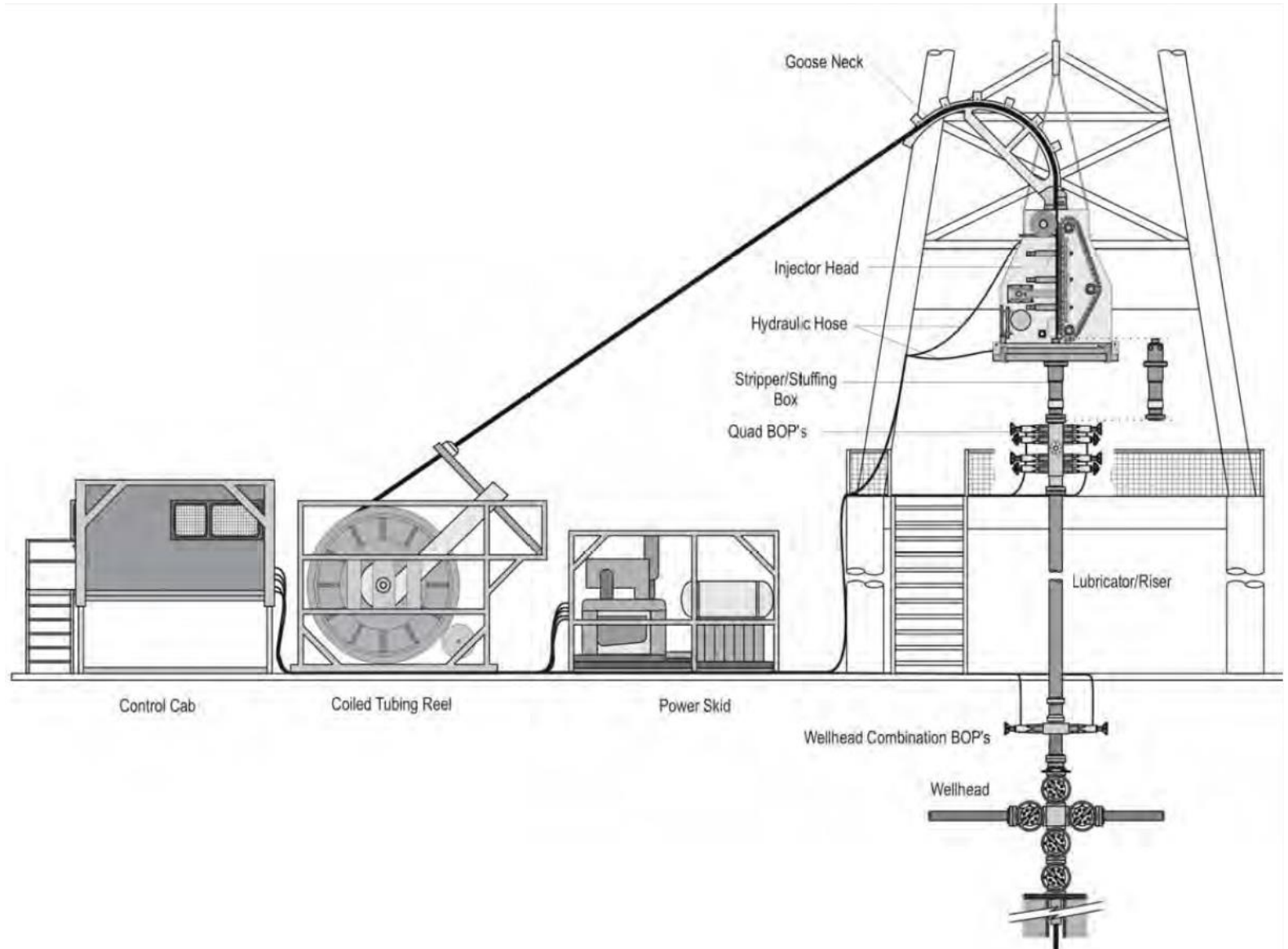
- Two Wireline BOP rams (in conjunction with a grease pump) that can close and seal around the wire.
- Xmas tree swab valve, if the wire is broken and ejected.

- C. Tertiary**

- Wireline cutting valve (usually UMV designed for Wire cutting).
- Shear/seal valve or BOP installed directly onto the top of the Xmas tree.
- In general, tertiary barriers are rarely used unless a heavy-duty Wireline operation is being carried out.

- **Barriers**

- **Sequence of Barrier Operation (Coiled Tubing)**



- **Barriers**

- **Sequence of Barrier Operation (Coiled Tubing)**

- ✓ External pressure control is provided by:

- A. Primary**

- Stripper.
- Xmas tree valves when installing into, or removing tools from, the riser.

- B. Secondary**

- BOP.

- C. Tertiary**

- Shear/seal BOP mounted directly on top of the Xmas tree.

- **Barriers**

- **Sequence of Barrier Operation (Coiled Tubing)**

- ✓ Internal pressure control is provided by:

- A. Primary**

- Two check-valves in the BHA.
- Some well interventions are conducted without BHA check valves as it is necessary to reverse circulate. In these cases **the primary inside well control** is the BOP shear rams and a shear/seal BOP becomes the secondary.

- B. Secondary**

- Shear and Blind rams incorporated within the BOP.

- C. Tertiary**

- Shear/seal BOP mounted directly on top of the Xmas tree.

Barrier Types

1. Mechanical Barrier:

stops well flow by closing off the flow route

1.1 Closed Type

Must be “Tested From Direction of Flow”

Must be “Leak Tight”

1.2 Closeable Type

API leakage criteria “400 cc/min or 900 scf/hr”

2. Hydrostatic Barrier

Liquids with about 200 psi over balance higher than formation pressure and lower than Fracture pressure

- Liquids may be used as barriers so long as they are controllable and monitorable
- Must be diligently monitored for a period of time to Ensure Thermal Expansion & Contraction Effects Have Ceased.

MECHANICAL BARRIER	
CLOSED TYPE	CLOSABLE TYPE
Stuffing Box	BOP
Grease Injection Head	Annular Preventer
Stripper	Xmas Tree
BHA Check Valve	Subsurface Safety Valve
HYDROSTATIC BARRIER	
Drilling Fluids	Fresh Water
Completion Fluid	Salt Water
OTHER BARRIER EXAMPLES	
WL Plugs	Ice Plugs
Bridge Plugs	Overbalanced Fluid

- It is good practice to have more than one barrier always available
- we can combine (use mixture of) mechanical and fluid barriers in the same well
- Fluid barrier is always primary barrier

Tubing Plugs

Tubing plugs can be divided into four categories:

- Plugs which hold pressure from below - pump through plug (check valves)
- Plugs which hold pressure from above - test plug (standing valve)
- Plugs which hold pressure from both directions (positive plug)
- Plugs which hold pressure to set values - pump out plugs (shear plugs)

You should

- Check that the pressure rating is correct.
- Check that the contingency exists if solids might settle on the plug top.
- Check that the plug is holding pressure after setting.

Tubing Plugs

1. **Blanking Plugs (wire line plugs):** run on wire line & set in a nipple. They seal off in a nipple & hold pressure from both directions (positive plug). Either one trip or two trips.
2. **Pump Open Plug:** The inside of the plug shears when predetermined pressure is applied from top of the plug (positive plugs with a limited pressure rating.)
3. **Pump Out Plug:** Similar in principle to pump open plugs but shears off & drops down hole
4. **Pressure cycle plug:** Require pressure above the plug to be cycled a fixed no. of times before the plug opens.
5. **Ice Plug:** Wrapping the equipment to be frozen in a coil through which chemicals like glycol are passed after being cooled to a predetermined temperature by liquid nitrogen.
6. **Bridge Plug:** Used to shut-off water producing zones by setting in casing and capping it with cement (expandable plug).
7. **Pump-through Plugs:** Hold pressure from below but allow pumping if the need arises.
8. **Back Pressure Valves (one-way or two-way valve):** It is a check valve where fluid can pass through it with 1 psi ΔP and it is set in tubing hanger threaded profile.

• Barriers

➤ Principles of Fluid barriers

- ✓ For a fluid to be suitable for use as a barrier, the following conditions must be satisfied:
 - Correct specification
 - Testing
 - Observation.
- ✓ Correct specification means that the fluid has the correct specific gravity.
- ✓ Testing refers to monitoring the specifications when the fluid is being weighed and pumped.
- ✓ Observation means that the well should be monitored for a period in order to satisfy ourselves that the pumped fluid is stabilizing it.
- ✓ Simply, we should select the correct kill fluid (weight and specifications) to minimize formation overpressure & ensure formation fluid compatibility

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- Risk Assessment (**All involved parties**)

- ✓ **RA process involves:**

- Identifying and ranking risk events, developing mitigation options to minimize probability and severity of outcome, and providing contingency procedures that can be implemented if the events occur.
 - The risk events are then re-ranked assuming that mitigations and contingencies (M &C) have been applied, which will show whether the M&C have changed these risks to an acceptable level.

- Role of Risk Assessment

- ✓ Occurrence of well control events is an ever-present risk in oil industry.
 - ✓ In many cases, it is not possible to completely eliminate these risks, so they must be managed to an acceptable level in terms of HSE impact and financial losses.

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- Circulating System

- ✓ Friction pressures act in the opposite direction to flow. It helps determine how fast fluid can move through the well
- ✓ Friction pressure is affected by the fluid viscosity, the flow area, and the flow rate.
- ✓ The pressure on the pump is actually the amount of friction that must be overcome to move fluid throughout the wellbore at a given flow rate.

Completion Operations

Outlines

- I. Overview
- II. Introduction to Well Control
- III. Introduction to Barriers
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- V. Circulating System
- VI. Testing**
- VII. Influx Characteristics & Behavior
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- **Testing Barriers**

- ❑ **A) Mechanical Barrier**

- ✓ Pressure tested in the direction of flow from well is required
 - ✓ If this cannot be realistically achieved, barrier can be pressure tested not in the direction of flow from well

- ❑ **Hydrostatic Barrier**

- ✓ Required to be monitored and maintainable to qualify as a barrier.
 - ✓ Some stable fluids may qualify as a barrier without maintenance (e.g. Water)

- **Testing Barriers**

- **Documentation of leak and function testing of well barriers**

- All well integrity tests shall be documented and **accepted by an authorized person.**
 - The chart and the test documentation should contain
 - type of test,
 - test pressure,
 - test fluid,
 - system or components tested,
 - estimated volume of system pressurized,
 - volume pumped and bled back,
 - time and date.

- **What are items needed to document a barrier?**

- a. Pressure rating
 - b. Sign off
 - c. Fluid weight
 - d. Fluid viscosity
 - e. Joint operation manual

- **Mechanical Barriers**

- ✓ Barriers should normally be tested for both **inflow and pressure integrity**.
- ✓ This condition can be achieved by direct application of pressure (pressure testing) and measuring its seal tightness or by decreasing the hydrostatic pressure on the downstream side of the “seal” (Inflow testing) and measuring its effectiveness. Both are pressure testing in the direction of flow.
- ✓ **Function testing of well barriers:**
 - after installation, after having been subjected to abnormal loads, after repairs, routinely
 - **Permanent & temporary ????**

• Mechanical Barriers

□ Pressure Integrity Test

- Pressure testing of a barrier is performed by increasing the pressure to a value higher than that on the lower side of the barrier. We then observe the pressure for a period of time in order to detect any leaks.
- The point of pressure tests is that they should reflect the lowest and maximum expected pressure in the phase concerned.

□ Inflow Testing Procedure

- An inflow test is performed by reducing the hydrostatic head above the item to be tested by circulating to a lighter fluid.
- The barrier is closed and the section downstream the barrier is bled down. The bled down section of the tubing or flow line is then closed against a tested barrier. The pressure in the section will be monitored for any pressure increase. If no pressure increase is monitored the barrier is successfully tested.

- **What item of equipment should be pressure tested before we rig up intervention equipment?**
 - a. Xmas tree
 - b. Tubing hanger
 - c. Packer
 - d. Annulus
- **Tree gate valves can be primary barrier?**
 - a. True
 - b. False
- **A low pressure and high pressure leak test are normally performed for component that is to be or may be exposed to well pressure. Which one of the following statement is correct?**
 - a. If no leaks are visible on the equipment, the test is OK.
 - b. The acceptance criteria should be found in relevant documentation (well program, API, NORSOK, or manufacturer recommendations).
 - c. 10% pressure drop is inside the acceptance criteria.
 - d. 25% pressure drop is inside the acceptance criteria.
 - e. 5% pressure drop is inside the acceptance criteria.

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- Well influx can consist of gas, water, oil, or any combination of these media.
- **Gas Expansion, Uncontrolled, Well Open**
 - If the volume of gas doubles, the pressure is reduced by half in the bubble (Boyles Law, $P_1V_1 = P_2V_2$).
- **Gas Migration, Well Shut In**
 - Pressure in actual bubble is usually at formation pressure.

- **Which of the following indicates that a gas cap may be forming in a live well that has just been shut in at the tree?**
 - a. Initial SIWHP slowly falls.
 - b. Initial SIWHP slowly rises.
 - c. Initial SIWHP remains steady.
 - d. Annulus pressure slowly falls.
 - e. Annulus pressure slowly rises.
 - f. Annulus pressure remains steady.

- **When shutting in a producing well, there will be rapid increase in pressure displayed on X/T gauge. Which one of the following is correct?**
 - a. The increase is mainly caused by gas migration.
 - b. The increase is mainly caused by temperature increase.
 - c. The increase is mainly caused by cross flow in the reservoir.
 - d. The increase is mainly caused by temperature decrease.
 - e. The increase is mainly caused by pressure build-up from the reservoir.

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- Shutting-in Xmas Trees

- **Correct shutting in**

- ✓ During well operations, the local hydraulic pump is connected to the DHSV and the hydraulically actuated master valve. This pump must be monitored continuously during the operation.
- ✓ When valves in the Xmas tree are being opened, we must first make sure that the pressure across the valves has been equalized. If we forget to do so, the valve may become “pressure locked”.
- ✓ When the well is to be shut in, this is done by first closing the valves on the line in the direction of the X/T; wing, UMV, and finally the DHSV.

- # Shutting-in Xmas Trees

- ✓ **When we open a valve that is pressurized on only one side;**

- The valve may suffer damage
- The valve will suffer a strong hydraulic shock
- There will be high probability of pressure lock
- The mechanical force required to turn handle can damage or break the stem
- The high differential pressure disables the gate movement when opening
- The sudden pressure surge on opening can damage piping equipment downstream.

- ✓ **When we shut down a well at the Xmas tree;**

- Close the Swab valve while counting turns if you need to shut in after pulling out of the hole with the intervention tool string
- Counting the turns as the valve is opened or closed can show any obstructions in front of the valve.
- The lower master valve is normally not in use
- The wing valve closes off all flow from the well
- The valves may be damaged if they are closed against a wireline, tubing, etc.
- The upper master valve is normally utilized
- Swab valve can be used to cut Slickline & Some master valves can be used to cut Slickline
- After closing, a manual valve handle should be backed out by part of a turn

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- ✓ The objective of the various kill methods is to circulate out any invading fluid and circulate a satisfactory weight of kill mud into the well without allowing further fluid into the hole.
- ✓ Ideally this should be done with the minimum of damage to the well.
- ✓ On killing, we may have loss from one zone while we continue to produce from another. This may require the temporary shut-in of the “thief” zone (with LCM) in order to permit circulation or reverse circulation.

METHODS OF KILLING A PRODUCTION WELL

- BULLHEADING (one or double TBG capacity, **problem or emergency**)
- FORWARD CIRCULATION (**little application**)
- REVERSE CIRCULATION (**planned killing**)
- LUBRICATE AND BLEED (**impossible to bull head**)

BULLHEADING

ADVANTAGES:

- Quick and easy method especially in smaller tubing sizes
- No hydrocarbons brought to surface

DISADVANTAGES:

- Scale or contaminants in the tubing are pumped against the formation
- Low formation fracture pressures may be exceeded causing accidental fractures
- Gas may slip up the tubing if the pump rate is not sufficient for larger tubing sizes

- ✓ Method involves pumping kill fluid directly into tubing to force tubing contents back into formation.
- ✓ Preferred when reverse circulation is not possible (stuck closed sliding sleeve or insufficient information is available to calculate a reverse-circulation kill).
- ✓ Used only if there is no sand or scale (perforations open)
- ✓ Used only if it does not causes any appreciable damage to formation (it is likely to have formation damage)
- ✓ Cannot be used if there is blockage (stuck plug in tail pipe or severely collapsed casing above perforations) or tight formation preventing bull heading.

Note: DHSV with serious leak in control line is not a blockage

- ✓ Calculation must be made for Max allowable surface pressure and tubing burst pressure.
- ✓ Pump rates must be restricted to keep pressure below fracture point.

✓ **A live well is to be killed by bull heading. Which of these factors limit the maximum allowable surface pressure?**

- a. Completion size.
- b. Maximum pump speed.
- c. SIWHP.
- d. Maximum safe working pressure of the surface equipment.
- e. Completion burst limits.
- f. DHSV operating pressure.
- g. Possible formation fracture.

✓ **The stack configuration is 5000 psi working pressure. The wellhead pressure during a test is 4860 psi. the well needs to be killed using bullheading method. The formation is tight. Is the stack pressure rating sufficient?**

- a. The bullheading can't be performed with this stack pressure rating
- b. The stack is sufficient within 10% overpressure limit
- c. The stack is sufficient within 5000 psi working pressure
- d. The stack pressure rating is irrelevant for the situation

FORWARD CIRCULATION

ADVANTAGES :

- Little or no damage to the formation by foreign fluids or contaminants from the tubing or annulus

DISADVANTAGES :

- Less safe as hydrocarbons are brought up through the hanger spool outlets and disposal is more difficult
- Gas and oil will mix with the packer fluid due to gravitation requiring special handling equipment
- High circulation pressures on the annulus
- No advantage at all over reverse circulation

REVERSE CIRCULATION

Advantages

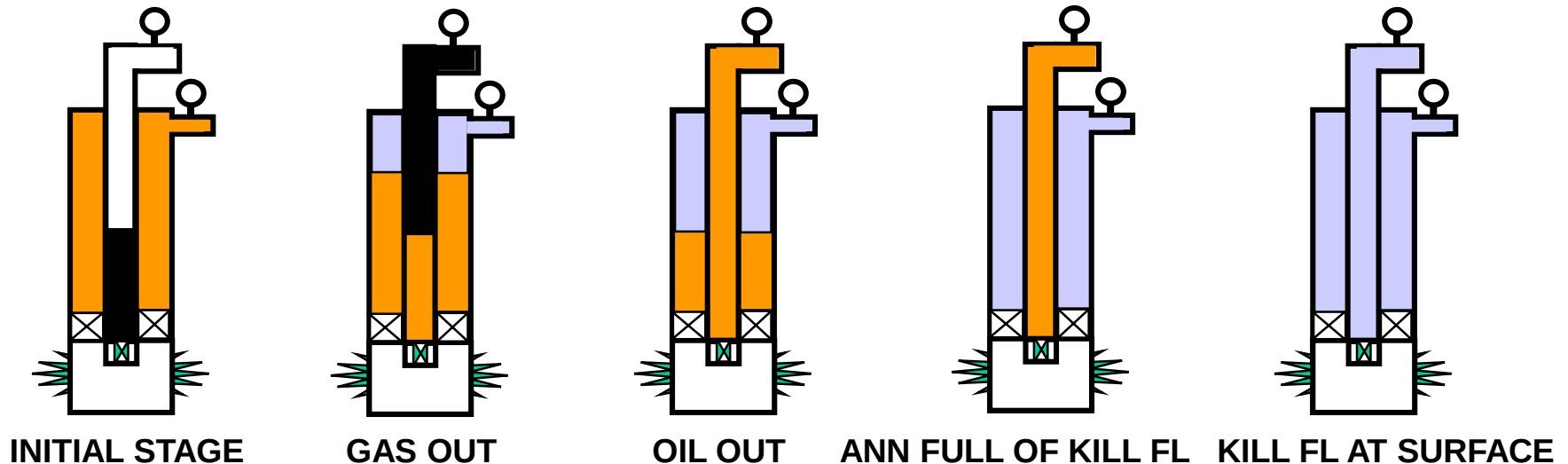
- It utilizes natural U tubing effect resulting in lower circulating pressures.
- Lesser chances of mixing as gas & oil are lighter and remain above the completion fluid while reverse circulation.
- More predictable as every thing comes out in order.
- Little risk of accidentally fracturing formation like in Bullheading.
- Little or no damage to formation by contaminants from tubing or annulus.
- The surface pressure is kept low
- The production tubing and annulus end up with pure killing fluid
- All wells can normally be killed using this method.

Disadvantages

- Slower than Bullheading
- We have to utilize wireline

REVERSE CIRCULATION

DIFFERENT STAGES



- It is important to install a wire line set plug below packer to isolate formation from completion & kill fluid.
- In case plug is not installed, maintain 200 psi extra pressure from surface through choke.

Q. NO. 1

✳ Find SITHP and SICP at different stages of reverse circulation.
Draw Tubing and Casing pressure graphs.

Tubing Capacity = 0.01458 bbl/ft

Gas = 2000 ft, 0.04 psi/ft

SSD = 6000 ft

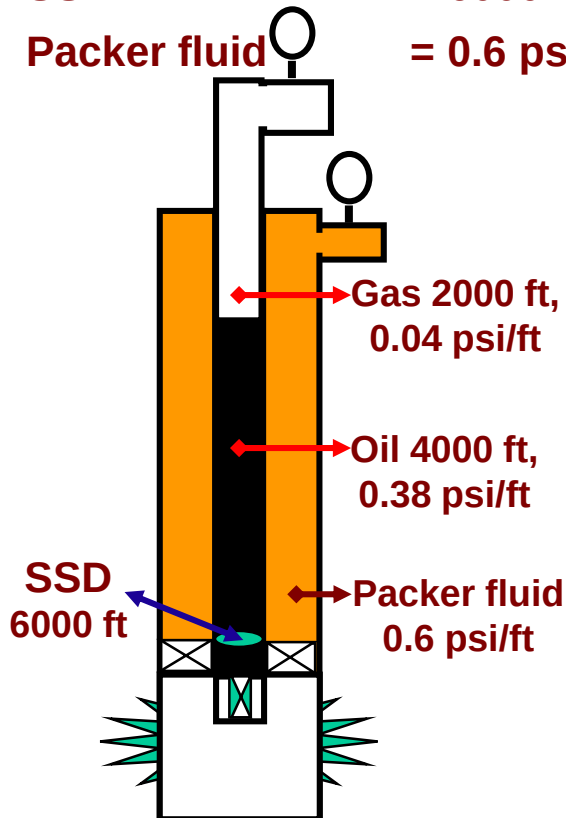
Packer fluid = 0.6 psi/ft

Annulus Capacity = 0.0252 bbl/ft

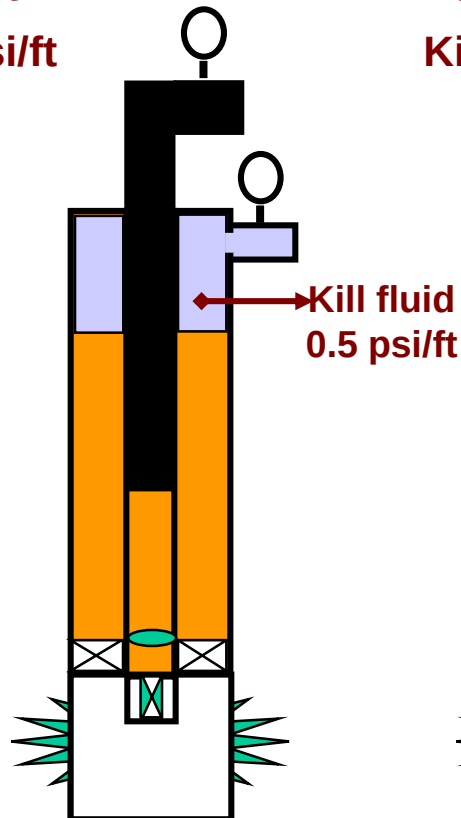
Oil = 4000 ft, 0.38 psi/ft

Pump output = 0.0157 bbl/stroke

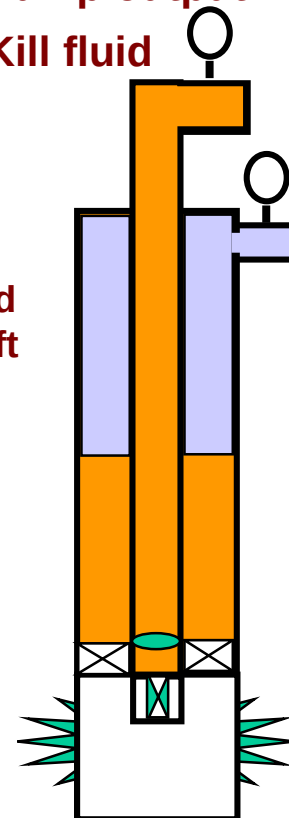
Kill fluid = 0.5 psi/ft



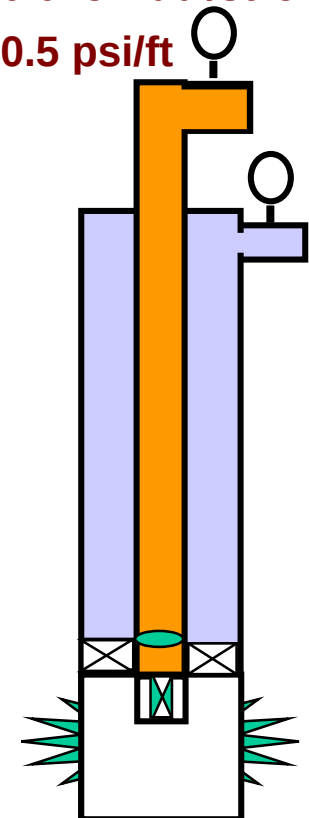
STAGE-1
INITIAL STAGE



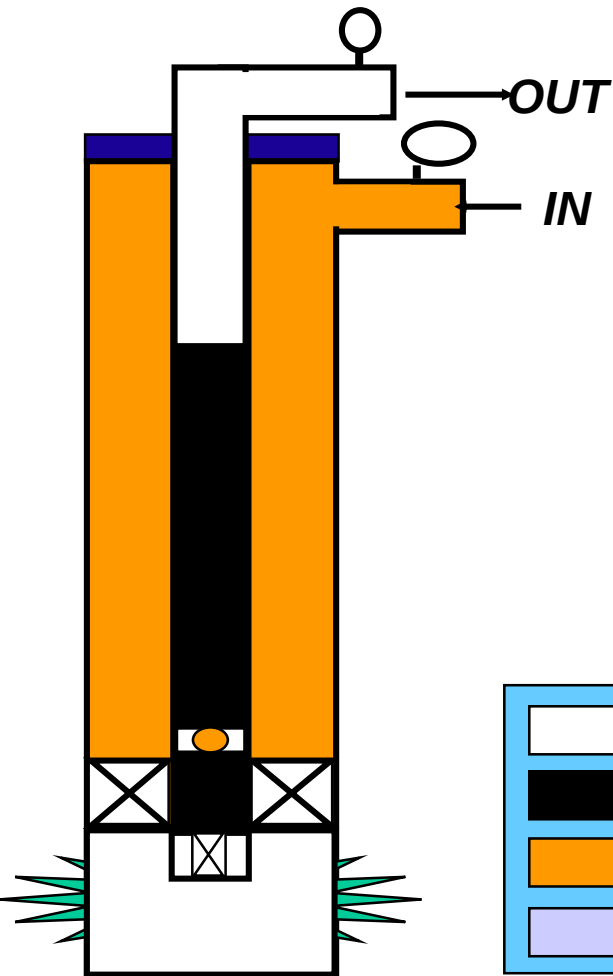
STAGE-2
GAS OUT



STAGE-3
OIL OUT



STAGE-4
ANN FULL OF KILL FLUID



(Assuming Formation is initially balanced by Brine)

Formation Pressure = BHP

$$= 6000 \times 0.6 = 3600 \text{ psi}$$

Initial Tubing Pressure

= Formation Pressure – Tubing Hyd Pr

$$= 6000 \times 0.6 - (2000 \times 0.04 + 4000 \times 0.38)$$

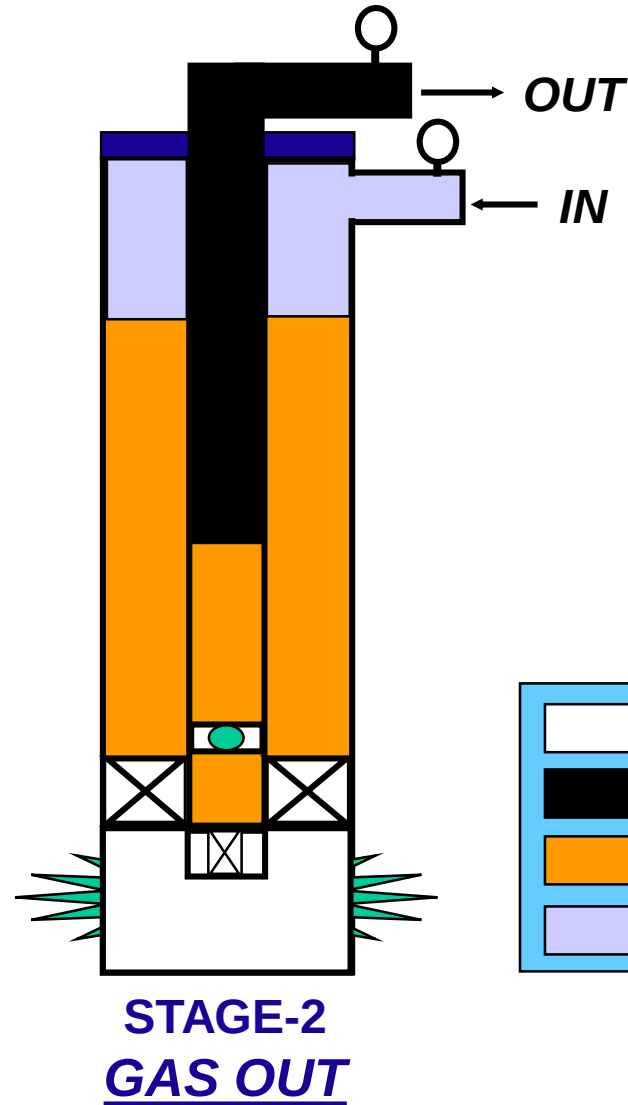
$$= 3600 - (80 + 1520)$$

$$= 2000 \text{ psi}$$

Initial Casing Pressure

$$= 0 \text{ psi}$$

STAGE-1
INITIAL STAGE



	GAS
	OIL
	PACKER
	KILL

• **Tubing Pressure when gas is out**

$$\begin{aligned}
 &= \text{Formation Pr} - \text{Tubing Hyd. Pr} \\
 &= 6000 \times 0.6 - (4000 \times 0.38 + 2000 \times 0.6) \\
 &= \mathbf{880 \text{ psi}}
 \end{aligned}$$

• **Kill fluid Vol Pumped in = GAS Vol out**

$$\begin{aligned}
 &= 2000 \times 0.01458 \\
 &= \mathbf{29.16 \text{ Bbls}}
 \end{aligned}$$

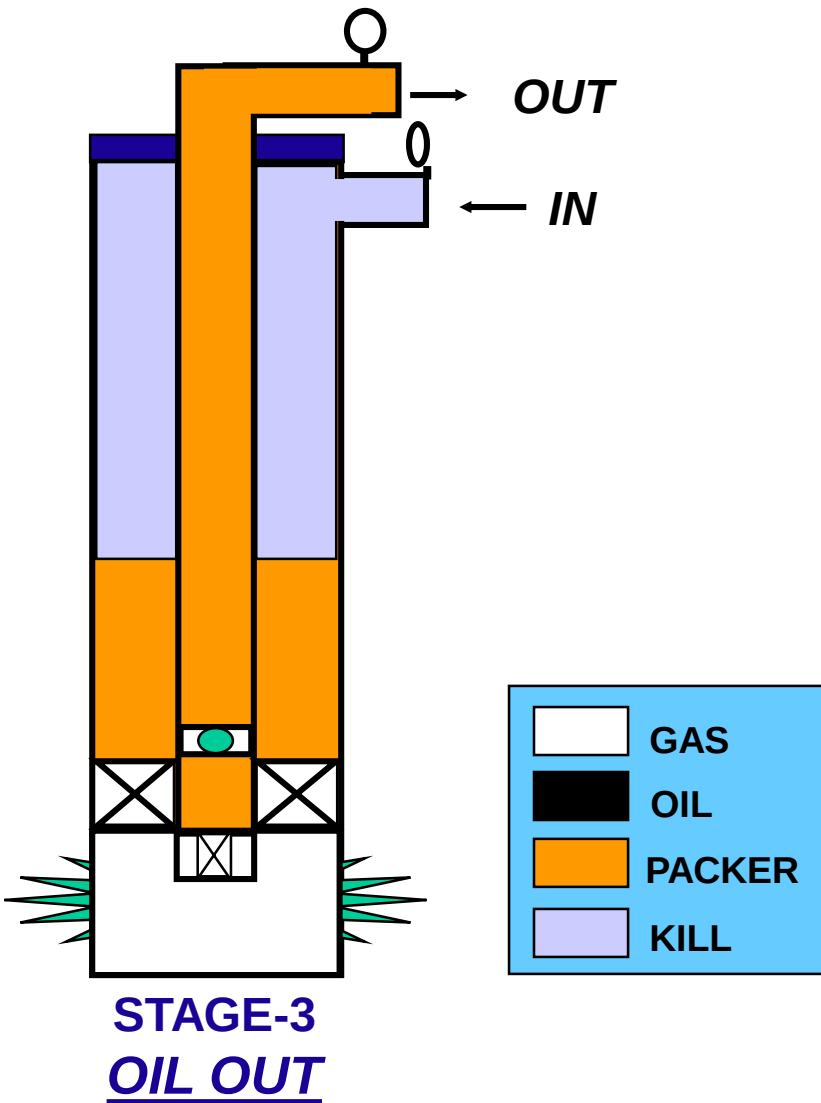
• **Kill fluid height = Kill fluid vol / Ann vol**

$$\begin{aligned}
 &= 29.16 / .0252 = \mathbf{1157 \text{ ft}}
 \end{aligned}$$

• **Completion fluid ht = 6000 – 1157 = 4843 ft**

• **Casing Pressure when gas is out**

$$\begin{aligned}
 &= \text{Formation Pr} - \text{Casing Hyd. Pr} \\
 &= 3600 - (1157 \times 0.5 + 4843 \times 0.6) \\
 &= \mathbf{115.7 \text{ psi}}
 \end{aligned}$$



• **Tubing Pr when brine at surface**

= Formation Pr – Tubing Hyd. Pr

= 6000x.6 – 6000x.6

= **0 psi**

• **Kill fluid Vol Pumped in = Tubing vol**

= 6000x.01458

= **87.48 Bbls**

• **Kill fluid height = Kill fluid vol / Ann vol**

= 87.48 / .0252 = 3471 ft

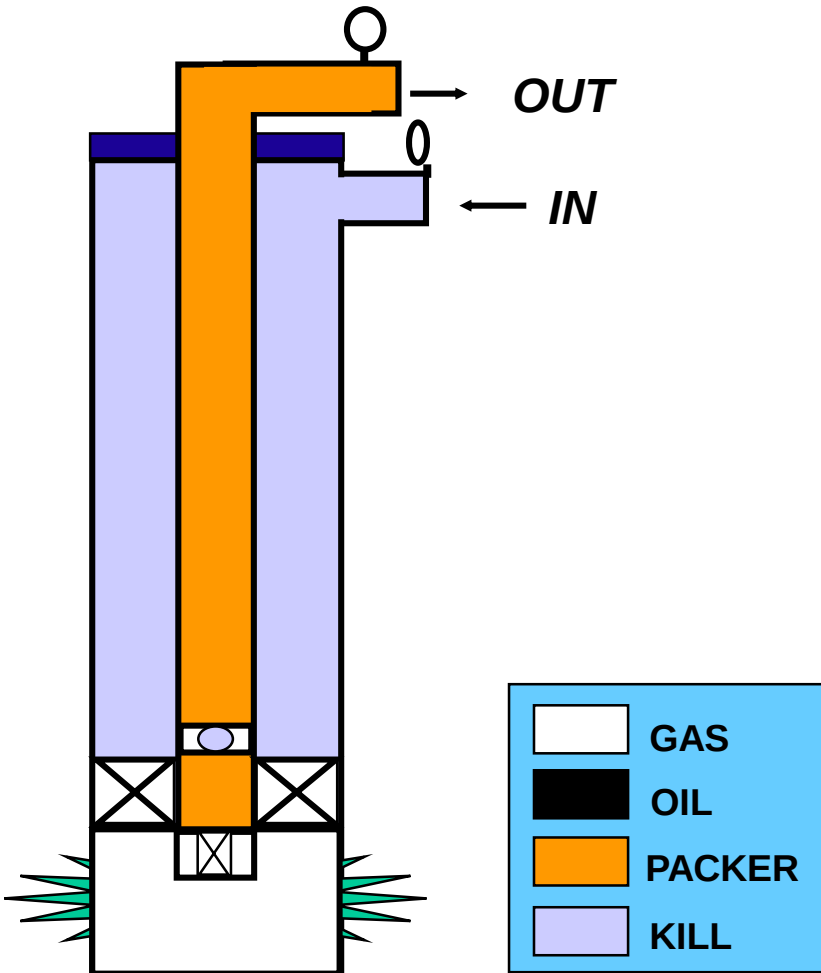
• **Completion fluid ht = 6000 – 3471 = 2529 ft**

• **Casing Pr when brine at surface**

= Formation Pr – Casing Hyd. Pr

= 3600 – (3471x.5 + 2529x.6)

= **347 psi**



STAGE-4

ANN FULL OF KILL FLUID

• **Tubing Pr when kill fluid enters tubing**
 $= \text{Formation Pr} - \text{Tubing Hyd. Pr}$
 $= 6000 \times 6 - 6000 \times 6$
 $= \mathbf{0 \text{ psi}}$

• **Kill Fluid Vol Pumped in = Annulus Vol**
 $= 6000 \times 0.0252$
 $= \mathbf{151.2 \text{ Bbls}}$

• **Casing Pr when kill fluid enters tubing**
 $= \text{Formation Pr} - \text{Casing Hyd. Pr}$
 $= 3600 - 6000 \times 5$
 $= \mathbf{600 \text{ psi}}$

Q. NO. 2

The well in the schematic is vertical and has a constant ID/OD. Friction pressure losses and any fluid losses to the formation are ignored. An overbalance of 200 psi is held over formation pressure at all times during the well kill.

Using the following data and reverse circulation kill graph, answer the questions below.

- **Tubing Data :**

Size 3 ½" in

Weight 10.3 lbs/ft.

Capacity = 0.0083 bbl/ft

- **Casing Data:**

Size 7 in

Weight 29 lbs/ft

Annulus Capacity = 0.0252 bbl/ft

- **Gradients**

Brine in annulus = 0.49 psi/ft

Kill fluid = 0.52 psi/ft

Gas in tubing = 0.12 psi/ft

Oil in tubing/casing = 0.35 psi/ft

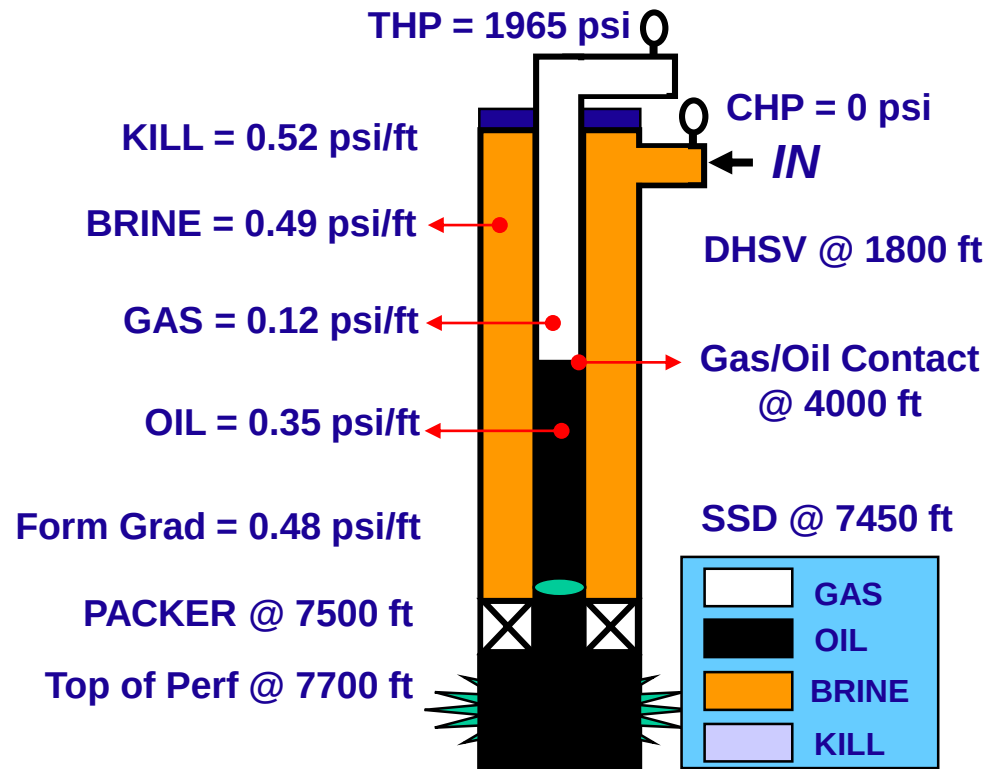
Formation Gradient = 0.48 psi/ft

Pump out put = 0.0899 bbl/stroke

SITHP = 1965 psi

Transition from gas to oil = 4000 ft

Sliding Side Door depth = 7450 ft



SICHP = 0 psi

Packer depth = 7500 ft

Top of perforation = 7700 ft

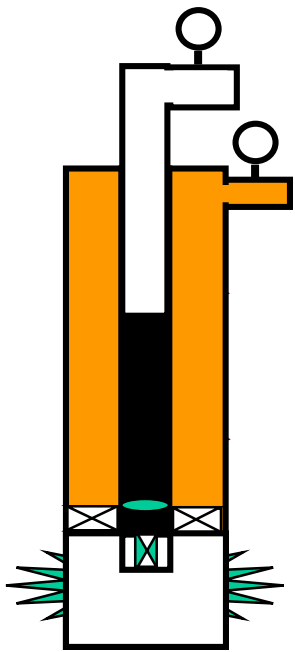


FIGURE (A)
INITIAL STAGE

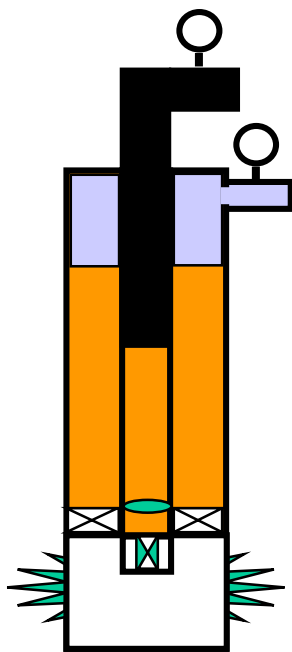


FIGURE (B)
GAS OUT

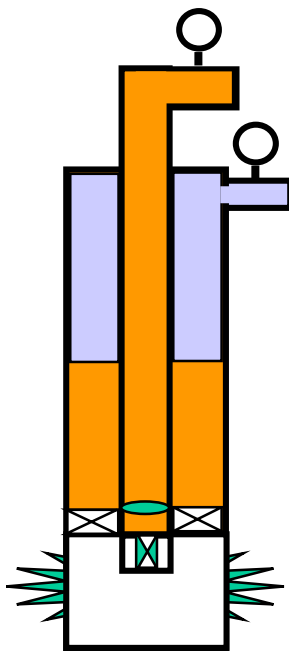


FIGURE (C)
OIL OUT

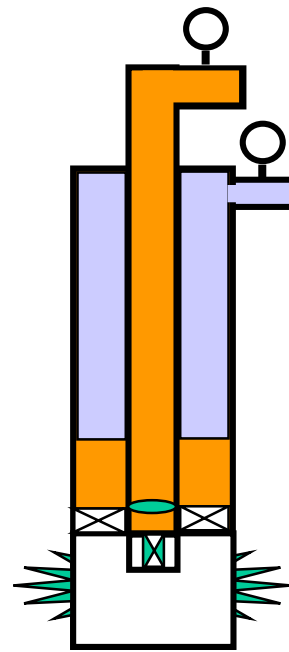


FIGURE (D)
KILL FLUID HEAVIER

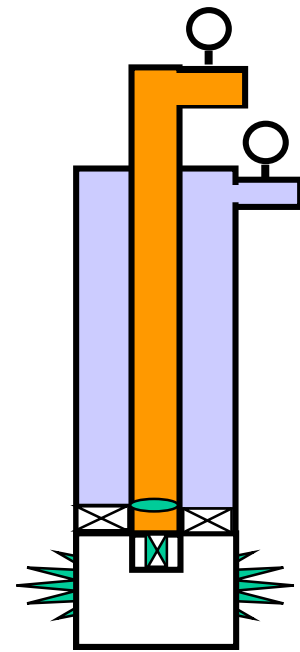
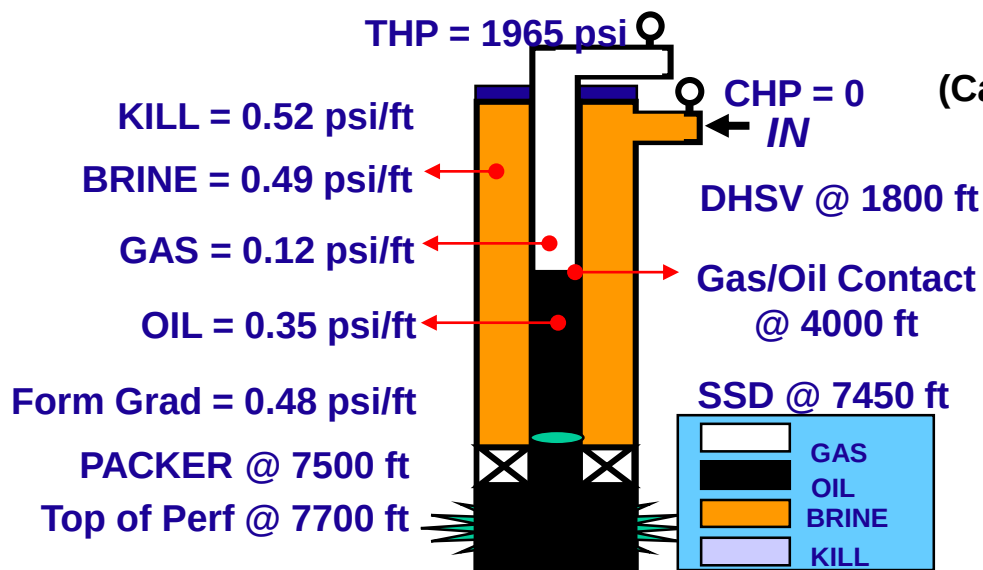


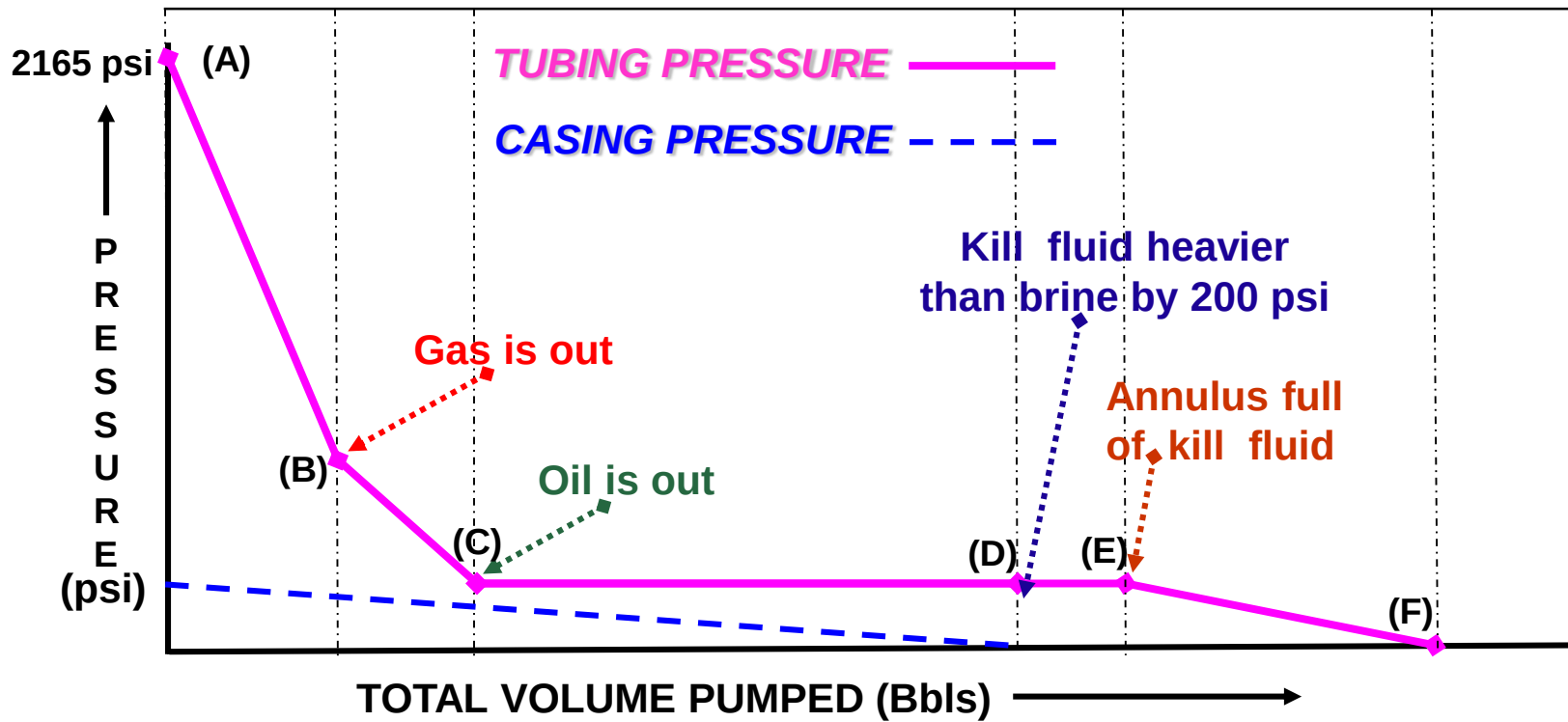
FIGURE (E)
ANNULUS FULL
OF KILL FLUID



THAN BRINE BY 200 psi
(Casing pr down from 200 to 0)

200 psi overbalance is held

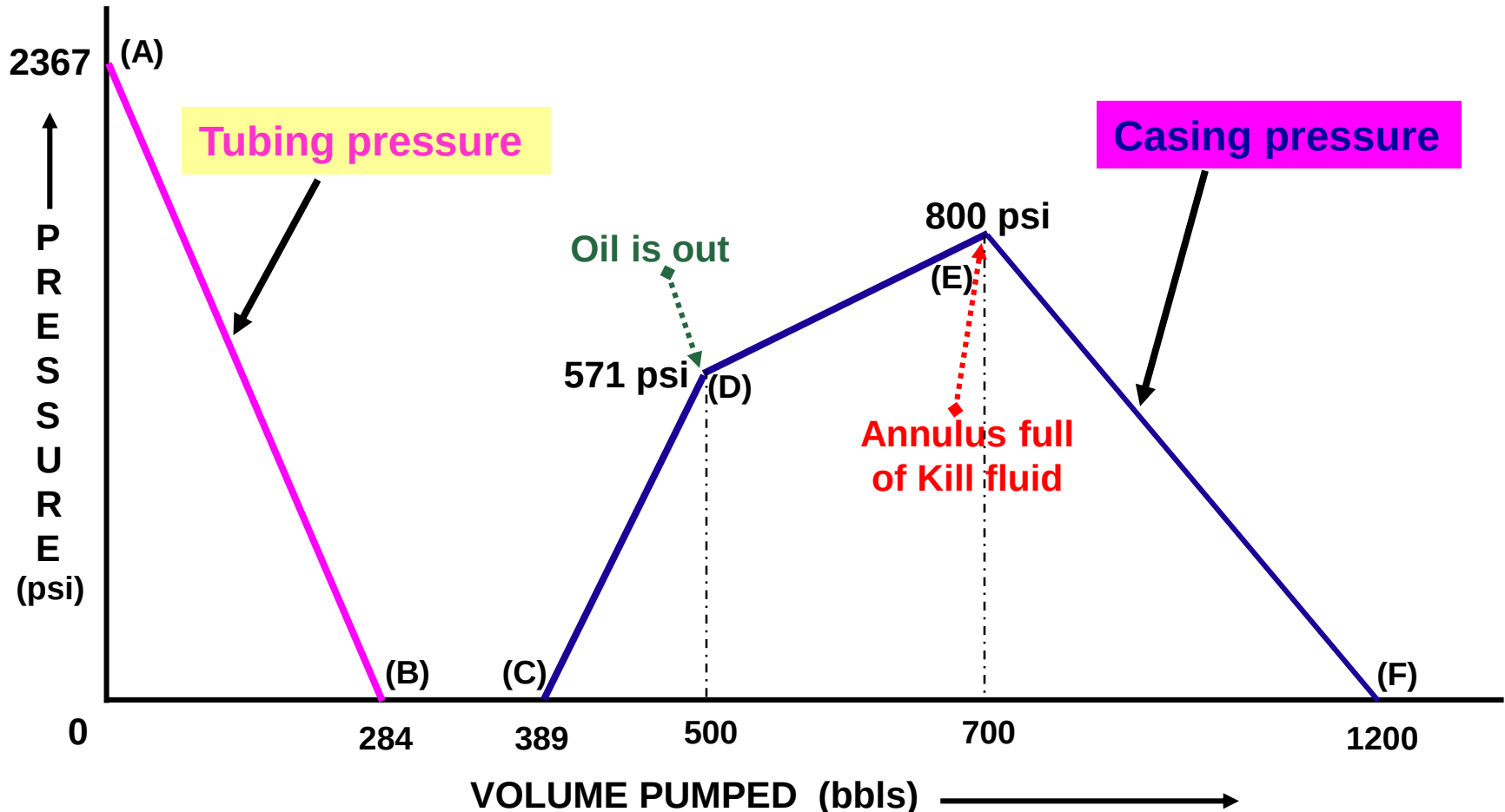
Kill fluid is heavier than brine



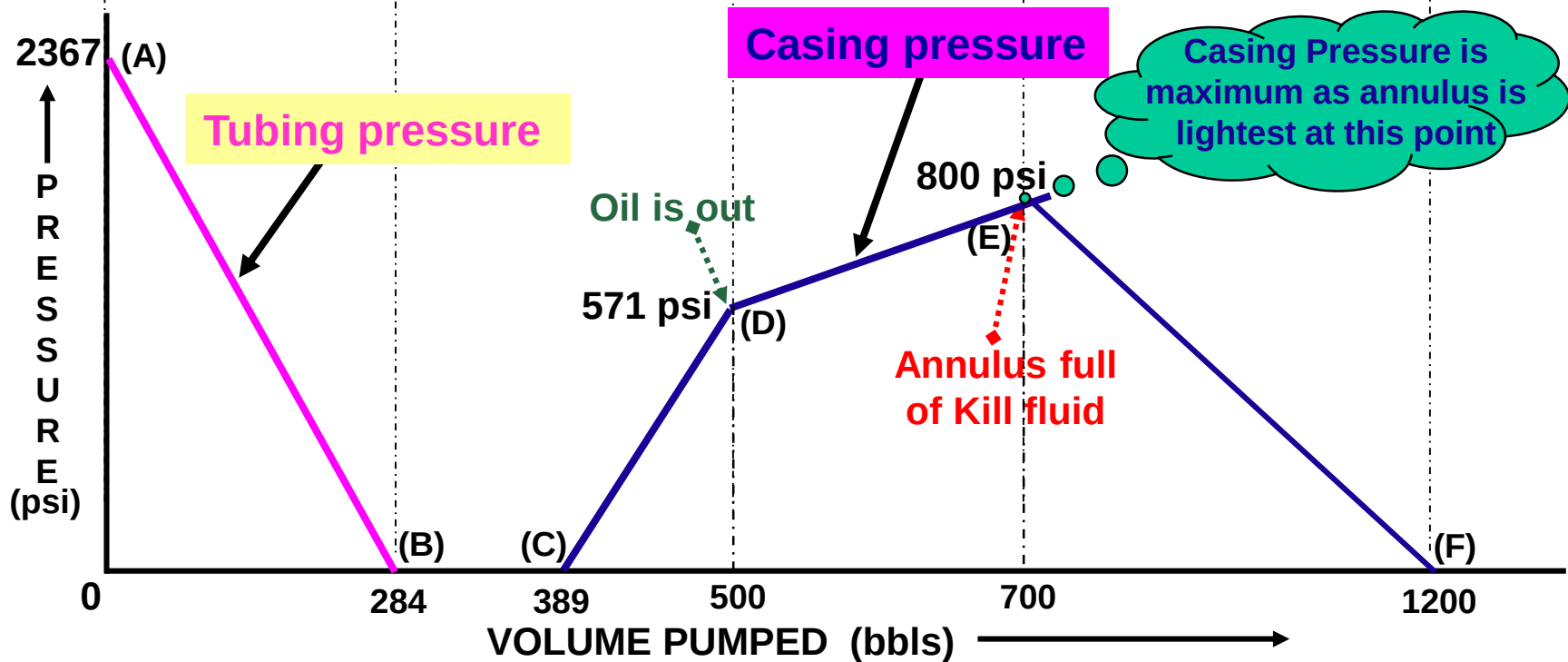
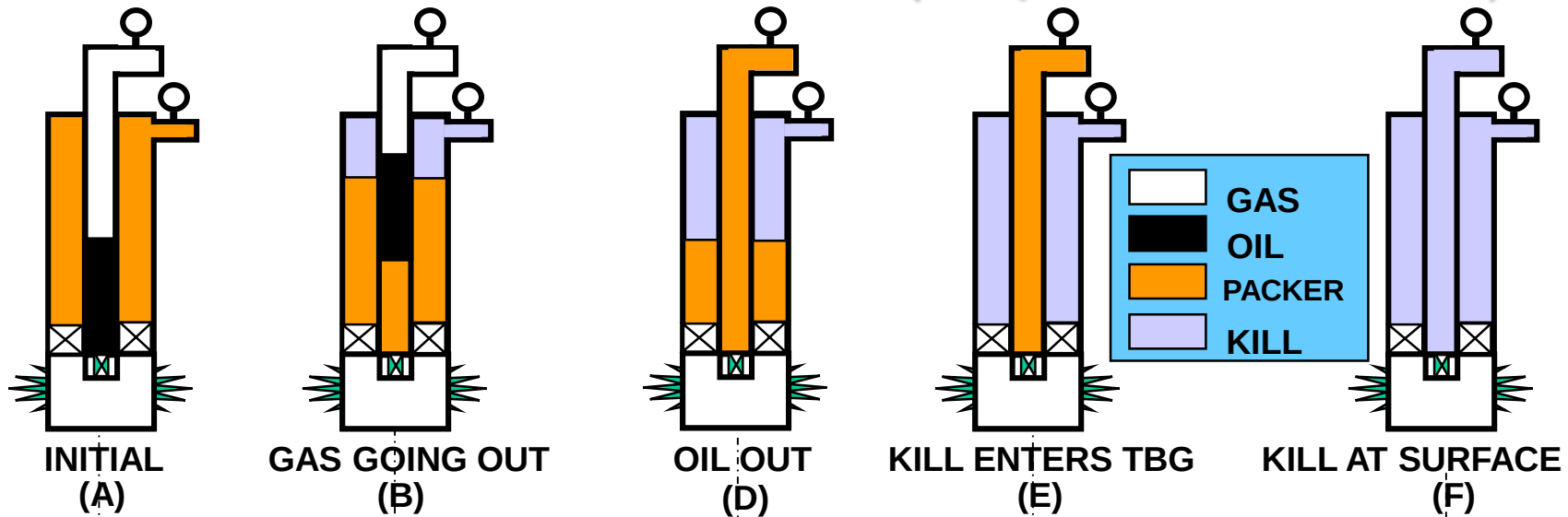
Position	A	B	C	D	E	F
Volume pumped (bbls)	0	33	62	170	188	250
THP (psi)	2165	685	200	200	200	0
CHP (psi)	200	162	128	0	0	0

Q. NO. 3

Figure below shows a reverse circulation kill graph that has been generated for a constant ID/OD tubing containing gas and oil with a heavy completion fluid in the annulus. The kill fluid is lighter density than the fluid installed in the annulus on completion.



(Completion fluid is heavier)



TUBING PRESSURE CURVE

- A-B** ➡ Completion fluid replaces gas from tubing. Tubing pressure decreases.
- Completion fluids is heavy, tubing pressure becomes 0 at B even before all gas is out of tubing. At B, tubing hydrostatic equals formation pore pressure.
 - Further as completion fluid displaces tubing fluid, tubing pressure continues to remains 0 as after B nothing is left on the tubing gauge to reduce.

CASING PRESSURE CURVE

0-C ➡ Casing pressure remains 0 due to heavy fluid in the annulus.

- C-D** ➡ Casing pressure shows increase due to
- Decrease in annulus hydrostatic pressure due to lighter kill fluid .
 - At D, oil is out of the tubing and completion fluid has reached the surface.

- D-E** ➡ Casing pressure continues to increase but at a lesser rate due to
- Decrease in annulus hydrostatic pressure due to lighter kill fluid
 - At E, completion fluid is completely displaced from annulus to tubing.
 - Tubing is maximum heavy & annulus is lightest. Casing pressure is maximum.

- E-F** ➡ Casing pressure starts to decrease due to
- Decrease in tubing hydrostatic as lighter kill fluid displaces heavier completion fluid.
- .At F, kill fluid reaches surface. Same fluid in tubing & annulus, casing pressure = 0.

LUBRICATE & BLEED

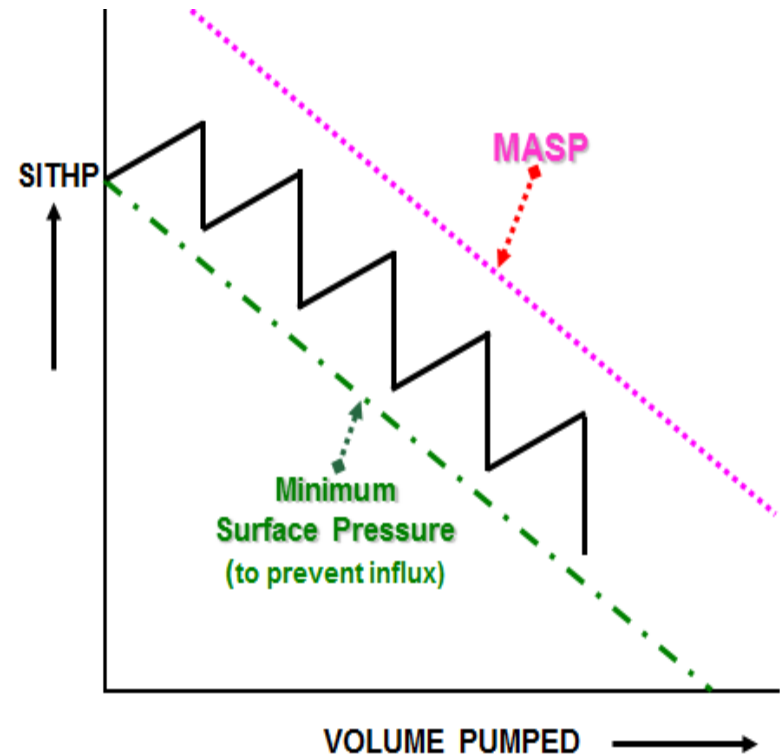
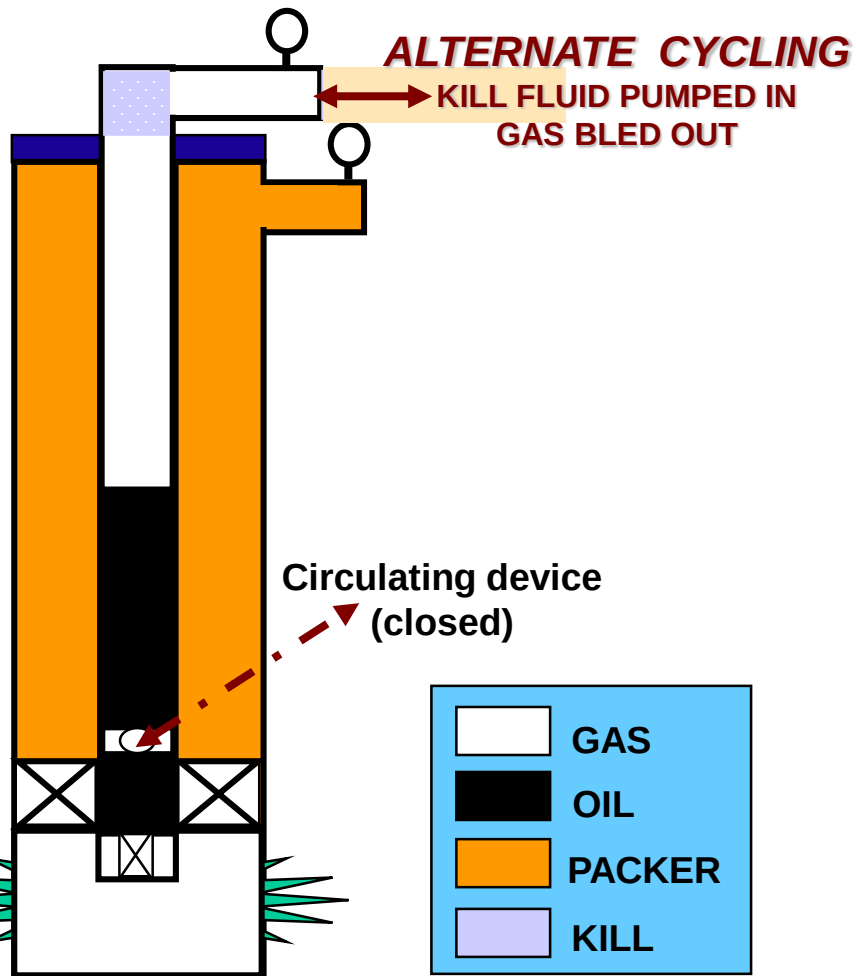
ADVANTAGES:

- Only method which can be employed under extreme circumstances

DISADVANTAGES:

- Very slow

LUBRICATE & BLEED



- ## Bleed-off/lubrication

- ✓ Bleed-off/lubrication is a method that can be used when:
 - circulation is impossible (we cannot enter the well as a result of a restriction)
 - Bullheading cannot be employed as it would result in excessive topside pressures.
- ✓ Lubrication & Bleeding involves pumping kill fluid into tubing, allowing it to lubricate down and bleeding gas out through choke.
- ✓ The process is repeated several times till gas is out and tubing pressure becomes 0
- ✓ Tubing hydrostatic pressure is increased due to pumping kill fluid and then BHP is maintained by reducing the tubing pressure through choke.
- ✓ Since the gas necessarily has to migrate through each injection of kill fluid sufficient time must be allowed for this to happen before we bleed off again and inject a new “pill” of kill fluid.
- ✓ Killing a well by the lubrication method may take several days.

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- Surface Problems

- 1. **Bleeding off gas (Joule-Thomson Effect)**

- ✓ When we have flow through a pipe or a valve we may find that the temperature of the flowing medium is lower than the ambient temperature. The difference between the ambient temperature and the temperature of the flow is known as the Joule-Thomson Effect.
- ✓ When gas pressure falls (through a pipeline or across a valve) the gas expands and loses energy. If the expansion of the gas takes place with constant enthalpy (H) where;

$$\mathbf{H = U \text{ (internal energy) + PV (pressure x volume) = constant}}$$

some of the energy will manifest itself as a reduction in temperature due to the lower pressure.

- ✓ Normally, temperature falls by about half a degree Celsius per bar of pressure drop, i.e. 100 bar pressure drop will produce a 50 degree fall in temperature (23 F/ 10 psi)
- ✓ Bleeding off gas at low temperature, with a large pressure drop and low outlet pressure may result in brittle fracture in pipelines and other equipment. Stainless steel may suffer from brittle fracture at temperatures as “high” as -20 C.

- # Surface Problems

- 2.

Blockages in the well

- 2.1

Blockage mechanisms in wells

- Formation sand (from sand-producing wells)
 - Scale
 - Mechanical blockages (collapsed casing, jammed valves & downhole strings)

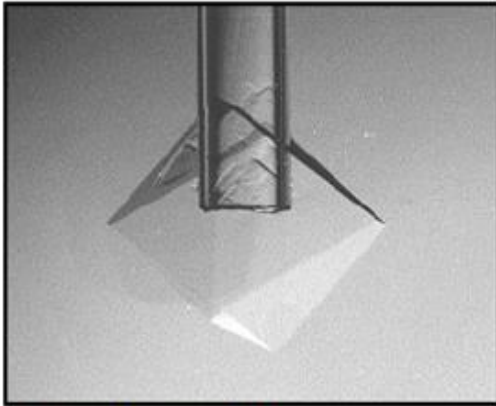
- 2.2

Removing blockages

- There are normally two ways of doing this: By mechanical methods or By means of chemicals
 - Mechanical methods require the use of intervention equipment such as wireline, coiled tubing (jarring or milling).
 - Chemical methods involve pumping chemicals down to the region of the restriction. We allow the chemicals to react for a while, and produce them by the same route up again. In order to prevent the process from becoming unstable, it may be necessary to add more chemicals at the surface before the original chemicals enter the process.

The flow rate from a naturally flowing production well has been slowly and steadily reducing. It has been checked that the tree valves and the DHSV are all fully open. It is though that there may be a blockage forming in the well bore. What is the first correct action to take?

- Hydrate Formation



Single crystal



From natural gas



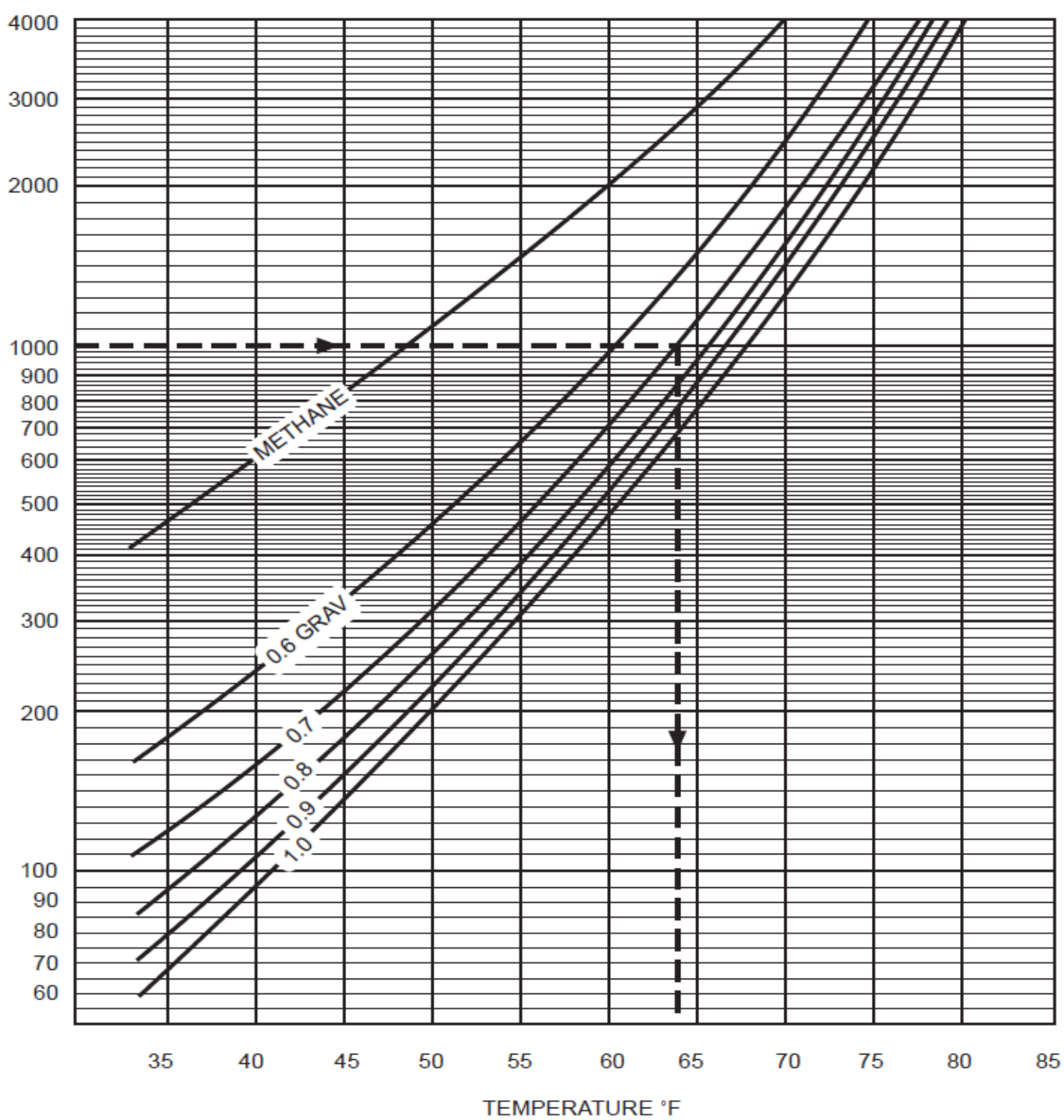
From black oil

• Hydrate Formation

➤ **Conditions under which hydrates form**

1. Presence of free water
 2. Presence of light gas molecules
 3. Relatively high pressure
 4. Relatively low temperature.
-
- ✓ Hydrates consist of water in a crystal lattice structure mixed with light gas components.
 - ✓ Hydrates do not require a pressure drop to form. However, the refrigeration effect from a small pressure drop, such as a stuffing box leak, may be sufficient to produce optimum pressure and temperature conditions for hydrate formation.
 - ✓ Hydrates can form under flowing or static conditions.
 - ✓ The first indication of them forming in the tubing or annular flow string is a drop in flowing wellhead pressure followed by an initially slow then progressively rapid drop in wellhead flowing temperature.

PRESSURE FOR HYDRATE FORMATION PSIA



• Hydrate Formation

➤ Methods of removing hydrates

1. Pressure reduction (lower pressure)

- ✓ In a well in which a hydrate plug has formed, reducing pressure can be risky. If the hydrate plug slips and there is gas between the plug and the surface system, the plug may shoot through the Xmas tree topside and cause a blowout.
- ✓ If we have hydrates in a pipeline or vessel, the pressure must be reduced by an equal amount on both sides of the plug in order to prevent it from moving as it melts.

2. Temperature rise

- ✓ We raise the temperature while maintaining constant pressure. The heat will break up the lattice structure in the plug.

3. Use of inhibitor

- ✓ Methanol, glycol and salt can all dissolve hydrates. When these substances come into contact with hydrates, the equilibrium curve in the area of contact will change, but at the same time, the inhibitory effect will be used up.
- ✓ If the effect is to be maintained while the hydrates are being dissolved, inhibitor must be added either continuously or batch-wise.

Glycol / Water (% v/v)	Freezing Point (deg C)	SG
100/0	-7	1.115
90/10	-28	1.109
80/20	-43	1.101
70/30	-60	1.091
60/40	-60	1.079
50/50	-44	1.068

Table I.1 - Freezing Points Of Mono-Ethylene Glycol/Water Mixes

- **The 4 things required for hydrate formation are:**

1. Water, methanol, high temperature, low pressure
2. Drill water, barite, diesel, pressure
3. Water, gas, high temperature, low pressure
4. Gas, water, high pressure, low temperature

- **True or False?**

1. Hydrates are likely to form at emergency blow-down lines.
2. Hydrates are likely to form at pressure relief valves.
3. Hydrates can only form in the presence of free (liquid) water.
4. Hydrates can damage well bore and intervention equipment.
5. Hydrates may cause damage if they become free with a high differential pressure.
6. Injecting brine into the flow stream can reduce the formation of hydrates.
7. Temperature below freezing point of water (0 °C) is necessary for hydrates formation?
8. Hydrates are less likely to form if injecting Glycol
9. For the same pressure, hydrates melt at the same temperature as they formed
10. Hydrates are common downstream of chokes
11. Hydrates will form at low rather than high pressures

- **Which of the following measures can prevent or remove hydrates?**
 - a. Rapid bleed-off of gas from the topside system
 - b. Use of water/glycol mixture during pressure testing
 - c. Pressure testing up to close-in pressure
 - d. Raising the temperature of the equipment used
 - e. Injecting methanol into the equipment.
 - f. Pull out of the well and fill the topside equipment with diesel oil
 - g. Check for the presence of external ice in order to locate the hydrates
 - h. Close BOP, bleed pressure above it, open connection above BOP and remove the hydrates
 - i. Try to warm up the hydrates using a high-pressure steamer
 - j. Work the string up and down while you bleed off the surface pressure.

Completion Equipment

Outlines

- I. Blowout Preventers
- II. Completion Equipment

• Blowout Preventers

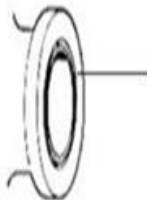
➤ General Blowout Preventer Stack Specifications

- ✓ New API metal rings are to be used each time a flange is assembled or a connection is broken. Flange grooves should be well cleaned and dry.
 - ✓ API RX or BX rings are required. Use of API R rings will not be permitted.
 - ✓ All preventer packing elements and gaskets are visually inspected at the time of installation and rejected if they aren't satisfactory
 - ✓ In operating side valves, the inside valves are considered master valves and would normally never be opened or closed when there is pressure unless the outside valve is closed.
 - ✓ Preventer assemblies will be dismantled periodically between wells to inspect for internal erosion or corrosion and to check flange bolts.
 - ✓ All testing shall be done with clear water.
-

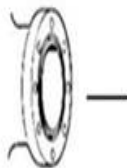
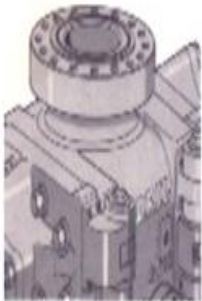
Flanges & Ring Gaskets



Studded connection



Clamped Hub



Flanged connection



Ring gasket



Flange

Ring groove

Ring gasket



Flanges and Sealing Rings

The most common flange types are

- API 6B Rate work pressure 2,000-5,000 psi
- API 6BX Rate work pressure 10,000-2,0000 psi

Flanges type API **6B** Use Sealing Rings API type **R** or API type **RX**

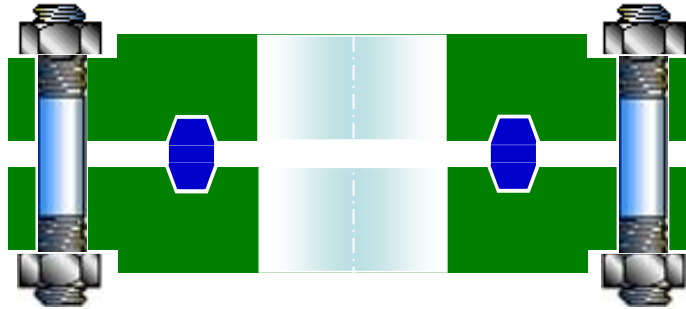
Flanges type API **6BX** Use Sealing Rings API type **BX**



- Type **RX** and **BX** gaskets provide a pressure energized seal but are not interchangeable.
 - It is recommended that a new gasket be used each time the joint is made up.
-

Type R and RX

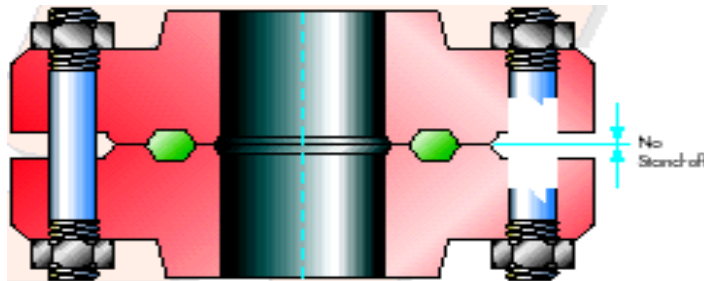
6B Flanges Are of The Ring Joint Type and **Are Not Designed** for Face-to-Face Makeup. The Connection Makeup Bolting **Force Reacts on The Ring Gasket.**



6B Flange

Type BX Rings

6BX Flanges Are Designed for Face-to-Face Makeup. The Connection Makeup Bolting Force Reacts Primarily on the Raised Face of The Flange.



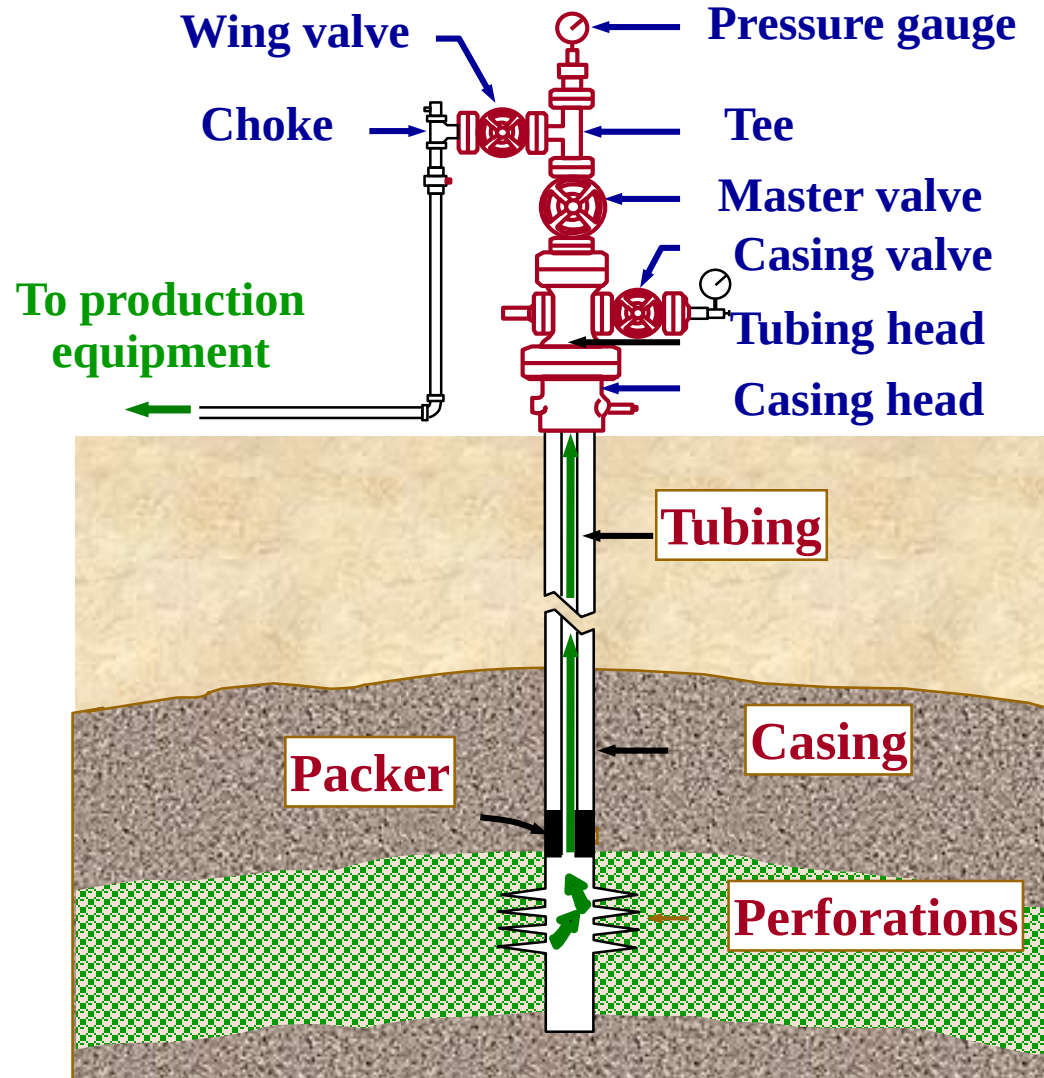
6BX Flange

Well Completion

Outlines

- I. Blowout Preventers
- II. Completion Equipment

Components of a Typical Well



• Completion Equipment

➤ Xmas tree

- ✓ The Xmas tree is an item of safety equipment placed on top of wellhead.
- ✓ The purpose of the X/T is to control the flow of hydrocarbons from the well and to allow access to the well during the operational phase. We call the Xmas tree a safety barrier.
- ✓ The following valves are normally installed on the Xmas tree:

☐ Manual operated master valve

- This is located closest to the wellhead and is the backup valve for the Hydraulic master valve (NOT working valve)
- It is normally impossible to cut wire lines with manually operated valves. The torque required would be so high that parts of the valve would be destroyed before the wireline was cut.

☐ Hydraulic operated master valve

- This is located above the Manual master valve. It is hydraulically actuated (from the platform control room) and is “fail-safe closed”. In the case of a hydraulic leak, the valve will revert to its closed condition

☐ Choke Valve

- Restrict, control or regulate flow of well fluids.
-

• Completion Equipment

➤ Xmas tree

✓ The following valves are normally installed on the Xmas tree:

❑ Hydraulic operated wing valve

- This is the “last” valve in the Xmas tree before the well-flow enters the flow line to the choke valve/process system. This valve is normally “fail-safe closed”.

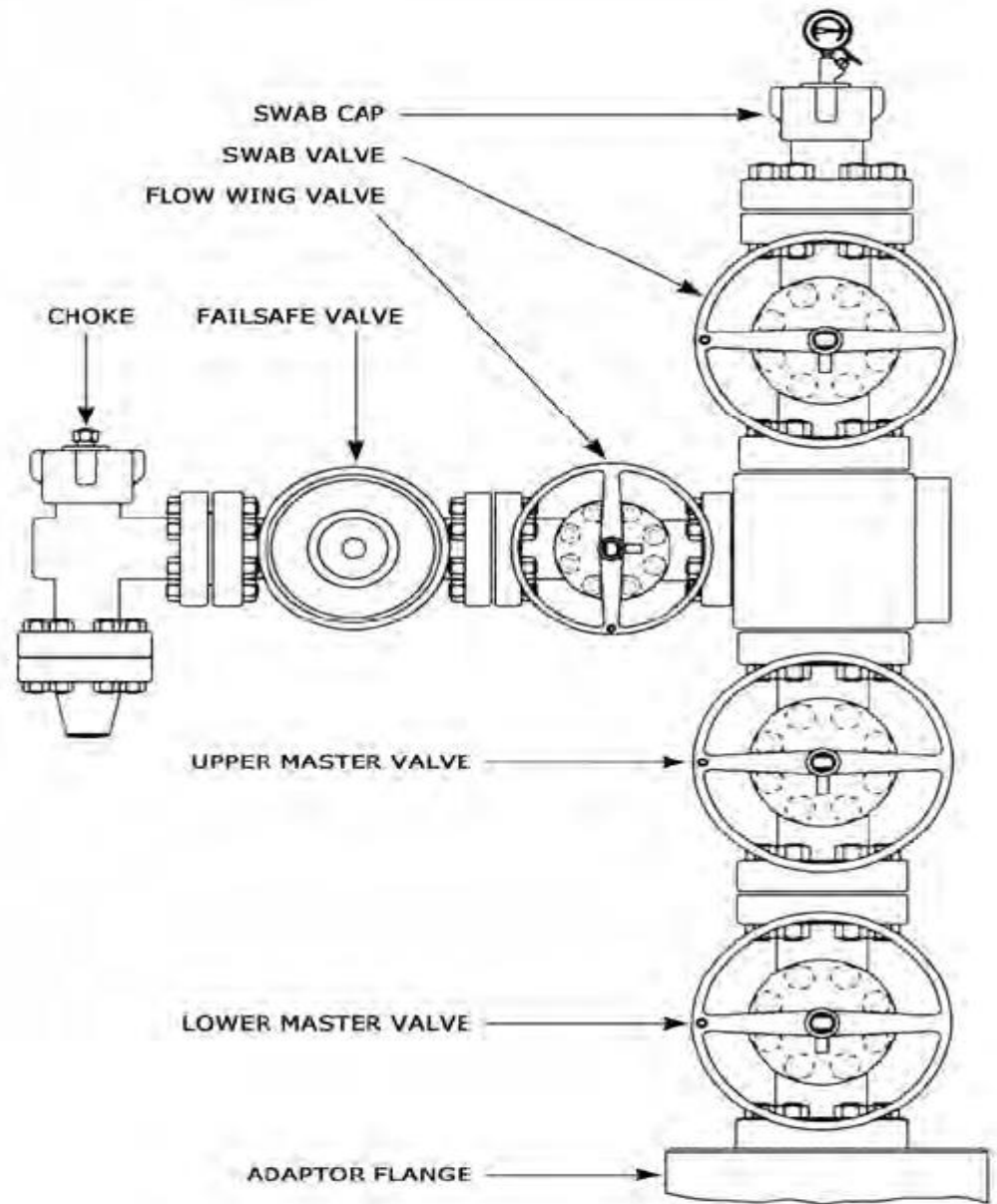
❑ Swab valve/ Crown Valve

- This is a manual valve that stands at the top of the Xmas tree under the “tree cap” (also known as the “lubricator adaptor”).
- This valve must be opened in the case of well intervention.
- On the outside of the swab valve we often find a cup that should be fitted with a needle valve so that any pressure on the other side can be checked using a pressure gauge and bled off trapped pressure before it is opened.

❑ Kill valve

- This valve is located next to the wing valve and is used during pumping and production or injection via neighboring wells. This valve is normally also fitted with a cup and “needle valve”.
-

- This counts as a barrier in most operations (production, pumping, wire line).
- In normal operation the Xmas tree is a primary barrier.



Conventional Xmas Tree

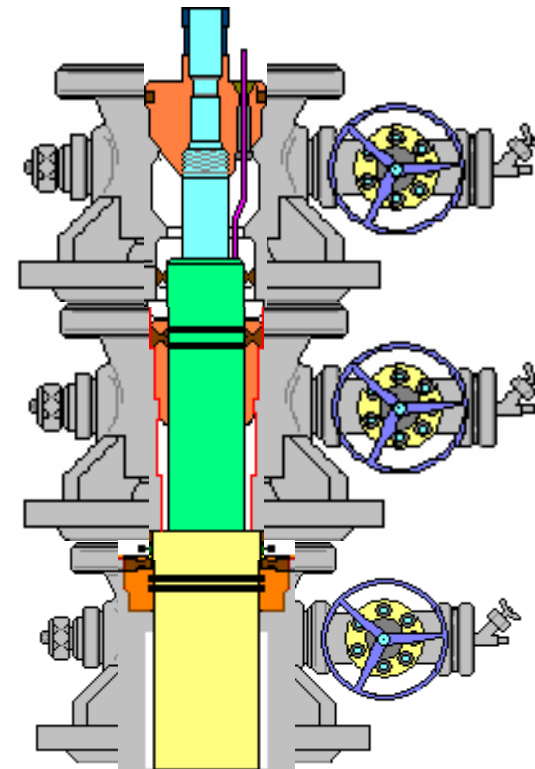
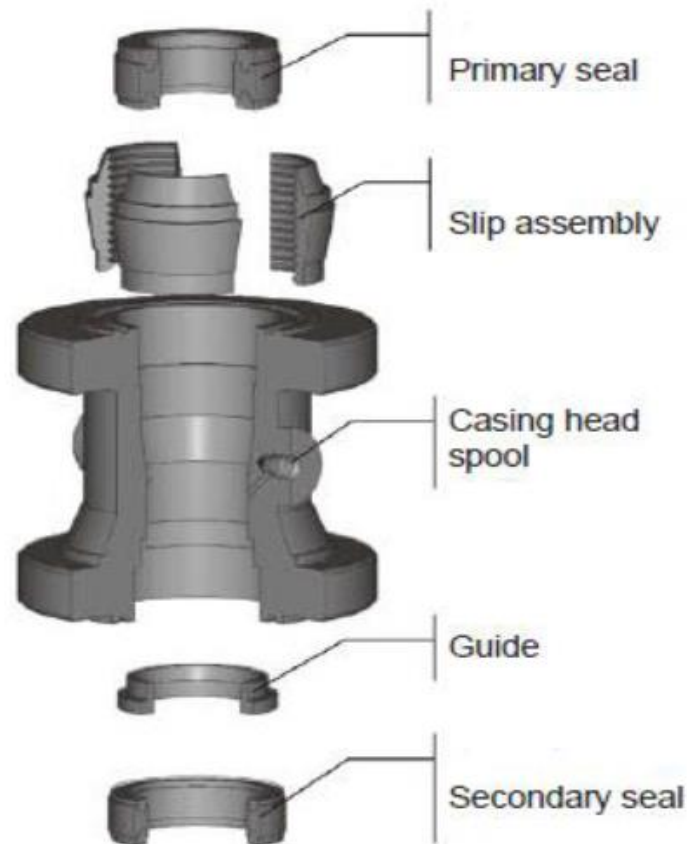
• Completion Equipment

➤ Wellheads

- ✓ The wellhead provides the following functions:
 - Anchoring/connections for each individual casing
 - Pressure isolation of individual casings
 - Connections for Xmas tree
 - Connections for completion string
 - Isolation of production annulus
 - Access to annulus for pressure monitoring and/or pumping.
-

Conventional Wellhead Features:

1. Starter Spool (Conductor Pipe)
2. Surface Casing Head
3. Surface Casing Hanger
4. Production Casing Head
5. Production Casing Hanger
6. Tubing Head
7. Tubing Hanger

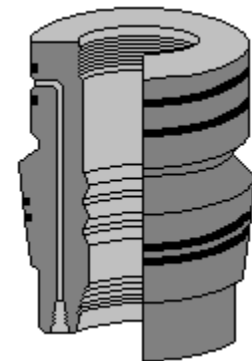
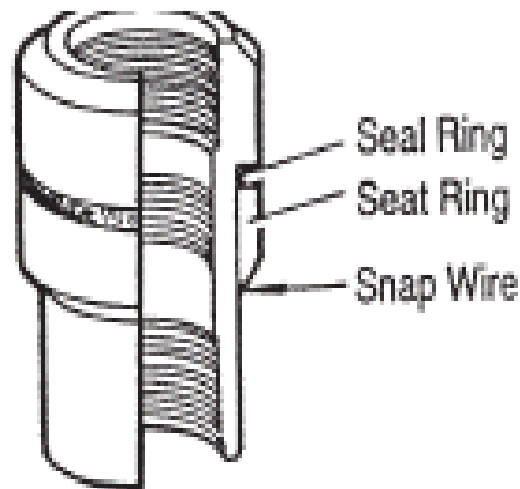


• Completion Equipment

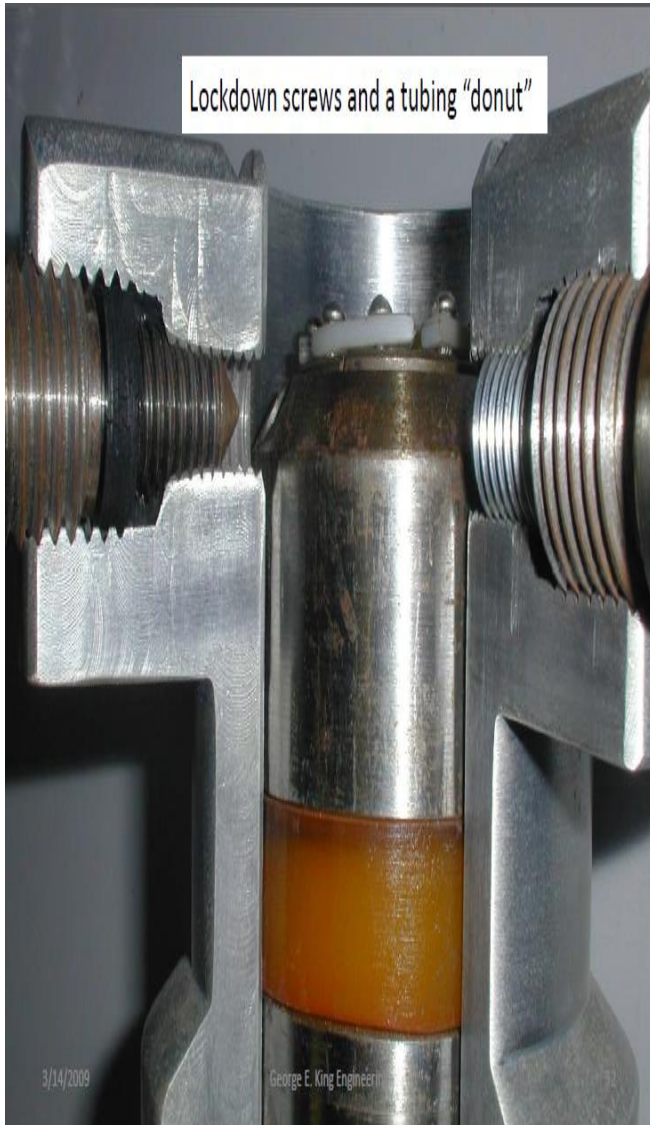
➤ Tubing Hanger

✓ Functions

- It is set in Tubing Head Spool to transfers (support) the weight of the top completion string to the wellhead
- It prevents annulus pressure from leaking past the tubing hanger to X/T or out to atmosphere or sea, or vice-versa).
- It is BPV receptacle as it allows an internal plug to be installed
- It provides a seal around hydraulic and electrical control lines.



Lockdown screws and a tubing "donut"





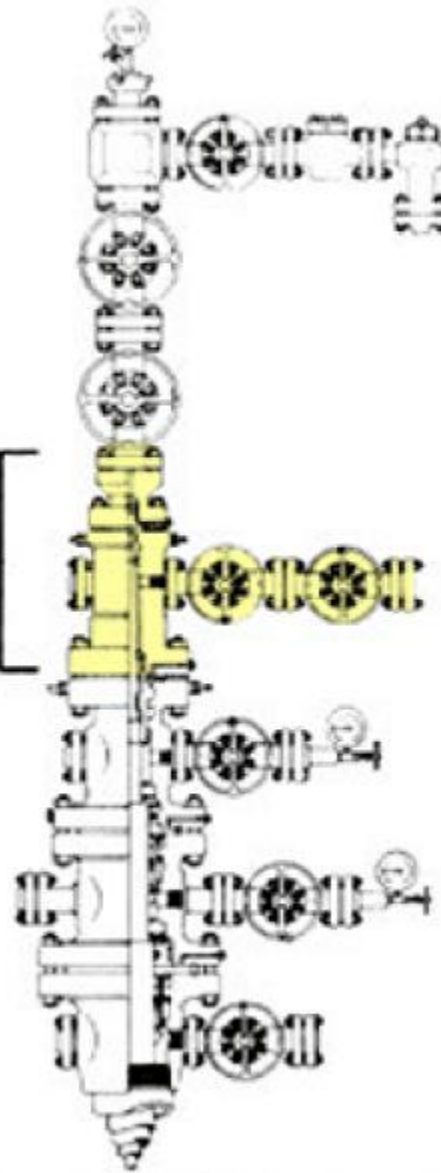
**Tubing Head
Adaptor**

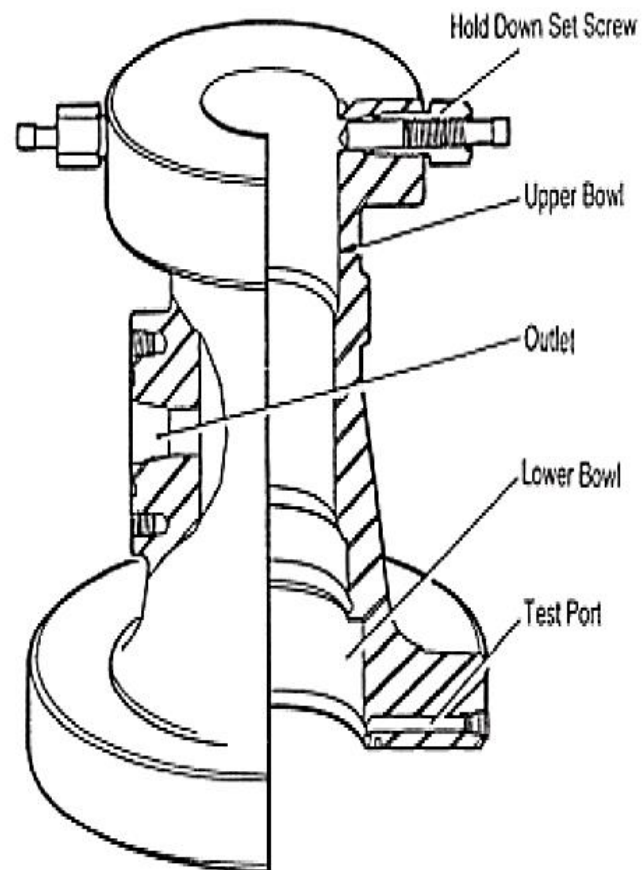


**Tubing
Hanger**



**Tubing
Head**





Well Completion

Well Completion design is dictated by the type of well to be completed.

- 1. Producer**
- 2. Injector**

Completion Functional Requirements

- 1. Provide optimum flowing conditions**
 - 2. Protect casing**
 - 3. Downhole emergency isolation**
 - 4. Permit downhole chemical injection**
 - 5. Enable the well to be killed**
 - 6. Permit routine downhole operations**
-

Completion Accessory

1. Anchoring Device

- Landing Nipples

2. Circulation Device

- SSD
- SPM

3. Isolation Device

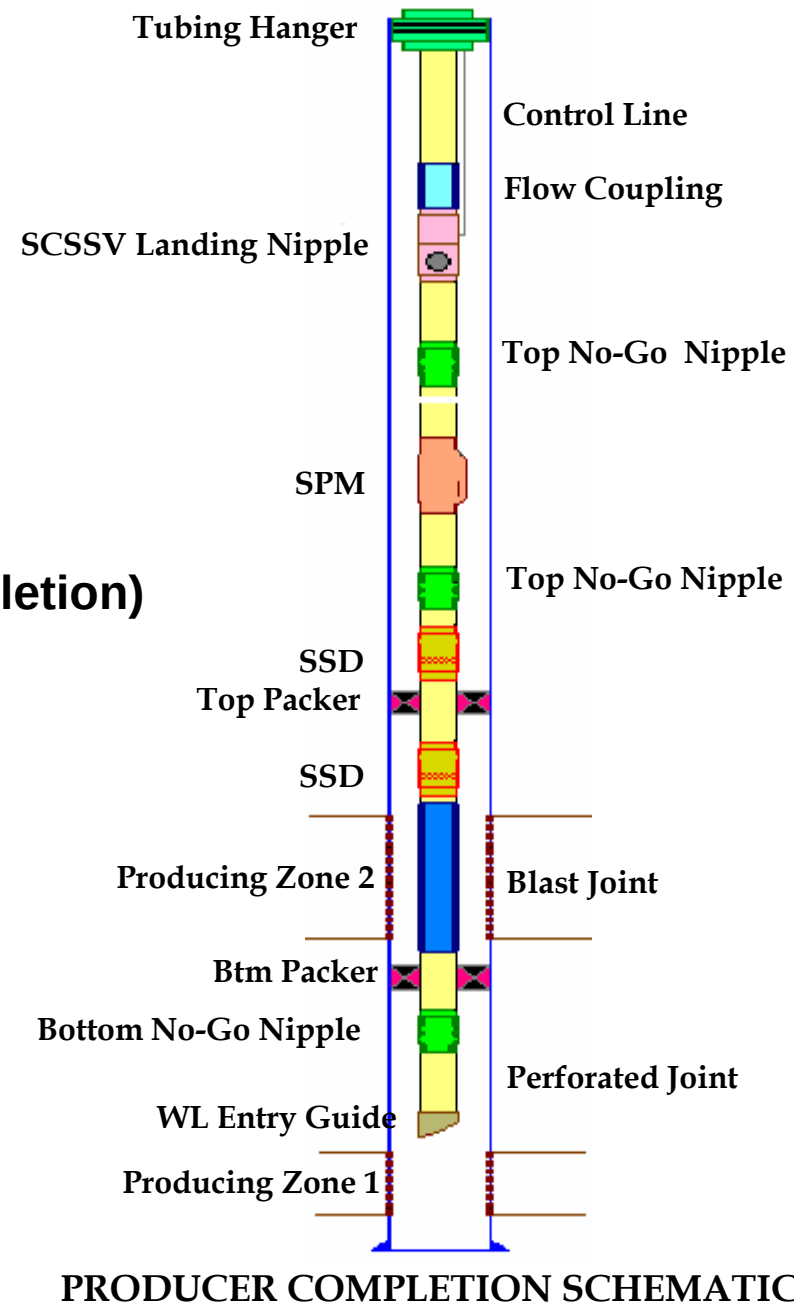
- Packer (between top & bottom completion)
- Tubing Seal Assembly, PBR

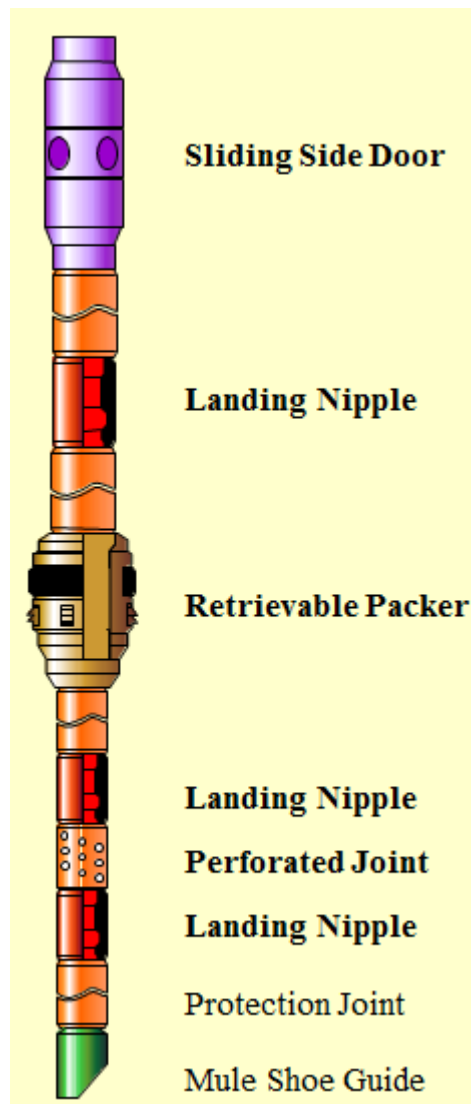
4. Anti-Erosion Device

- Blast Joint
- Flow Coupling

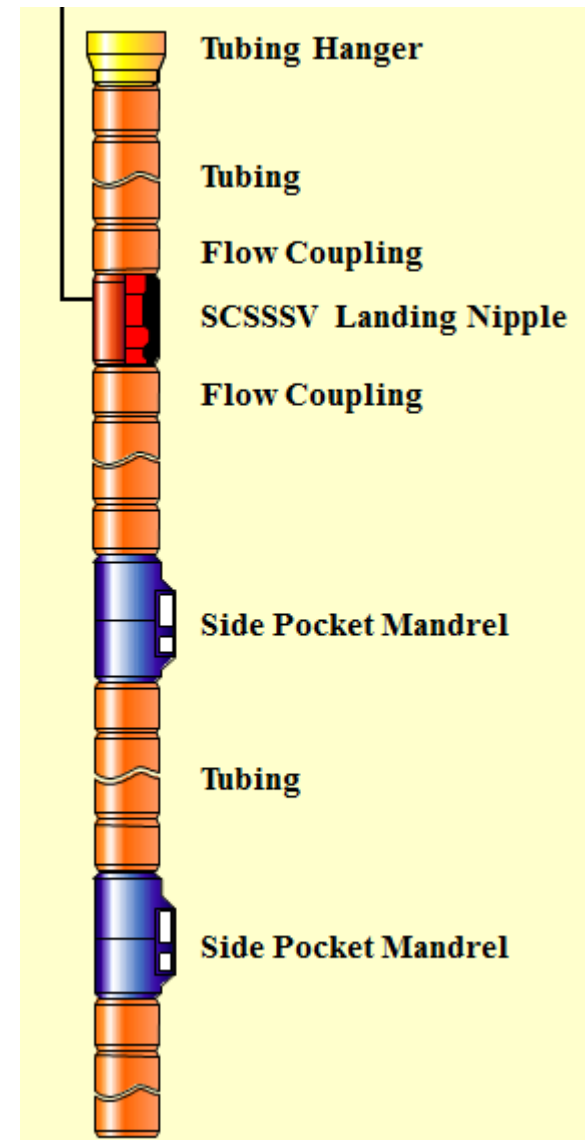
5. Others

- WLEG
- Perforated Joint
- Control Line
- Tubing Hanger





**Lower Completion
Equipment**



**Upper Completion
Equipment**

• Completion Equipment

➤ Down Hole Safety Valve

- ✓ Close the well in an emergency
- ✓ Downhole safety valves (DHSVs) are installed in the well in order to prevent blowouts if:
 - the topside Xmas tree becomes damaged (e.g. by collision with a crane lift)
 - surface equipment is sabotaged
 - a neighboring well suffers a blow-out (in order to prevent a domino effect)
 - drilling collision from neighboring well.
- ✓ They are set below the depth where they can be damaged by surface impact explosion
- ✓ Such valves are of the “Fail-safe Closed” type. If hydraulic pressure to the valve is lost, it will immediately close.
- ✓ These valves can also be pumped through from the surface even when they are closed.
- ✓ Such valves open and close by effect of **pressure**. There exist two **closure mechanisms**:
 1. Flapper valves
 2. Ball-valves.

• Completion Equipment

➤ Down Hole Safety Valve

- ✓ Internationally, the DHSV is not regarded as a well control barrier. This is because API permits a certain amount of leakage through the DHSV 3.6 BPD liquid . Some countries regard DHSV as a barrier after an approved inflow test which verifies that leakage rates are not exceeding 3.6 BPD.
- ✓ They can be divided into two categories :
 1. SSCSSSV (Sub-Surface Controlled Sub-Surface Safety Valve)
 2. SCSSSV (Surface Controlled Sub-Surface Safety Valve)

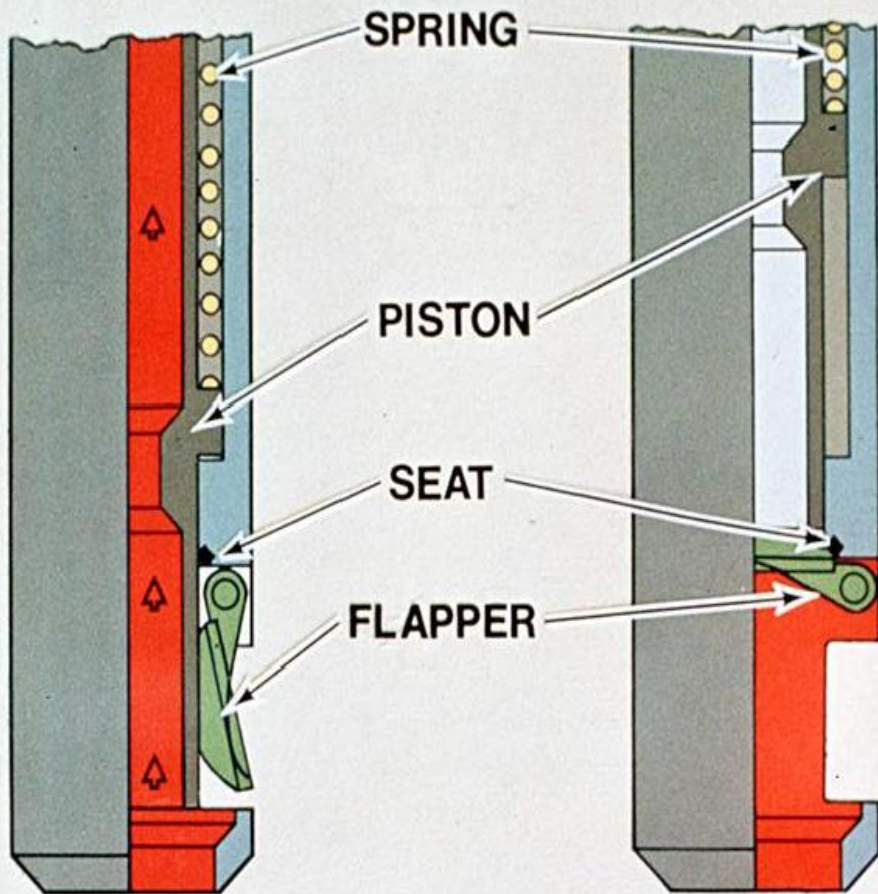
❑ Sub Surface Controlled Sub Surface Safety Valves :

- Sub-surface controlled DHSV re-opened by pressuring up above the valve and overcoming any differential pressure
 - They can be subdivided into two categories :
 1. Differential pressure controlled (velocity / excess flow):

valve is held open by a spring. When the flow rate through the valve becomes so high that the resistance force collapses the spring and the force of the reservoir holds the valve shut.
 2. Ambient pressure controlled (absolute pressure):

Uses a pre charge pressure chamber, the valve closes when flowing pressure drops below pre-charge pressure.
-

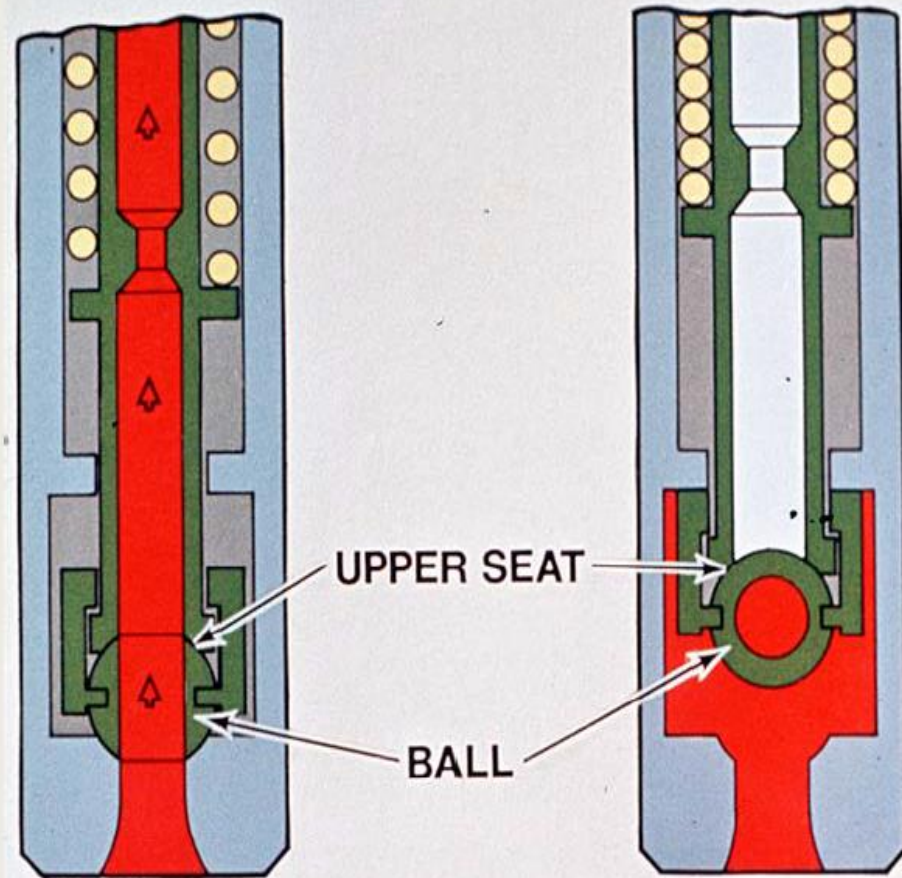
SSC SSSV



OPEN

CLOSED

FLAPPER-TYPE CLOSURE



OPEN

CLOSED

BALL-TYPE CLOSURE

Storm Choke



• Completion Equipment

➤ Down Hole Safety Valves

❑ Surface Controlled Sub Surface Safety Valves :

1. Wireline retrievable DHSVs

- Safety valves of this type are installed in a nipple profile in the completion string.
 - Wireline-retrievable DHSVs often produce a restriction in the well which may result in local erosion as well as choking the well-flow.
 - If intervention in the well is required below the location of the DHSV, this must be retrieved ahead of the intervention.
 - If a problem with a DHSV arises, it is a simple matter to **retrieve it and install a new one**.
 - Run in the open position With a **prong** on the wire line running tool
-

• Completion Equipment

➤ Down Hole Safety Valves

❑ Surface Controlled Sub Surface Safety Valves :

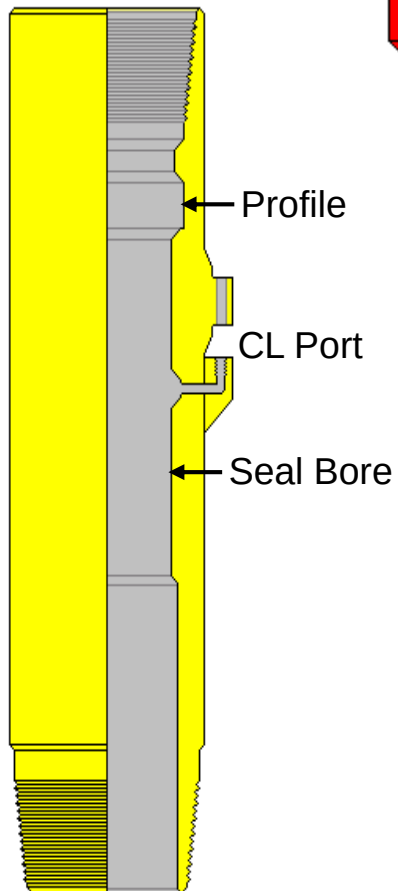
2. Tubing retrievable DHSVs

- It is installed as an integral part of the completion string. In case of problems with such valves the completion must be pulled.
 - Some valves also have a nipple profile for a wireline-retrievable DHSV as a back-up solution in case problems should occur.
 - The advantage of this type is its large ID.
 - It is important that the valve should remain open throughout a well intervention.
 - Some DHSVs are fitted with an integrated pressure-equalization mechanism.
-

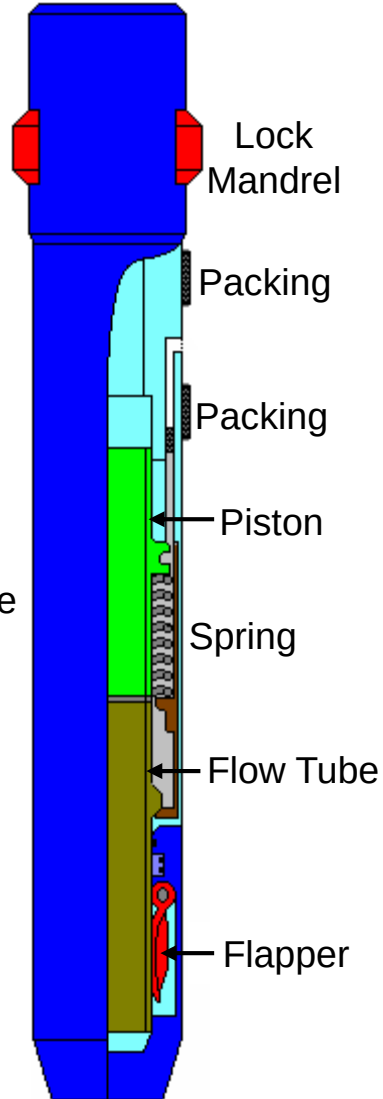
W/L SCSSSV

WRSV Set In SVLN

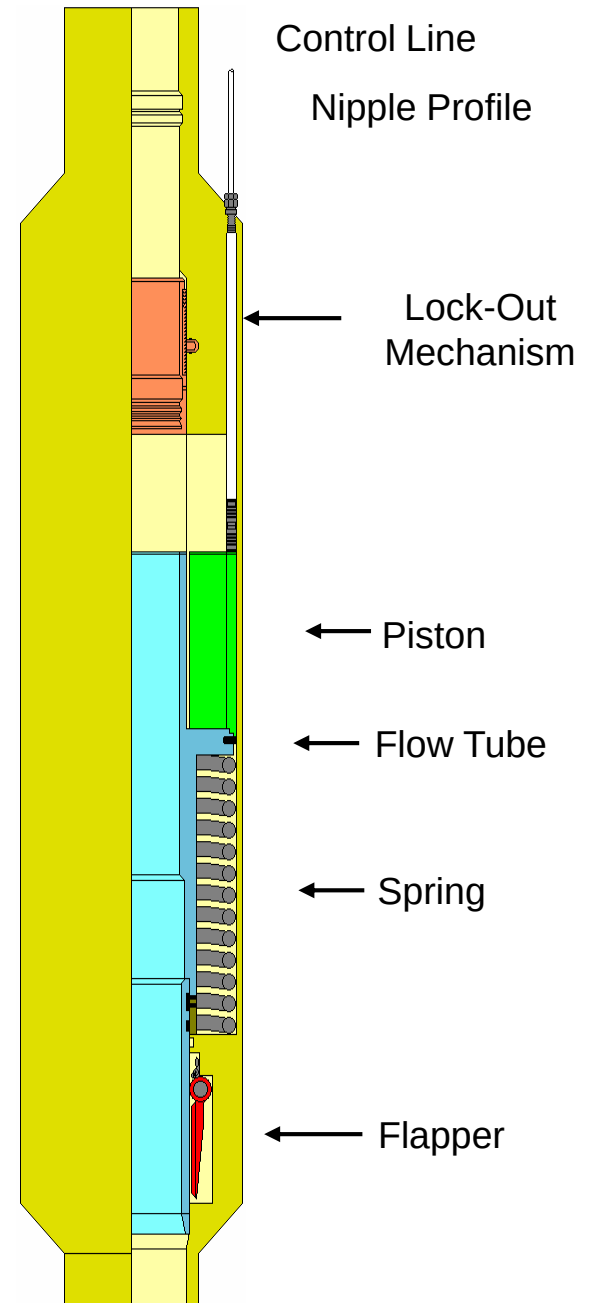
SVLN

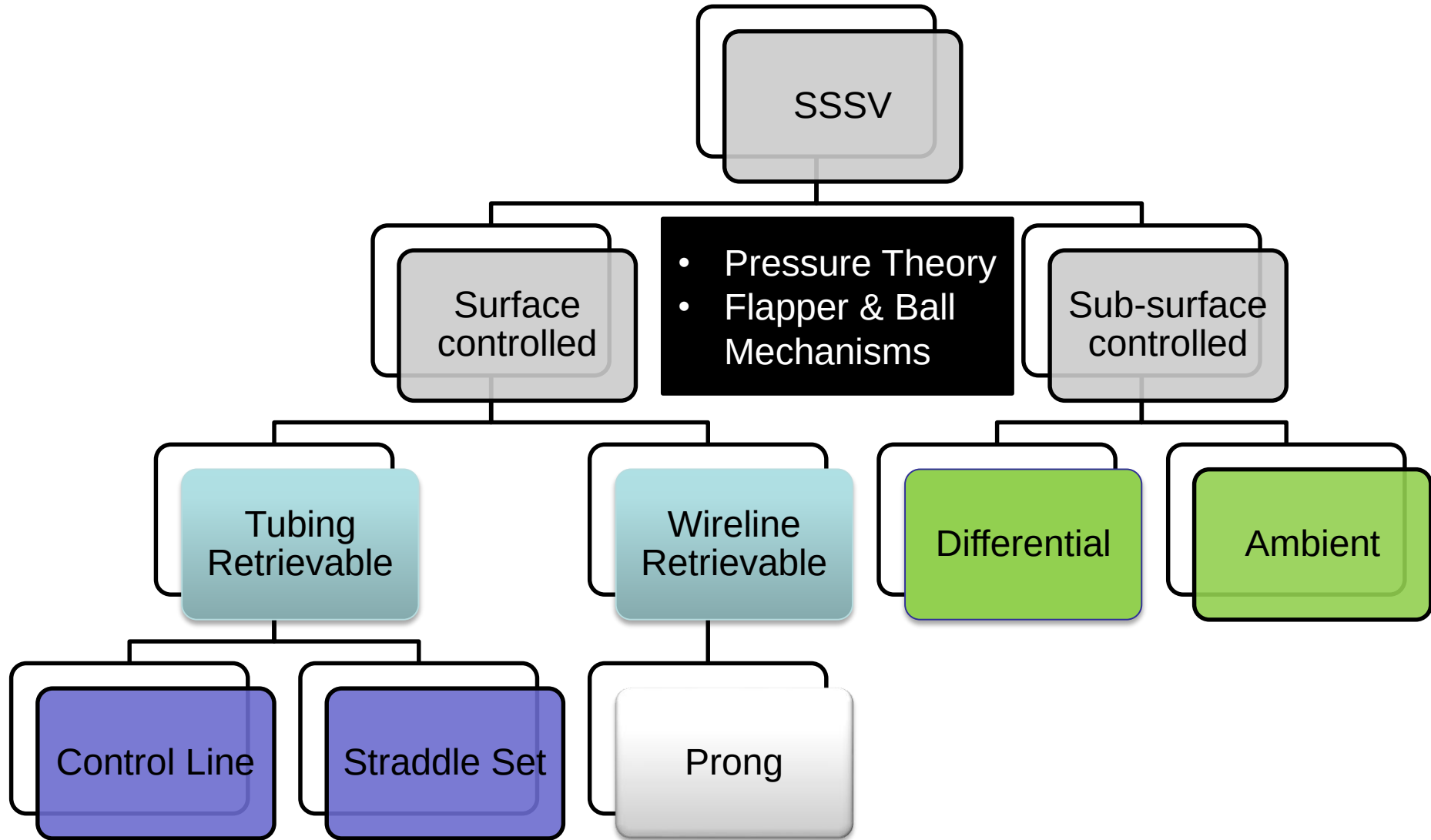


WRSV

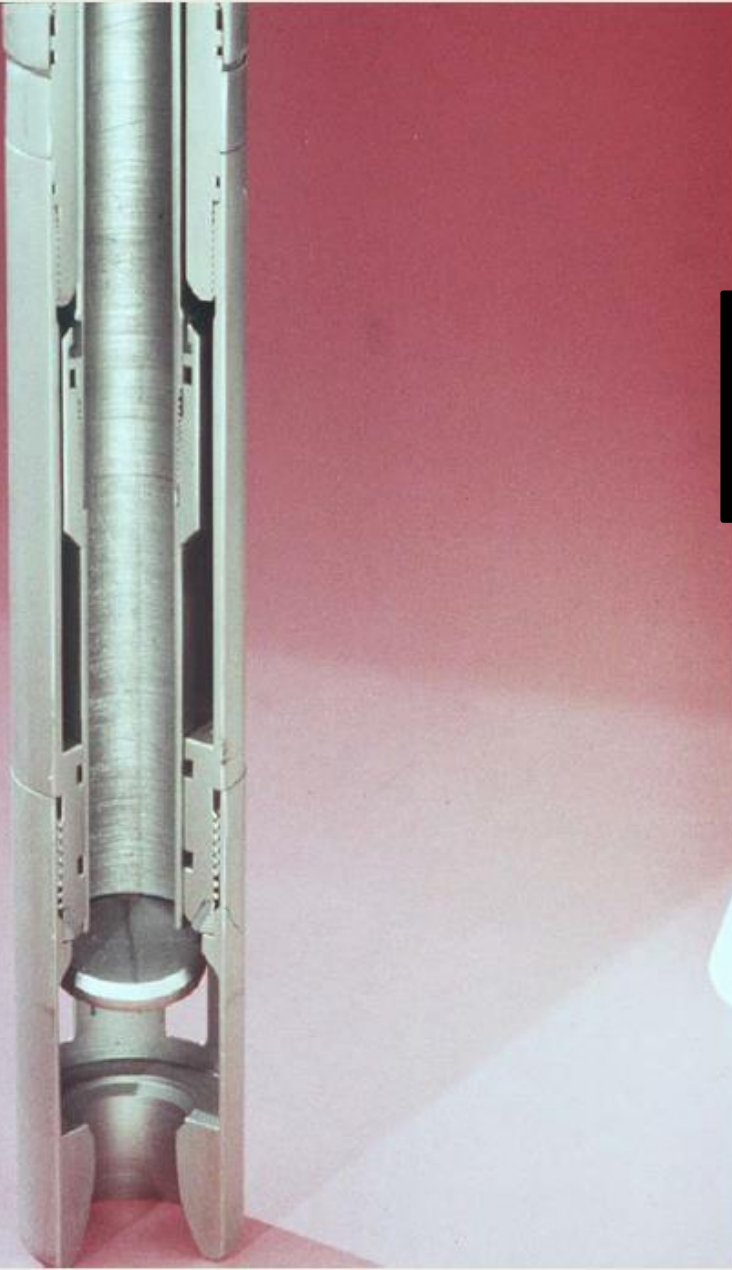


TRSV



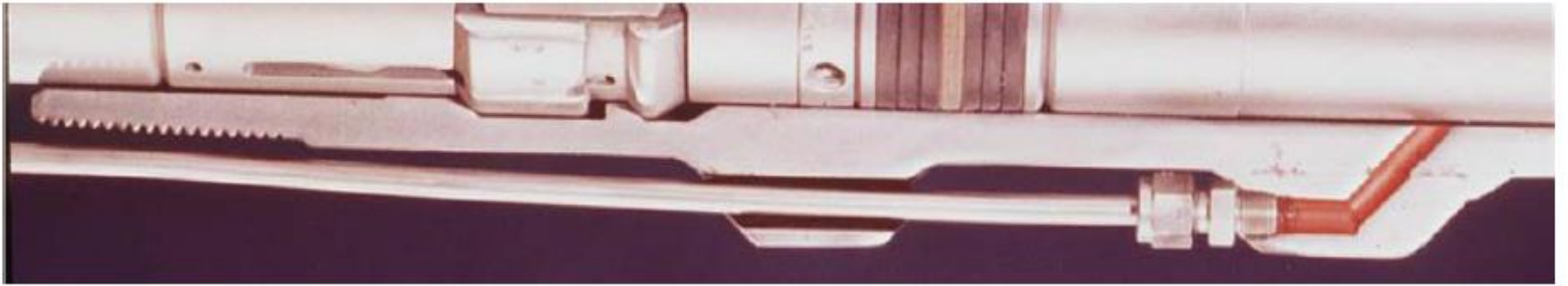


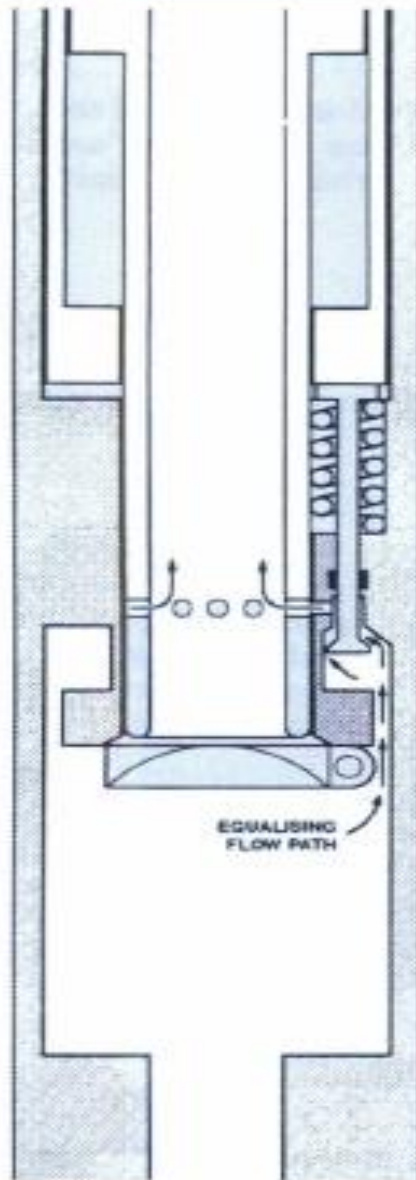
Flapper and Ball Valves



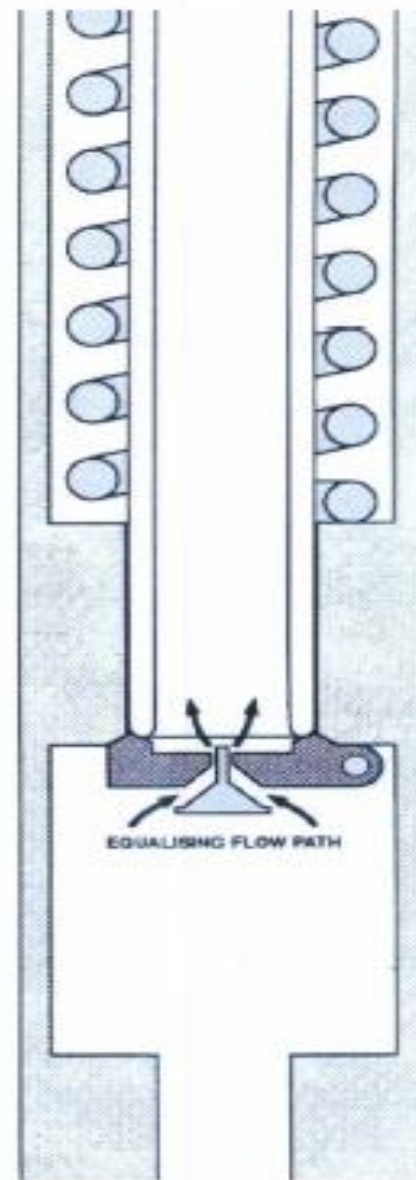
SCSSSV







EQUALISING PATH WITHIN
THE VALVE HOUSING



EQUALISING PATH IN
FLAPPER BODY

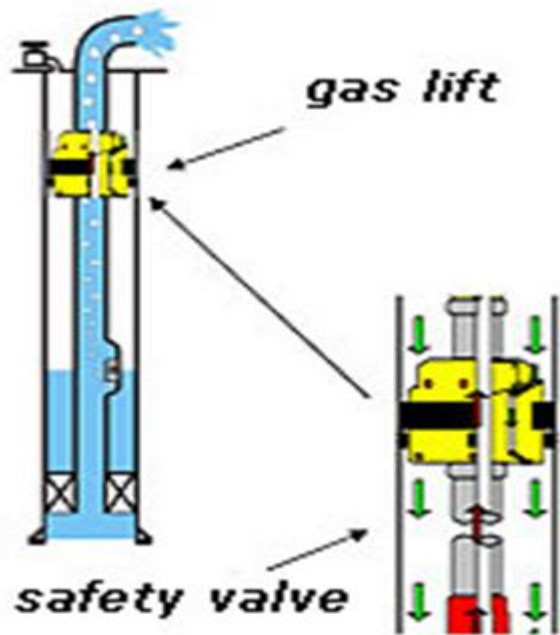
- **When a Tubing retrievable DHSV is run into the well we need to lock it open**
 - a. False
 - b. True
 - **Would it be an acceptable practice to lock open and set an insert valve to reinstate production in a well with a failed tubing retrievable DHSV?**
 - a. No
 - b. Yes
 - **What is the correct action to take to re-instate production in a well with a failed tubing retrievable DHSV?**
 - a. Workover the well
 - b. Lock open DHSV
 - c. Lock open DHSV and install an insert valve
-

• Completion Equipment

➤ Down Hole Safety Valves

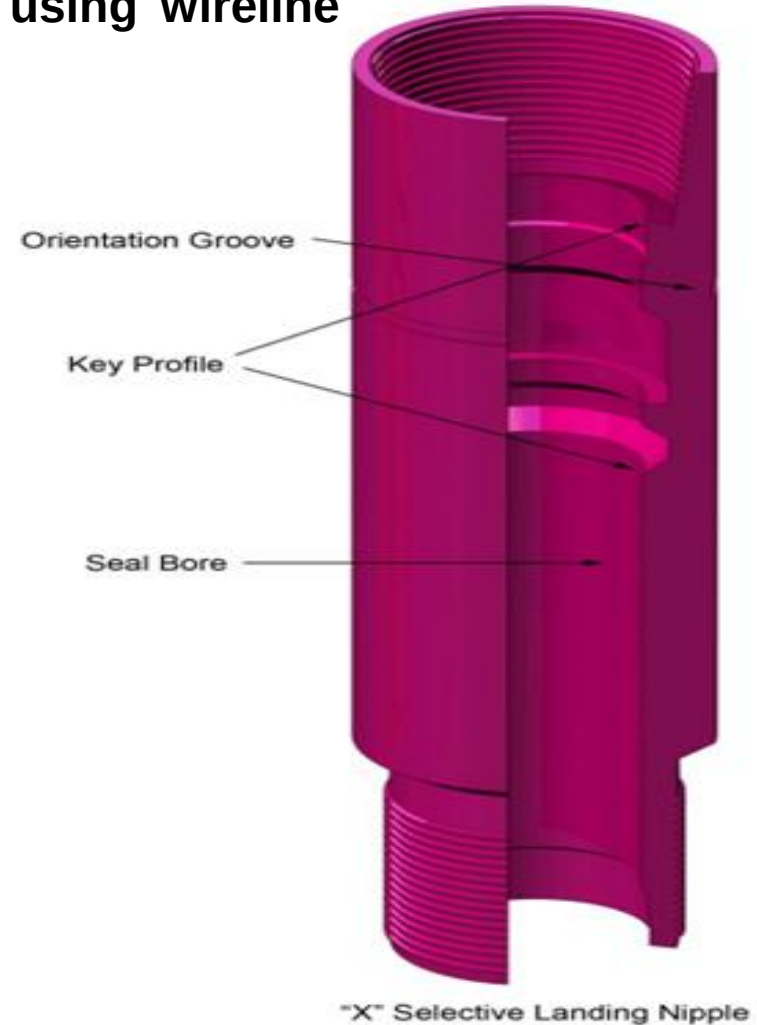
❑ Annulus safety valve

- Some countries require annulus safety valves to be installed in gas-lift wells on fixed installations.
- The idea is the same as for DHSVs, i.e. to prevent an uncontrolled reverse backflow of gas from the annulus, if any of the wellhead and/or Xmas tree equipment should be damaged.

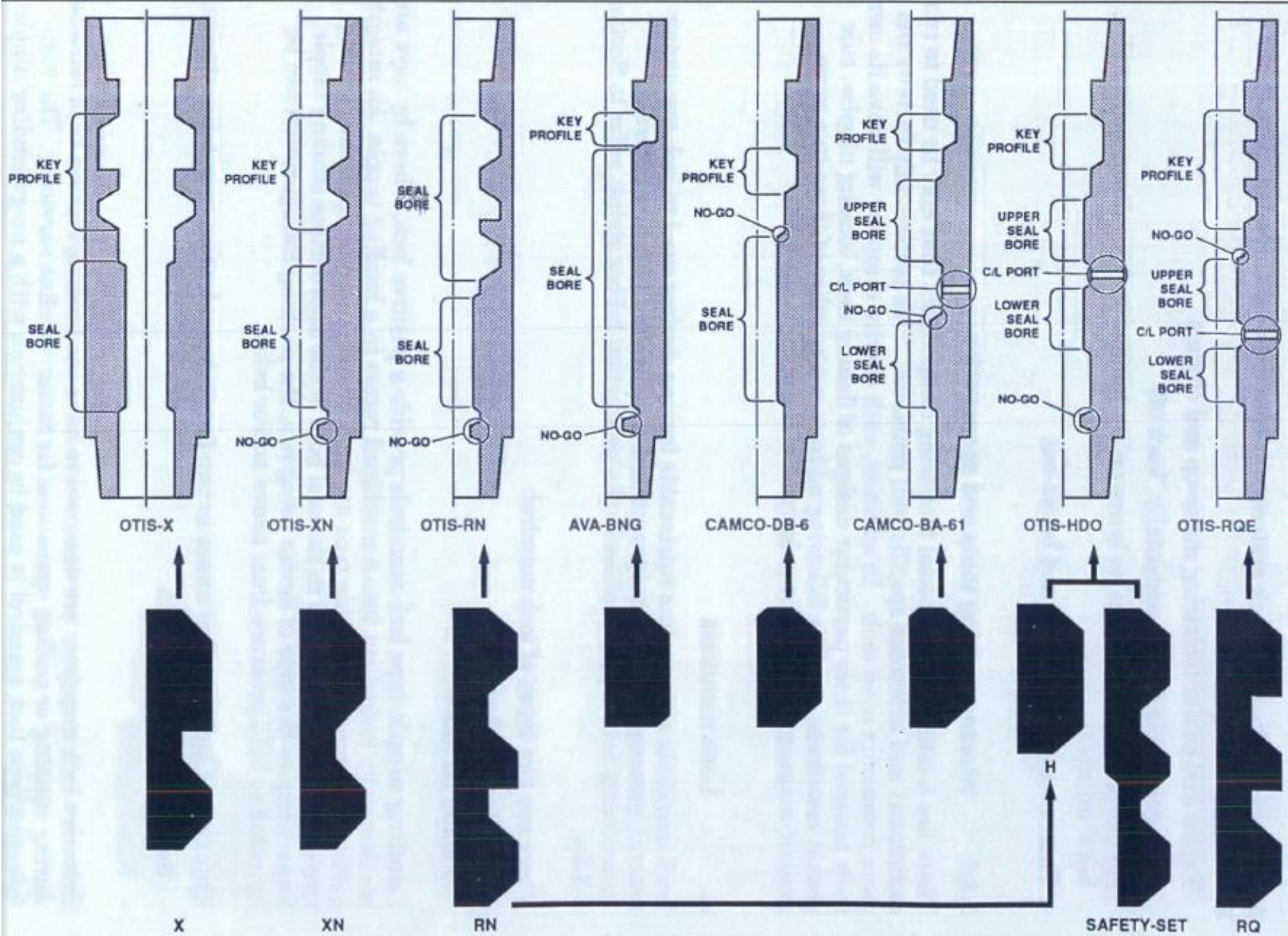


➤ Landing Nipples

- Used to receive flow control devices
- Short tubular with 3 main features; Shoulder, Seal Bores, and Profiles. It can accommodate and secure a locking device called a lock mandrel run usually using wireline well intervention equipment.
- Types
 - Selective
 - Non-selective or no-go

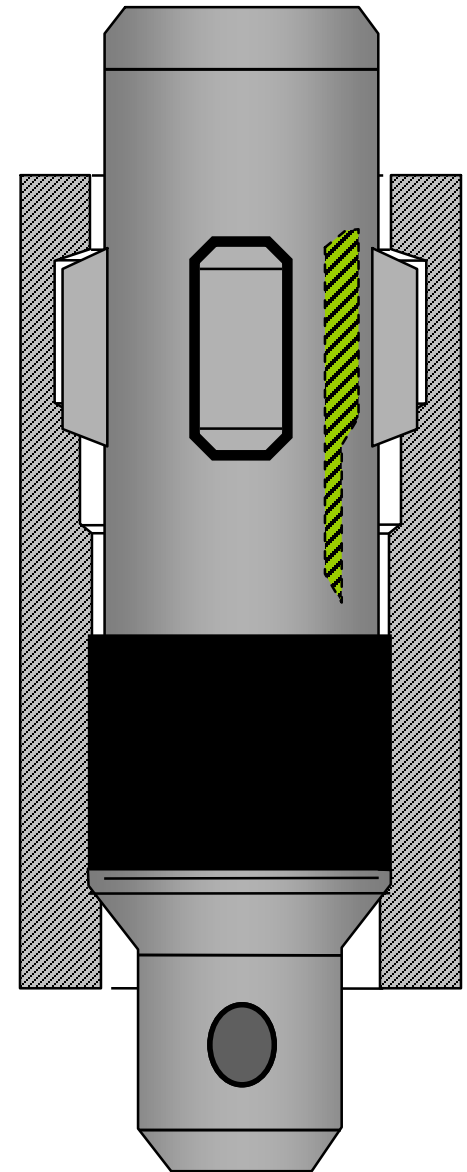






Selective Nipples

- Main advantage is that **all seal bores are the same size.**
- No tapering of tubing size.
- Selective nipples have a locating profile that is “found” by the dogs on the lock.



Non-selective or no-go nipples

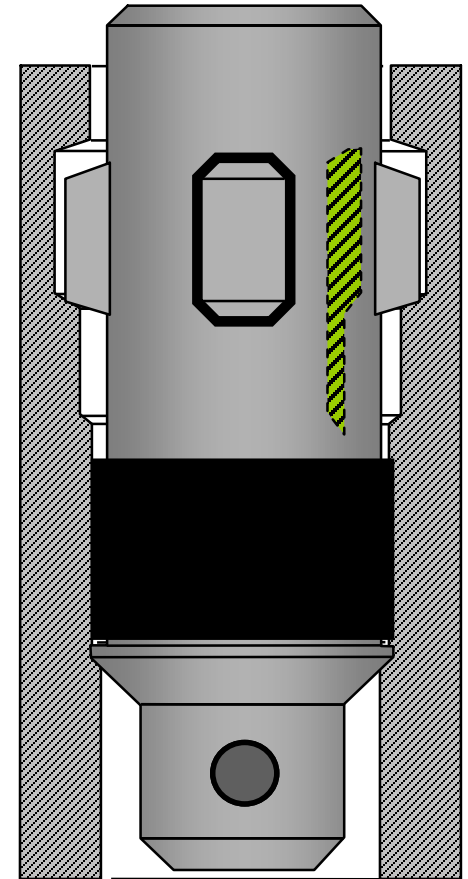
➤ Rely on nipple having a smaller ID (no-go) than the lock

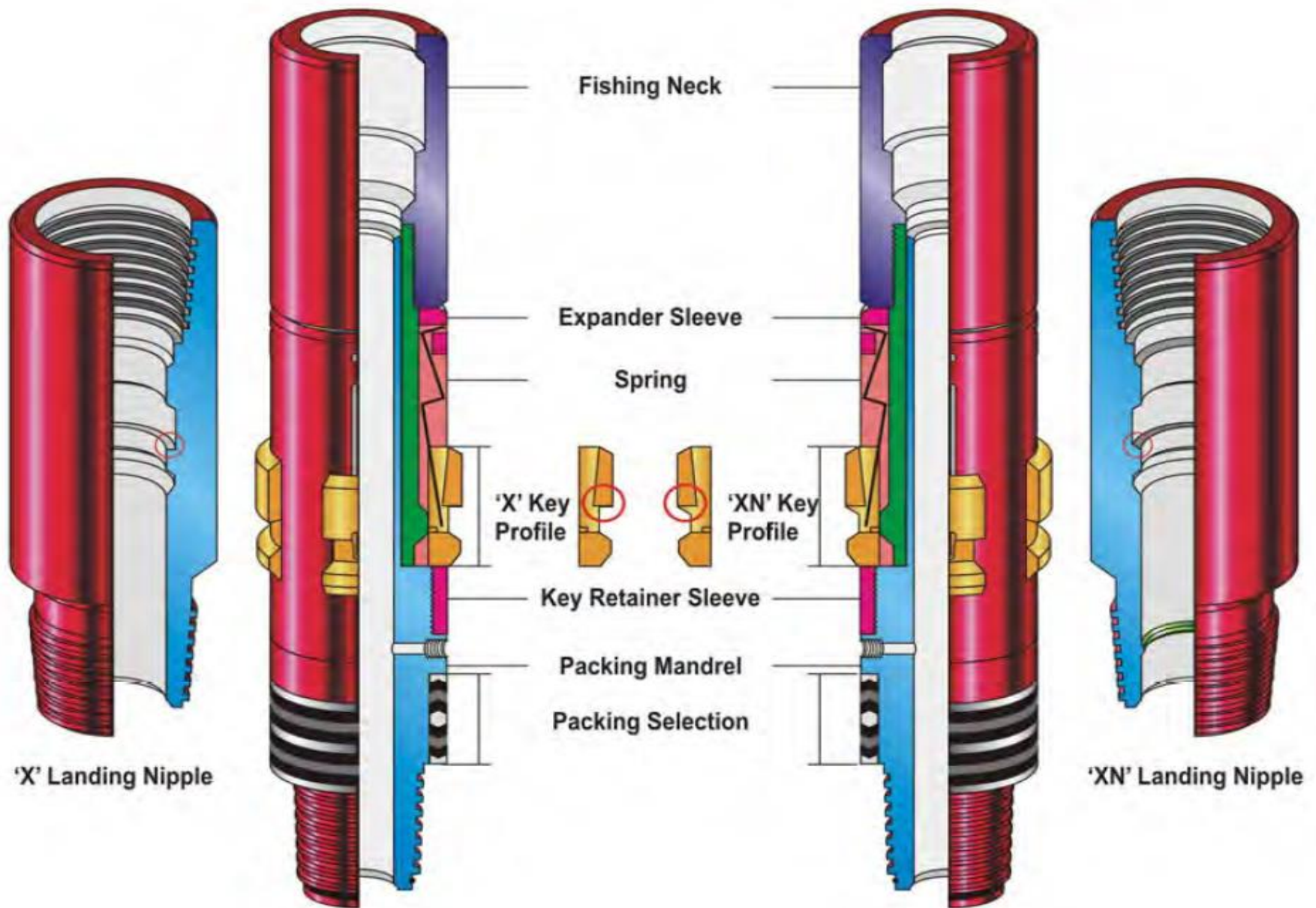
✓ **No-go is used to**

1. To locate the correct lock mandrel
2. To prevent the tools falling out of the bottom of the tubing

✓ Only one No-Go landing nipple of the same size can be used in completion string, the lowermost nipple being No-Go nipple.

✓ More than one No-Go landing nipple can be incorporated in a completion string provided that a step down in No-Go shoulder size is observed.





Landing Nipples

• Completion Equipment

➤ Circulation Equipment

- ✓ The term “circulation equipment” refers to the following items:
 - Side Pocket Mandrels (SPM)
 - Sliding sleeves
 - Nipples with port(s)/sleeves
- ✓ Circulation equipment is employed to:
 - Provide gas-lift (SPM with gas-lift valves)
 - Create an under-balance in the well in advance of perforation (SPM, sliding sleeve and nipple)
 - Kill the well (SPM, sliding sleeve and nipple).

- **Completion Equipment**

- **Circulation Equipment**

- ❑ **Side Pocket Mandrels**

- SPMs are used either for gas-lift purposes or for temporary communication between the annulus and the completion string (to create an under-balance in the well, to kill the well, etc.).
 - An important point to remember before plugs or valves in the SPM can be pulled is to equalize the pressure across the GLV.
 - Equipment that can be located in side pockets:
 - Circulation or orifice valve
 - Injection valve
 - Pressure and temperature sensors
 - Plug (dummy valve)
 - Gas-lift valve.

Side Pocket Mandrel

- **Gas Lift Valves (Nitrogen charged or spring loaded)**
 - Different designs are available. They may be orifice valves or pressure operated type valves.
 - They all contain **a check valve** to prevent tubing to Annulus flow These check valves may leak with time and should not be considered as well control barriers.
 - **Dummy Valves**
 - These valves can be installed in SPM for the purpose of tubing & casing isolation. They also facilitate pressure testing from both tubing & annulus.
 - It is also possible to leave the valve in the side pocket mandrel for continued circulation. This is accomplished by shearing down on the inner core with a special tool.
 - **Circulating Valve (Shear orifice valves / Differential kill valves)**
 - These can be installed in SPM whenever tubing/casing circulation is to be carried out. They enable circulation through SPM without damaging the pocket
 - Circulation is achieved by shearing a pin or disc when defined differential pressure is applied from the annulus to tubing.
 - **Chemical Injection Valve (Nitrogen charged or spring loaded)**
 - These valves facilitate injection of chemicals from annulus to tubing & include Reverse flow check valves to prevent flow of injected chemical from tubing to annulus
-



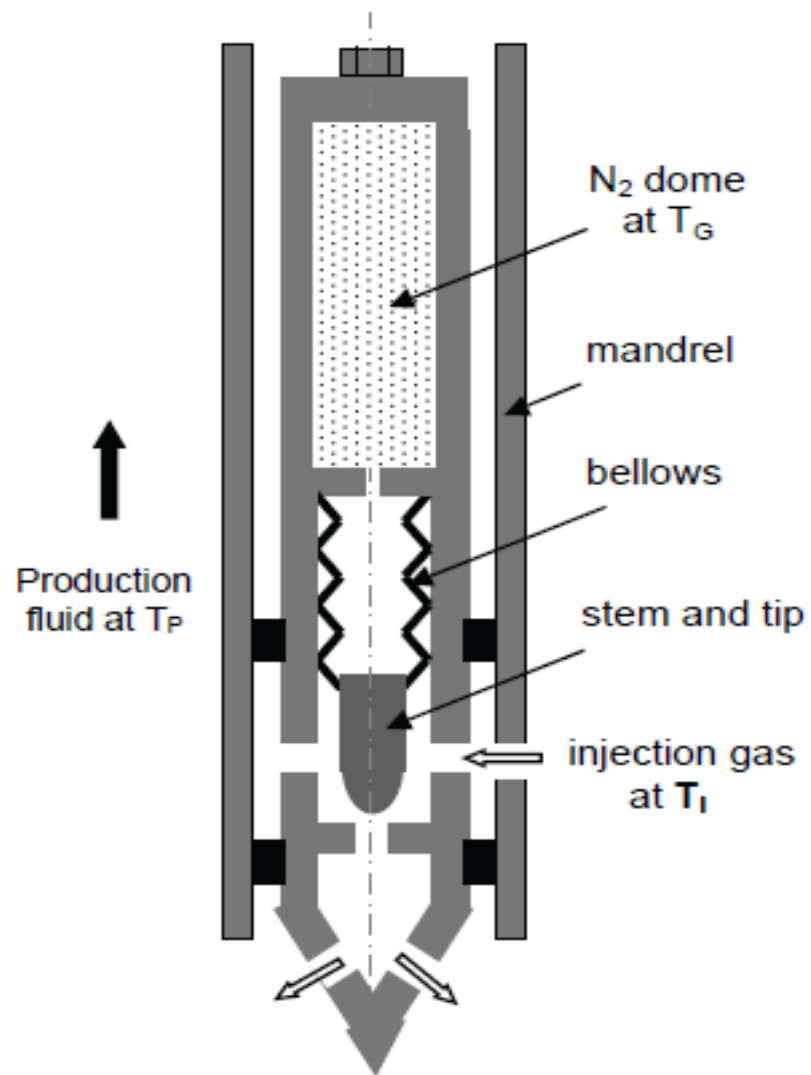


Figure 2. Schematic view of the GLV in the side pocket mandrel.

• Completion Equipment

➤ Circulation Equipment

❑ Sliding sleeves

- Sliding sleeves can also be used for gas lift purposes.
- A sliding sleeve consists of a movable inner sleeve with ports and a fixed outer part. We can move the inner sleeve up and down with the aid of a shifting tool which is run on a wireline or CT.
- **SSD is normally positioned**
 1. Immediately above the top production packer
 2. Between 2 packers
- **You should**
 1. Check that the SSD is fully open before starting circulation
 2. Check that the pressures across the SSD are equalized before opening

❑ Nipple with port/sleeve

- This is a nipple that has ports between the inside and the outside. When the port that provides access to the annulus is not to be used, a sleeve with packers in the nipple profile is installed.
 - Equipment of this sort is seldom utilized.
-



Open



Closed

SSD

• Completion Equipment

➤ Production Packers (Isolation Device)

- ✓ packer is used to provide a seal between the casing and the tubing in a well.
 - ✓ Packers are installed just above the top perforation
 - ✓ The purpose of using a packer in a well completion is:
 - To isolate the casing from reservoir fluids (provide seal between the tubing & completion annulus)
 - To protect the casing from the effects of flowing pressures by isolating the annulus above the packer from casing below packer & tubing
 - To isolate various producing zones.
 - Packers can be used as service tools to perform acid/fracture jobs
 - ✓ In general, packers are constructed of hardened slips, which are forced to bite into the casing wall to prevent upward or downward movement while a system of rubberized elements contact the casing wall to effect a seal.
-

• Completion Equipment

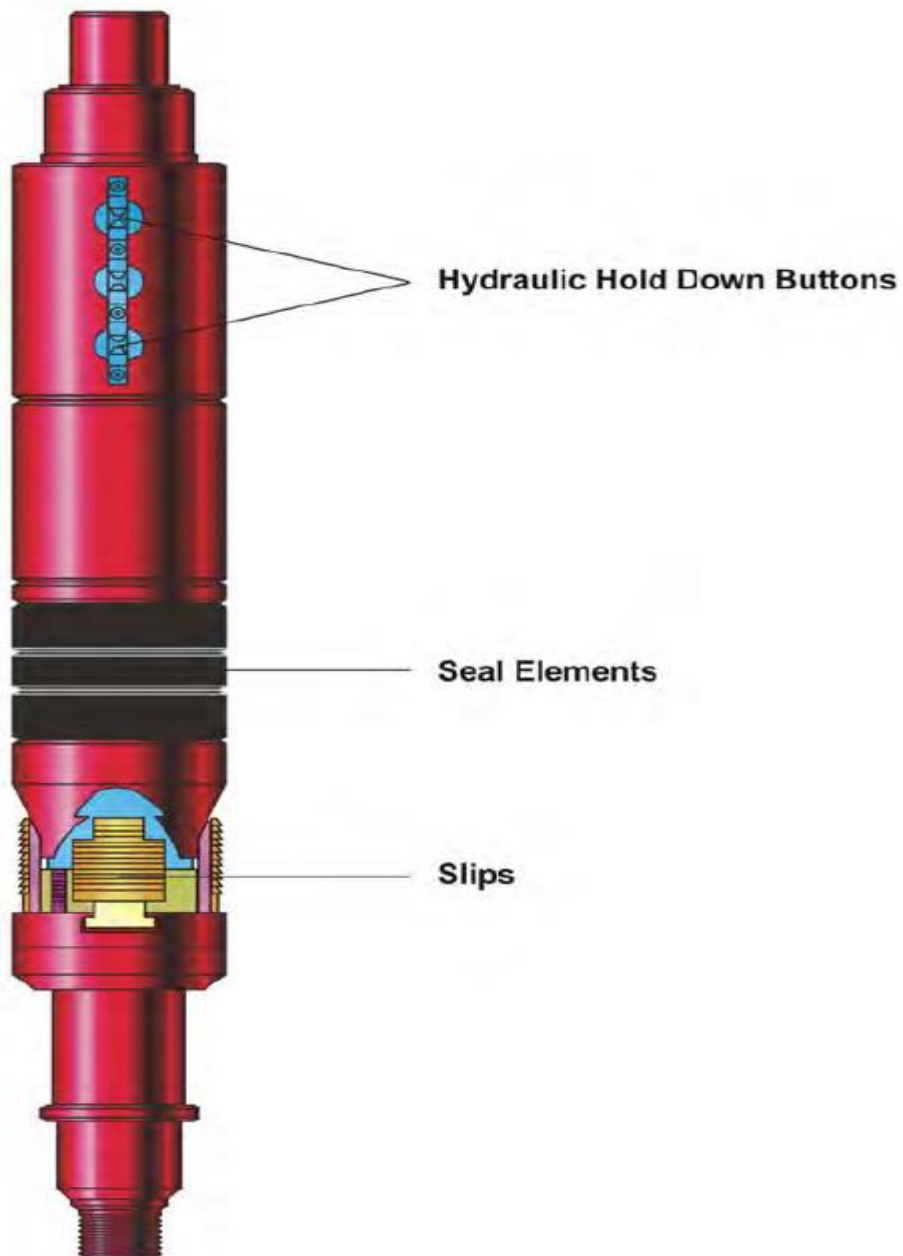
➤ Production Packers

❑ Retrievable Production Packers

- Are run on the tubing string and may be set mechanically or hydraulically.
- Can be retrieved with completion string or with special pulling tool on drill pipe

❑ Permanent Production Packers

- These may be run in a variety of ways and become an integral part of the casing once set but can only be retrieved by milling
- A permanent packer may be run as follows:
 - on electric wireline
 - on pipe and set hydraulically by application of pipe pressure.
 - latched onto the completion tubing and hydraulically set by the application of tubing pressure.



Example of a Retrievable Packer

Example of Hydro-Set Permanent Packer

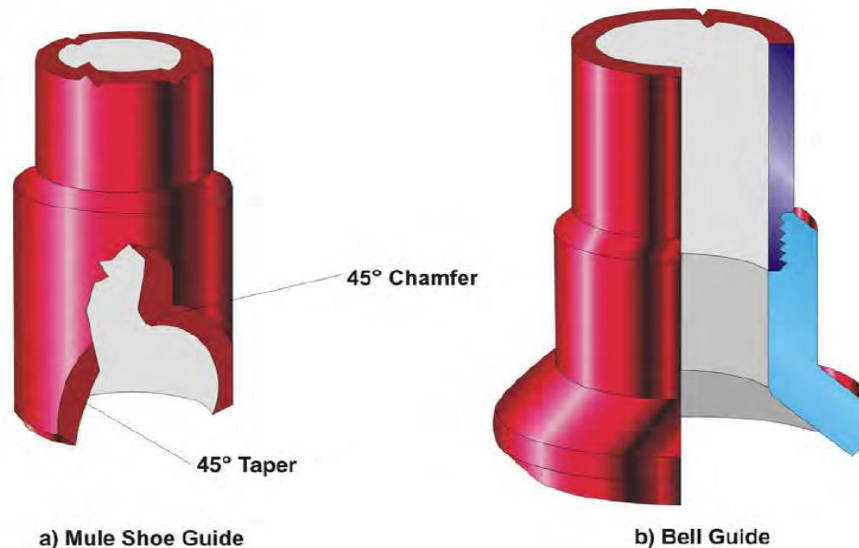


Example of a Permanent Packer

• Completion Equipment

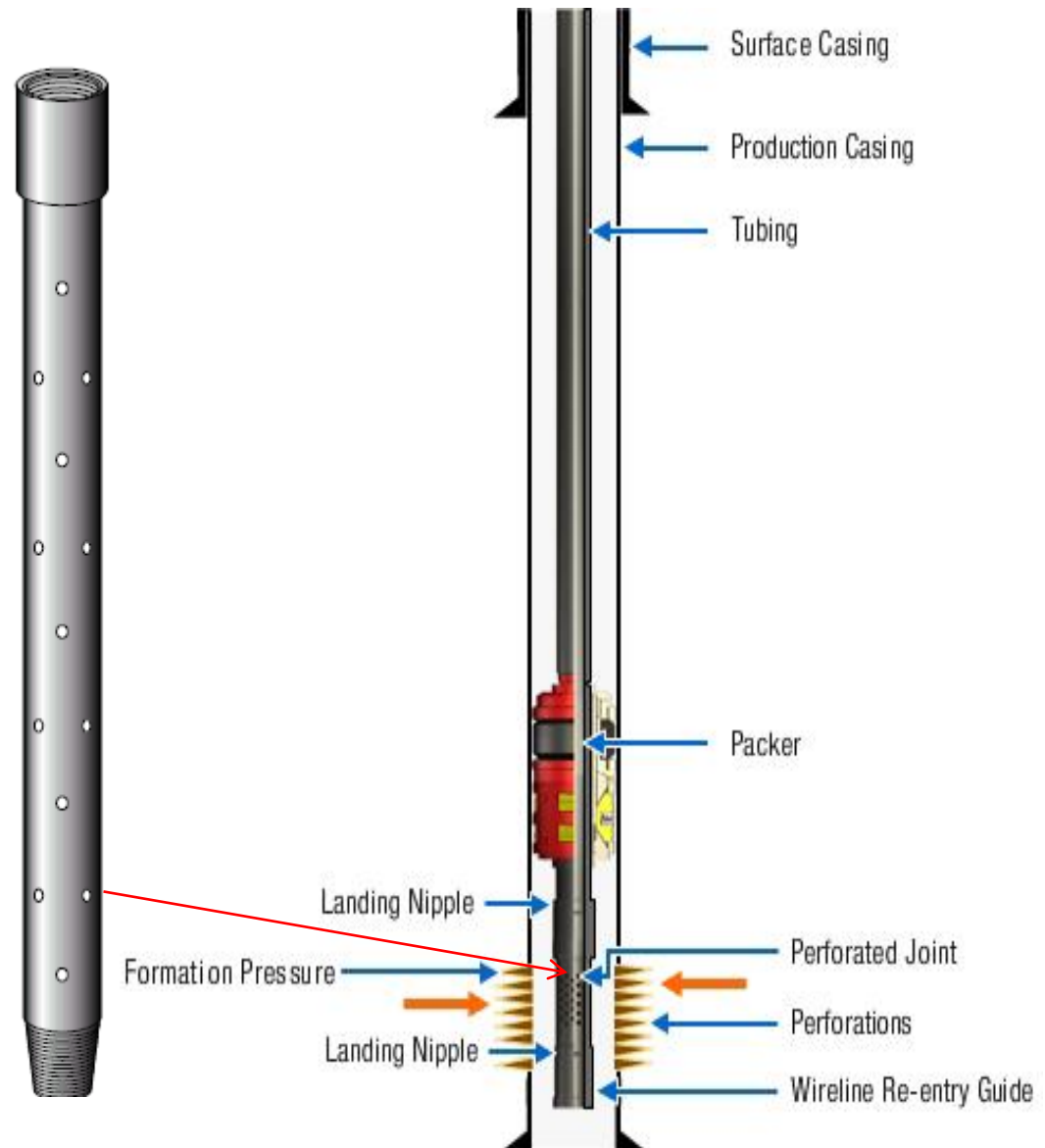
➤ Re-entry Guide

- ✓ A re-entry guide generally takes one of two forms:
 - Bell guide
 - Mule shoe.
- ✓ The bell guide has a 45° lead in taper to allow easy re-entry into the tubing of well intervention tool strings (i.e., wireline or coiled tubing).
- ✓ The mule shoe guide is essentially the same as the bell guide with the exception of a large 45° shoulder.



➤ Perforated Joint

- Tubing pup joint perforated with greater flow areas than tubing.
- Placed in front of perforations
- Subjected to high velocity
- Acts as a sacrificial joint
- Designed to provide true temperature / pressure flow readings & prevent vibrations to gauges installed in high velocities flowing production tubing.



➤ Anti-Erosion Device

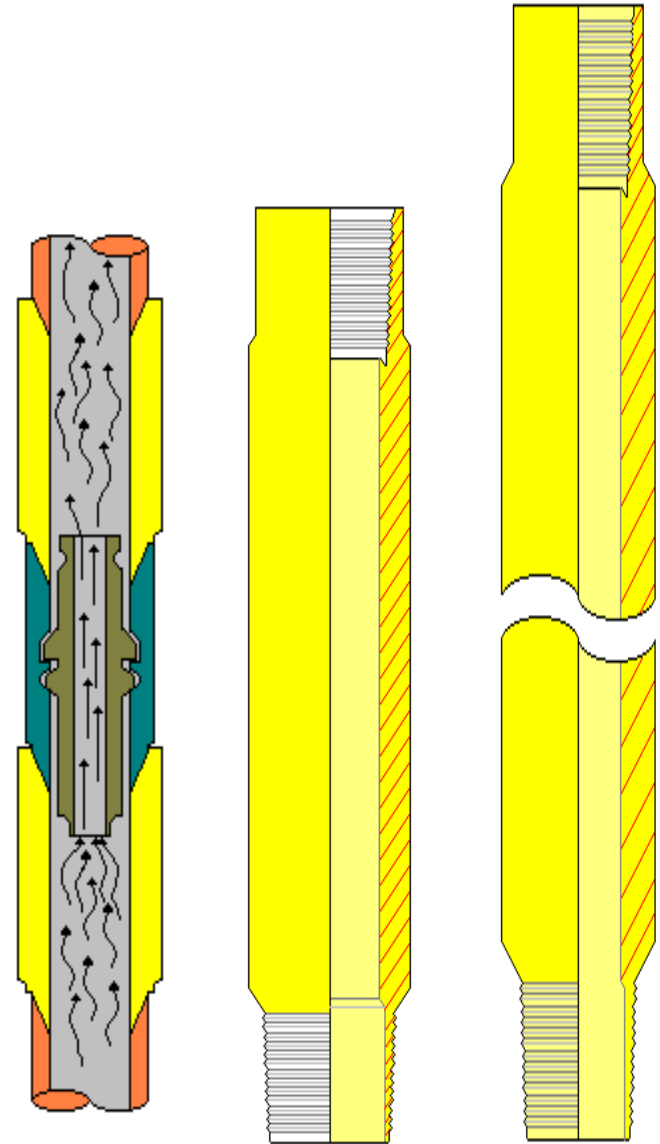
Anti-Erosion Device

Blast Joint

- Heavy-walled tubular.
- Installed opposite perforations where abrasive action & external cutting occurs caused by formation fluids or sand.

Flow Coupling

- Heavy-walled tubular.
- Installed down stream FCD which caused turbulence e.g. Nipple, SSD, SCSSV.
- High velocity & high pressures wells installed both up-stream & down-stream.



Wire Line

Outlines

- I. Introduction to Wireline
- II. Typical Slickline Rig up
- III. Typical Braided Line Rig up
- IV. Managing A Leak or Malfunction on Surface
- V. Operational Considerations

• Introduction to Wire Line

- ✓ Wire line operation is a method used to lower and raise various tools and down-hole controls, in and out of a production well .
- ✓ In addition, it is also used to set and retrieve down hole controls.
- ✓ The setting and retrieving operations are achieved by means of string which is attached to the wire.
- ✓ Wire line operation can be carried out in dead or live wells. However, it has its limitation on highly deviated wells.
- ✓ Generally there are three types of wire line commonly in use: Slick line, Braided line, and Electric line.
- ✓ The solid single strand slick line is commonly described as: Piano wire, Solid wire, or Measuring line.

- **SLICKLINE**

- ✓ Single strand wire with wire OD range from 0.105", 0.108", 0.125" & 0.190 (0.092", 0.082", 0.072" & 0.066").

- **BRAIDEDLINE**

- ✓ Multi-strands wire cable OD range from 3/16", 7/32", 1/4" & 5/16".
- ✓ Application:
 - Retrieval of downhole tools with slick line failure.
 - Fishing job

- **ELECTRICLINE**

- ✓ Multi-strands wire cable range from 3/16", 7/32", 1/4" & 5/16", 7/16" & 3/4" OD with single (mono) or multiple conductors.
- ✓ Application;
 - SRO Pressure & Temperature Recorder
 - SRO Electronic Logging Instrument for open hole & cased hole.
 - Packer setting
 - Perforations etc.

• **WIRELINING EQUIPMENT**

1 Surface Equipment

1.1 WL Unit

1.2 Power Pack

1.3 Hoisting Unit

2 PCE

2.1 Quick Union

2.2 Stuffing Box / GIH

2.3 Lubricator

2.4 BOP

2.5 Tree Adapter

3 Subsurface Equipment (SSE)

3.1 Toolstring

3.2 Service Tools

3.3 Tubing Conditioning Tools

3.4 Running Tools

3.5 Pulling Tools

Optional PCE

2.6 Tool Trap

2.7 Tool Catcher

2.8 Injection Sub

2.8 SCU SAFETY CHECK UNION

2.9 Pumping Tee

3.6 Shifting Tool

3.7 Kick Over Tool

3.8 Overshot

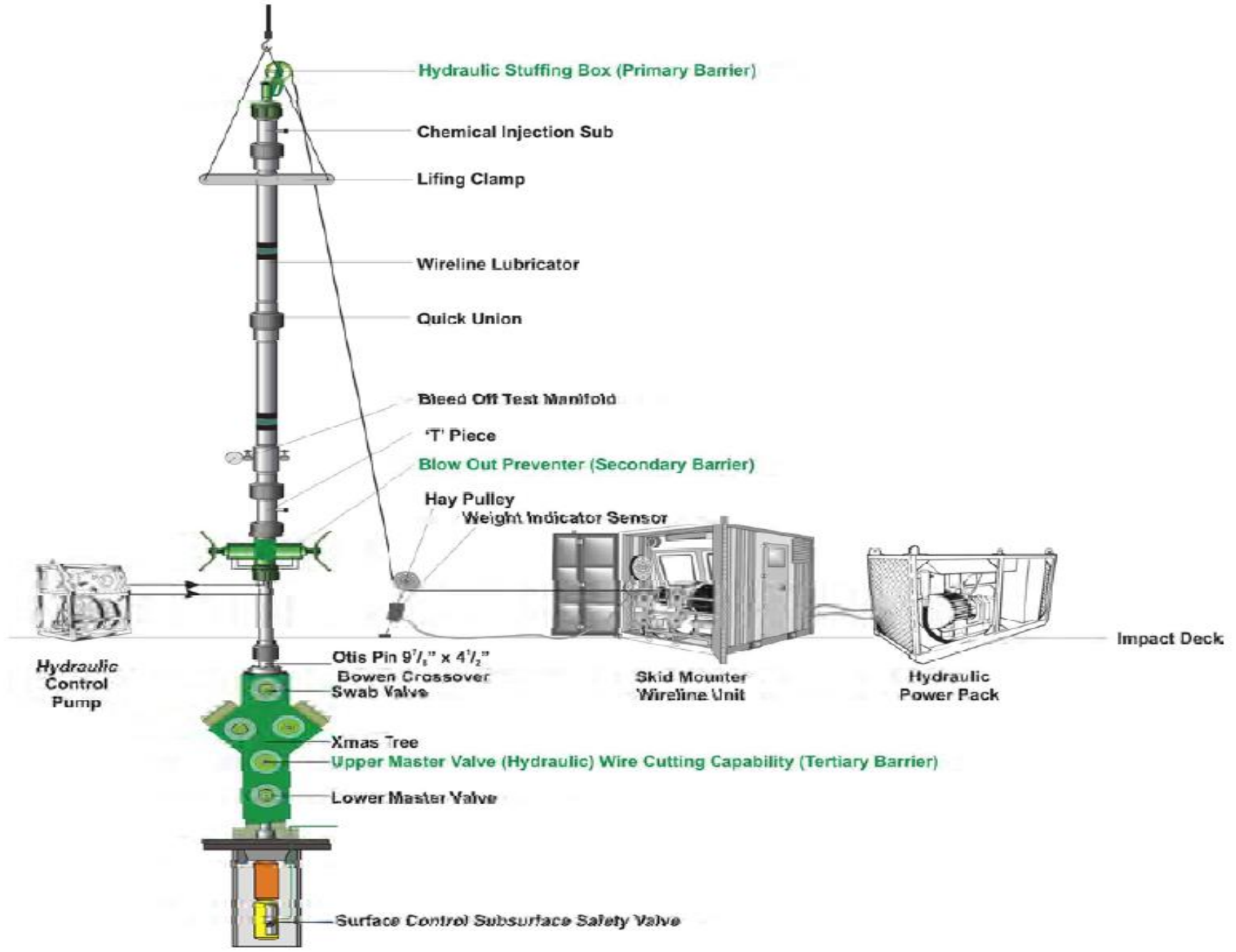
3.9 Spear

**3.10 Flow Control
Device**

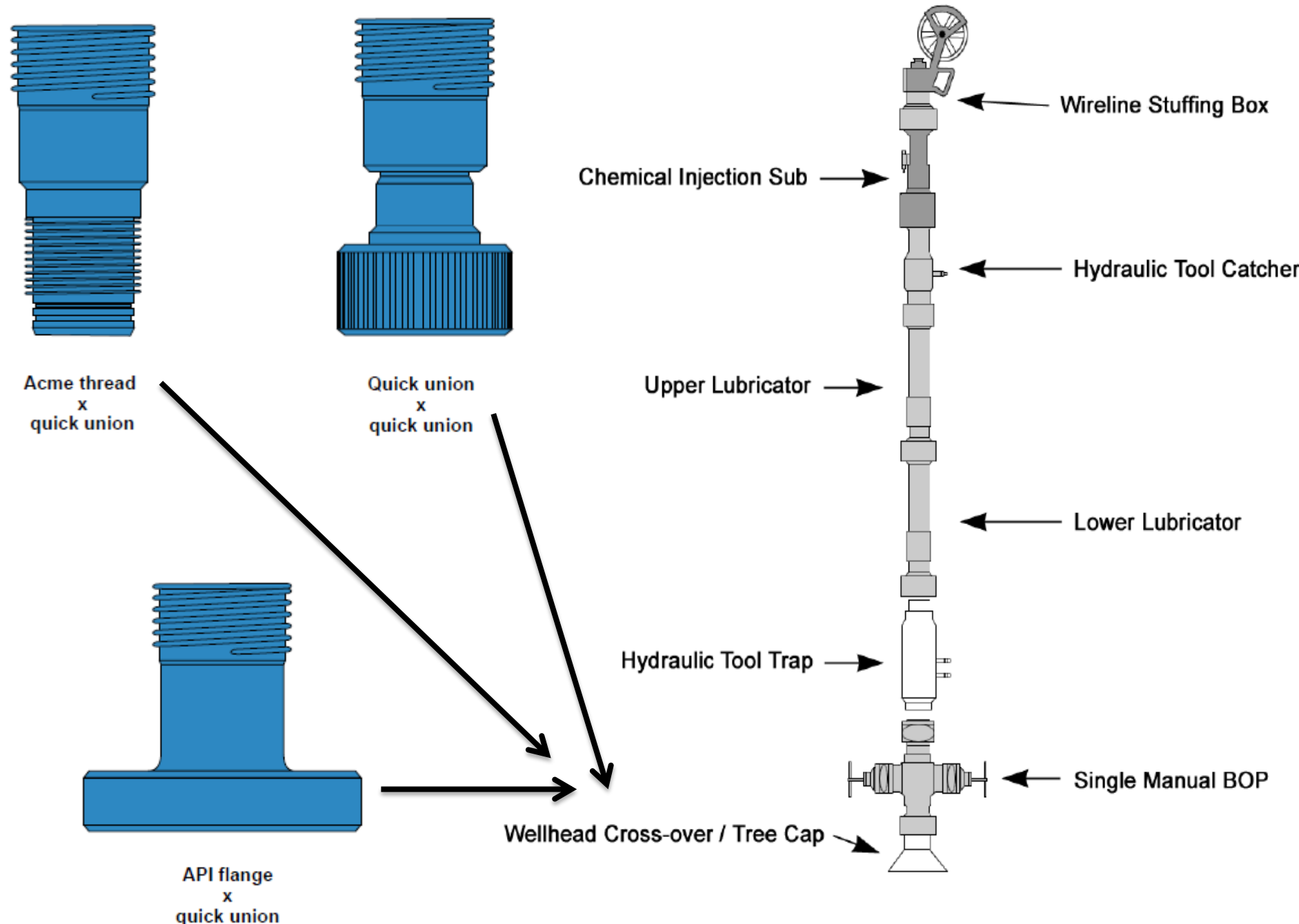
Wire Line

Outlines

- I. Introduction to Wireline
- II. Typical Slickline Rig up
- III. Typical Braided Line Rig up
- IV. Managing A Leak or Malfunction on Surface
- V. Operational Considerations



Typical Slickline Rig up



Slickline Pressure Control Stack

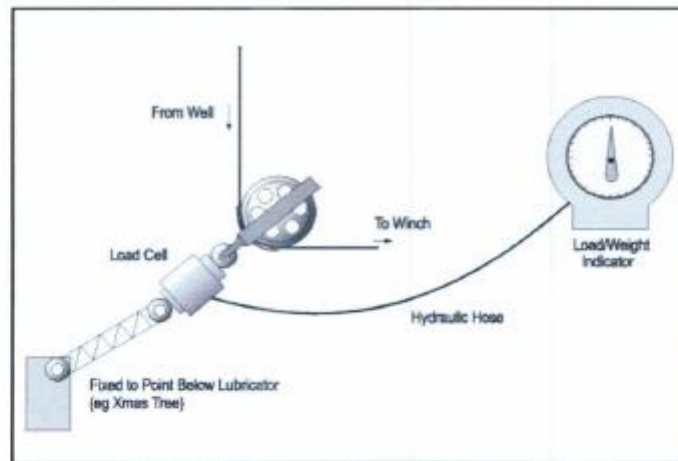


Figure: 10.1.9 Weight indicator hook-up

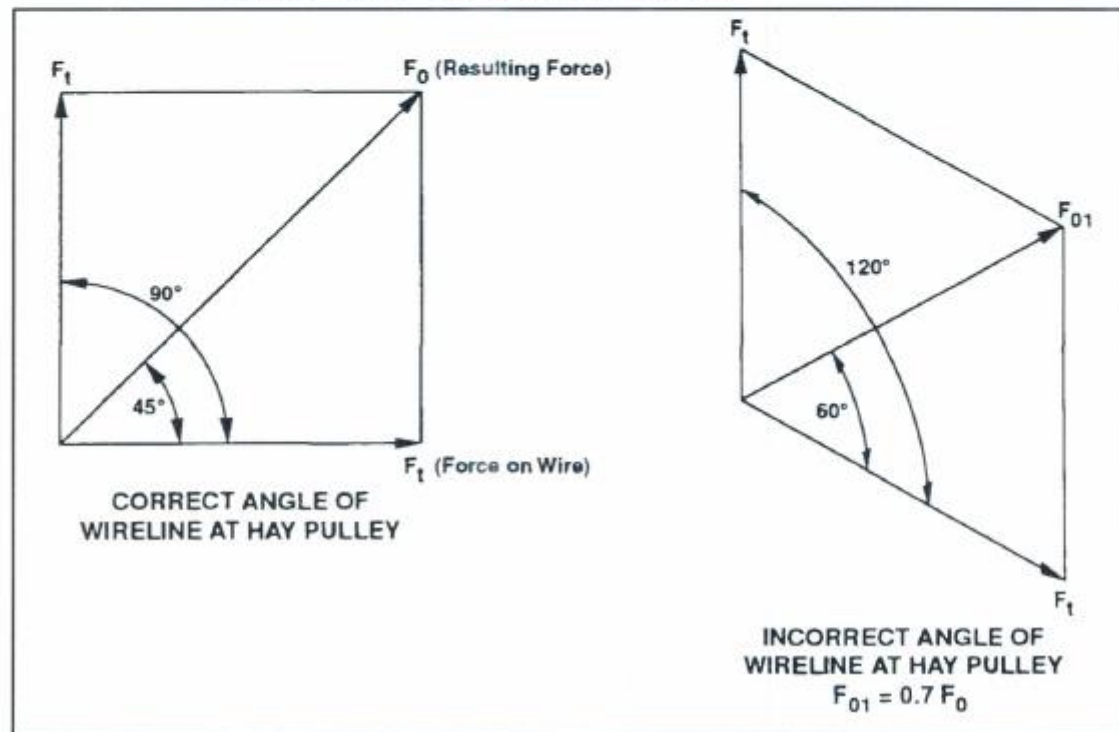
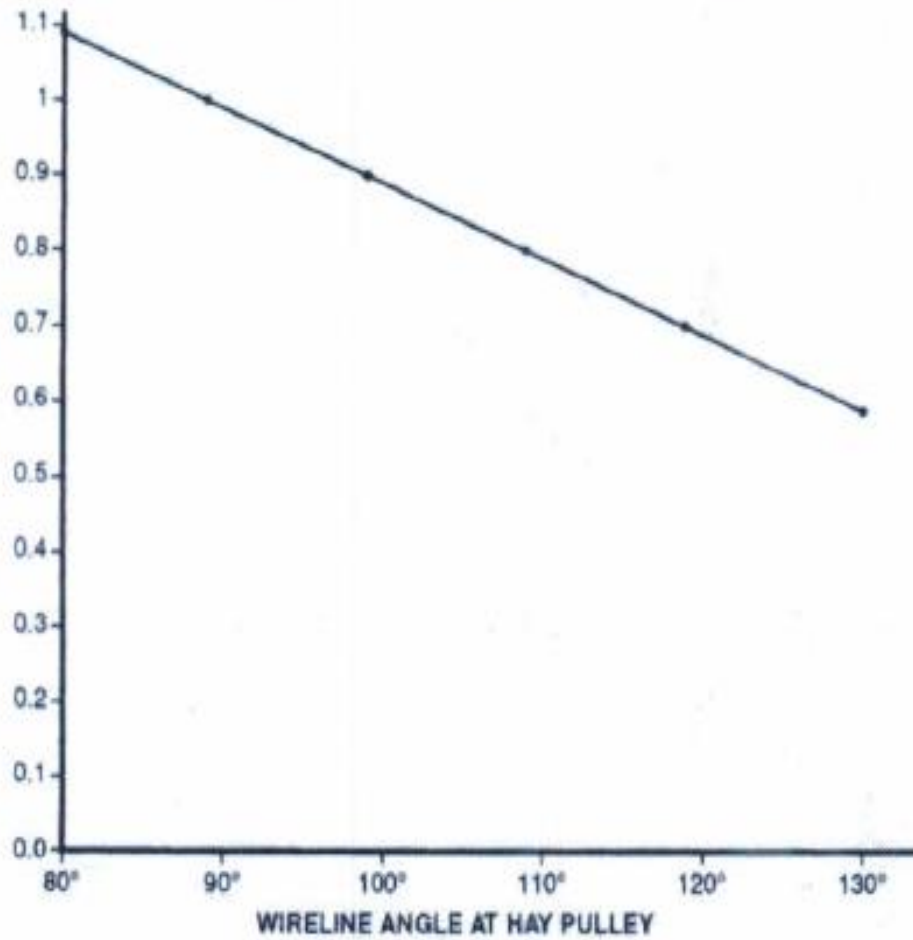


Figure 10.1.10: Wire angle at Hay Pulley

CORRECTION FACTOR
APPLIED TO INDICATED
WEIGHT TO OBTAIN
TRUE WEIGHT ON WIRE



• Typical Wireline Rig up

➤ Stuffing box

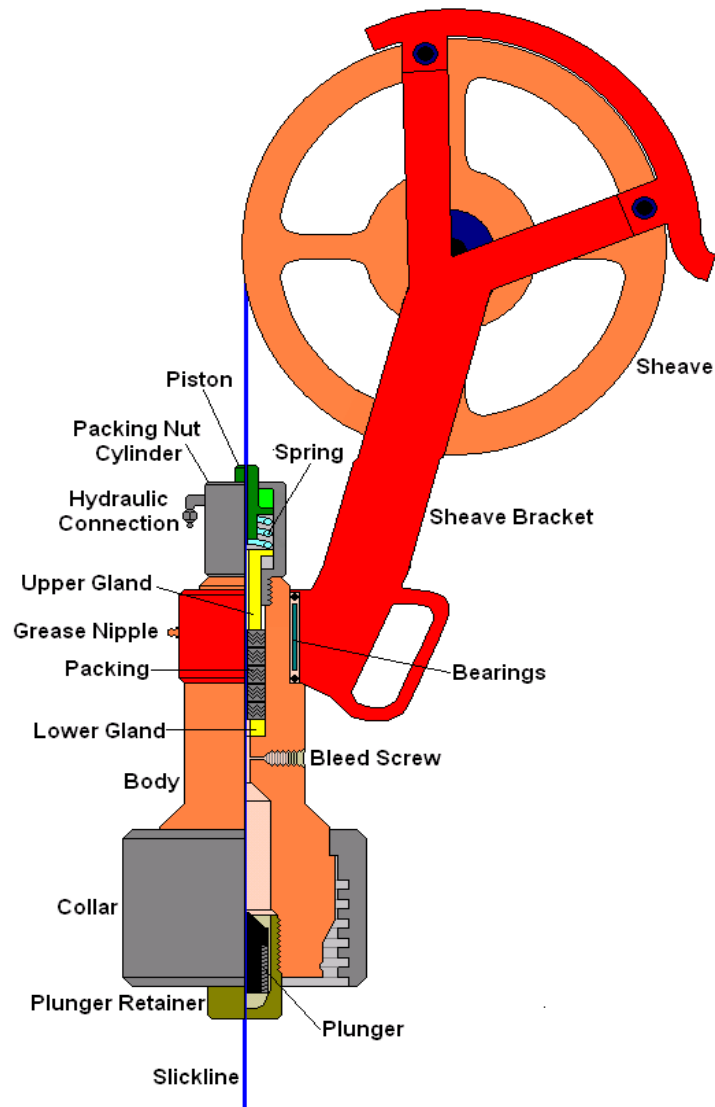
- ✓ The stuffing box is the primary barrier during a slickline operation. Main Function is to seal around the wire at upper end of lubricator during slickline operations , in both static and dynamic conditions.
- ✓ All stuffing boxes incorporate a blow-out valve which will close fluid flow if the wireline breaks and is blown out of the stuffing box.
- ✓ With no wire in place ,the flow causes the BOP plunger to activate& well bore pressure is isolated until the BOP or Swab valve can be closed to fully secure the well.
- ✓ A swivel- mounted sheave wheel & guard to Guide the wire as it enters or leaves the well Bore (having 360° free movement).
- ✓ The Sheave guard on the stuffing box is designed to trap wire, if a break occurs at surface.
- ✓ Before an operation we should check the stuffing box and determine whether the set of packers is in good condition. If there is any sign of wear it should be replaced with a new set.

• **Typical Wireline Rig up**

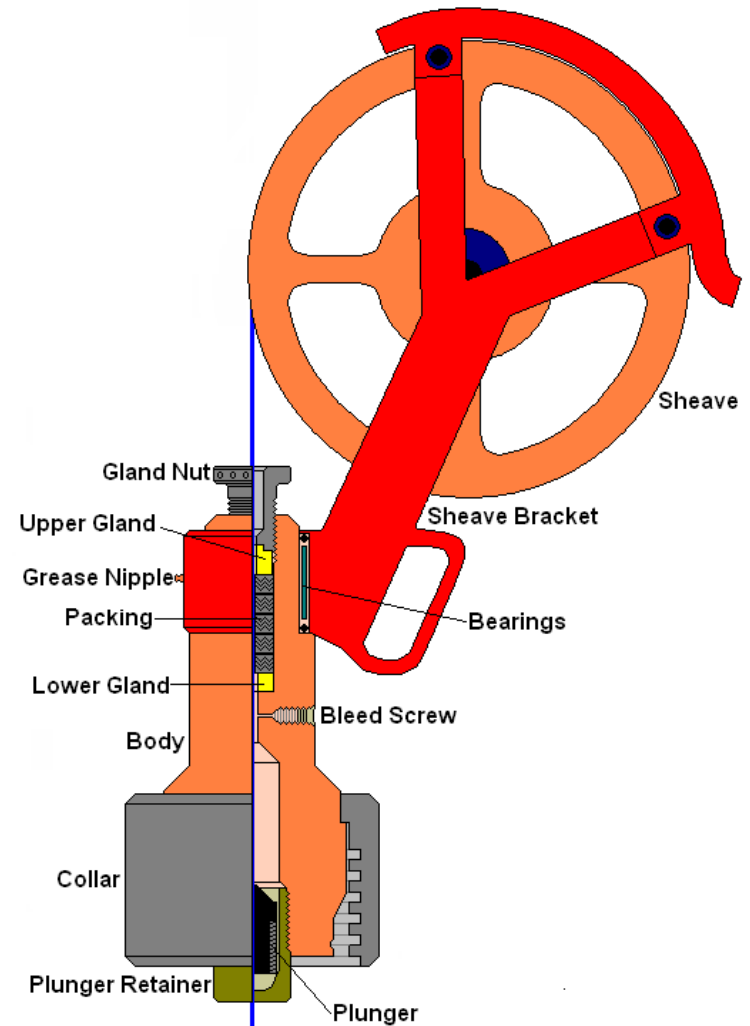
➤ **Stuffing box**

- ✓ We should make sure that the blowout plug has been inserted the right way up, i.e. with it conical end pointing upwards.
- ✓ Injection Port (Bleed Port) can be used for following purposes:
 - Injection of glycol or similar antifreeze fluid in low temperature environments or when working on high –pressure gas wells where hydrates may be a formed.
 - Injection of a chemical inhibitor to protect slickline in corrosive well conditions (H₂S)
 - Bleed of pressure to activate the BOP plunger if packing has to be changed under pressure
- ✓ The use of a remote hydraulically activated packing nut enables the packing forces to be adjusted from a safe distance. This feature is particularly useful when:
 - Stuffing box cannot be easily reached during operation.
 - On high- pressure wells when close proximity to a pressurized stuffing box is not advisable.
 - On wells containing dangerous levels of H₂S.

PCE - STUFFING BOX



Hydraulic Stuffing Box



Manual Stuffing Box

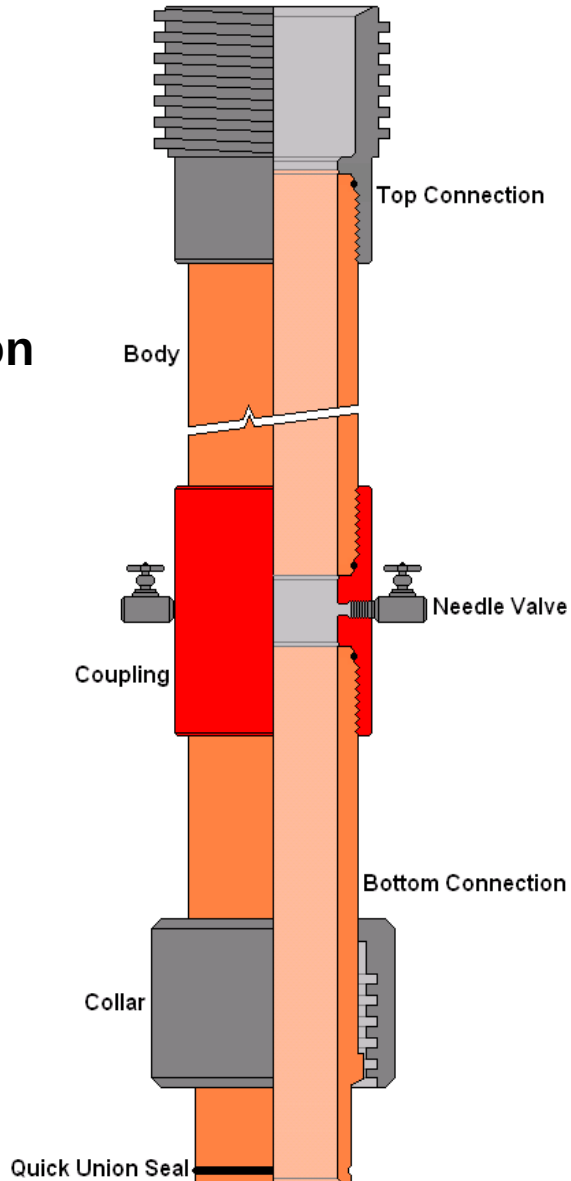
• Typical Wireline Rig up

➤ Lubricator

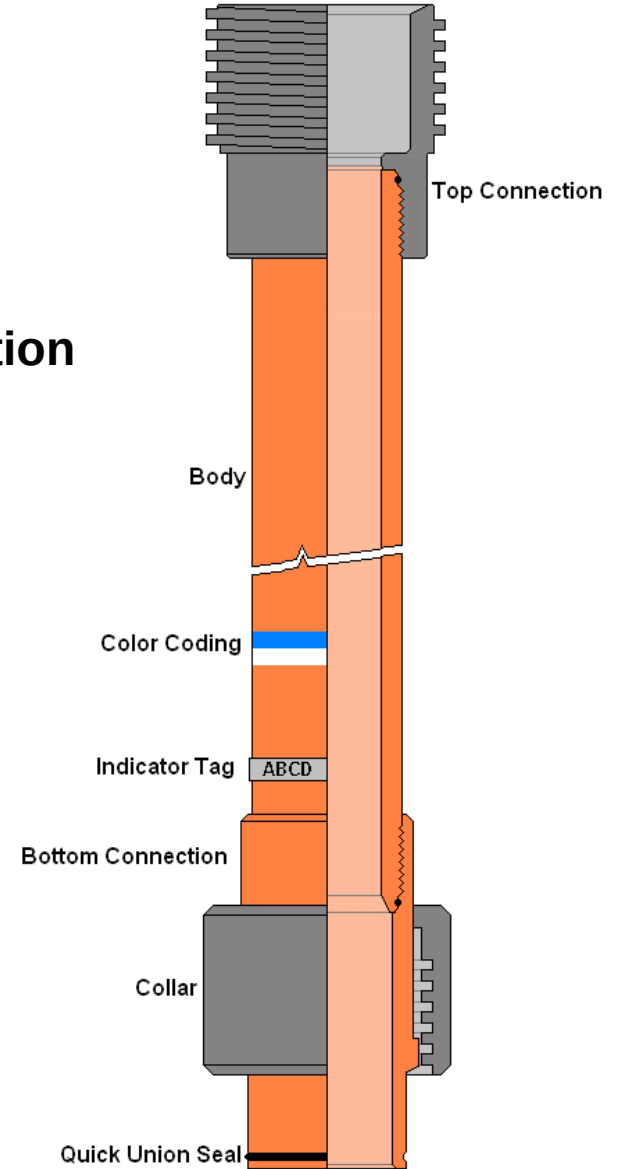
- ✓ Lubricator sections enable the slick line tool string and equipment to be inserted and removed from a well under pressure
- ✓ A lubricator consists of cylindrical pipes connected by quick-release joints as needed to make up the required length.
- ✓ The quick-release joints are fitted with O-ring seals. Every time of R/U, we should check that the O-rings are in good condition.
- ✓ At the lower end of the lubricator we find a **bleed-off port** on which to mount the lubricator manifold, which normally consists of a needle valve for pressure venting.
- ✓ Lower section must be of sufficient ID to accommodate the tools or equipment being run.
- ✓ Upper sections may be smaller as only the tool string is housed inside.

PCE - LUBRICATOR

Lower Section

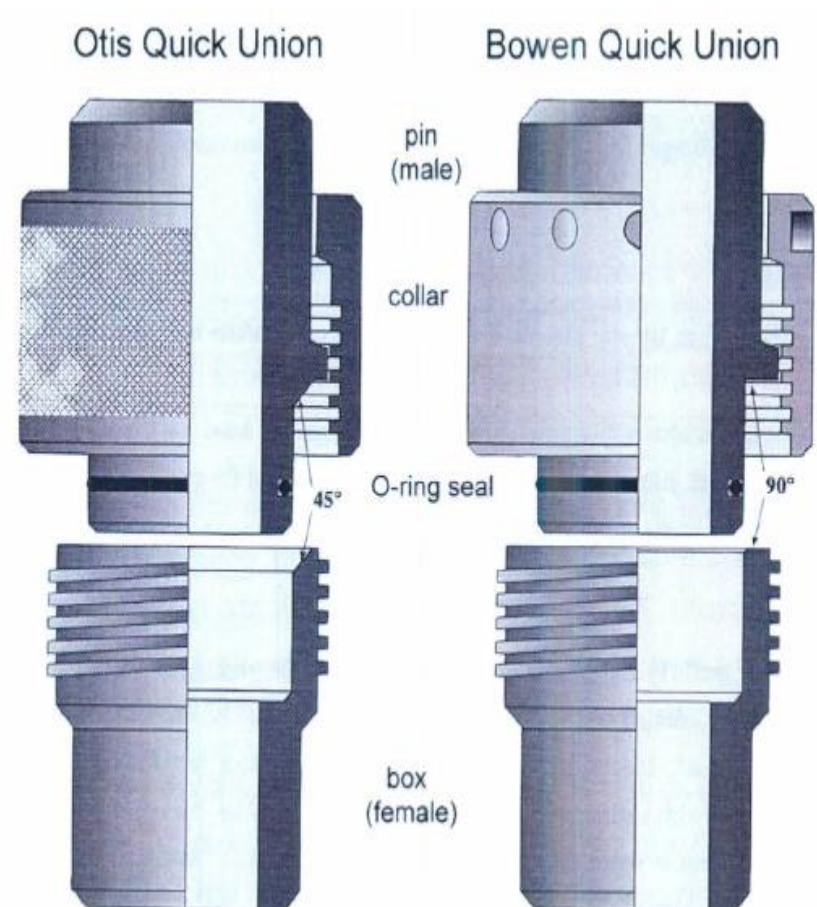
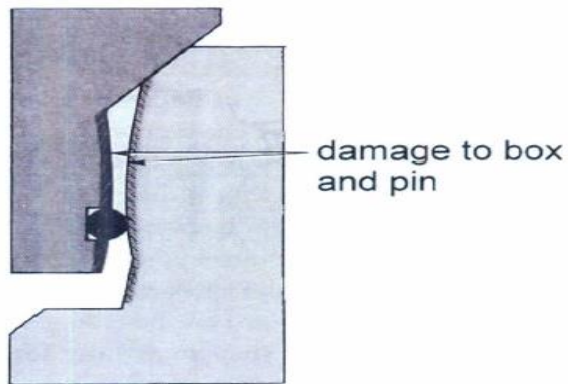


Top Section



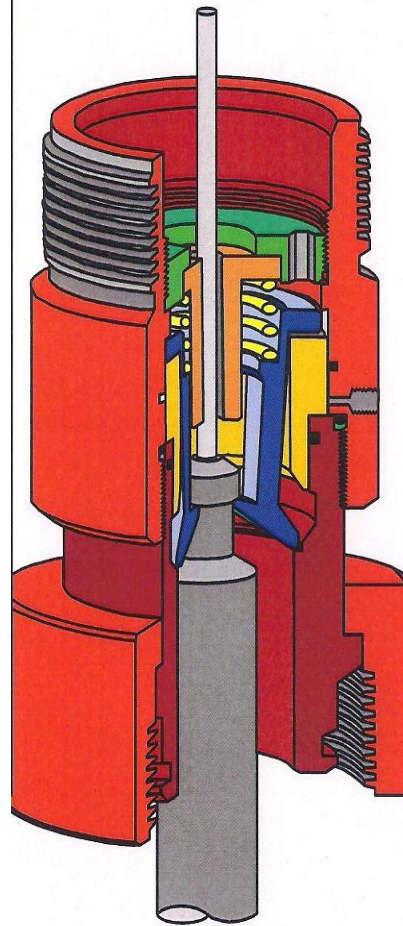
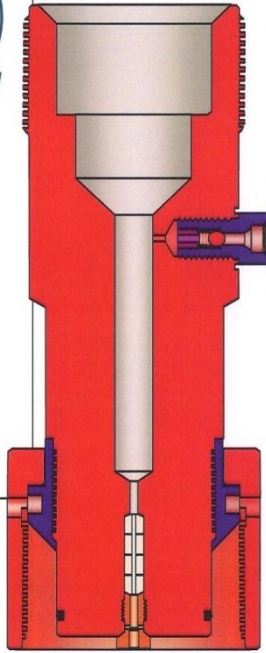
PCE - QUICK UNION

- ACME Thread Connection, Hand Tight, NO Pipe wrenches, chain tongs or hammers
- O-Ring seal: To be kept clean at all time.
- Check thread & O-ring before make-up.
- O-Rings need lubricating with oil or grease prior to assembly.
- Worn connections, where the gap between the pin and box exceeds tolerances, will lead to failure.
- Lubricator sections with screwed on quick unions should not be used.



- Liquid Chamber Sub
- (Chemical Injection Sub)

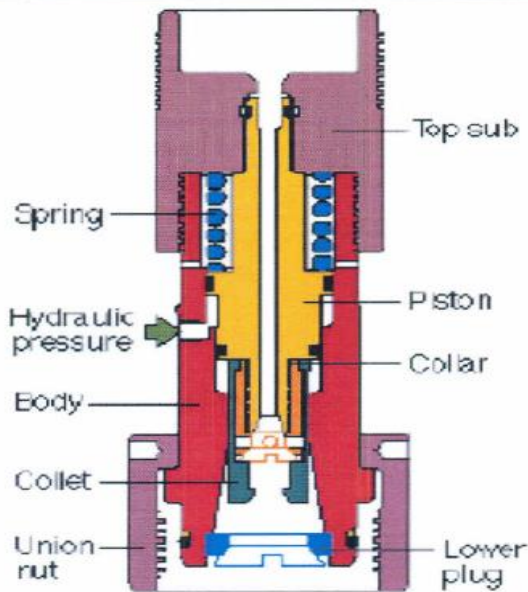
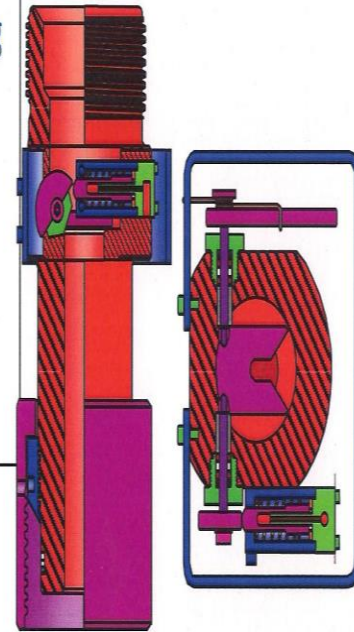
- Located at lubricator stack's top below stuffing box
- Optional
- Lubricates wireline
- Corrosion inhibition
- Prevents hydrate formation
- Sour conditions
- Sandy conditions



Hydraulic Tool Catcher

- Tool Trap

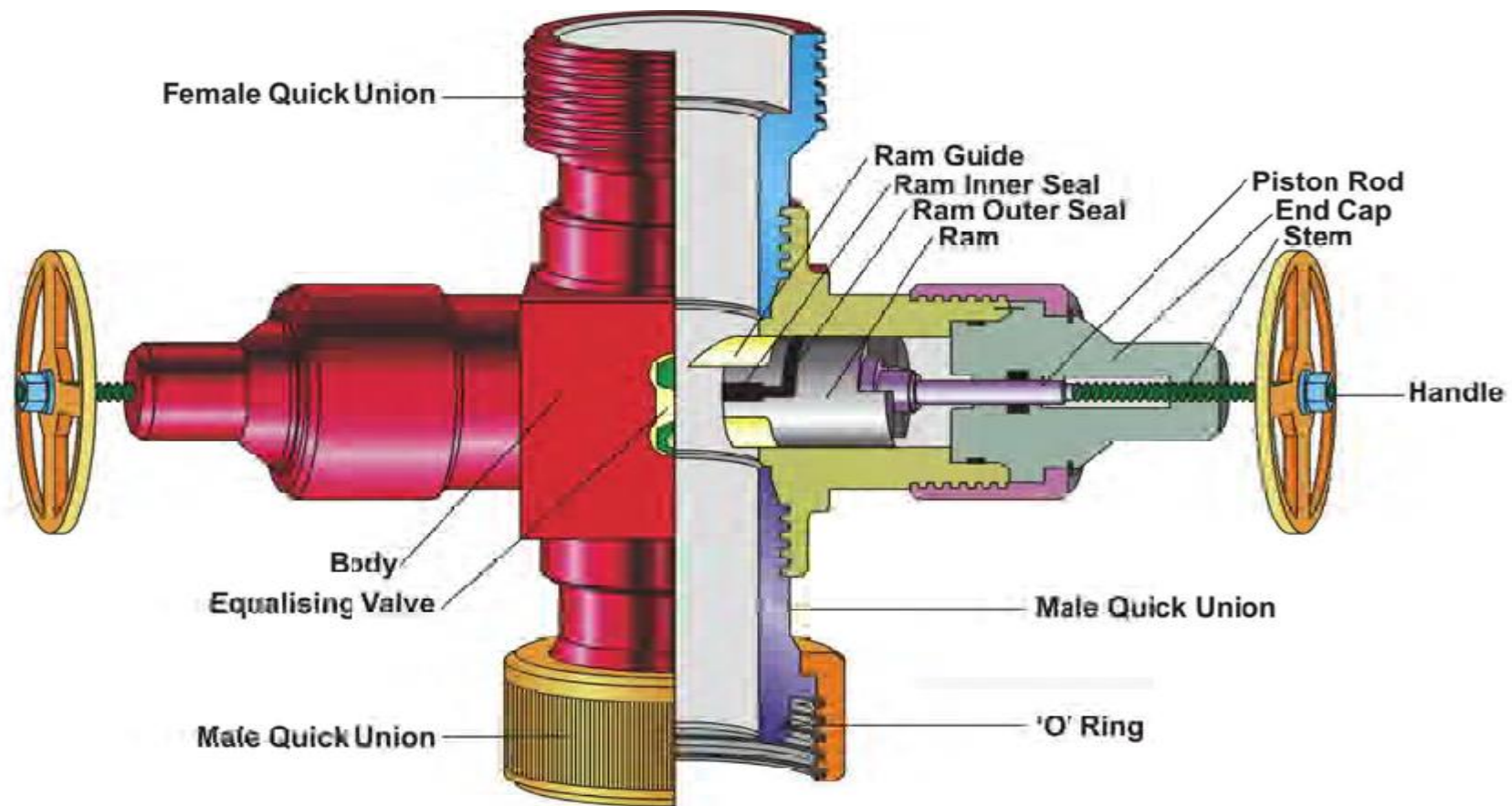
- Placed above BOPs at lubricator stack's bottom
- Hinged flapper closes when tool string moves above it to prevent tool string from falling back out of lubricator



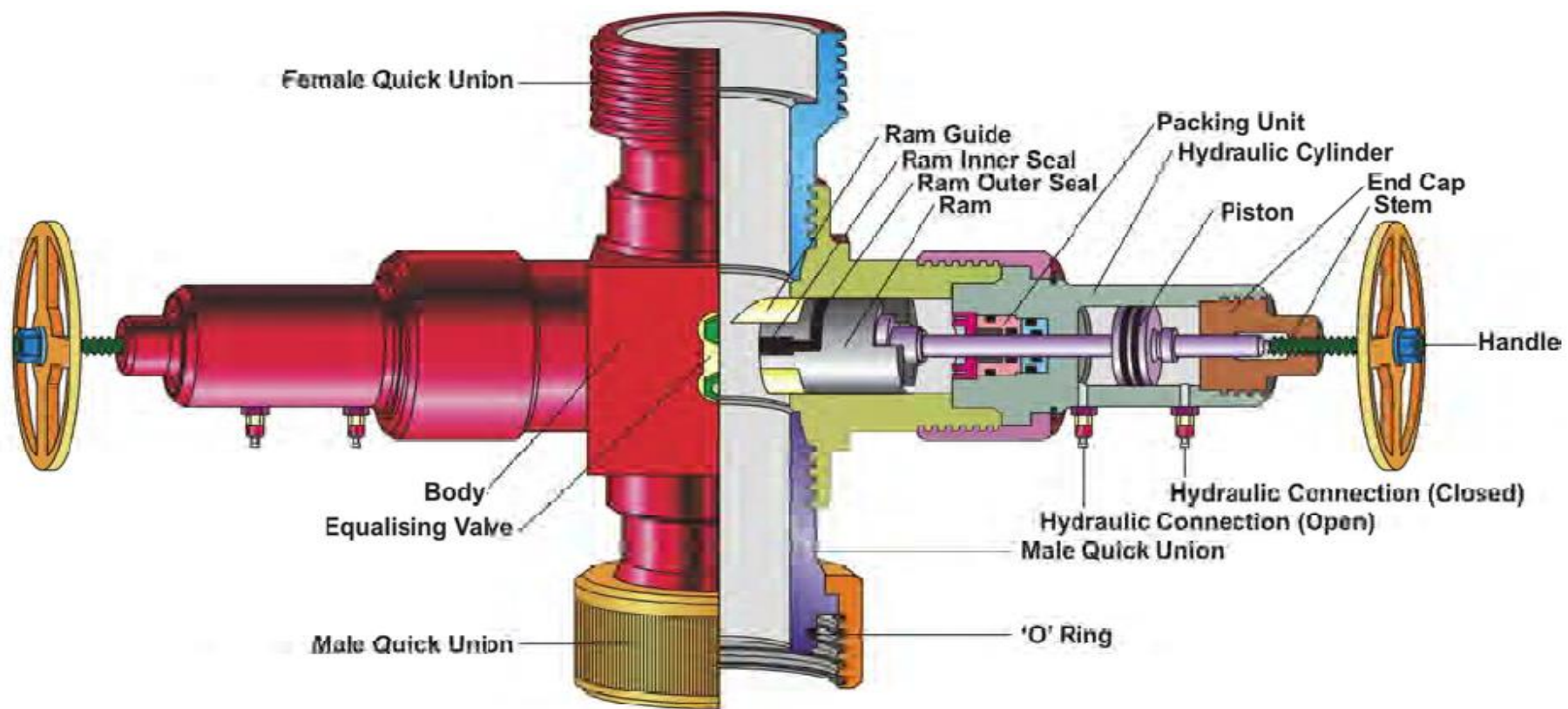
• Typical Wireline Rig up

➤ Blowout Preventers (BOPs)

- ✓ A Slick line BOP (also known as a Wireline valve) is generally installed between the tree connection and lower lubricator section.
- ✓ The BOP provides facilities for contingency and emergency procedures and must be included in all rig-ups.
- ✓ Primary purposes of BOP are to :
 - Enable well pressure to be isolated without cutting the wire by closing the master valve.
 - Allow a wire line cutter to be prepared and dropped if tool string becomes stuck in the well.
 - Enable stripping of wire through closed rams, only when necessary
- ✓ The only exceptions by which a BOP is not rigged up next to the tree connection are:
 - When installing or retrieving BPVs (back pressure valves). Check that this provides sufficient length to close the rams on the wire, i.e., above the rope socket.



Manual BOP



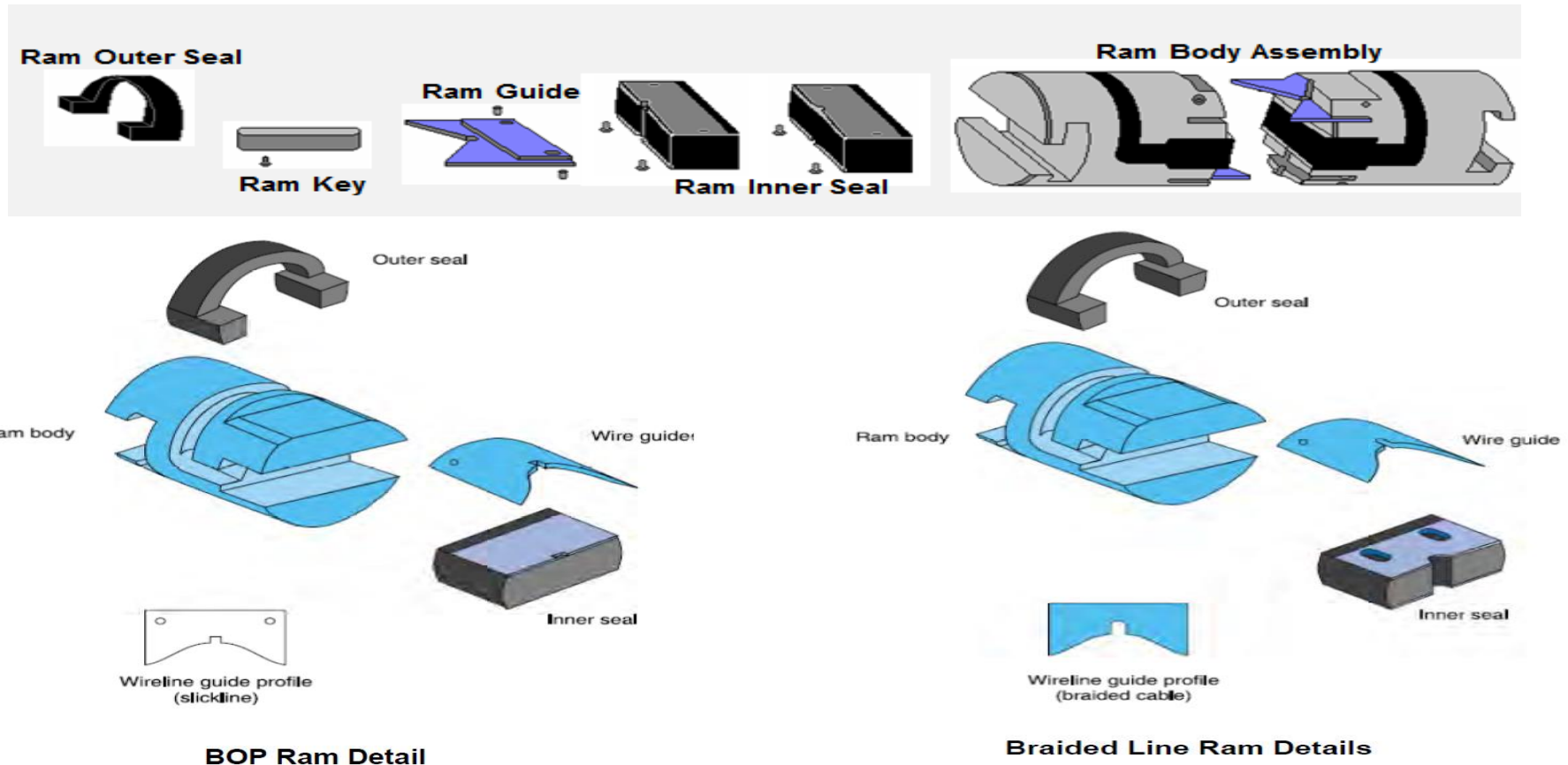
Single Ram Hydraulic Slickline BOP

• Typical Wireline Rig up

➤ Blowout Preventers (BOPs)

✓ Several types or styles of ram are commonly encountered:

- **Slick line rams** - use blind rams with rubber inserts on the sealing faces to seal with or without wire across the rams.
- **Braided line** - use rams with a semicircular groove in the seals to match the line diameter.



- **Typical Wireline Rig up**

- **Blowout Preventers (BOPs)**

- ✓ Slick line BOPs are available in manual or hydraulic operating modes and are used in the following **configurations**:
 - **Single BOP** - Installed between the tree connection and lower lubricator.
 - **Dual BOP** - Double or twin ram BOP are primarily used with braided line. Generally hydraulically operated, a dual BOP is generally configured in a single housing or body containing two sets of rams. To obtain a seal against braided line, a grease injection point is provided between the ram sets.

- **Typical Wireline Rig up**

- **Explosive Decompression in BOP Rams**

- ✓ Gases in a producing well are capable of causing an additional effect on the elastomer. When operating in an environment with high free gas, the elastomer absorbs gas up to a saturation value.
- ✓ If a decompression process occurs and the gas is liberated too rapidly, then the elastomer suffers internal **blistering**, tearing or cracking due to the abrupt gas expansion. This phenomenon is commonly known as explosive decompression and occurs frequently in environments with a high concentration of CO₂.
- ✓ Elastomer permeability, degree of cross-linking as well as its mechanical properties define material resistance to explosive decompression.
- ✓ The use of a slow decompression rate and a minimum elastomer thickness helps in preventing this phenomenon.

- **Typical Wireline Rig up**

- **Blowout Preventers (BOPs)**

- ✓ The key design and operating features (**BOP Operation**) of Slick line BOPs include:

- ❑ **Equalizing Valve**

- All BOPs have a means of equalizing the pressure across the closed rams.
 - Any pressure differential acting on the cross-section of the rams creates a force that makes opening the rams extremely difficult.
 - Attempting to open the rams without equalizing may result in internal damage.
 - The equalizing valve should normally be kept closed so that in an emergency, BOP will hold pressure as soon as the rams are closed.

- **Typical Wireline Rig up**

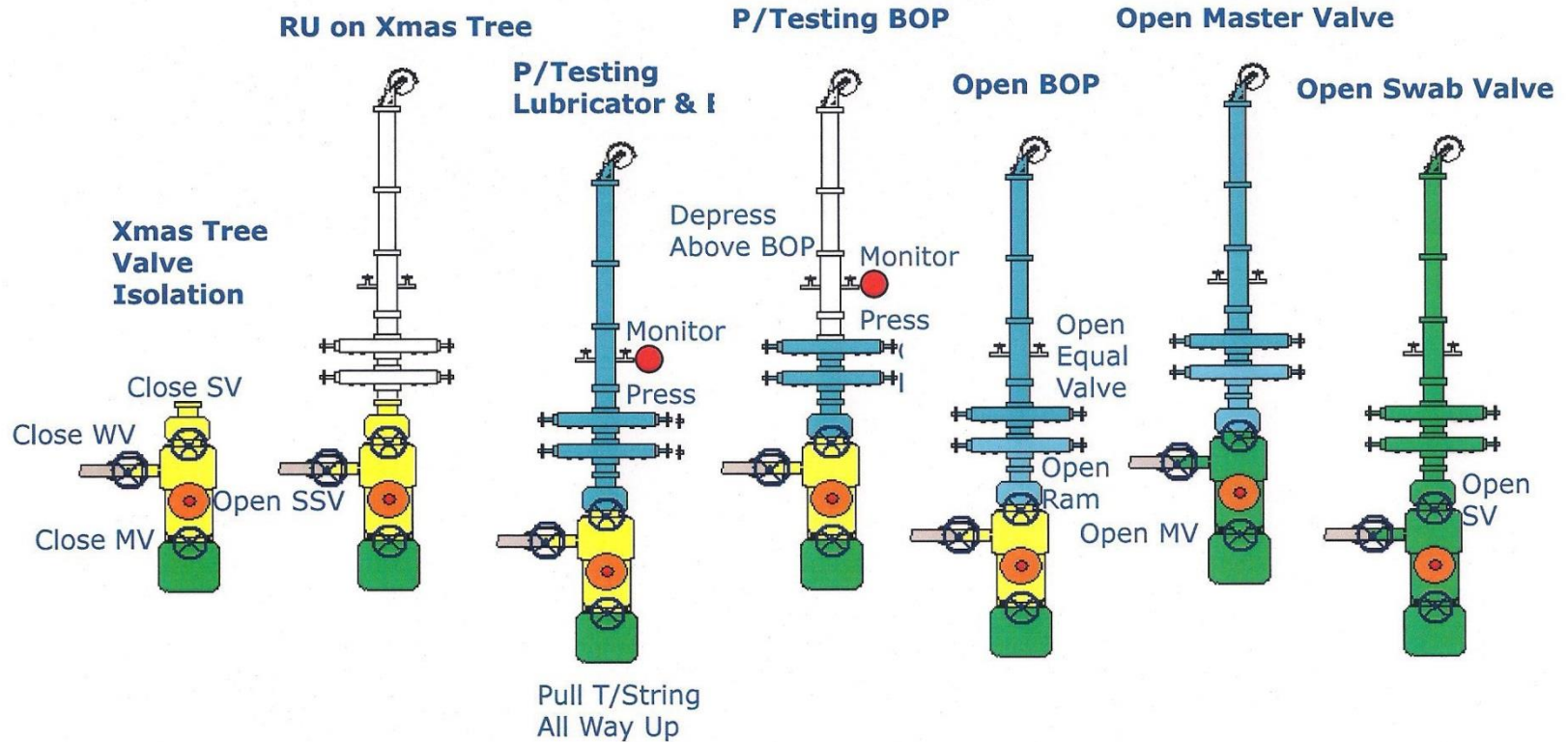
- **Blowout Preventers (BOPs)**

- ✓ The key design and operating features (**BOP Operation**) of Slick line BOPs include:

- ❑ **Function and Pressure Testing**

- Testing performed from below as blind rams are not designed to hold pressure from above.
 - All BOPs should be tested on a regular basis. Full body tests are conducted with rams open & applying 150% of working pressure.
 - Working pressure tests are performed with rams closed, applying 100% of the working pressure to test the ram seal against the wire diameter. A test rod of the same size as the wire to be used should be inserted between the rams.
 - The force created by pressure acting on the cross-sectional area of the test rod will act to push the rod out of the rams. For this reason, it is not recommended that any locally made devices be used unless they have been subjected to recognized inspection and certification.

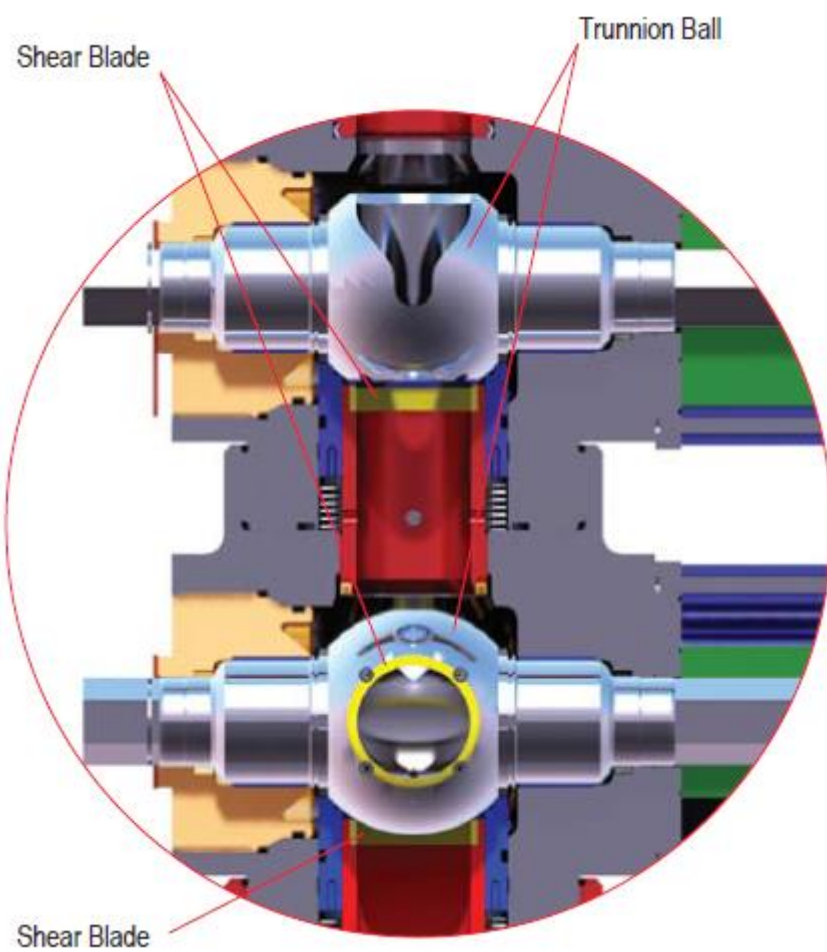
PRESSURE TESTING



- **Typical Wireline Rig up**

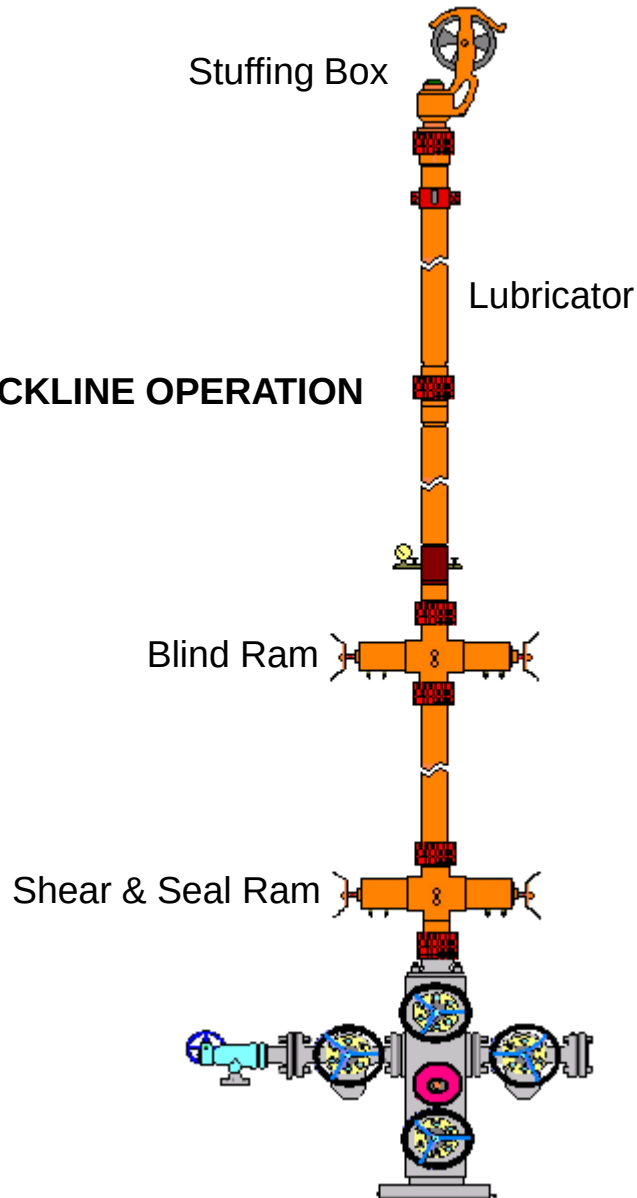
- **Shear/ Seal BOP (Remote control unit)**

- ✓ The shear/seal BOP is installed between the well's Xmas tree and the wireline BOP.
- ✓ It comprises the third barrier in the wireline operated pressure control system.
- ✓ The Shear-seal BOP is hydraulically actuated, but it can also be operated manually.
- ✓ The valve stack is equipped with a shear ram which, when closed, also functions as a seal BOP. The valve is actuated when there is a need to cut the wire at the surface, and to avoid having to utilize the Xmas tree, which might lead to damage to the valves.

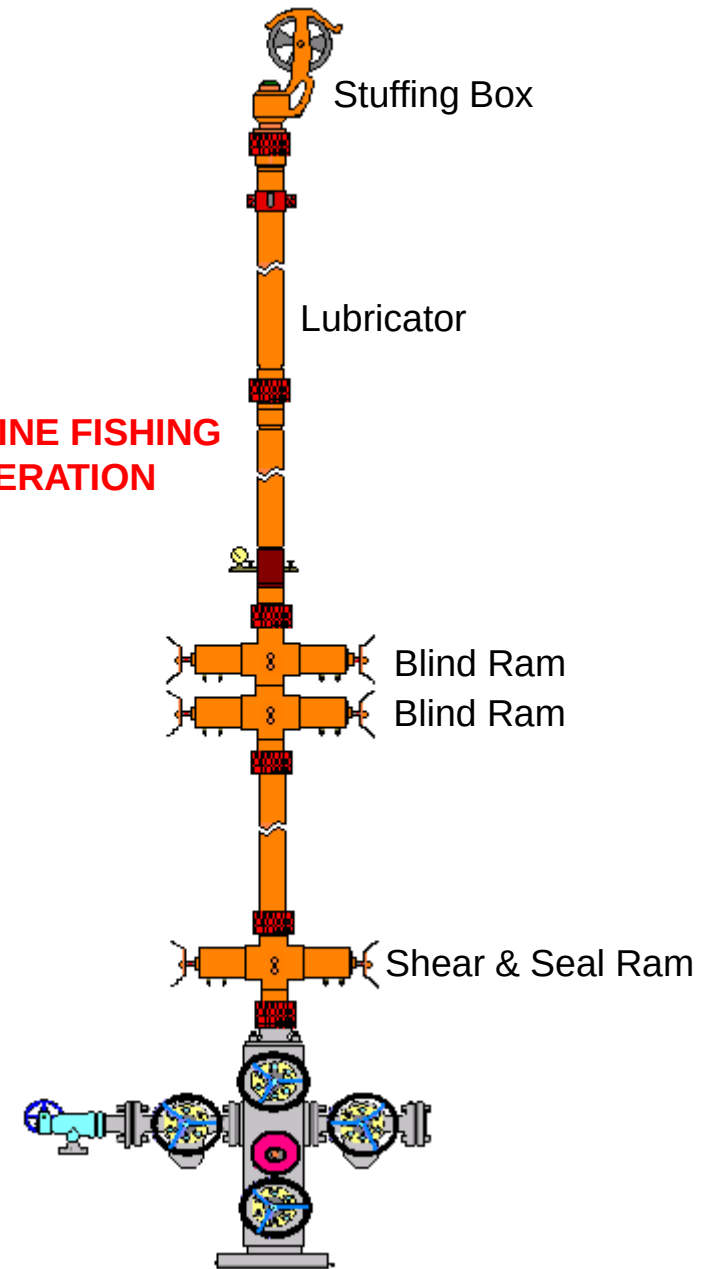


Slickline PCE

SLICKLINE OPERATION



SLICKLINE FISHING OPERATION





Dual Drum

Wire Line

Outlines

- I. Introduction to Wireline
- II. Typical Slickline Rig up
- III. Typical Braided Line Rig up
- IV. Managing A Leak or Malfunction on Surface
- V. Operational Considerations

Grease Control Head

Lubricator

Dual BOP

T-Piece

Shear/Seal BOP

Riser

Xmas Tree
Upper Master Valve

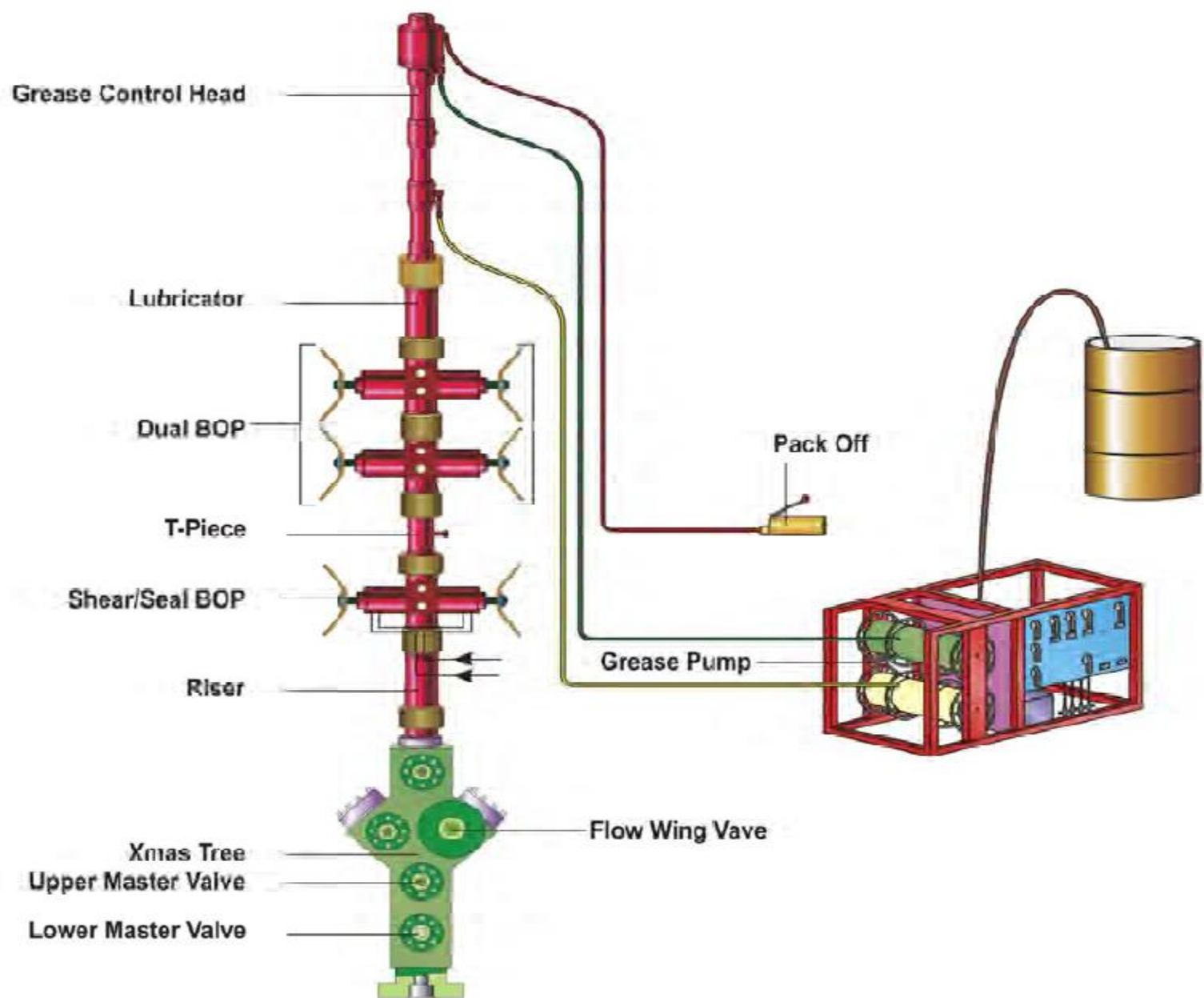
Lower Master Valve

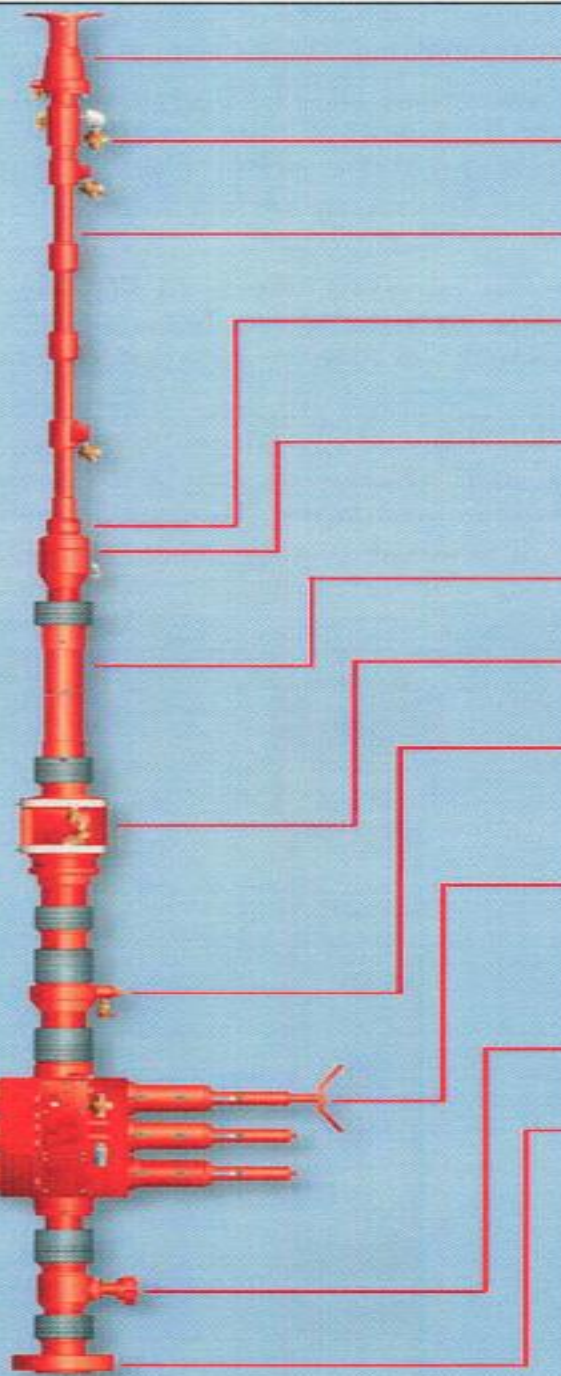
Pack Off

Grease Pump

Flow Wing Vave

Braided Line Rig up



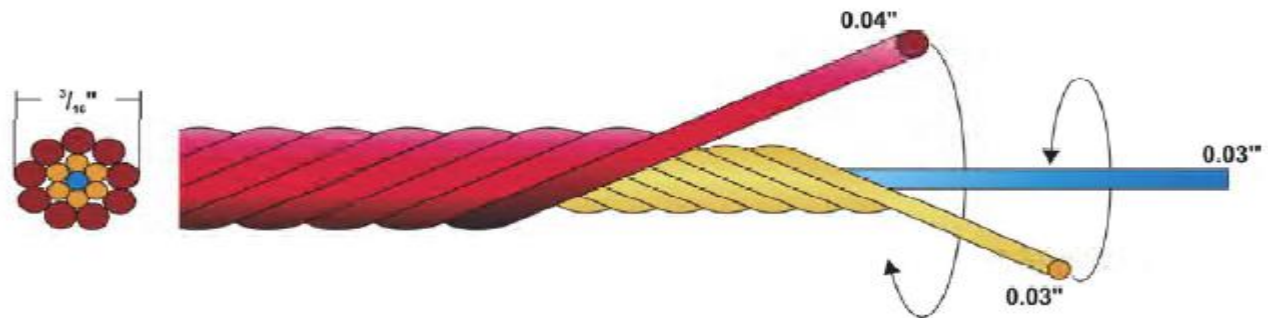


- **Line Wiper:** Wipes the moving cable clean of excess grease when pulling-out of the well.
- **Wireline Stuffing Box:** Packs-off around a stationary cable if grease seal is lost.
- **Grease Injection Head:** Controls and contains well pressure while allowing cable travel in/out of a well.
- **Ball Check Valve:** The Ball Check Valve is a safety device for installation below the grease injection control head. It will seal off the well in the event of the cable being broken/pulled out of the Grease Injection Head.
- **Tool Catcher:** The Tool Catcher is used to catch and hold the tool safely both during pressure testing and in the event of the cable being inadvertently pulled off at surface thus preventing a possible fishing job.
- **Lubricator Sections:** The Lubricators are used to insert and retrieve a tool string on a well under pressure.
- **Tool Trap:** The Hydraulic Tool Trap with External Indicator protects the wellbore from inadvertent tools pull-off.
- **Quick Test Sub:** The Quick Test Sub is designed to save substantial rig time while pressure testing the Wireline Pressure Control Equipment String in multiple run operations.
- **Wireline Valve:** The Wireline Valve is a mono-block construction and is designed for use in single, dual, triple or quad configuration - Multiline ram seals allow the use of the same seal configuration for electric line and slickline.
- **Pump-in Sub:** The Pump-in Sub is designed to allow introduction of high volumes of fluid into the well.
- **Wellhead Adapter Flange:** The Wellhead Adapter Flange provides a means of connecting in a safe manner the Wireline Pressure Control Equipment String to the wellhead.

• Typical Braided Line Rig up

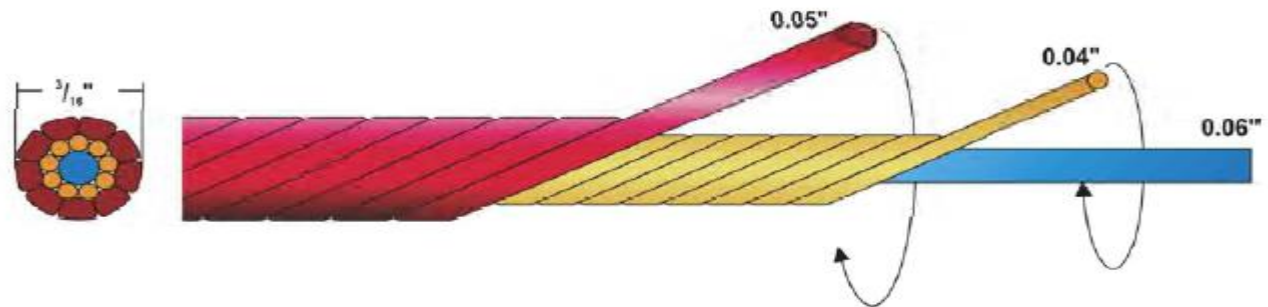
- ✓ Braided lines with no internal conductor are called sand lines and a rig usually has one on a special winch used for fishing etc.
- ✓ They are much stronger than electric line, usually at least double the breaking strength.
- ✓ Electric line has either single or multiple conductors.

- Left over right hand lay
- No twisting



Component Strands of Conventional Braided Line

- Right hand lay
- Dyform
- Better sealing

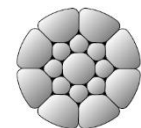
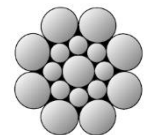
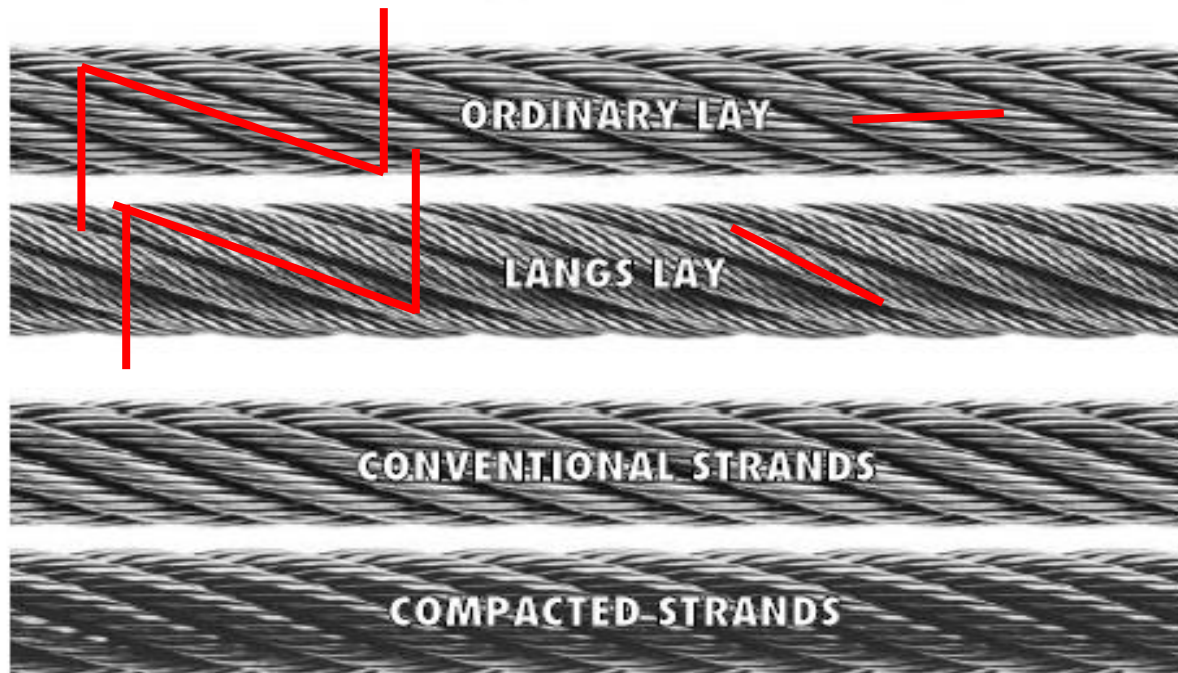


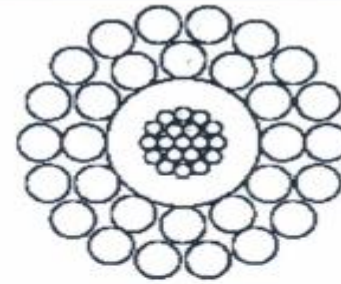


S



Z





Example

Number of Electrical Conductors

Armor Package

(Number Inner Wires/Number Outer Wires)

B - 12/24	J - 24/24
C - 18/24	K - 15/15
E - 12/15	L - 12/12
F - 11/15	N - 12/18
G - 10/16	Q - Other
H - 18/18	

Nominal Cable Diameter In Hundredths of an Inch

Type of Electrical Copper Strand

R - 6 wires around 1 central wire
S - 6 wires around 1 central wire Nickel Coated
P - 12 wires around 6 wires around 1 wire
W - 12 wires around 6 wires around 1 wire Nickel Coated
T - 18 wires around 12 wires around 6 wires around 1
Y - Others

Type of Electrical Insulation

P - Propylene Copolymer
T - Teflon*
Z - Tefzel*
X - Camtane
A - PFA*

*Dupont trade mark

1 N 32 P P



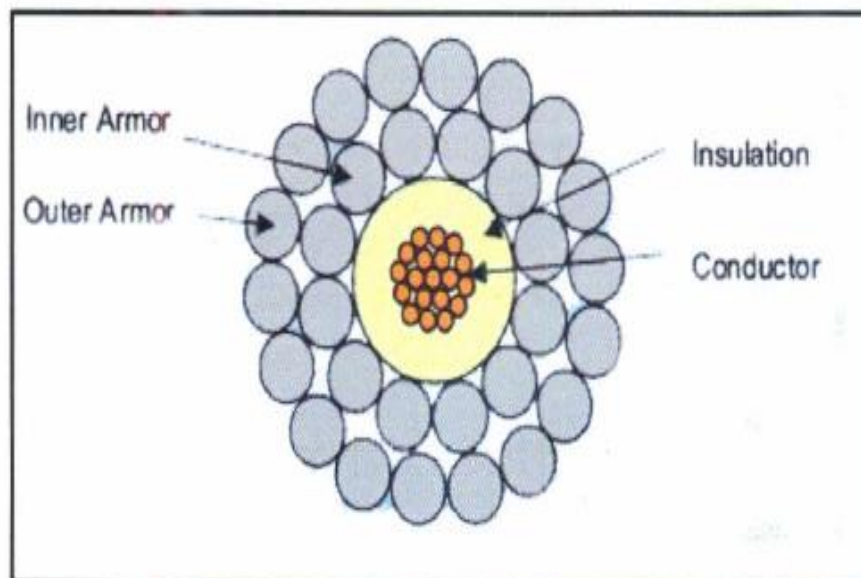
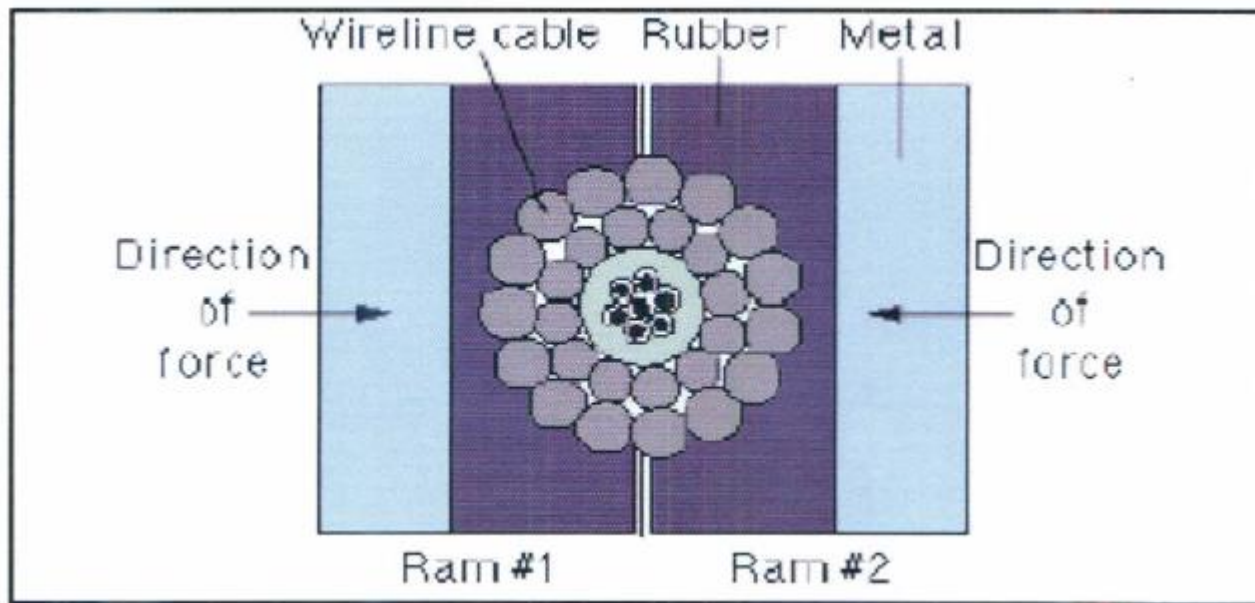


Figure 10.2.55: Mono-Conductor Cable

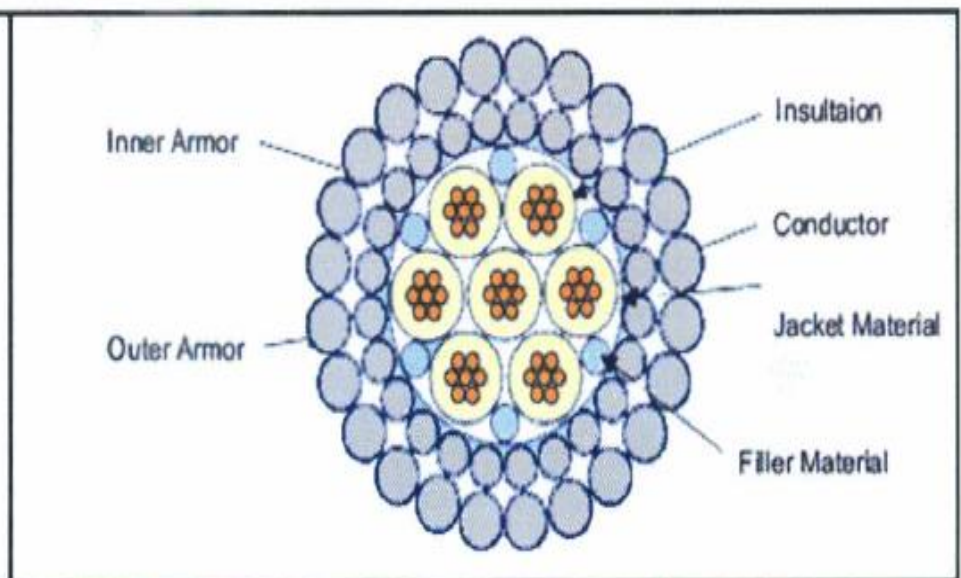
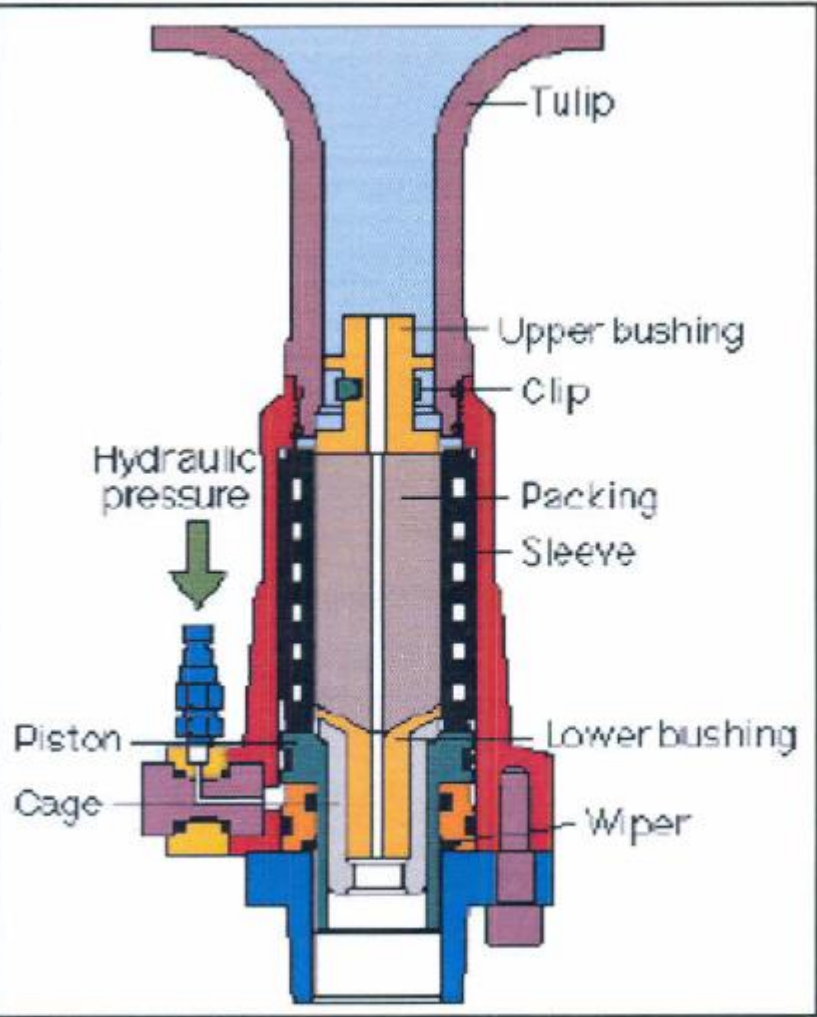
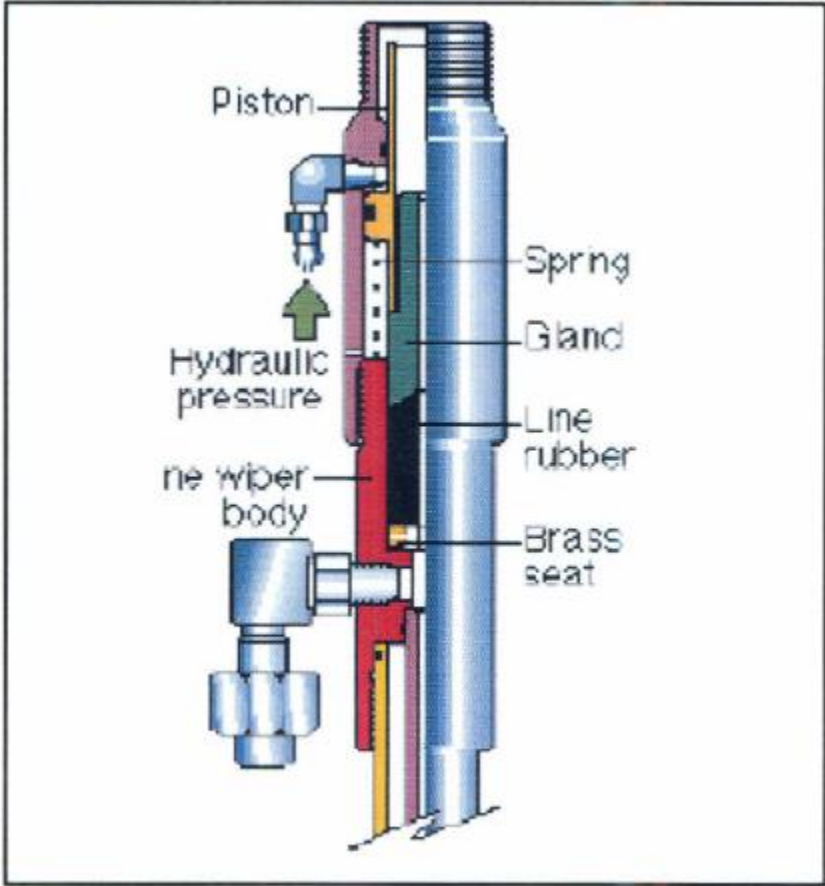


Figure 10.2.56: Seven Conductor Cable



Line Wiper



Pack-off

• Typical Wireline Rig up

➤ Grease Injection Control Head

- ✓ The grease injection control head has the same functions as a stuffing box when braided cable is being utilized.
- ✓ Grease is pumped through the lower nipple of injection port and up through the wireline and the seal tubes.
- ✓ A minimum of three lengths of seal tubing is essential on an injection head to reduce the pressure.
- ✓ The pressure is controlled from a separate panel and is usually connected to the unit by means of an air-driven pump. The grease is withdrawn through a return nipple below the packer unit.
- ✓ Before the start of operations we should check that the sealing tubing (flow tube) is suitable for the dimensions of the wireline. Excessively large sealing tubing or a worn wireline will mean excessive grease consumption.
- ✓ We should also make sure that sufficient grease is available for the complete operation before we start working.

PCE – GREASE INJECTION HEAD

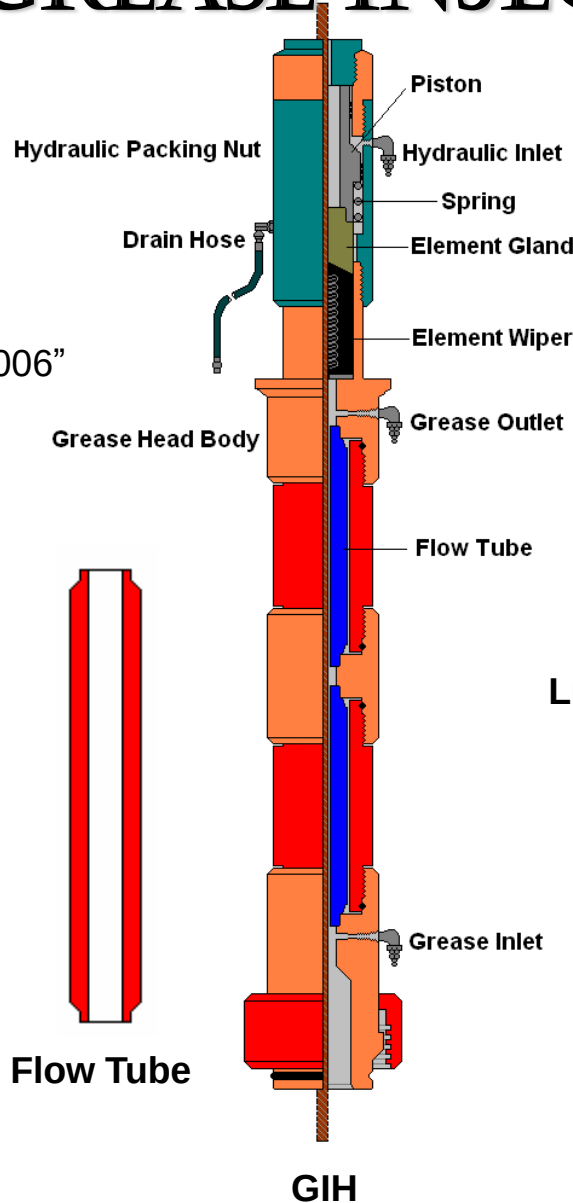
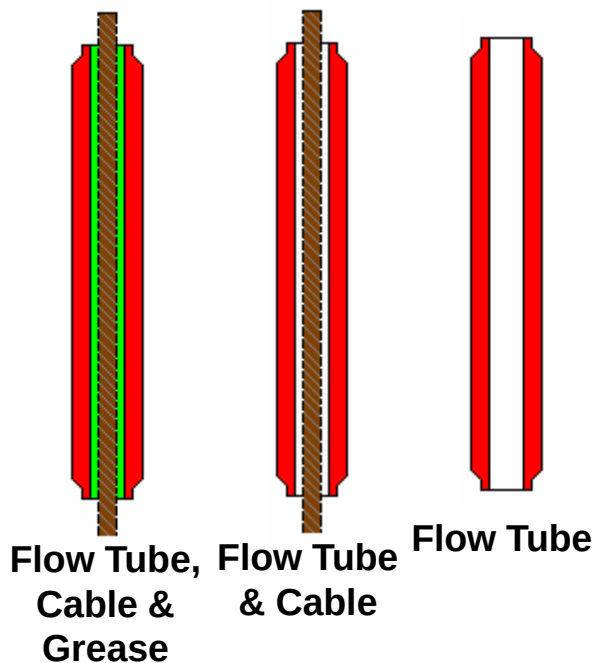
Flow Tube Range:

2 Tubes: <2K psi

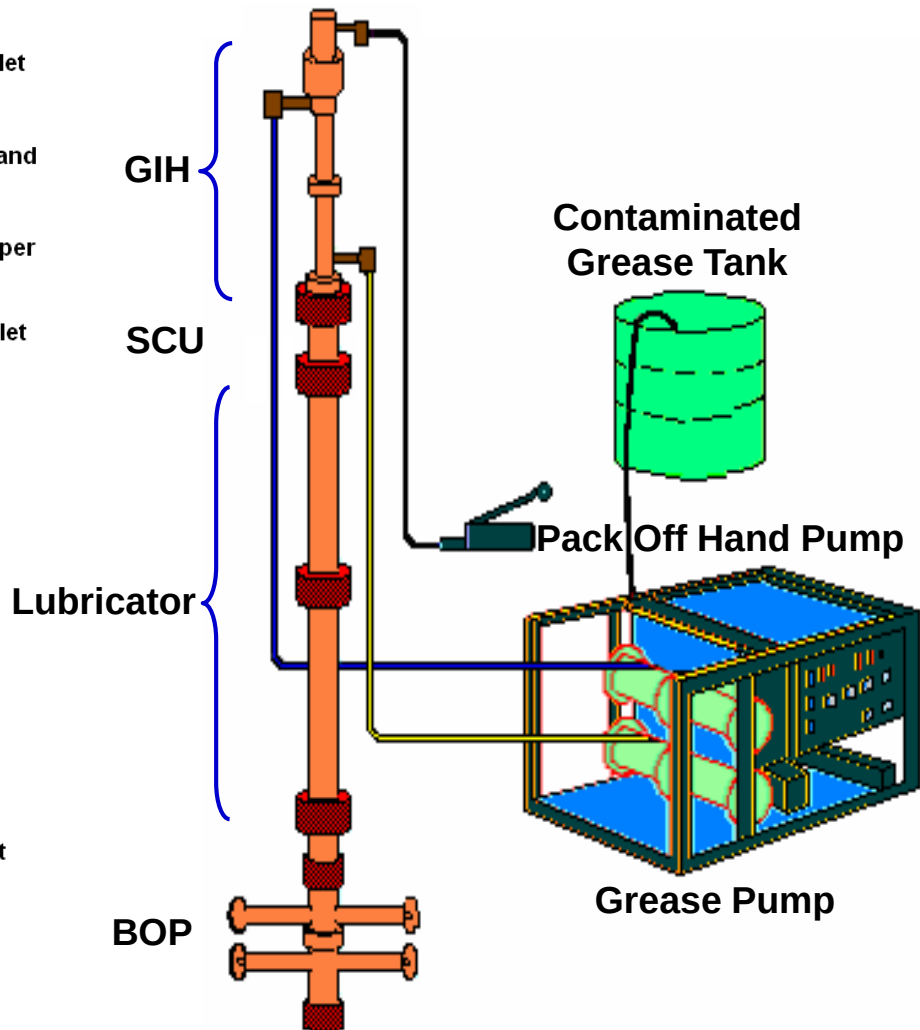
3 Tubes: <10K psi

4 Tubes: <15K psi

Clearance: 0.004" – 0.006"

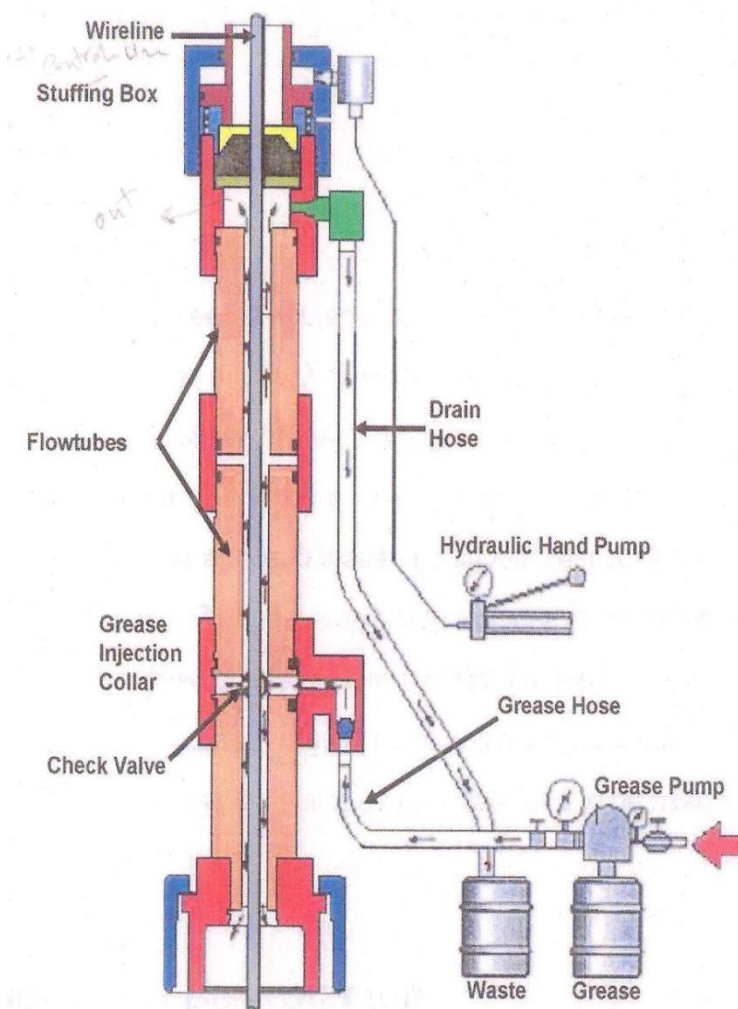


GIH

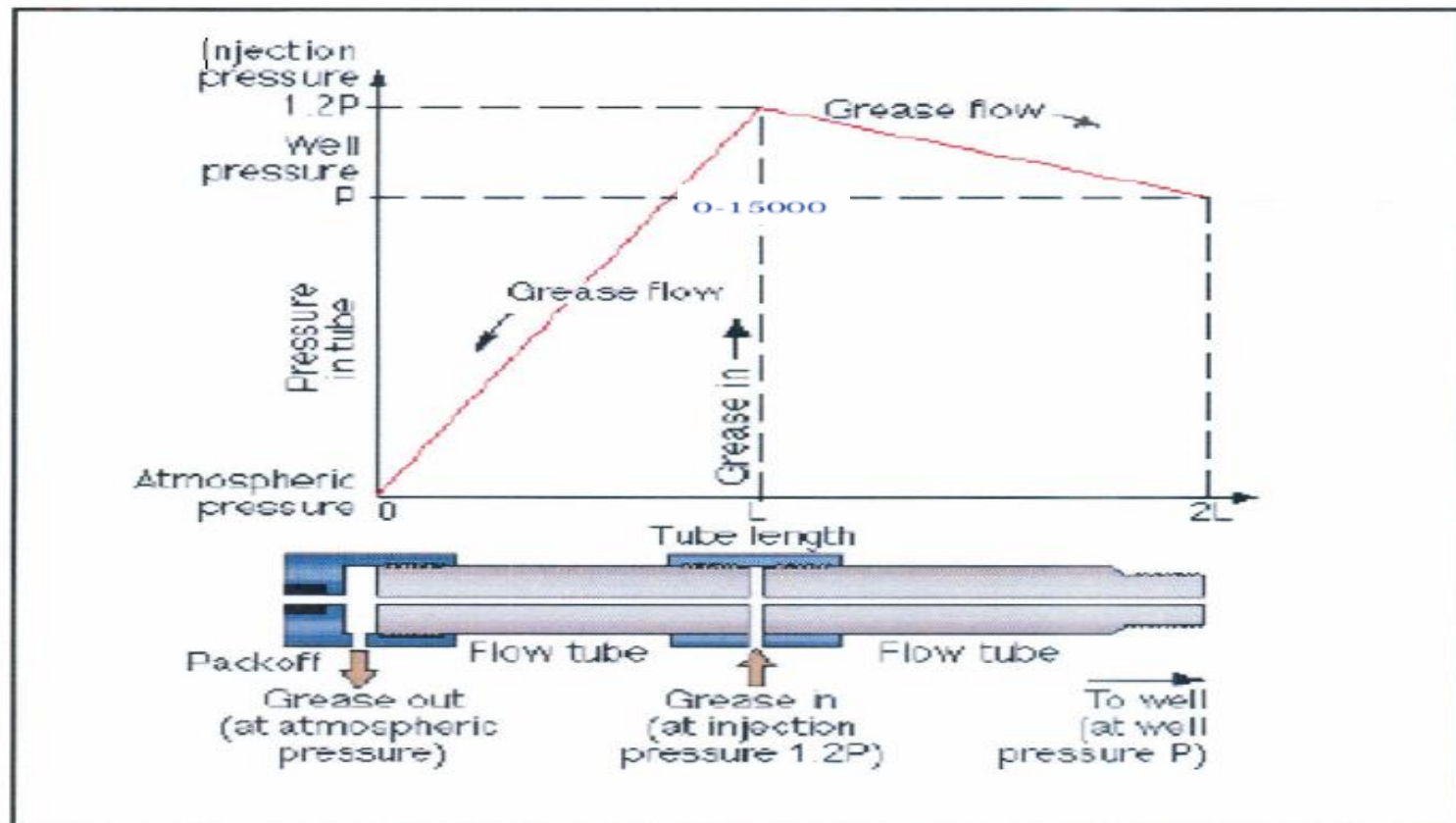


GREASE SYSTEM

Well Pressure	Type	No. Flow Tubes
0-5000psi	Liquid	3
0-5000psi	Gas	4
5-10000psi	Liquid	5
5-10000psi	Gas	5-6
10-15000psi	Liquid	6
10-15000psi	Gas	6 or more



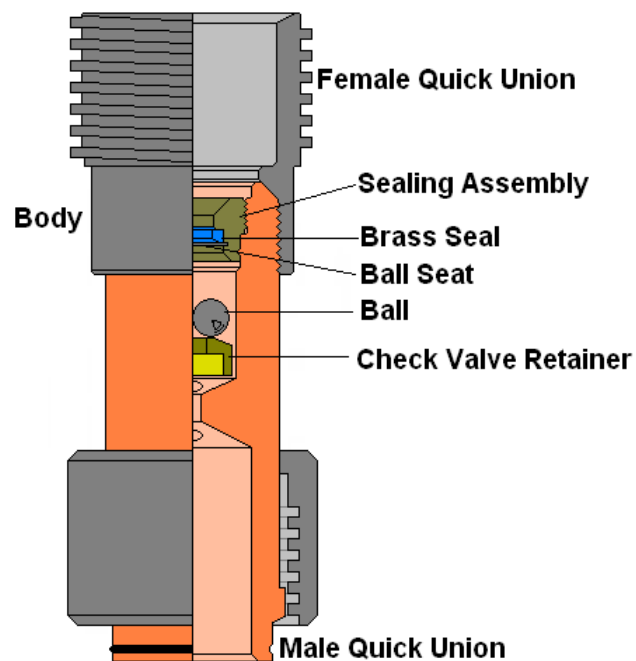
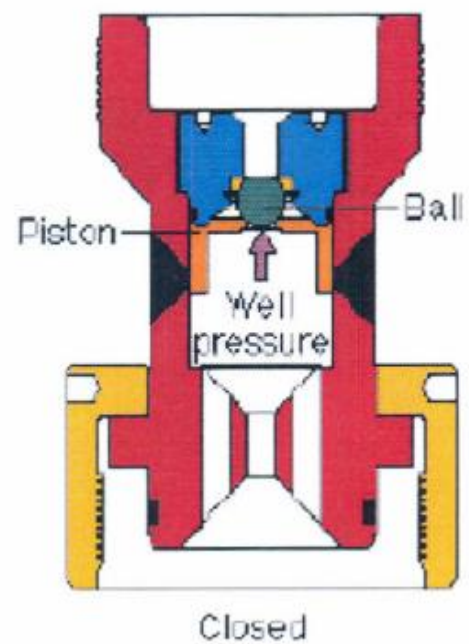
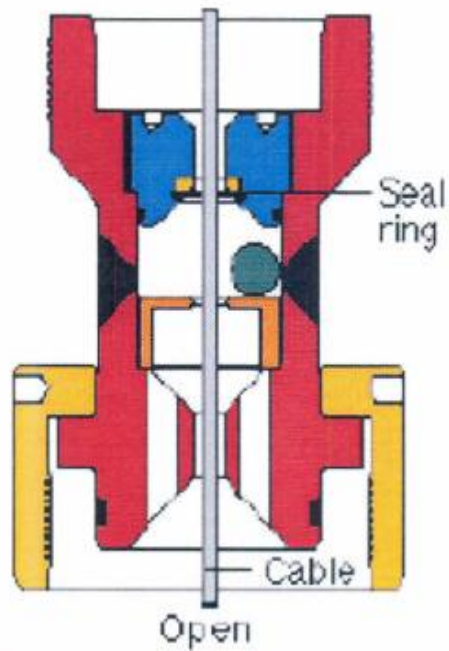
	WORKING PRESSURE (Psi)	OPERATING PRESSURE (Psi)	OPERATING SYSTEM
LINE WIPER	0-15000	500-1000	PUMP CLOSE SPRING OPEN
PACK OFF	0-15000	0-3000	PUMP CLOSE SPRING OPEN
GREASE INJECTION HEAD	0-15000	1.5 X WELLHEAD PRESSURE	PUMP TO PRESSURE



- **Typical Wireline Rig up**

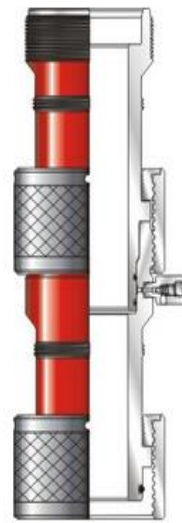
- **Safety check union/sub**

- ✓ The safety check union/sub is intended to retain the pressure if the wireline breaks or is pulled out of the wireline head.
- ✓ It is installed just below the injection head and is a check valve in the form of a ball which is pressed against a non-extrusion ring/seating.
- ✓ Before installation and use, the valve should be checked, and if any damage is found the ball and its brass seating should be replaced.
- ✓ The use of SCU is compulsory during W/L operations in sour gas & oil wells.
- ✓ SCU must be installed between upper section of lubricator & injection Nipple



- **Typical Wireline Rig up**

- **In-Situ Pressure Test sub**



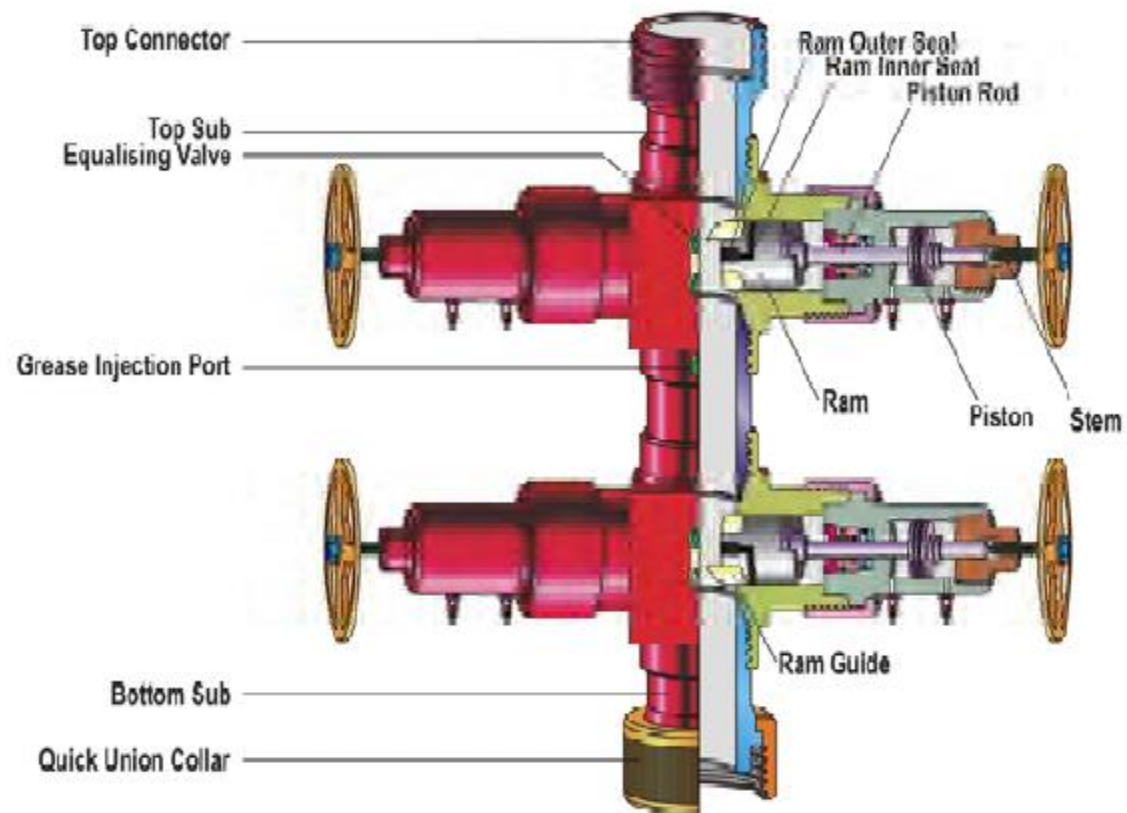
The In-Situ Pressure Test Sub (sometimes called Quick Test Sub) was originally designed to reduce the time spent pressure testing a lubricator rig up, prior to exposing it to live well pressure. In situations where the lubricator has to be tested each time the bottom connection is broken, much time can be lost.



• Typical Braided Line Rig up

➤ Braided Line BOPs

- ✓ Ram type BOPs are designed to seal with the cable static, and cable movement should be stopped before closing the rams.
- ✓ Braided line dual ram BOPs are configured with the lower set of rams inverted, and with a grease injection port in between the rams. This allows the two sets of rams to trap a cavity full of grease between them of higher pressure, preventing escape up or down.
- ✓ By filling the cavity at a pressure higher than wellhead pressure, the grease fills the spaces and prevents escape.
- ✓ This is mandatory in gas wells since gas will migrate up the cable between the inner and outer armor.

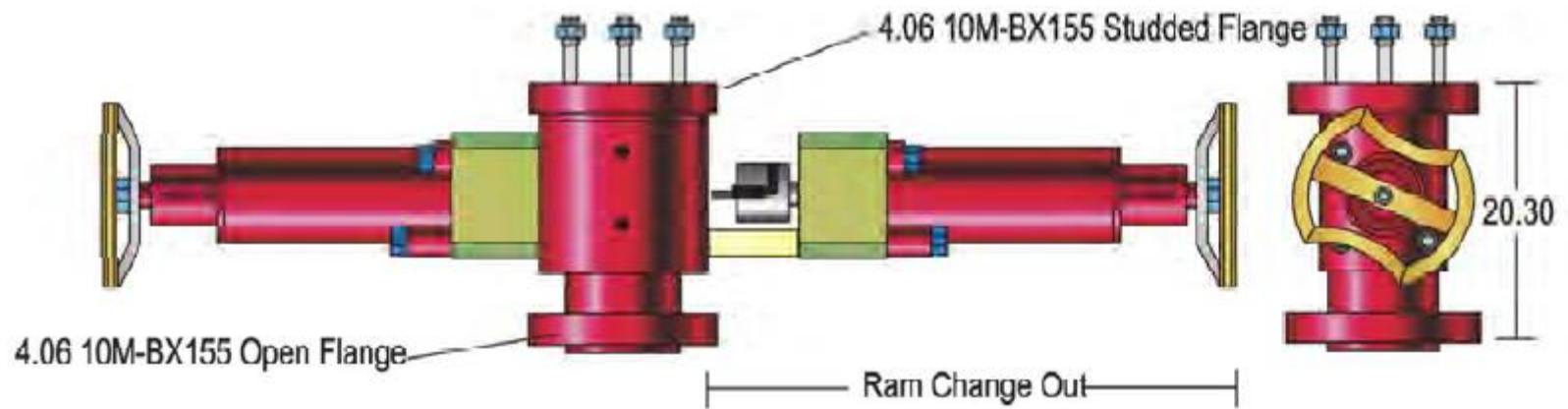
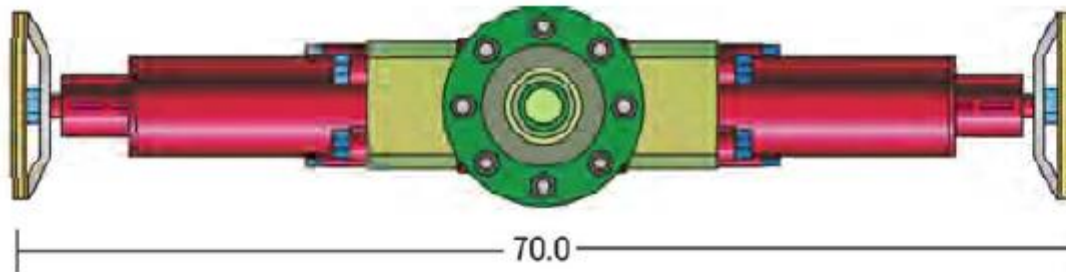


Braided Line Dual Hydraulic BOP

• Typical Braided Line Rig up

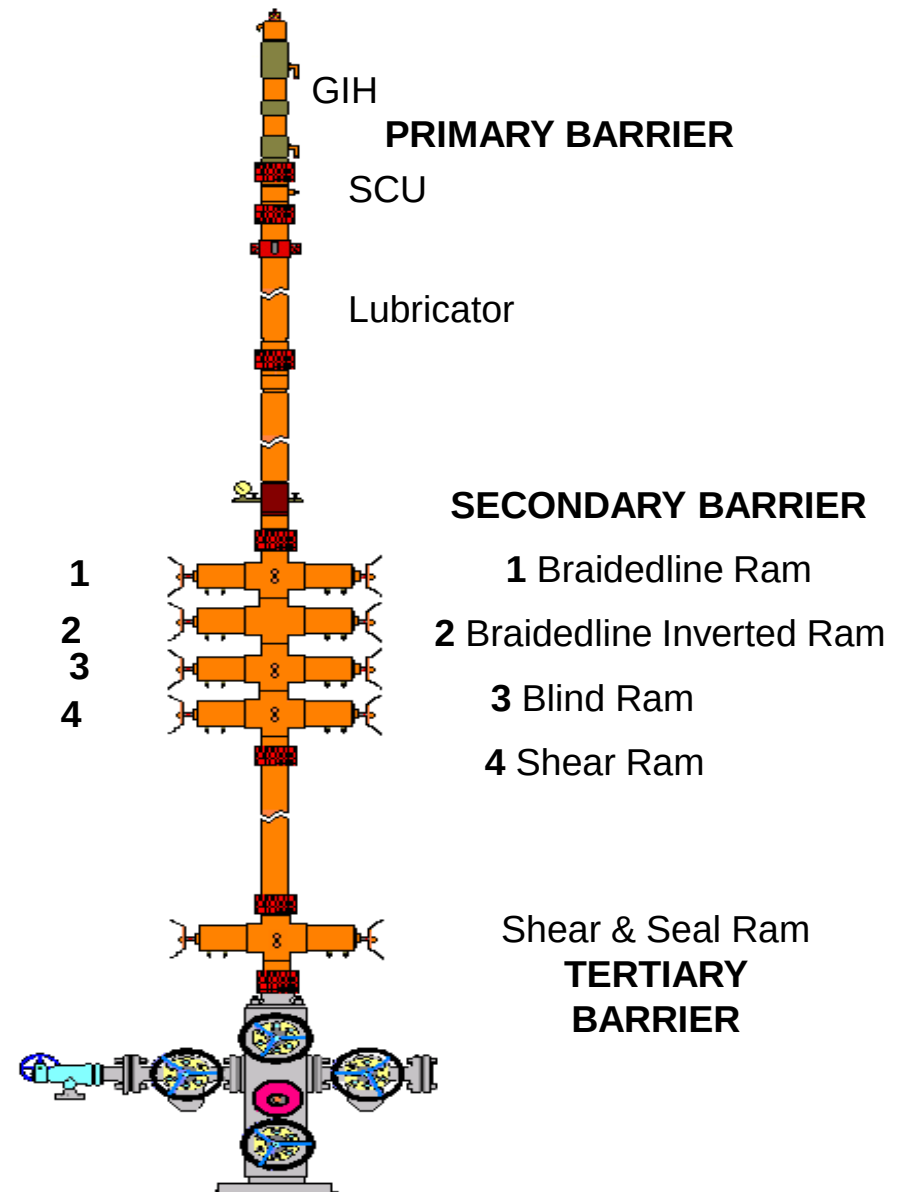
➤ Shear Valves (Tree/Stand-Alone)

- ✓ During W/L operations it may become necessary to cut the wire.
- ✓ Some Xmas Tree valves are designed to do this without damaging the valve. In some cases, e.g. when fishing with heavy duty wire, it may not be advisable to do so.
- ✓ In such case, it may be necessary to include a purpose designed shear valve, mounted just above the tree.
- ✓ It is a ram type shear/seal BOP as it has superior cut capacity.
- ✓ The valve should cut the wire or the BHA. It has the additional advantage of sealing at the bottom of the riser.
- ✓ **Shear valves should be considered when:**
 - The DHSV is locked permanently open using a sleeve. This means that it can't be used as a barrier with W/L out of the well.
 - The lubricator is not long enough to contain the whole tool string. If X/T valves leak, the lubricator could not be isolated.
 - An extra barrier is required due to the nature of the operation, or the equipment configuration.



Shear Valves

BRAIDEDLINE PCE



• Typical Braided Line Rig up

➤ Pressure Testing

- ✓ Each time the tool string is loaded into the lubricator the system should be pressure tested prior to opening up the well and running in hole.
- ✓ The test value should be decided beforehand, and would commonly be 110% of SIWHP, rather than design pressure.
- ✓ It is unnecessary to test to several thousand psi when the SIWHP of a depleted reservoir may only be a few hundred psi.
- ✓ A typical test will consist of pressuring up the lubricator (and riser, if used) against the closed tree swab valve, to the working pressure of the lowest pressure rated item of equipment in the rig up.

• Typical Braided Line Rig up

➤ Pressure Testing

✓ When using braided line:

- Pump at a sufficient rate to vent air via the grease injection head. As the liquid begins to leak out at the grease injection head slow the pump rate down and increase the grease injection pressure to effect a seal.
- Continue to pressure up with the test pump and hold stabilized pressure for the prescribed time.
- The grease injection system should always be pressure tested to its maximum, irrespective of the well pressure expected.

✓ When using slick line:

- While pumping, vent air through lubricator bleed off line
- When water appears at bleed off line reduce pump rate and close in needle valve
- Continue to pressure up with the test pump and hold stabilized pressure for the prescribed time.

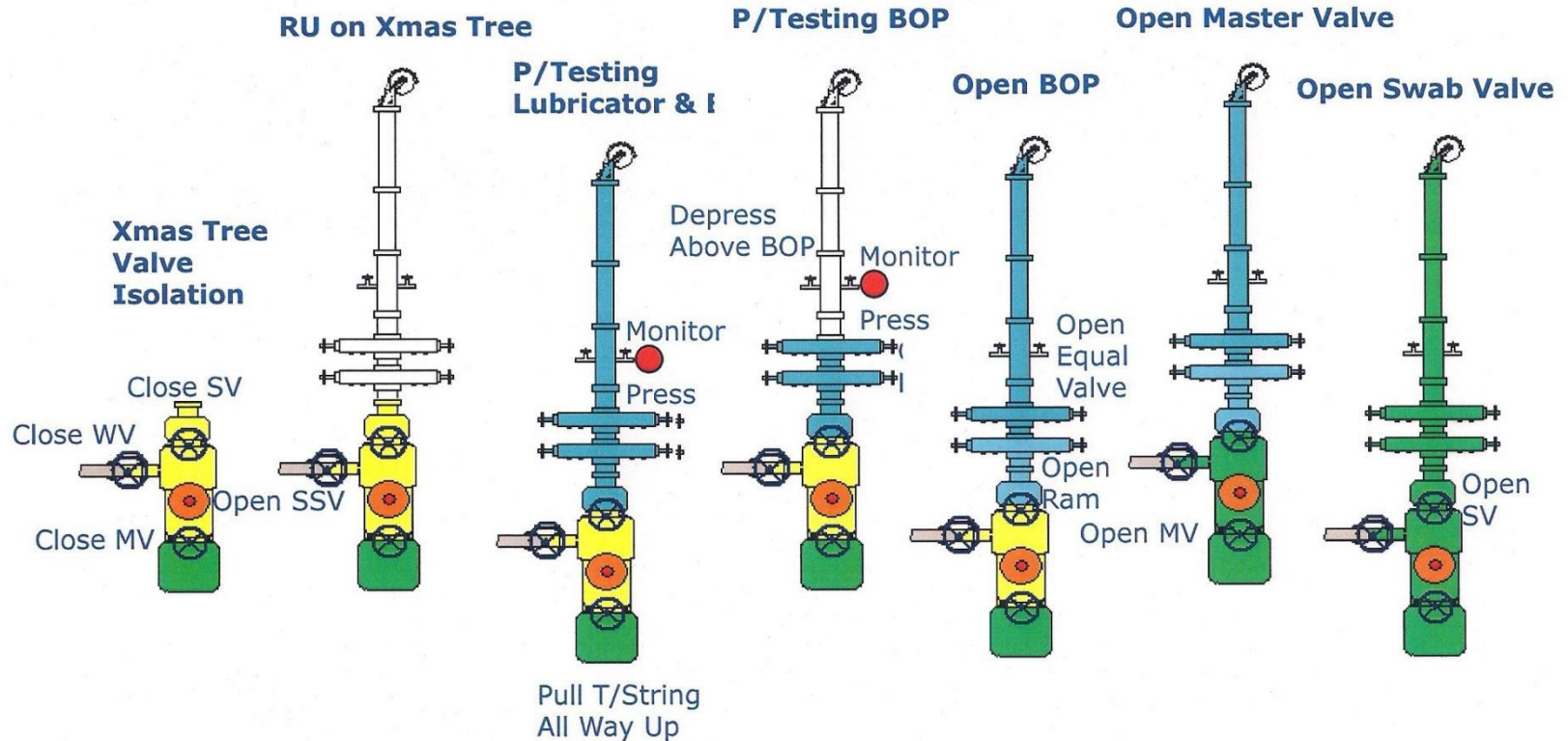
• Typical Wireline Rig up

➤ Pressure Test Procedure

❑ Primary Pressure Test

1. After completing rigging up with the tool string in the lubricator and the Xmas tree valves closed, connect the test pump to the lubricator.
2. With the lubricator bleed off valve open and the stuffing box packing de-energized, pump water (dosed with glycol if hydrate formation is a risk; 60/40 or 50/50) and fill the lubricator.
3. Close the bleed off valve and energize the stuffing box.
4. Slowly bring the pump pressure up to the test pressure according to the well program but never to exceed the lubricator assembly WP (which is the WP of the lowest rated component).
5. Hold the pressure for 10min observing that there is no leakage.
6. Bleed off the pressure, record the test in the daily report and begin well operations.

PRESSURE TESTING



Max. SITHP + 500 psi safety factor (in stages)

• Typical Wireline Rig up

➤ Pressure Test Procedure

❑ Subsequent Pressure Tests

- Subsequent tests may be performed test using the well fluid pressure before making any further runs.
- The pressure test will be carried out as per the following procedure:
 1. With the lubricator installed and the bleed off valve close, slowly open the swab valve until pressure is observed on the gauge on the bleed off port. Stop the opening the valve and allow the pressure to stabilize.
 2. Verify the SITHP pressure on the gauge.
 3. Check for any leaks on the adapter flanges, quick unions or stuffing box.
 4. If there is no leakage, fully open the swab valve.
 5. Continue with operations.

Wire Line

Outlines

- I. Introduction to Wireline
- II. Typical Slickline Rig up
- III. Typical Braided Line Rig up
- IV. Managing A Leak or Malfunction on Surface**
- V. Operational Considerations

- **Managing A Leak or Malfunction on Surface**

- **Stuffing Box Leak (Slickline)**

- ✓ This leak is easily avoided by:
 - Correct packing nut setting (not over tightened)
 - Regular inspection of packing and changing as required.
- ✓ Leaks are most often cured, by simply increasing the compression accordingly.
- ✓ If tightening the packing nut does not cure the leak at that point, particularly in oil wells at moderate pressures, small leaks may be acceptable in order to pull out of hole and make repairs. Alternatively, it may be acceptable to close one set of BOPs and strip through them.

• Managing A Leak or Malfunction on Surface

❑ **Stuffing Box Leak (Slickline)**

- ✓ Otherwise, the stuffing box will have to be repaired by replacing the packing. Normally this repair is made by cutting the wire. To do this:
 1. Stop cable movement
 2. Close both sets of wire line BOPs and bleed down lubricator pressure
 3. Inflow test BOPs to ensure they are sealing. This is done by observing that the lubricator pressure stays at zero after closing the bleed valves after bleeding down
 4. Open the bleed screw in the stuffing box to ensure that no pressure remains
 5. Unscrew the gland nut completely to expose the packing elements
 6. Clamp and cut the wire
 7. Withdraw the packing elements and remove them from the wire
 8. Ream the new packing element with a piece of wire and place them on the wire and Rejoin the wire.

- **Managing A Leak or Malfunction on Surface**

- ❑ **Stuffing Box Leak (Slickline)**

- ✓ A temporary repair (if it is not possible to cut wire) may be made by splitting the packing with a knife diagonally to allow them to be placed on the line without cutting it.
 - ✓ This should be considered a last resort emergency measure only, and this temporary packing should be changed as soon as the tool is retrieved.

• Managing A Leak or Malfunction on Surface

❑ Grease Seal Leak (Braided Cable Operations)

- ✓ A leak past the grease injection head may be due to:
 - Pulling out (or running in) too fast, particularly on the first run (dry cable)
 - Setting the grease injection pressure too low
 - An increase in wellhead pressure, e.g. after perforating
 - Not enough grease in the supply tank
 - Restrictions in the grease supply system
 - Low grease pump air supply pressure
- ✓ However a leak could also be as a result of one or a combination of the following factors:
 - Grease type incompatible with conditions, e.g. ambient temperature or freezing due to gas escape
 - Contaminated grease becoming thinner
 - Incorrect flow tube insert ID or worn flow tubes
 - Insufficient flow tube length.

• Managing A Leak or Malfunction on Surface

❑ Grease Seal Leak (Braided Cable Operations)

- ✓ It may be possible to re-establish the grease seal in the following ways;
 - Increase grease injection pressure
 - RIH slowly to:
 - 1) Help the grease flow into the well rather than out
 - 2) Pass a well-greased section of wire through the flow tubes.
- ✓ The stuffing box may be used to slow the leak and help the grease seal re-establish itself as following;
 - Stop all cable movement then Close stuffing box
 - Close the valve at the end of the grease return
 - Once the leak has been contained, re-open the grease return line and allow grease to circulate before continuing the operation.

• Managing A Leak or Malfunction on Surface

❑ Grease Seal Leak (Braided Cable Operations)

- ✓ In Gas Wells; during braided line operations in gas wells, if the seal is lost it will prove harder to re-establish because:
 - The increased volume of gas compared to oil will tend to strip the grease out of the flow tube
 - Lube oil based grease becomes contaminated and thins (loses viscosity)
 - The cooling effect of gas escape thickens the grease.
- ✓ The escaping gas must be immediately controlled to prevent freeze and hydrate formation.
 - Stop cable movement,
 - Close BOP, inject grease between rams & bleed lubricator pressure
 - Once the escape is stabilized, a number of corrective steps can be taken in an attempt to re-establish the grease seal. These are:
 - 1) Wait for the freeze up to thaw, or assist with a steam hose.
 - 2) Pump heavy oil first to re-establish grease circulation.
 - 3) Circulate grease for some time to remove all thin grease or oil remains.
 - 4) Change grease type if possible.

- **Managing A Leak or Malfunction on Surface**

- ❑ **Grease Seal Leak (Braided Cable Operations)**

- ✓ If none of the above actions helps to abate the leak, the cable may have to be cut to allow the seal assembly to be inspected and repaired/reconfigured.
 - ✓ There are a number of possibilities which become possible once the cable has been cut:
 - Worn flow tubes replaced, or smaller ID inserts used
 - More flow tubes added
 - Additional grease injection points added.

• Managing A Leak or Malfunction on Surface

❑ Grease Seal Leak (Braided Cable Operations)

✓ The procedure for cutting the cable is as given below.

- Stop cable movement
- Close both sets of wire line BOPs and bleed down lubricator pressure
- Inflow test BOPs to ensure they are sealing
- Disconnect lubricator and secure wire above BOPs with fishing clamp
- Pull a few wraps a wire off cable drum and cut wire close to winch unit, pull wire out of stuffing box and grease seal assembly
- Make the necessary inspection and or/adjustments to the sealing assembly
- Rethread wire through stuffing box and knot free end at winch unit, spool loose wire back onto drum
- Remove fishing clamp and reconnect lubricator
- Equalize pressure and open BOPs
- Pull out of hole and replace knotted section of wire if sufficient is remaining on drum.

• Managing A Leak or Malfunction on Surface

❑ Leak In Lubricator

- ✓ Although the lubricator will have been pressure tested prior to commencing operations, the action of jarring etc. can induce bending in the lubricator and hence cause leaks at the connections probably as a result of 'O' ring failure.
- ✓ It is good practice to visually inspect and replace the 'O' ring at the connection used every time the lubricator is broken open.
- ✓ It may be sufficient to remove the damaged 'O' ring, split a new one diagonally with a knife, place over the cable, join it with a adhesive from seal manufacturers then place in the groove.

❑ Loss Of Power

- ✓ Clamp wire at the vertical section close to the lower sheave/pulley and apply manual brake to cable drum
- ✓ Close BOPs (optional)
- ✓ Repair or replace power pack
- ✓ Open BOPs, unclamp wire and recommence operations.

• Managing A Leak or Malfunction on Surface

❑ Broken Strand

- ✓ Breaking a single strand of a braided cable should be discovered by:
 - Fluctuations in the line weight, caused by the lower end of the broken strand.
- ✓ A broken strand is more likely to occur after closing the BOP on the cable.
- ✓ It can also be caused by crimping the wire, e.g. when using clamps, by excessive bending, etc.
- ✓ In all broken strand/birds nest situations, there is a danger that strands of wire can cause BOP not to seal or be damaged when actuated.

• Managing A Leak or Malfunction on Surface

❑ Broken Wire

- ✓ Broken slickline will fall and usually coil in a neat helix and can be normally easier to fish than braided line.
- ✓ However all fishing operations carry a high element of risk that the fish can't be recovered. This might lead to killing the well by bull heading, leading to loss of production.
- ✓ Broken braided line can form a bundle, which can sometimes be fished with difficulty, but usually which will plug the tubing.
- ✓ This may require the well to be killed (which itself could be difficult with plugged tubing) and the completion retrieved. For this reason braided line should always run with a weak point at the tool.
- ✓ Slickline tends to part more easily than braided line for a number of reasons, including:
 - Fatigue at pulleys caused by excessive jarring at the same point
 - Embrittlement of the wire due to the presence of H₂S
 - Accidental (or otherwise) closing of valve on wire
 - Pulling in excess of the wire's yield strength.

Wire Line

Outlines

- I. Introduction to Wireline
- II. Typical Slickline Rig up
- III. Typical Braided Line Rig up
- IV. Operation Planning
- V. Managing A Leak or Malfunction on Surface
- VI. Operational Considerations

• Operational Considerations

➤ Lubricator

- ✓ When conducting wire line operations on gas wells or high GOR oil wells it is important that the fluid used to pressure test the lubricator is inhibited, to prevent the formation of hydrate plugs around the wellhead. Typically, a mixture of glycol and water is used.
- ✓ Prior to opening X/T valves it is important that lubricator is approximately equal to or slightly above, the SIWHP. This is to:
 - Reduce the chance of damaging the valve seals when opening.
 - Reduce the chance of differential surges damaging the tool or wire.
- ✓ When RIH the tree valve opening sequence should be
 1. Hydraulic master valve (HMV)
 2. Swab valve
- ✓ So that the HMV does not have a differential pressure across it when opening, when pulling out the shut-in sequence should be:
 1. Swab valve (Check the correct number of turns)
 2. HMV

• Operational Considerations

➤ The Stuffing Box

- ✓ During slick line operations, the stuffing box needs to provide a leak tight seal around the wire while still allowing the wire to move.
- ✓ However the stuffing box should not be over tightened, since:
 - This causes excessive packing wear
 - More weight needs to be used to overcome the extra friction
 - The extra weight brings line tensions closer to limits
 - The tool string is lengthened, requiring a longer lubricator.
- ✓ On high wellhead pressure wells, note that extra stem is required not only to overcome the extra upwards force on the cable, but to overcome the extra stuffing box friction.

• Operational Considerations

➤ DHSV And Tree Valve Control

- ✓ During wire line operations, the normal tree valve control system is overridden and operation of the valves is achieved through a separate bypass control unit.
- ✓ This ensures that valves are not accidentally closed by the process control ESDs, etc. while wire line is in the hole.
- ✓ This unit should not be placed in the well bay area, unless it is permanently manned by someone in adequate communication with the wire line operator/supervisor.
- ✓ When conducting W/L operations below the DHSV, the control line can be closed in at the X/T, to lock in pressure, ensuring that the valve can't be accidentally closed on the cable.
- ✓ Removing a DHSV can permit the well fluids access to the CL. In the case of unexpected pressure in the well, a build-up can escape via this line. This can cause:
 - External leaks
 - Pressurization of components & hydraulic oil reservoirs beyond their rating.

- **Operational Considerations**

- **DHSV Protection**

- ✓ If the well contains a wire line retrievable DHSV, the valve may have to be pulled in order to provide access for the tool string.
- ✓ Protection sleeves are available for setting in the DHSV landing nipple that prevent damage of the seal bore by wire line cutting.
- ✓ When wire line is in the well in the hole below the DHSV, it is not normally considered to be a barrier, whether or not it is capable of cutting the wire.

• Operational Considerations

➤ Tool String Weight

- ✓ The weight of the tool string should be increased to:
 - Overcome sealing assembly friction and well pressure on the cross-sectional area of the wire line
 - Ensure sufficient weight to provide required jarring force.
 - Ensure sufficient weight to keep tool string stationary if flowing the well for logging purposes.
- ✓ Adjusting the tool string weight is simply achieved by adding or removing lengths of stem. However there are limits to the amount of weight that can be added because of:
 - Tool length limit
 - Lubricator length limit.
 - Down hole pick-up weight constraints (cable strength limit).

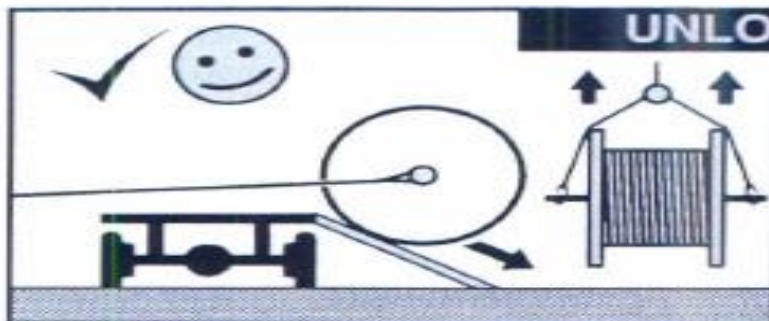
• Operational Considerations

➤ Flowing Wells

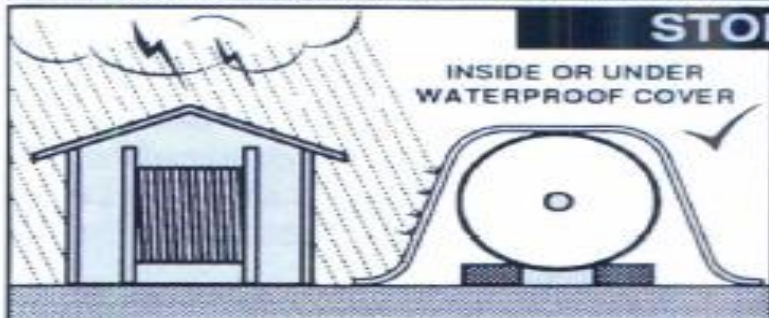
- ✓ Flowing the well with tools in the hole is carried out when
 - Production logging (braided line)
 - Pressure / temperature gradient / MPLT surveys (slick line).
- ✓ To prevent the tools from being carried up the hole by the force of the wellbore fluid, with the potential for the tools stuck;
 - Ensure sufficient tool string weight
 - Open well slowly, after positioning tool string below the fluid level
- ✓ When conducting slick line logging it is sometimes difficult to detect a loss in weight. With braided line the tool position can be monitored by means of the CCL.
- ✓ When pulling plugs, etc. it is important to ensure that the device is **equalized**, i.e. has the same pressure above and below.
- ✓ If the pressure above is too high, it may be impossible to pull the prong/device, perhaps eventually leading to a broken wire.
- ✓ If the pressure below is too high, the surge may cause the prong/device to be blown up the well, also leading to tangling and possibly broken wire.

CARE IMPROVES LIFE !!

UNLOADING



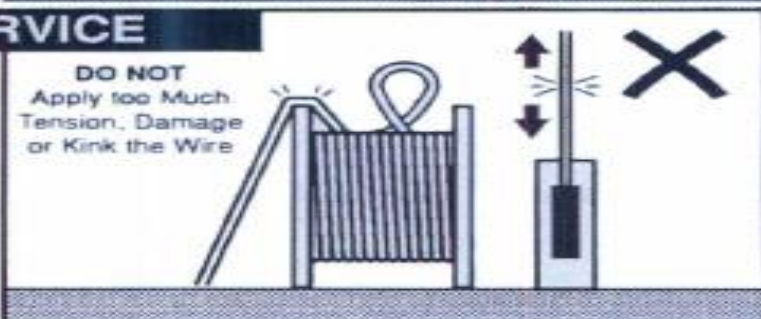
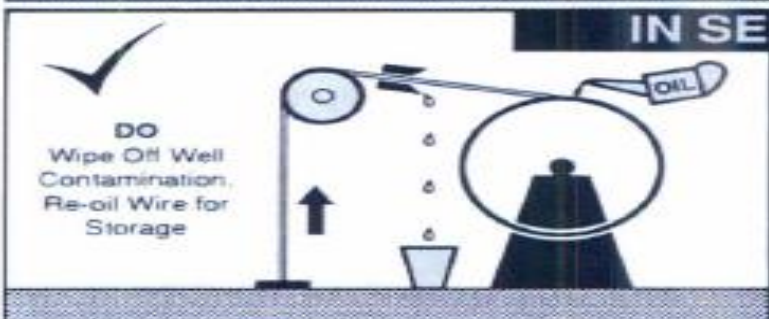
STORAGE



RESPoolING



IN SERVICE



Coiled Tubing

Outlines

- I. Coiled Tubing Description
- II. Description of Fatigue
- III. Pressure Control Equipment
- IV. BOP Test Procedures
- V. General Rig Up Procedures
- VI. Contingencies

CT EQUIPMENT



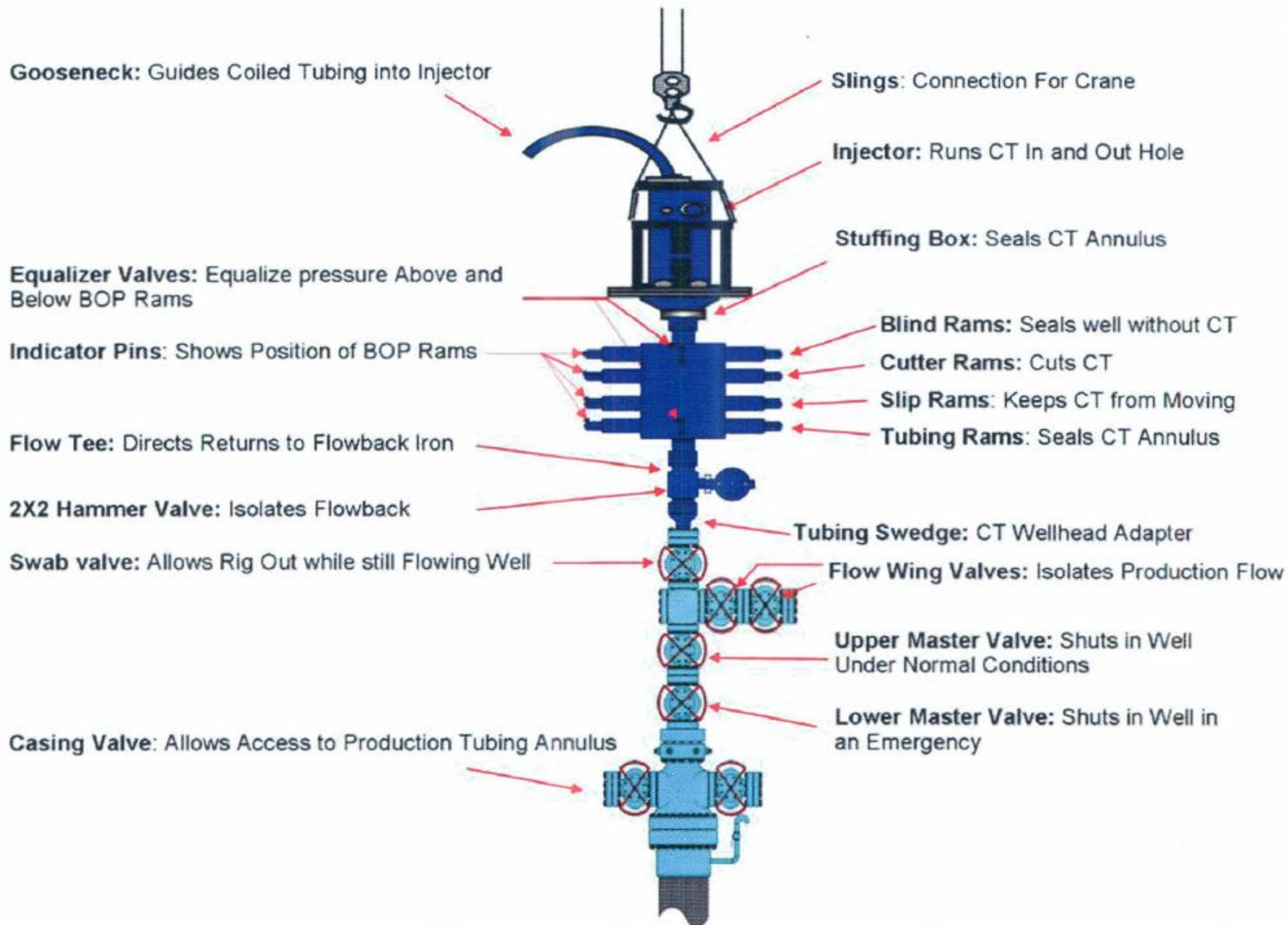
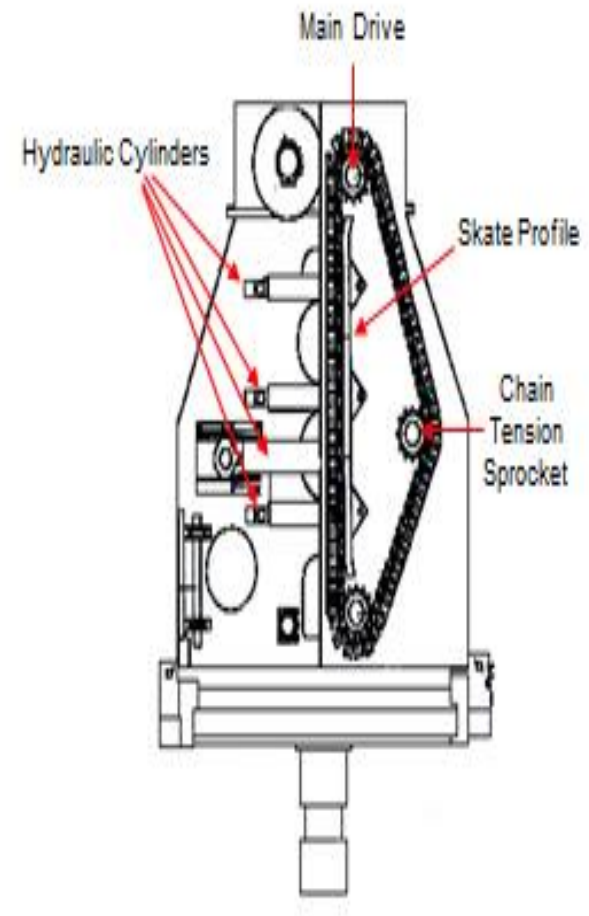


Figure 9.3.28: Typical C/T Rig up over Christmas Tree

➤ Tubing Injector Head

- ✓ The injector head is mounted above the BOPs and stripper and drives the tubing to be run into and out of the well under pressure.
- ✓ **Profile Gripper Blocks:** hydraulic adjustable friction grips CT without slippage & crimp CT.
- ✓ **Hydraulic Drive Chains:** 2 hydraulic unison motors drive chains & disk brakes to hold CT weight in stationary position.
- ✓ Face-to-face Gripper Block backed by hydraulic force Skate Profile grip CT in motion or stationary position.
- ✓ Accumulator compensate CT OD abnormalities.



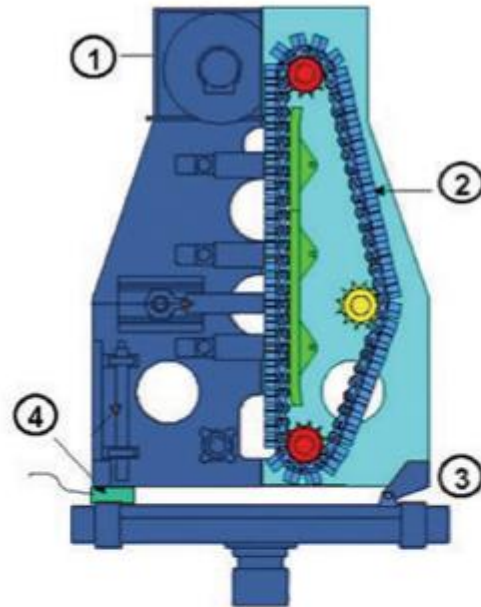
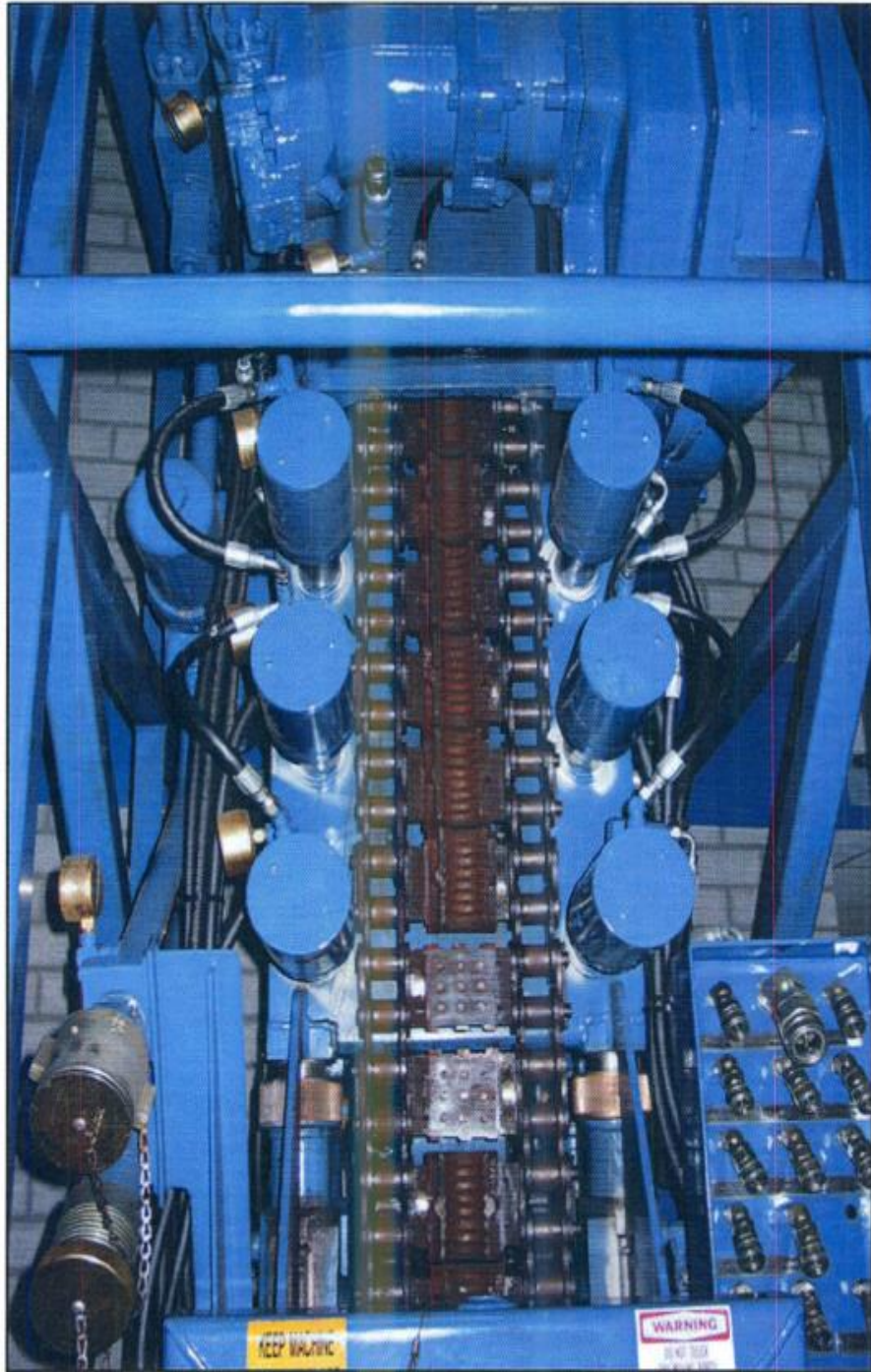
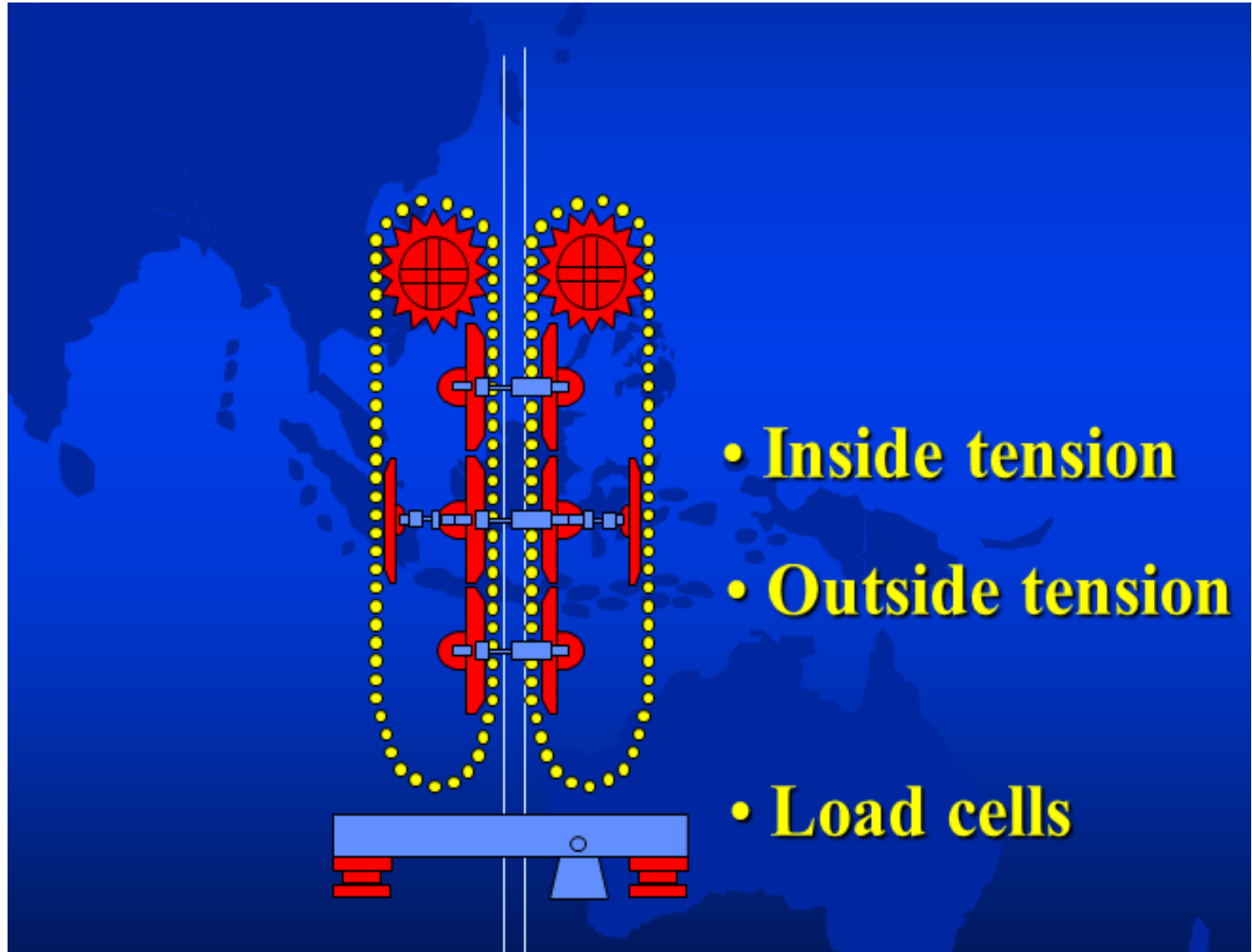


Figure 6-3: Injector assembly

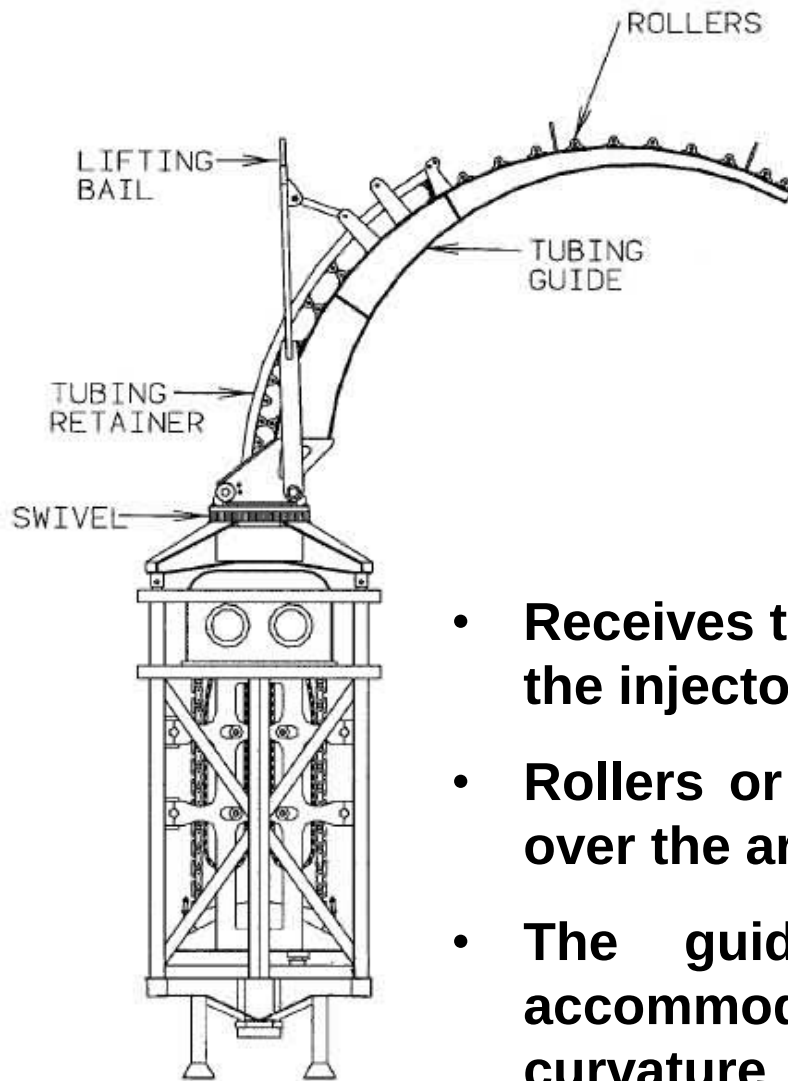
The injector head essentially consists of hydraulic motors (1) connected to chains (2) that move the tubing. The motors actually turn the chains through a set of timing gears to synchronize the movement of both chains.

The injector basically consists of two separate frames.

- The inner frame which is attached to the base with a hinge on one side (3) and a dual-acting electronic load cell (4) on the other side to measure the tensile and compressive forces on the injector head.
- The outer frame which mainly consists of the crash frame attached to the same base as the inner frame, but by rigid connections. The gooseneck is attached to the outer frame to avoid interference with the load cell readings.



➤ Tubing Guide Arch (Gooseneck)



- **Receives the coiled tubing from reel guiding it into the injector.**
- **Rollers or Teflon pads support the coiled tubing over the arch.**
- **The guide has the ability to swivel to accommodate fleet angle changes. The radius of curvature should be at least 30 times that of the coiled tubing OD.**

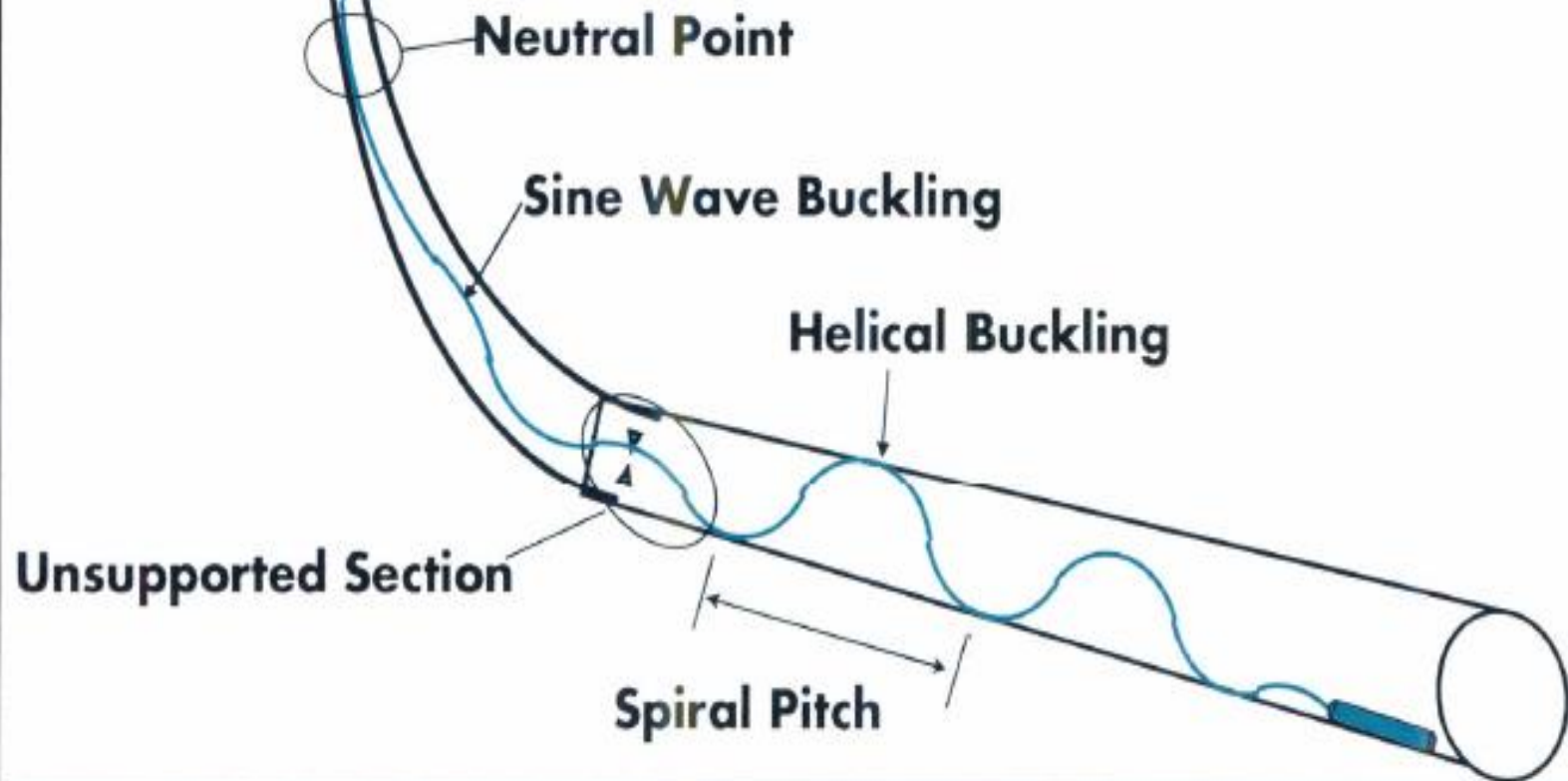
Coiled Tubing

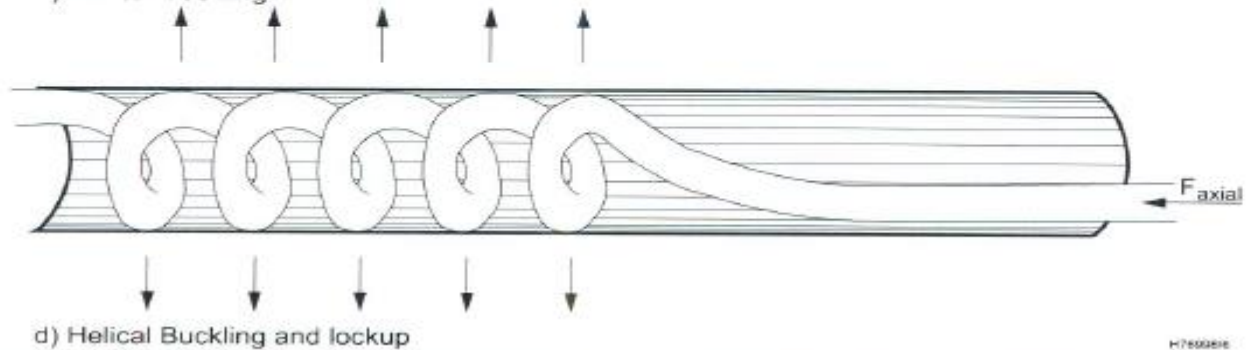
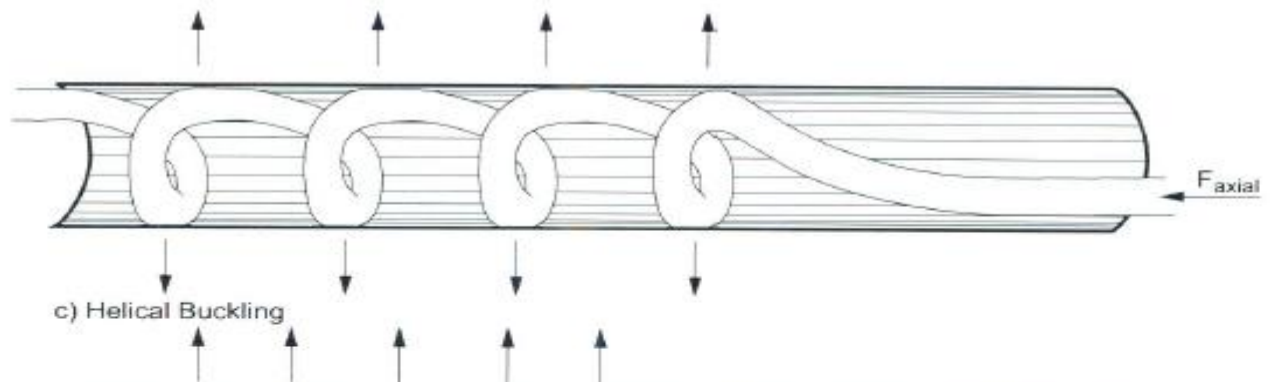
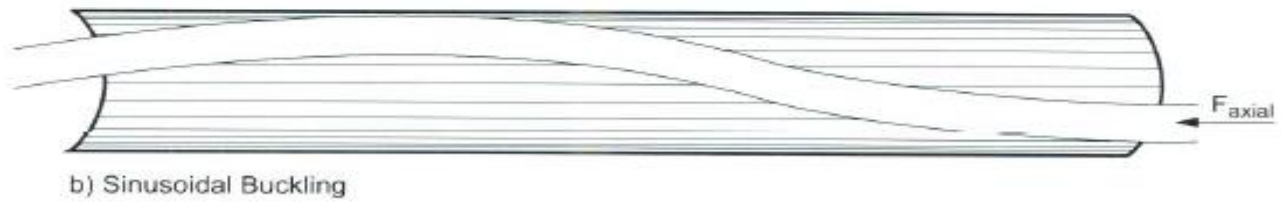
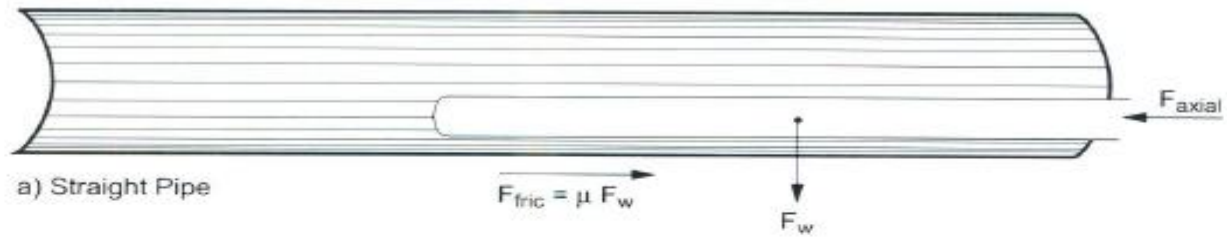
Outlines

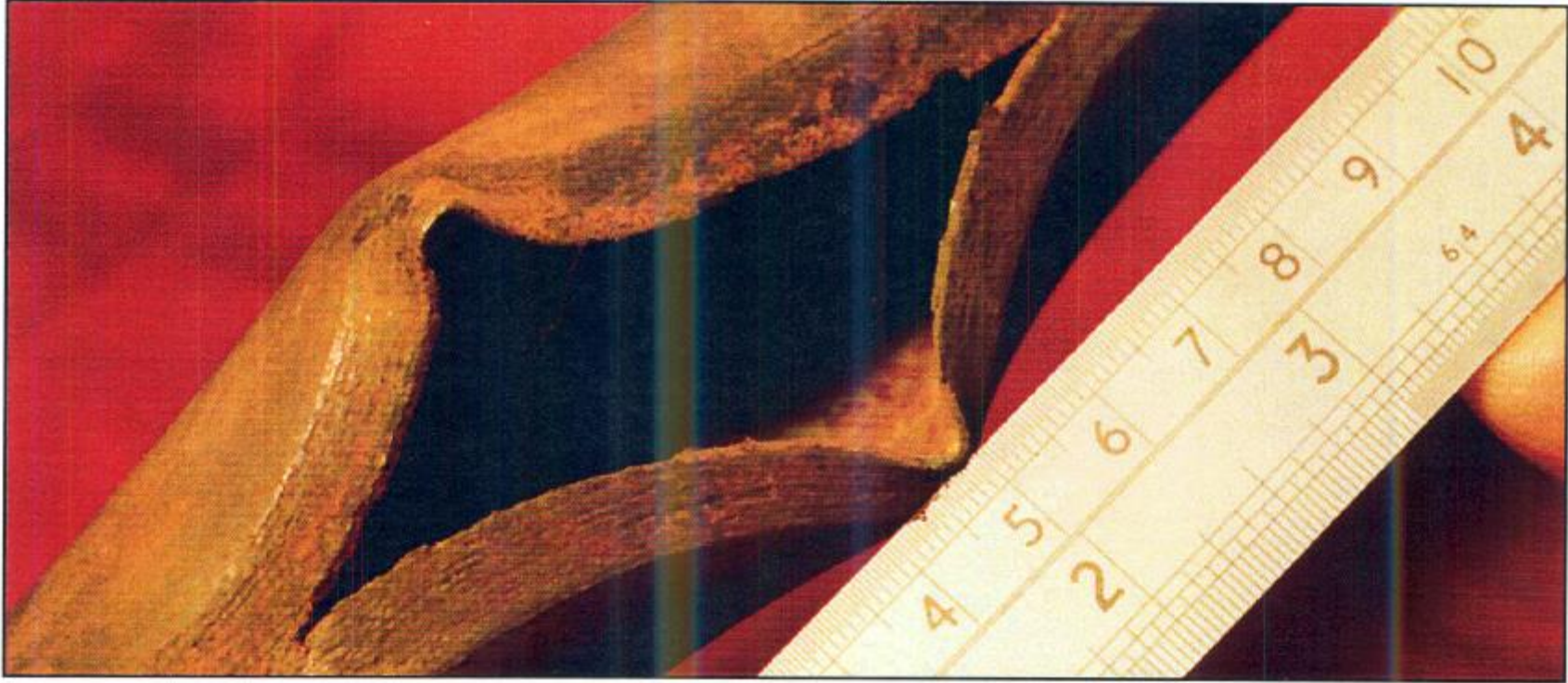
- I. Coiled Tubing Description
- II. Description of Fatigue**
- III. Pressure Control Equipment
- IV. BOP Test Procedures
- V. General Rig Up Procedures
- VI. Contingencies



Coil running in Hole





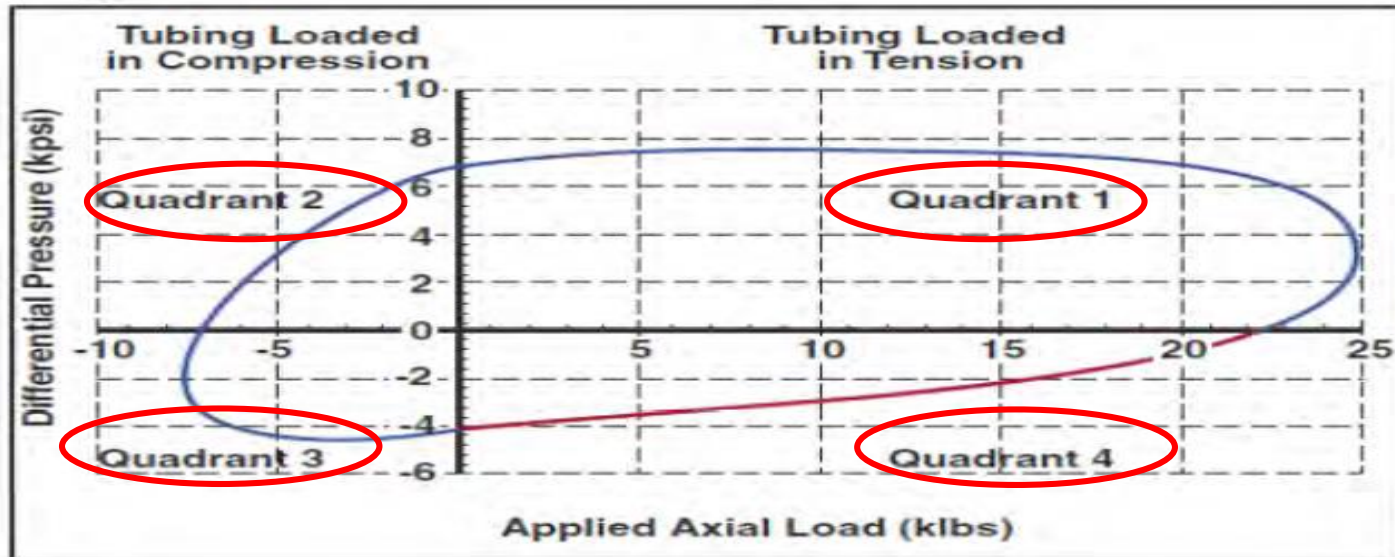


• Coiled Tubing Description

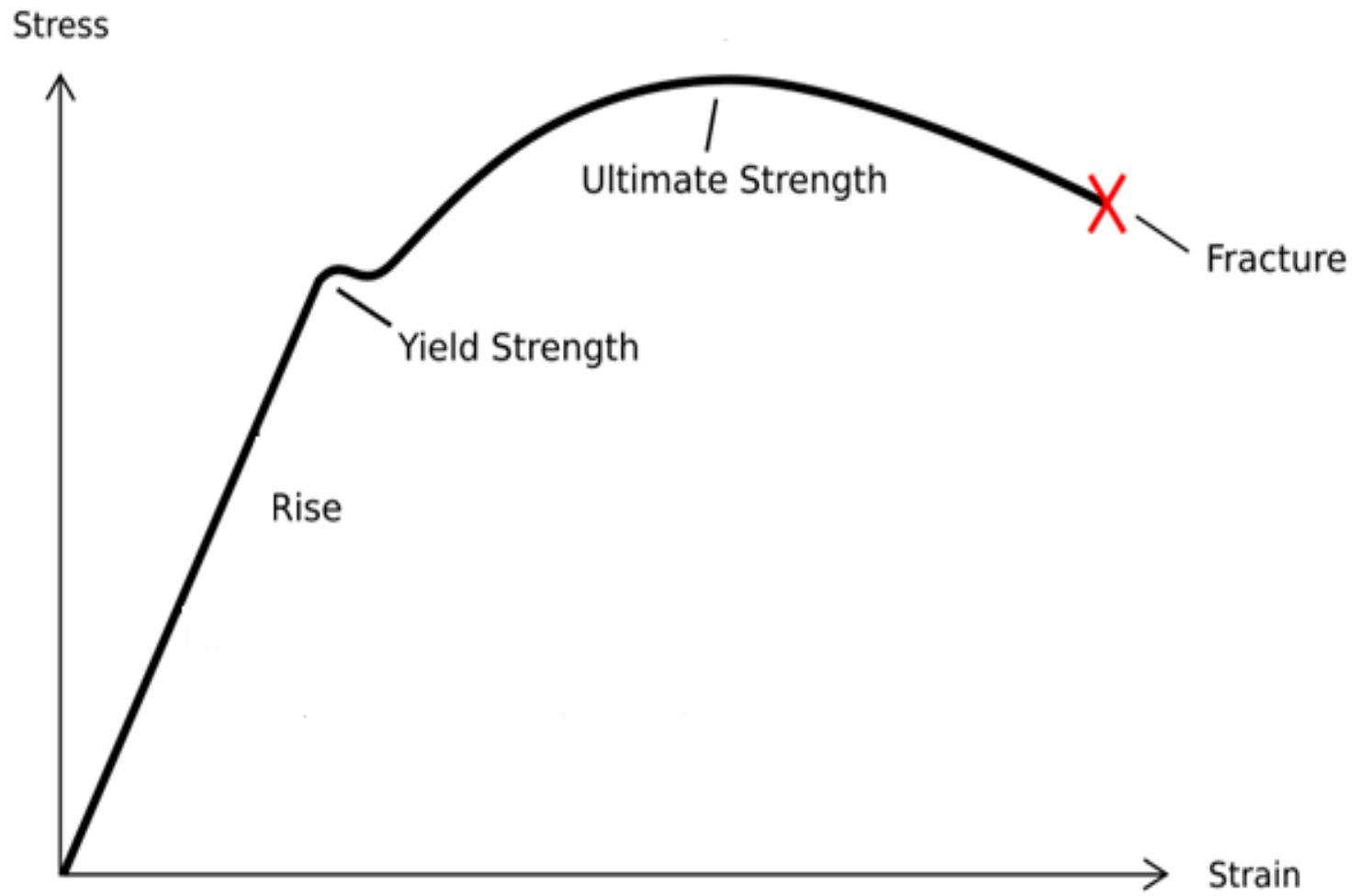
✓ The working life of the Coiled Tubing may be defined as the duration of service a continuous-tubing string can deliver when subjected to the following:

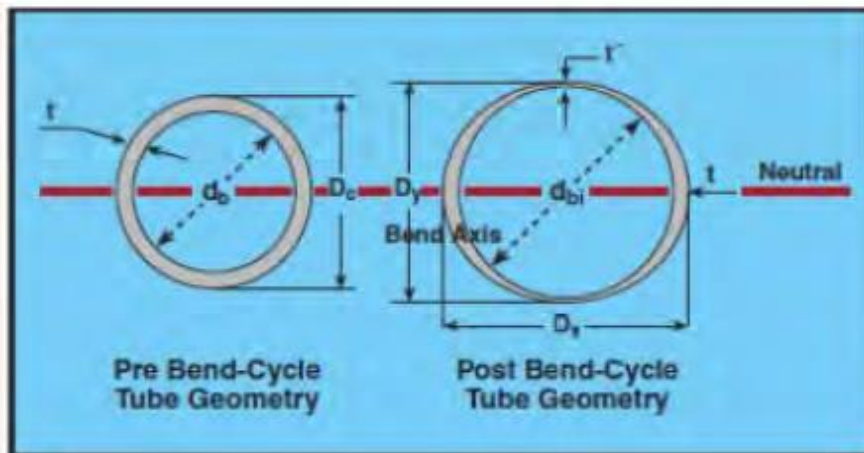
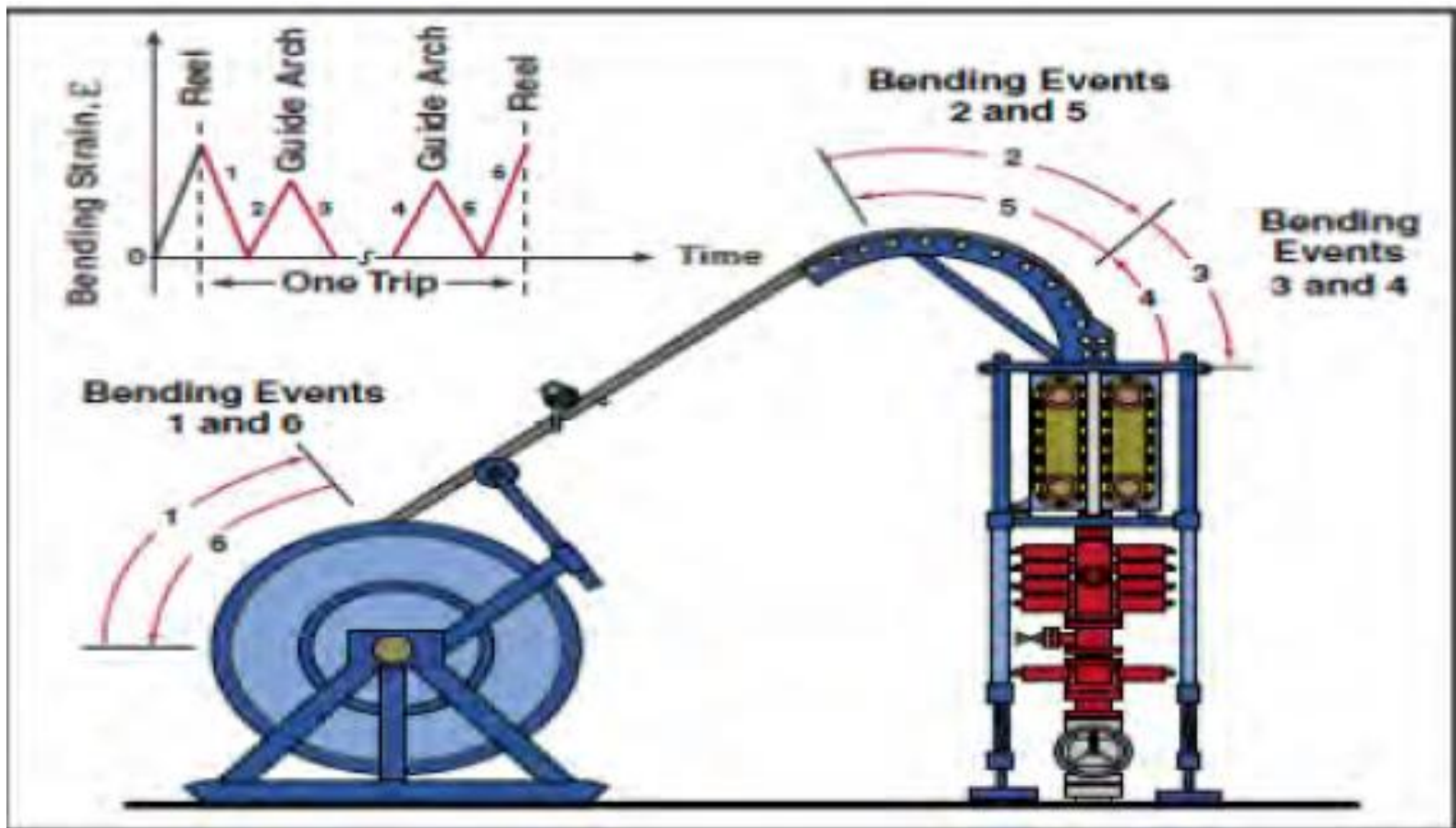
- Bend Cycle Fatigue
- Internal Pressure Loading
- Applied Axial Loading
- Corrosion
- Mechanical Damage

Coil LIMIT curve of pressure and load



Several Factors affect the operational limits of Coiled Tubing





Coiled Tubing OD (in.)	Typical Reel Core Radius (in.)	Typical Guide Arch Radius (in.)
0.750	24	48
1.000	20-30	48-54
1.250	25-36	48-72
1.500	30-40	48-72
1.750	35-48	72-96
2.000	40-48	72-96
2.375	48-54	90-120
2.875	54-58	90-120
3.500	65-70	96-120

- Beyond this minimum bending radius the steel will be stressed beyond its elastic strain limit.
- When coiled tubing is initially spooled plastic deformation will take place.

• Description of Fatigue

- ✓ The Coiled Tubing is subjected to repeated bending and straightening events, commonly referred to as bend-cycling.
- ✓ Bend cycling Coiled Tubing with internal pressure loading dramatically reduces the fatigue life of the pipe.
- ✓ A given bend-cycle load applied later in the tube working life causes greater fatigue damage than the equivalent bend-cycle load applied earlier in the tube working life.
- ✓ For predicting the fatigue condition of the coiled tubing, use;
 - Running Feet Method
 - Trip or empirical method
 - Theoretical method
- ✓ The working life of the Coiled Tubing can be extended by
 - increasing the bend radii of the tubing guide arch and reel core.
 - selecting a tube with a thicker wall.
 - increasing the material yield strength of the tube.
 - Decrease OD size of the Coiled Tubing.
 - Apply lower pumping pressures

Coiled Tubing

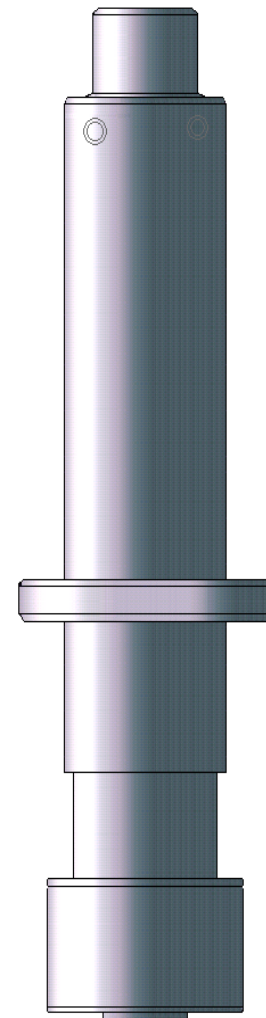
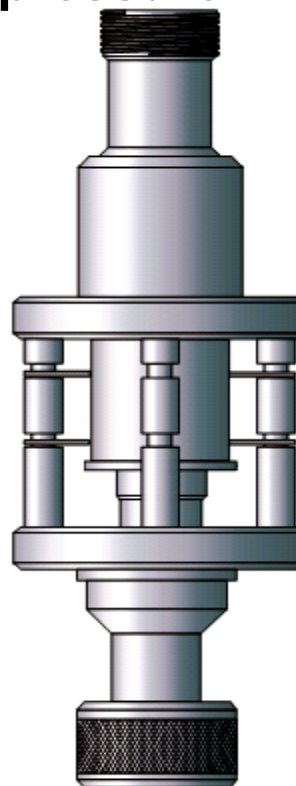
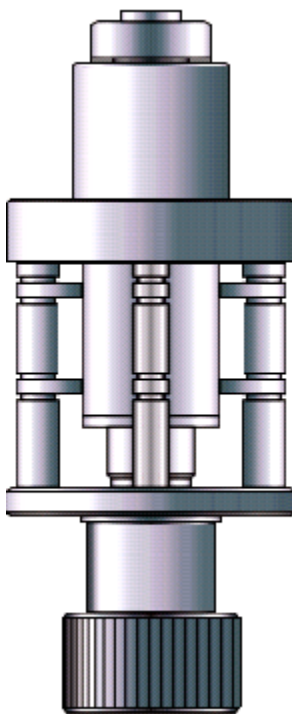
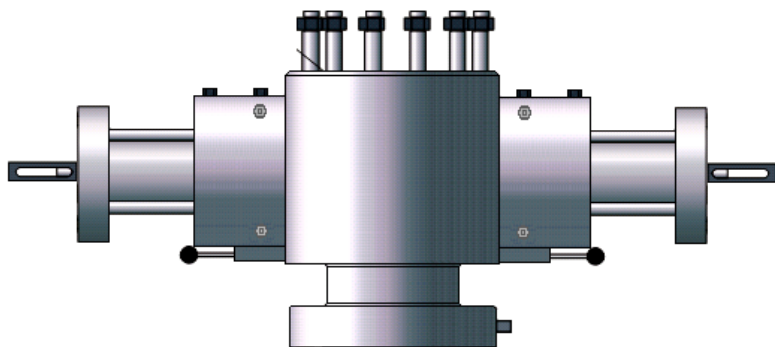
Outlines

- I. Coiled Tubing Description
- II. Description of Fatigue
- III. Pressure Control Equipment**
- IV. BOP Test Procedures
- V. General Rig Up Procedures
- VI. Contingencies

Stripper CT EQUIPMENT

- Pressure containment device – Primary Barrier in well control system.
- Installed above BOP & below IH.
- Dual assembly used for extreme well conditions.
- Hydraulic activation from Control Cabin.
- Can be redressed during operation with pressure isolated

1. Conventional Stripper
2. Tandem Stripper
3. Side Door Stripper
4. Radial Stripper



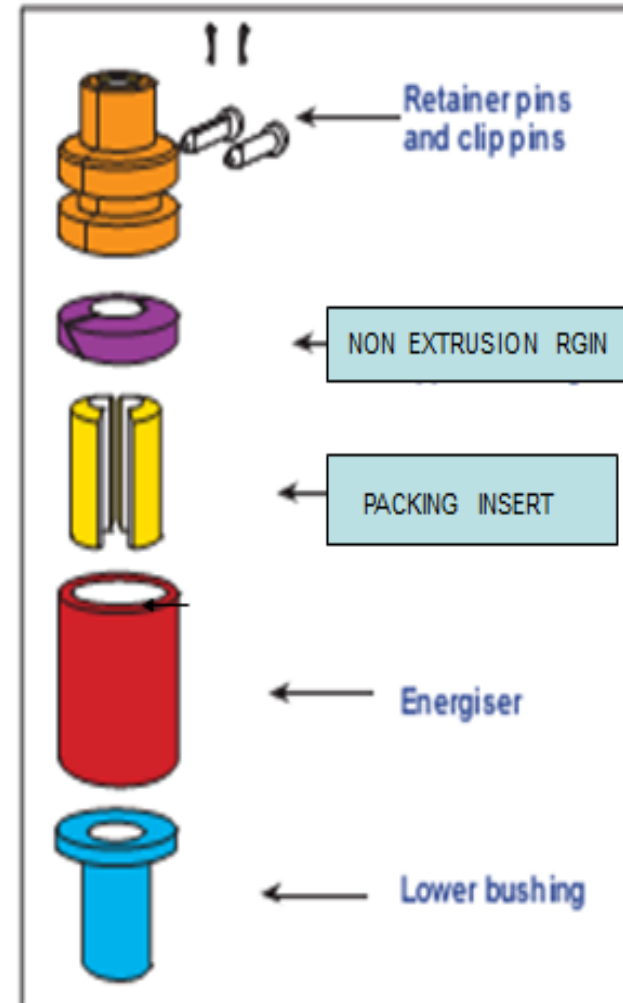
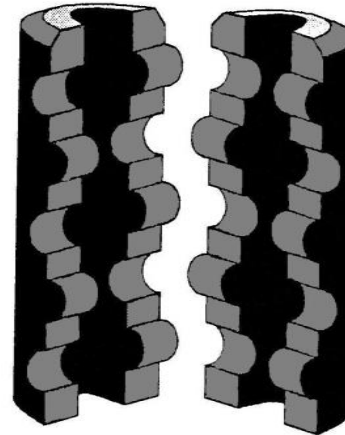
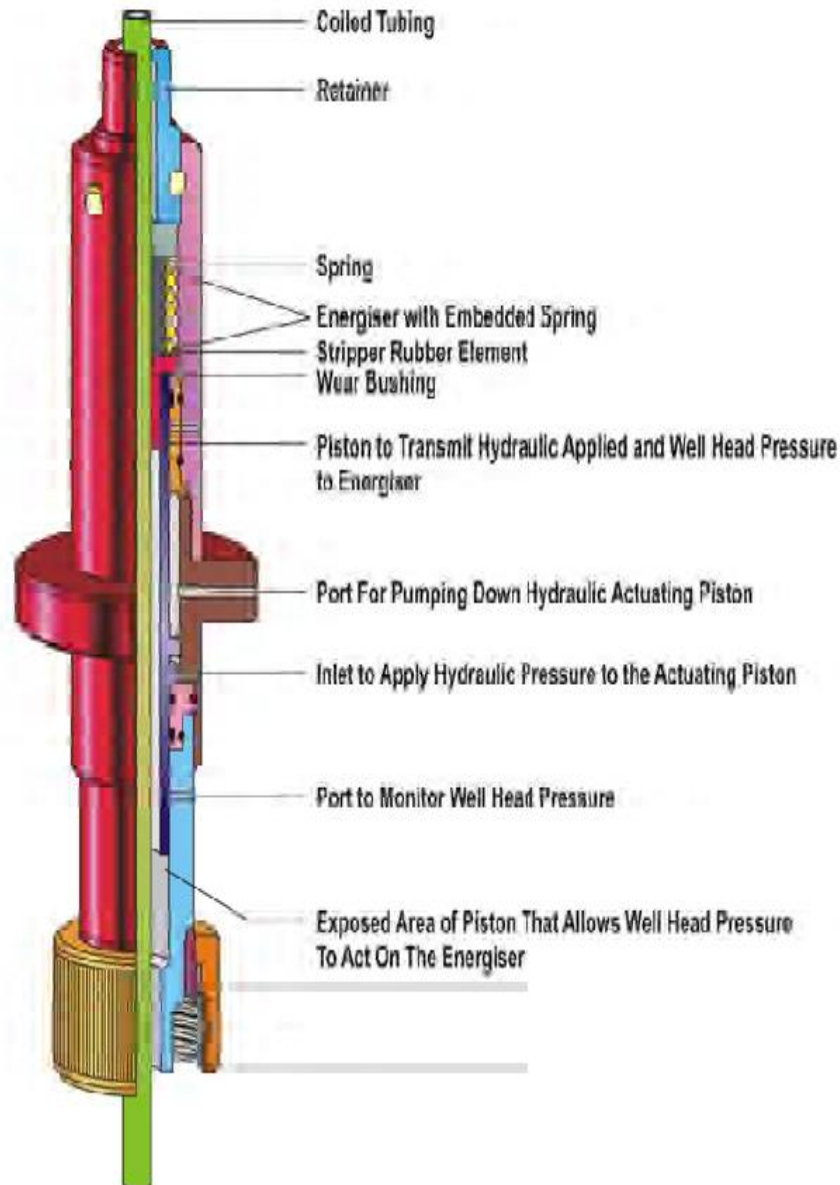
• Pressure Control Equipment

➤ Stripper Packer

❑ Conventional Stripper

- Hydraulic pressure applied through Pack Port moves Lower Bushing up & energize stripping element. De-energize is achieved through Retract Port.
- Energizing stripper is called “packing.” When packed, wellbore pressure will further energize stripper
- Access to insert replacement is from the top, within the injector head frame requiring a greater distance between its top and the gripping chains than other type of strippers. This is an important consideration for buckling at higher pressures.
- 45 min – 1 hr redressing period.

Conventional Stripper

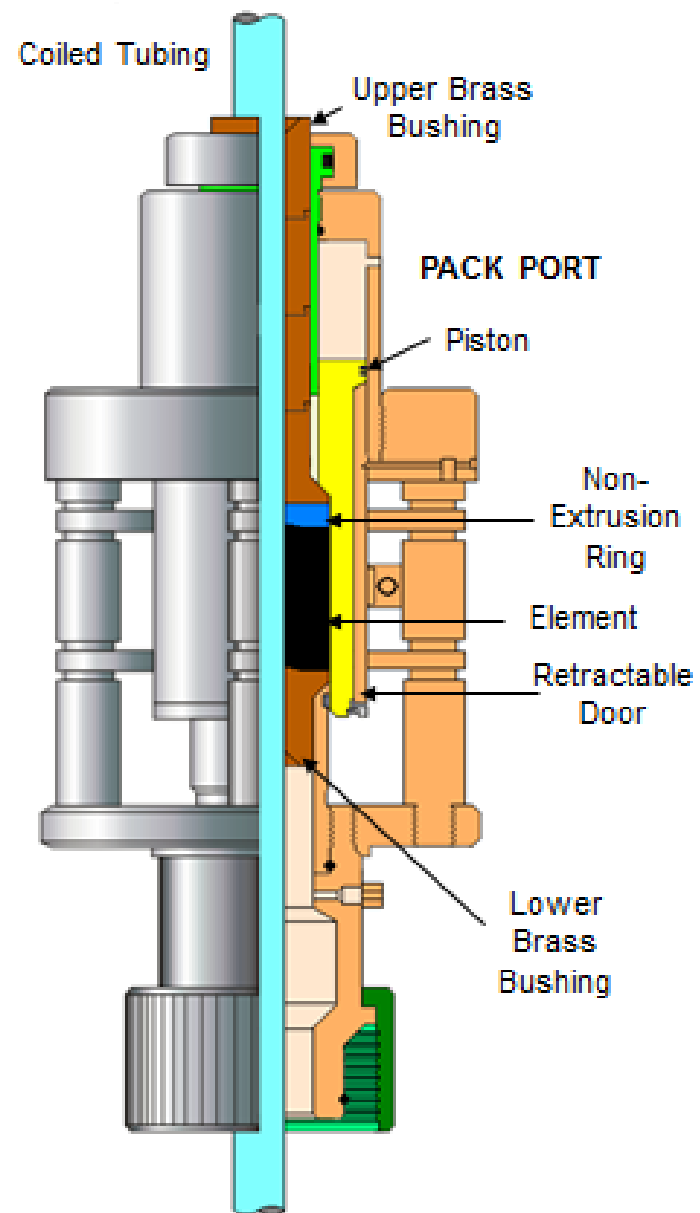


• Pressure Control Equipment

➤ Stripper Packer

☐ Side Door Stripper

- The side door stripper was developed to permit **easier access to the stripper packer arrangement.**
- The seal elements may be removed through the open window below the injector simplifying removal while tubing is in place.
- Since access to the stripper from above is not required, the assembly can be mounted closer to the injector head chains, **minimizing the chances of buckling** between the chains and stripper.
- Hydraulic pressure applied to packing Port moves Upper Bushing down & energize stripping element.
- Well pressure provide gives no impact in energizing element.
- 5 – 10 min redressing period.

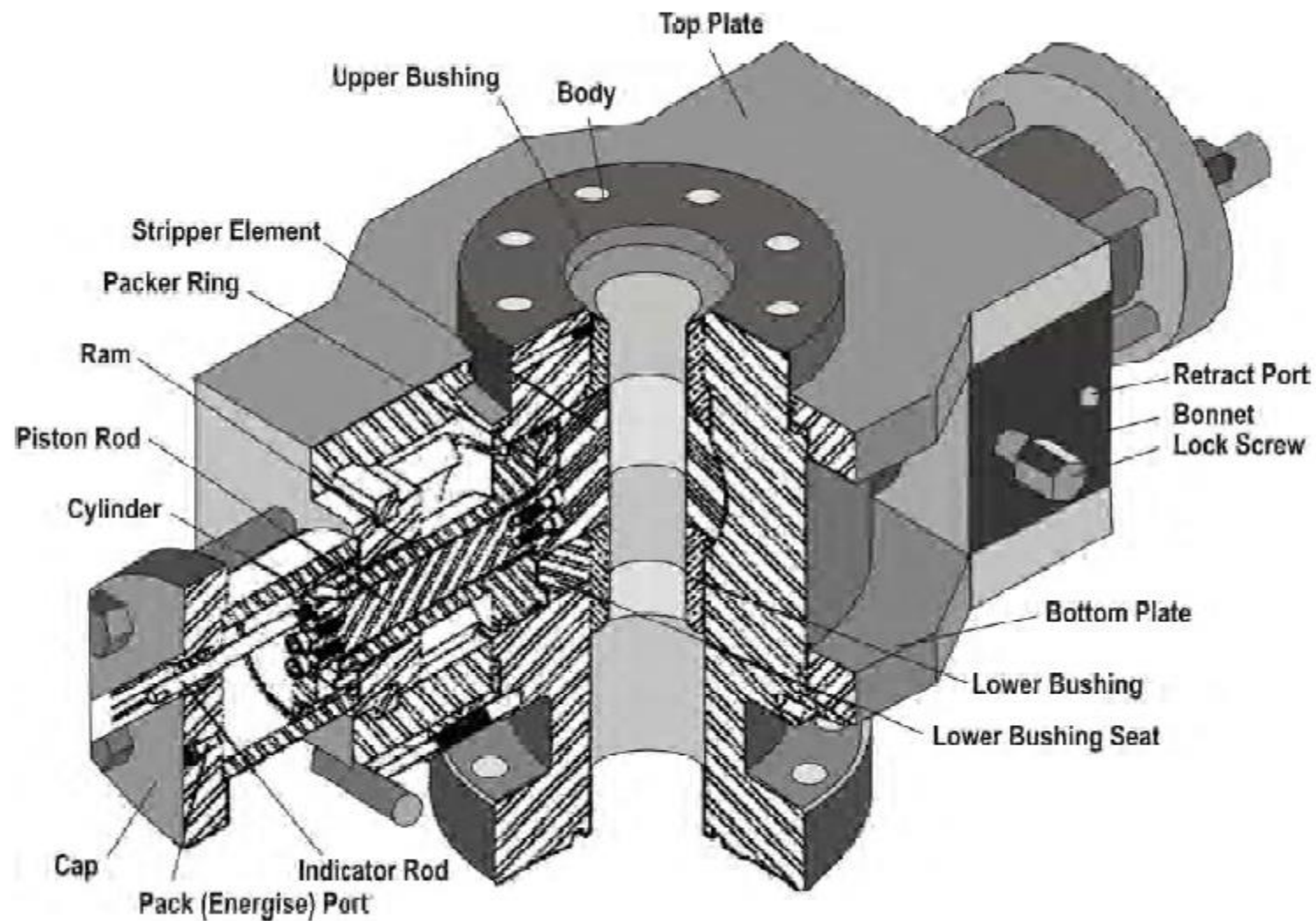


• Pressure Control Equipment

➤ Stripper Packer

☐ Radial Stripper

- The radial stripper is similar in design to a **pipe ram** with hydraulic fluid forcing opposing rams around the coiled tubing to effect the seal.
- The hydraulic actuator pistons also incorporate a visual indicator to indicate the piston's position and to indicate the degree of wear of the seal elastomers.
- Since the actuators are horizontal, a **reduction in overall height** over conventional strippers may be realized. Change out of the seal elements is performed from the open space below the injector.
- Generally used for high pressure well & back-up system.
- 45 min period redress



• Pressure Control Equipment

➤ Stripper Packer

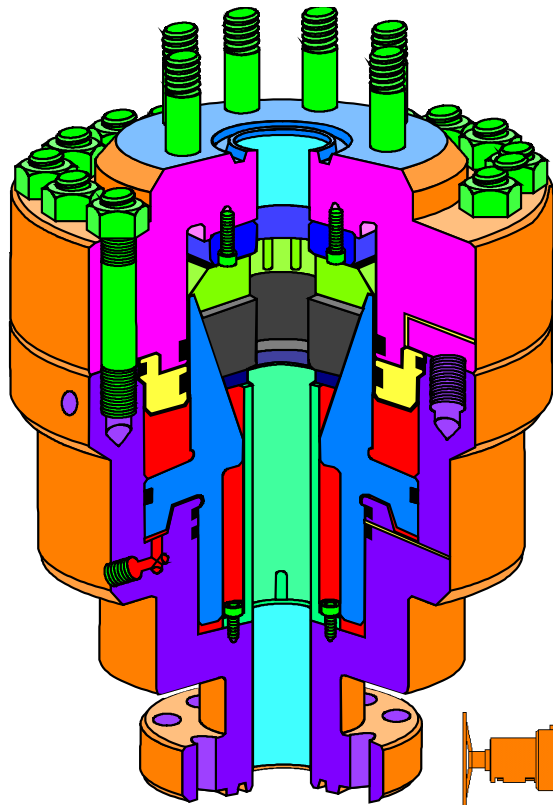
□ Tandem Stripper

- In higher pressure applications a tandem arrangement stripper arrangement may be used. It is similar in principle with Side Stripper
- One way to use this arrangement is to use both simultaneously to distribute the pressure differential between the strippers which may increase the service life of the elements.
- The upper stripper/packer, whether of side door or conventional design is typically a standard unit mounted as usual.
- The lower stripper/packer is mounted directly underneath. The upper stripper packer of the tandem arrangement is typically used as the primary packer.
- The lower packer is not energized and kept in reserve. Should the upper packer become worn, the lower can be energized to effect the seal. The operation may continue, or the upper packer replaced.

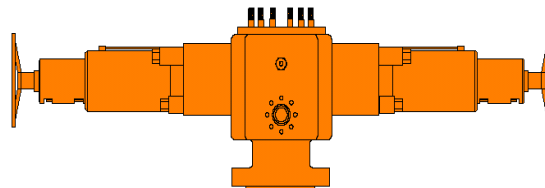
• Pressure Control Equipment

➤ BOP System

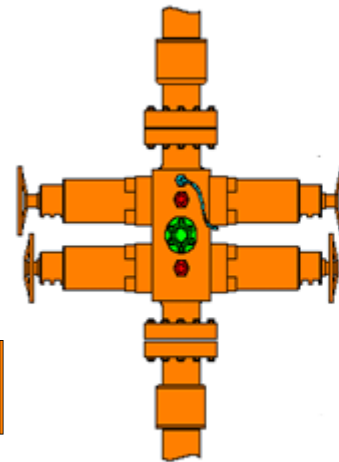
- ✓ BOP's are “secondary” barriers in coiled tubing operations. The stripper is used during normal operations and when tubing is in motion. Rams are not designed to close when tubing is moving.



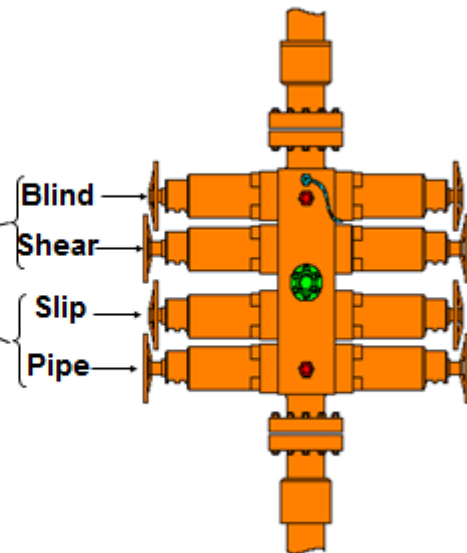
**ANNULAR
BOP**



SHEAR/SEAL BOP



COMBI BOP



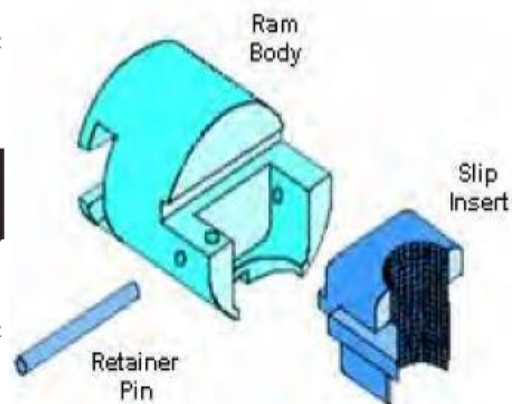
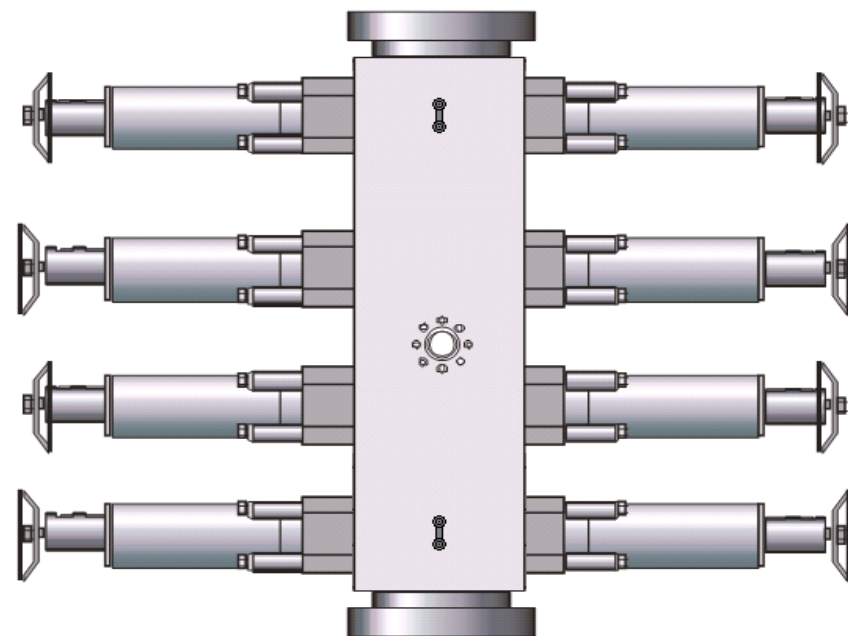
QUAD BOP

• Pressure Control Equipment

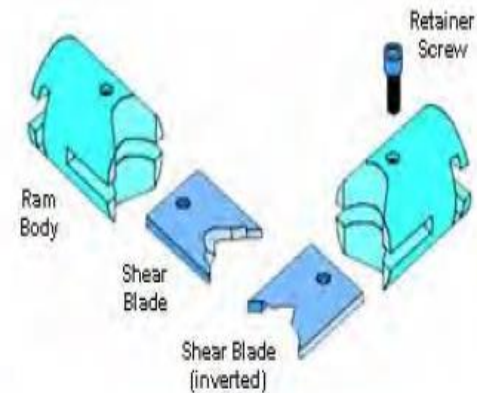
❑ Quad BOP

- A quad BOP has four pairs of ram actuators with the following functions, in order, from the top down

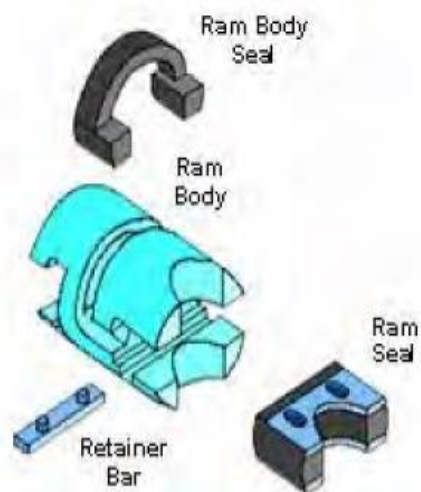
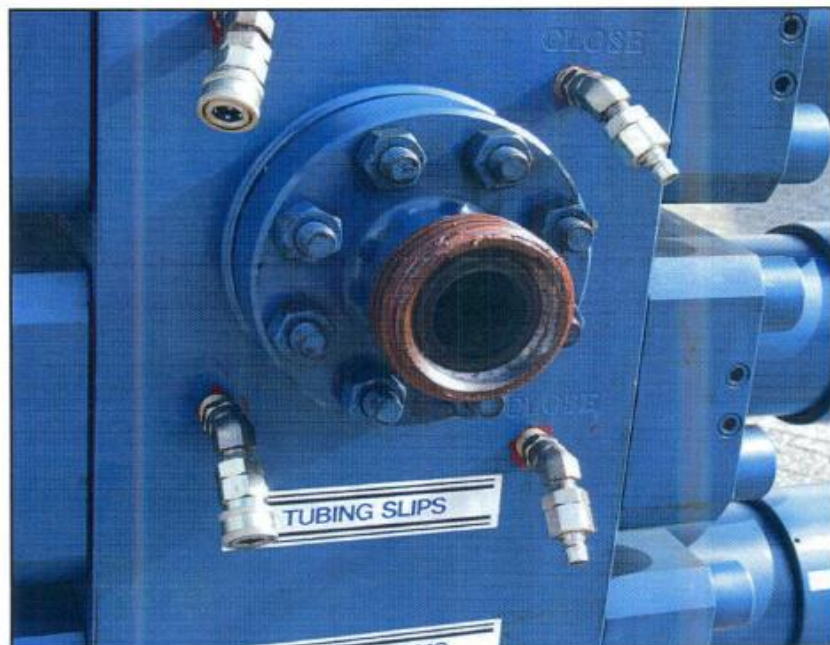
Blind rams	Used to seal the well bore off at surface when well control is lost. Sealing of the blind rams is achieved when the elastomeric elements in the rams are compressed against each other. For the blind rams to seal correctly the tubing must be removed. The rams are designed to hold pressure from below only.
Shear rams	Used to cut coiled tubing in an emergency. Rams have replaceable blades specifically for coiled tubing applications. As the shearing plates are closed on the coiled tubing, the forces imparted mechanically yield the body of the tube to failure. The cut will leave the tubing open ended so that circulation is still possible.
Slip rams	Designed to hold the tubing and prevent upward or downward movement. Rams have replaceable inserts for changing tubing size. To prevent damage of the tubing, by the slips, longer inserts are available adding 75% to the contact area. In order to break up the stress risers (caused by circumferential slip marks) the teeth have vertical grooves cut to interrupt the slip marks on the tubing.
Pipe rams	The pipe rams are equipped with elastomeric seals sized to the diameter of the tubing in use. When closed on the tubing they isolate the well annulus below the rams. Guide sleeves fitted to the ram assembly centralise the coiled tubing as the rams close.



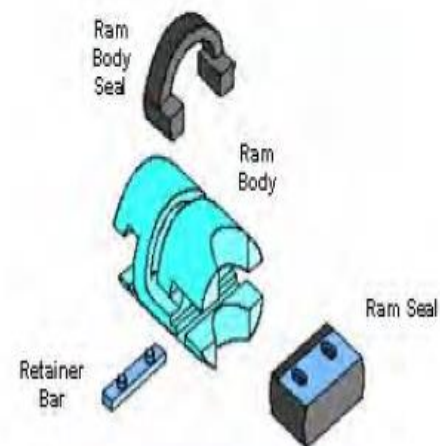
Typical Slip Ram Assembly



Typical Shear Ram Assembly



Typical Pipe Ram Assembly



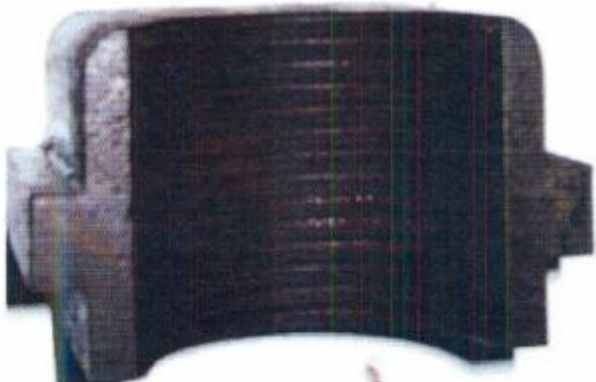
Typical Blind Ram Assembly



Slips



Pipe Rams



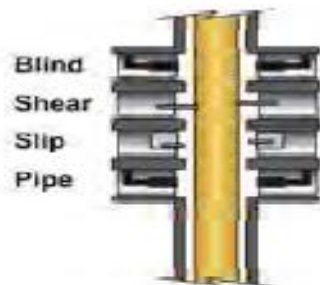
Shear Rams

Figure 9.3.36: C/T Ram inserts

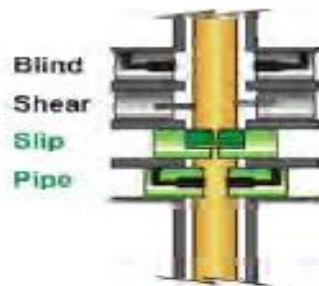
• Pressure Control Equipment

❑ Quad BOP

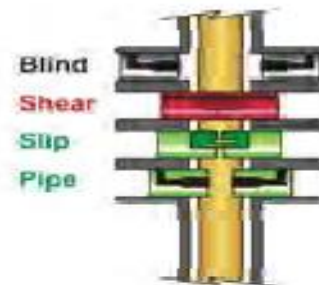
- The blind rams and shear rams are separated from the slip rams and pipe rams by a flanged outlet in the BOP body which is used as a kill line during well control. The kill port allows pumping down annulus or down tubing after shearing.
- Kill line can be used to reverse circulate fluids however it is not recommended as the pipe rams and slip rams would be exposed to debris which could impair their operation.
- The pressure above and below the blind ram and pipe ram must be equalized before they are opened. The equalizing valve must be left in closed position during normal operations.
- Most rams are designed to have wellbore pressure assist. They should not be opened until differential pressure is equalized.
- The hand wheel can be used to close only & also for ram lock.
- The weep hole and vent are used for detection and prevention of communication between the wellbore and hydraulic system.
- The ram position indicator provides a visual indication of ram travel and position.
- Slip rams are “bi-directional,” and hold the coiled tubing in place against upward and downward forces.



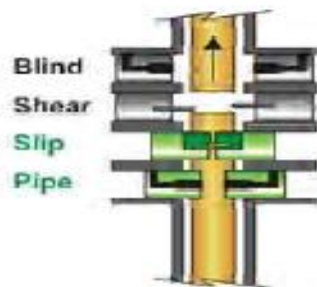
BOP Rams All Open



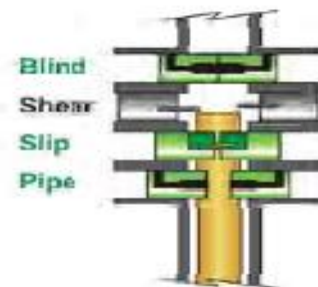
Close Slip and Pipe Rams



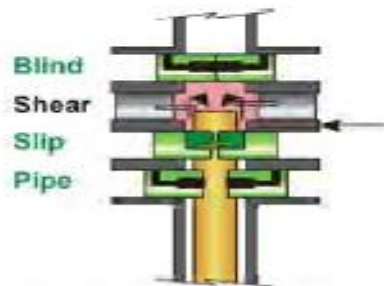
Cut Coiled Tubing With Shear Rams



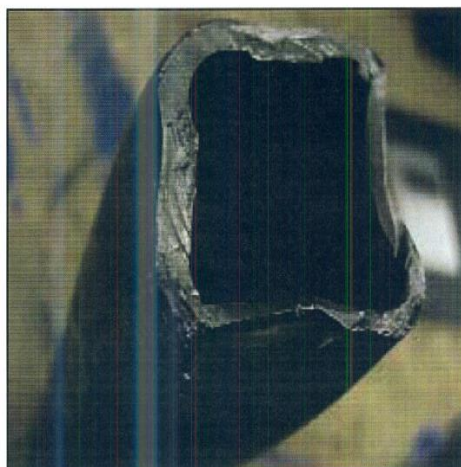
Pull The Remaining Coiled Tubing Out With The Injector



Close Blind Rams



Circulation Is Now Possible Through BOP Circulating Port

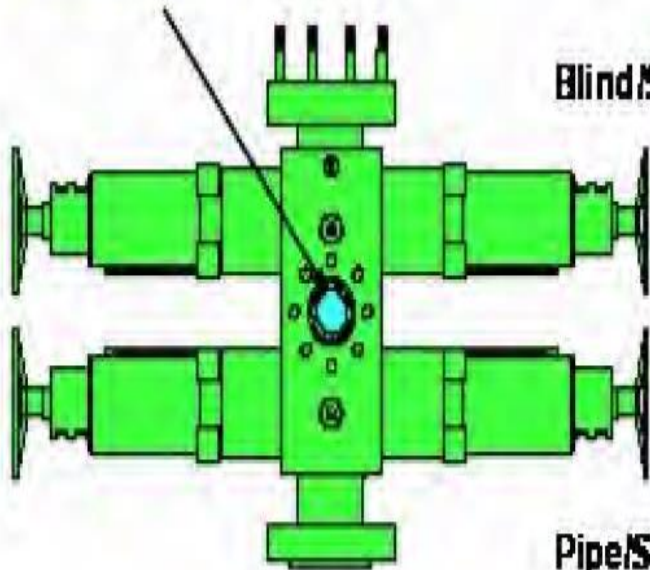


- **Pressure Control Equipment**

- **COMBI BOPS**

- The combi BOP has the same features as the quad BOP but combines the functions of two rams in one actuator:
 - Quad BOP blind and shear rams become combination shear/seal rams
 - Quad BOP slip and pipe rams become combination pipe/slip rams.
 - “Combi” BOP’s have **2 main advantages** over single function BOPs:
 - Less height, weight and number of hydraulic hoses.
 - fewer steps for emergency shut-in procedures.

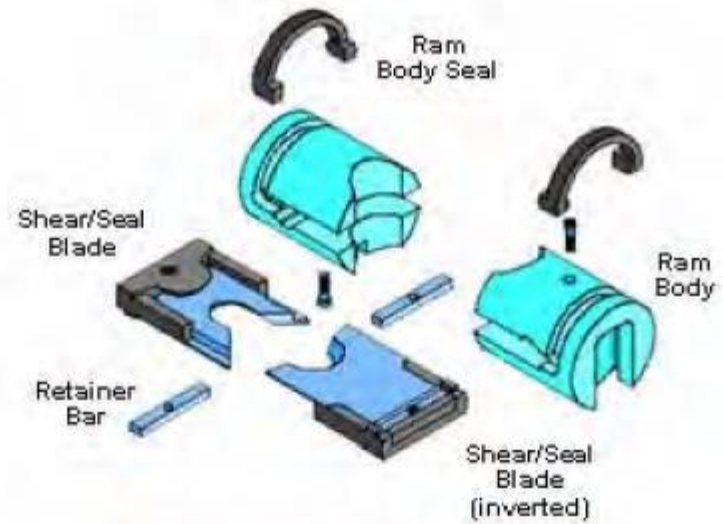
Kill Nipple



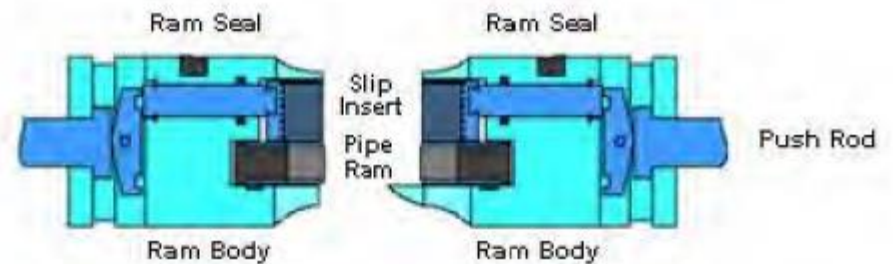
Blind/Shear Rams

Pipe/Slip Rams

Typical "Combi" BOP



Typical Shear/Seal Ram Assembly

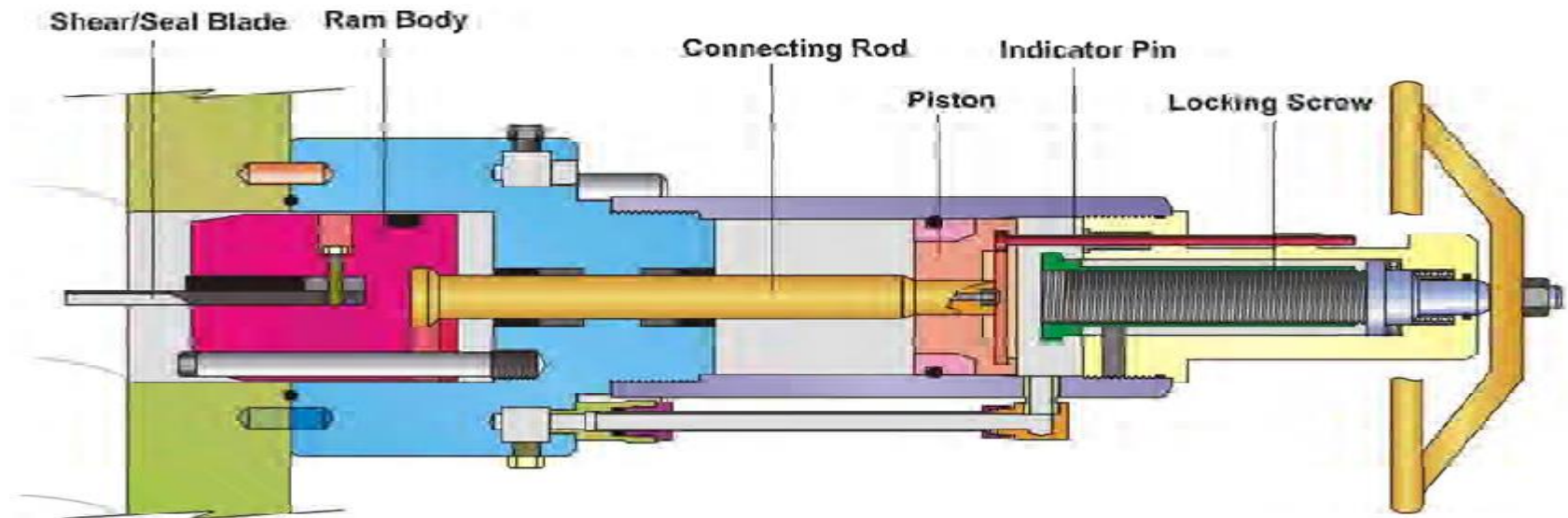


Typical Pipe/Slip Ram Assembly

• Pressure Control Equipment

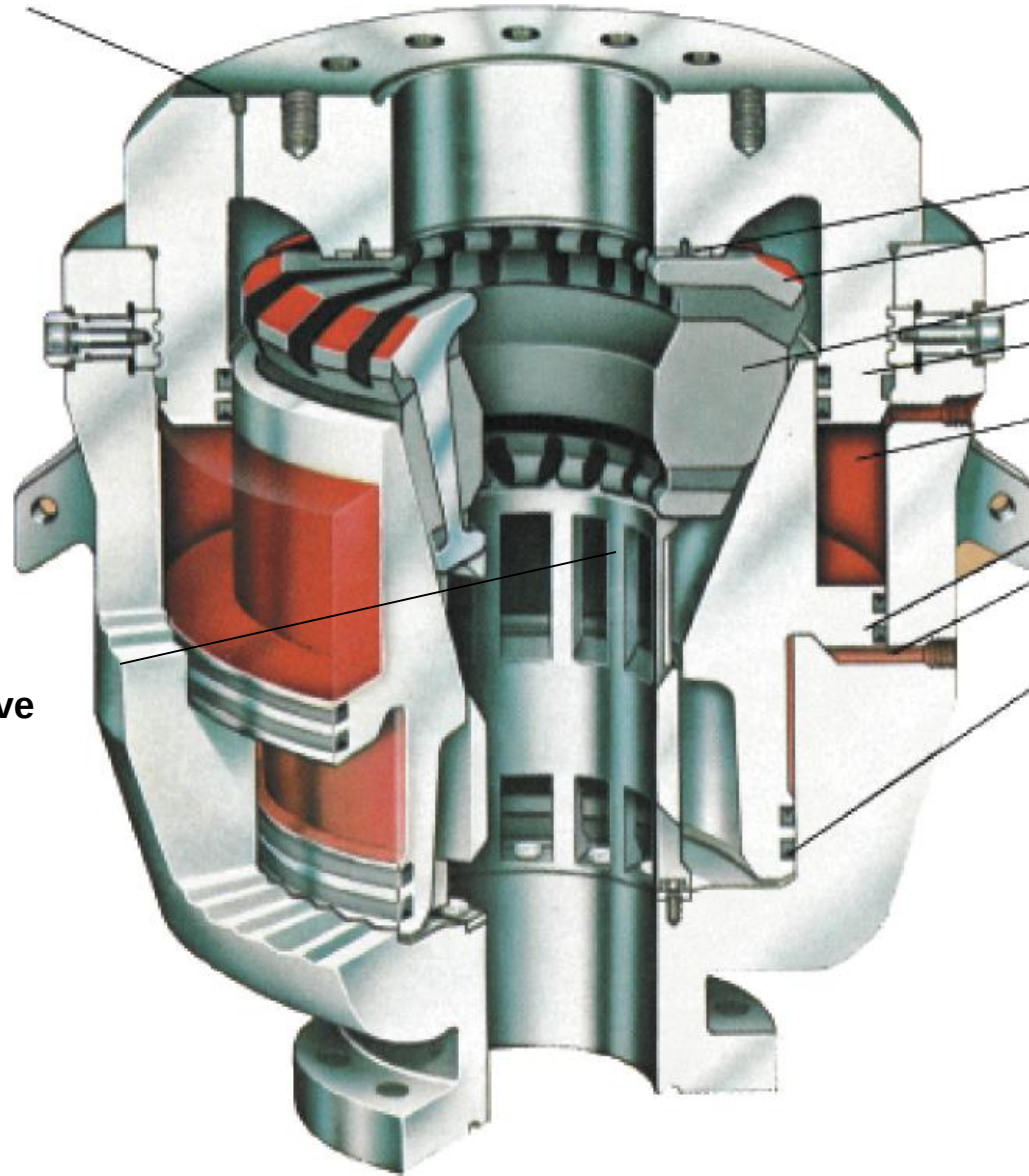
❑ Separate Shear/Seal BOP (remote control unit)

- Shear/seal rams are “tertiary” barriers provide emergency shearing capability. It is essential for emergency shutdown situations.
- The shear blades are designed for minimum deformation to the tubing so kill fluid can be pumped through the suspended coiled tubing after the cut.
- This item sometimes referred to as the **safety head** is rigged up directly on to the Xmas Tree.
- Always function test the ram before executing the CT operation & the blade/cutter is only good for one-time use.



CT ANNULAR PREVENTER

Piston indicator hole



Wear Plate
Element Segment

Packing Element
Latching head
Opening chamber

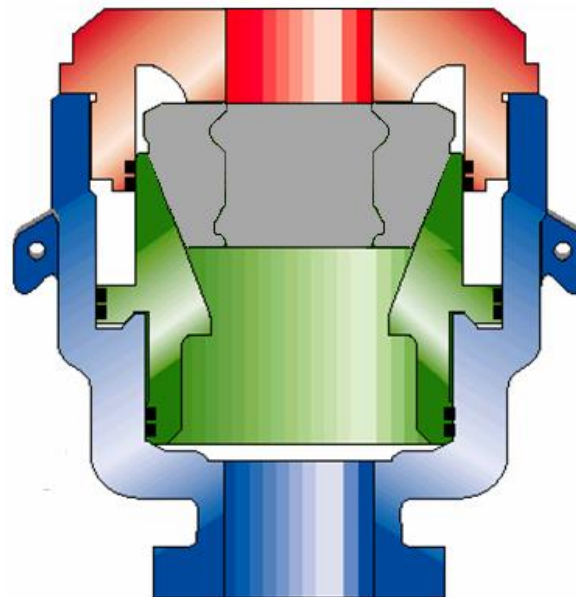
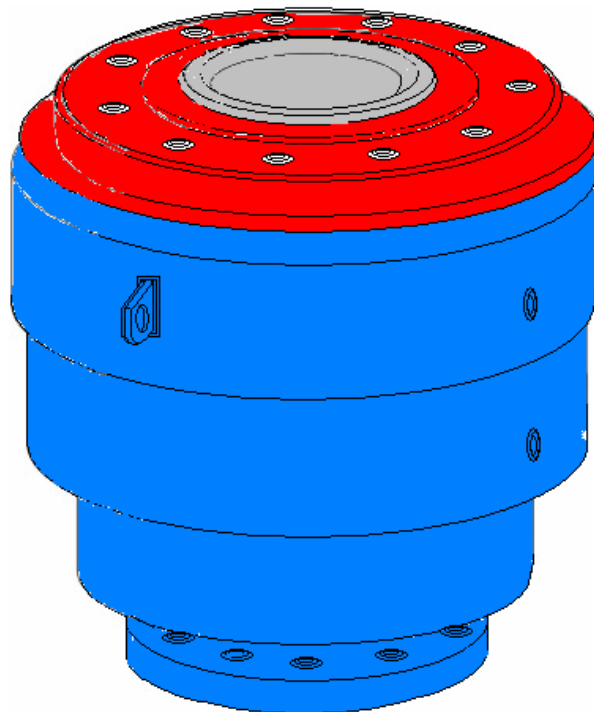
piston

CLOSING CHAMBER

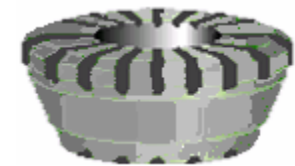
Piston Seals

Wear sleeve

- Seal with tubing or without tubing.
- When there is a need to seal around tools which have a different outside diameter than the CT
- Essential precautions should be taken when stripping through an annular BOP
 - 1. Closely monitor annular operating pressure and weight indicator reading
 - 2. Consult fishing (dimensional) diagram of BHA



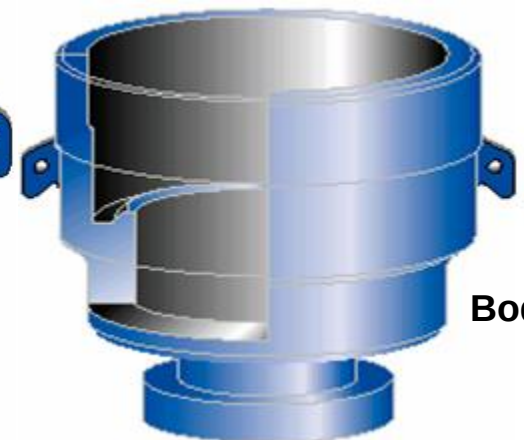
Head



Element



Piston



Body

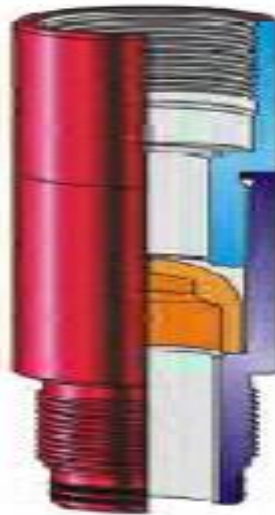
• Pressure Control Equipment

➤ Check Valves

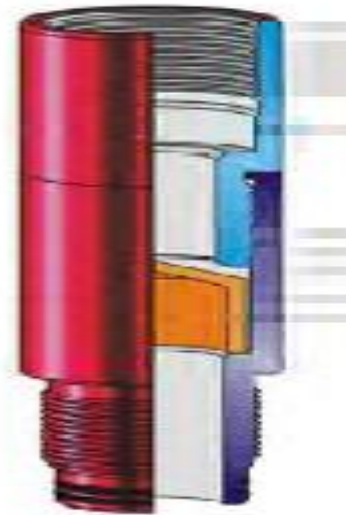
- ✓ Its function is to guard against a blow out if the tubing leaks or parts at surface by preventing flow back up the string.
- ✓ There are four types of check valve commonly available.
 - Ball
 - Dome
 - Dart
 - Flapper (lower pump pressures)
- ✓ The flapper valve is designed for use in conjunction with ball operated tools because the dart valve will not allow the passage of a ball.
- ✓ Flapper type check valves are preferred over ball & seat type as balls, plugs & darts can be pumped through



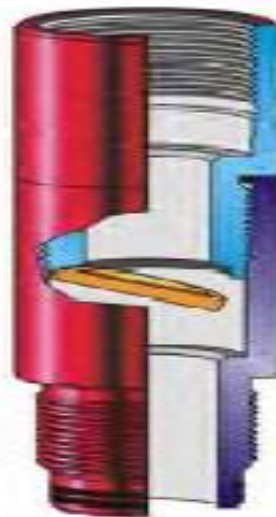
Ball Valve



Dome Valve



Dart Valve



Flapper Valve



Removable Cartridge
Flapper Valve

Coiled Tubing

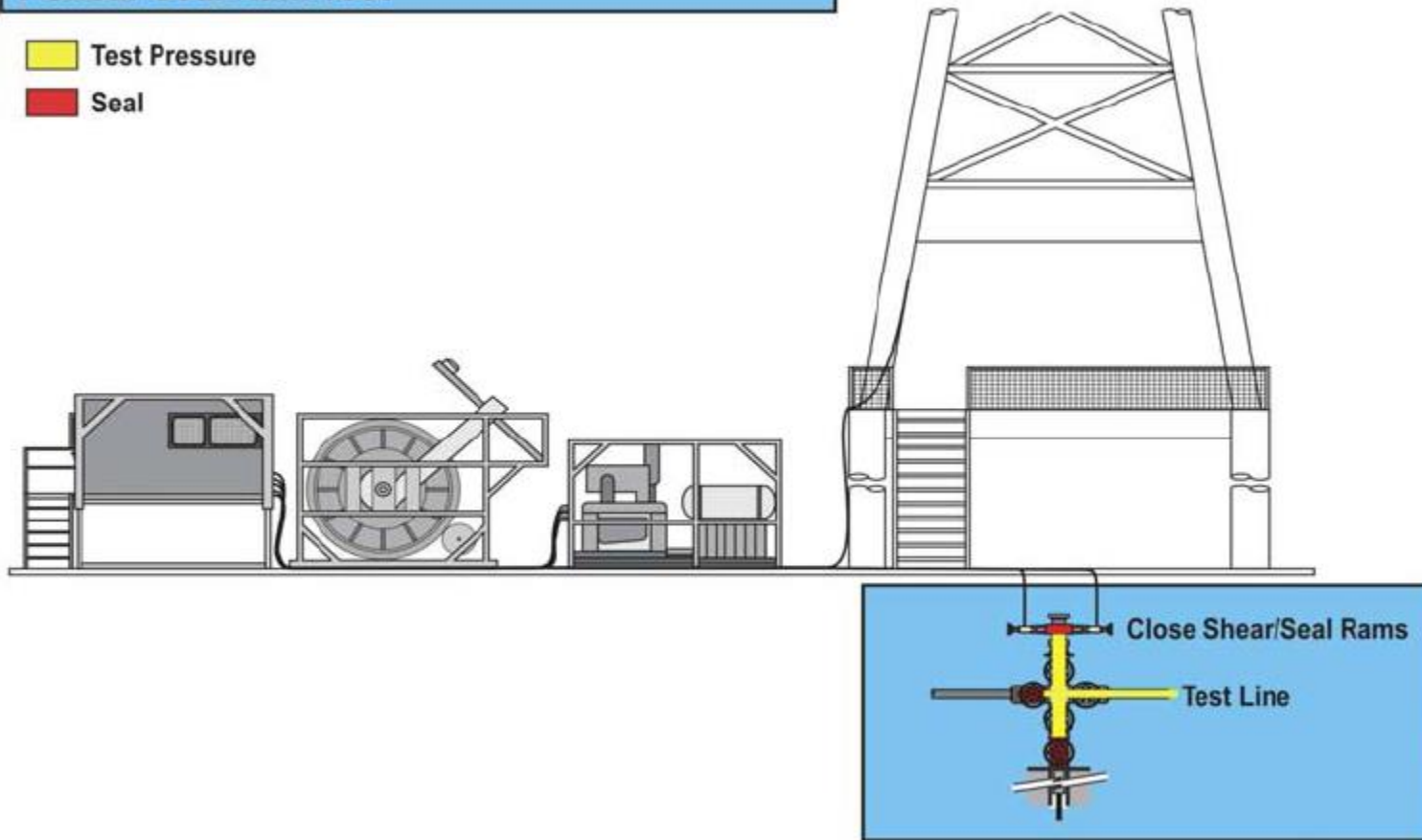
Outlines

- I. Coiled Tubing Description
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• BOP Test Procedures

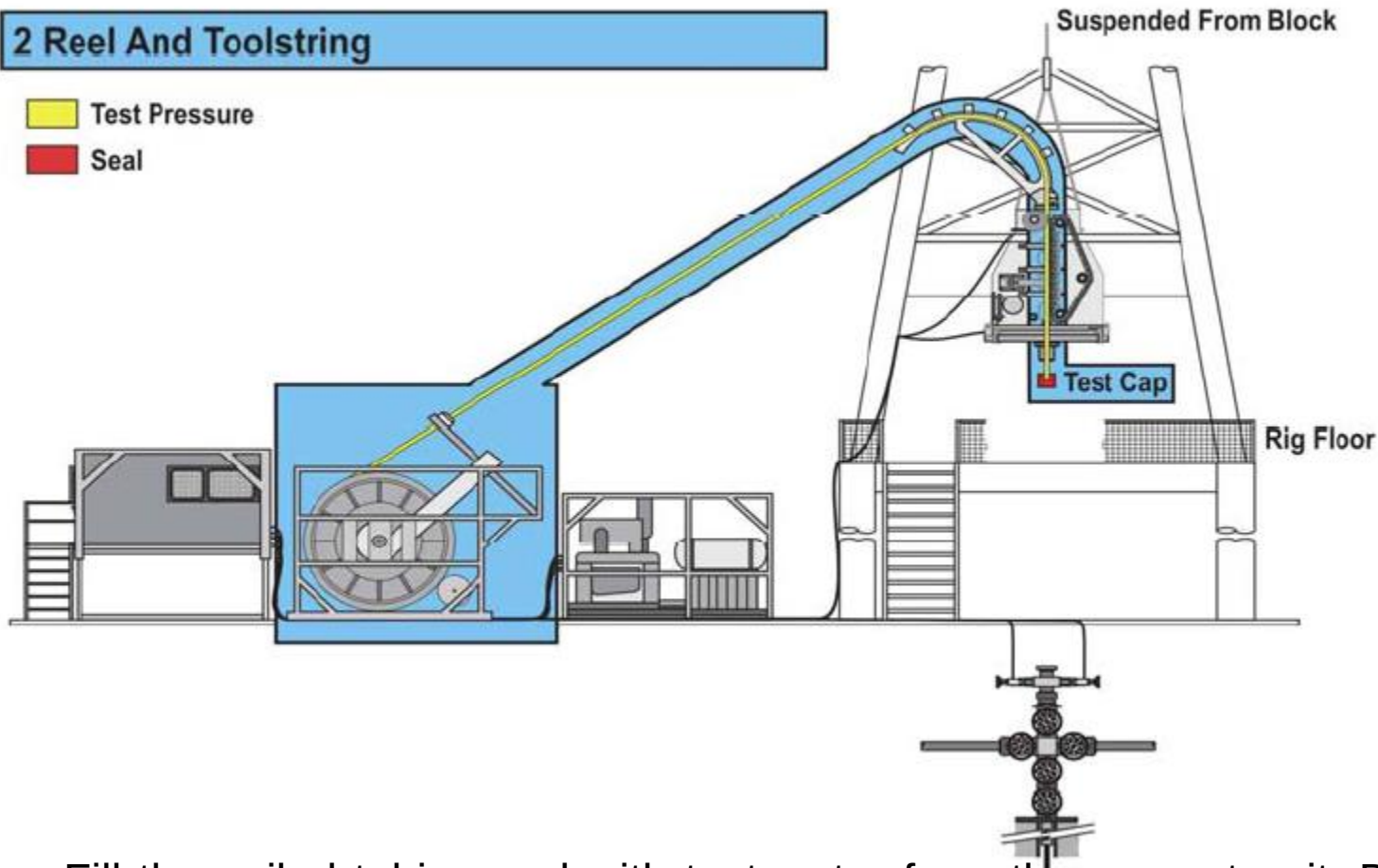
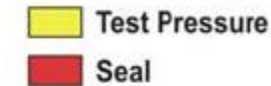
- ✓ Before starting a job, the BOP stack must be tested to insure that it will hold up to the maximum anticipated pressures from the job.
- ✓ Most BOPs are checked out and tested at the maintenance facility of the contractor prior to shipment to the job site.
- ✓ The onsite test includes function, low and high pressure tests. The results of each individual test must be recorded. The low pressure test must be performed first and should be conducted at pressures between 200 to 300 psi .The high pressure test should be the lesser of the maximum expected surface pressure, minimum burst pressure of the well's tubing, or the rating of the stack.
- ✓ All pressure tests should be recorded and held for a sufficient period of time to demonstrate that the component is effectively holding pressure.
- ✓ Test all components that will be exposed to pressure.

1 Shear Seal Ram Test



Fill up the riser and BOP via the test line to the tree valve. Close the blind rams. Increase pressure in 500 psi increments to maximum and hold for the prescribed time. Record on a chart.

2 Reel And Toolstring

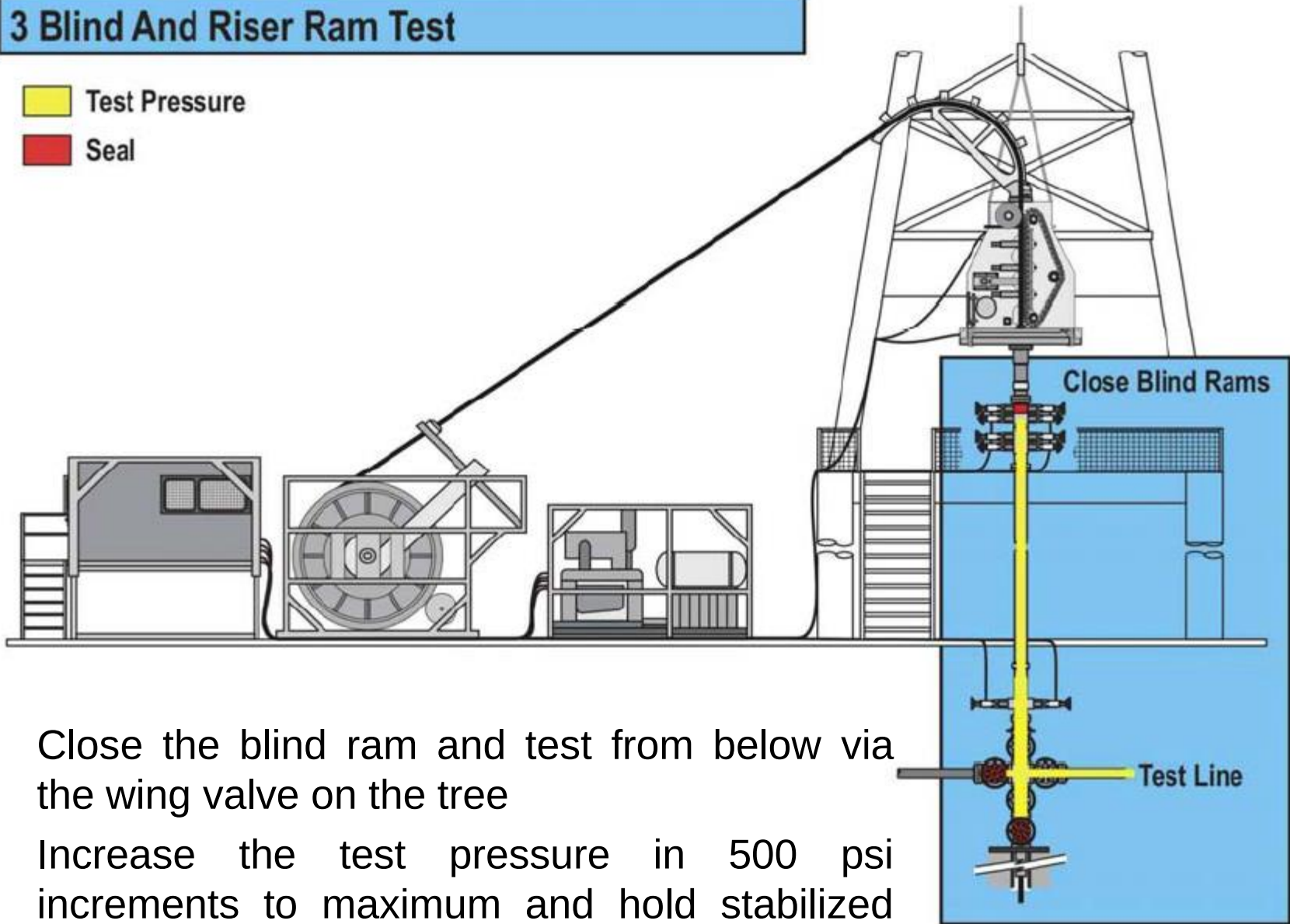


- Fill the coiled tubing reel with test water from the cement unit. Displace at least twice the tubing volume. During this circulation a pig and stainless steel ball can be used
- Attach the coiled tubing tools & close the test cap and pressure up in 500 psi increments.
- Hold stabilized pressure for the prescribed time. Record on a chart.
- The end of the string should be as close to deck as possible.

3 Blind And Riser Ram Test

Test Pressure

Seal



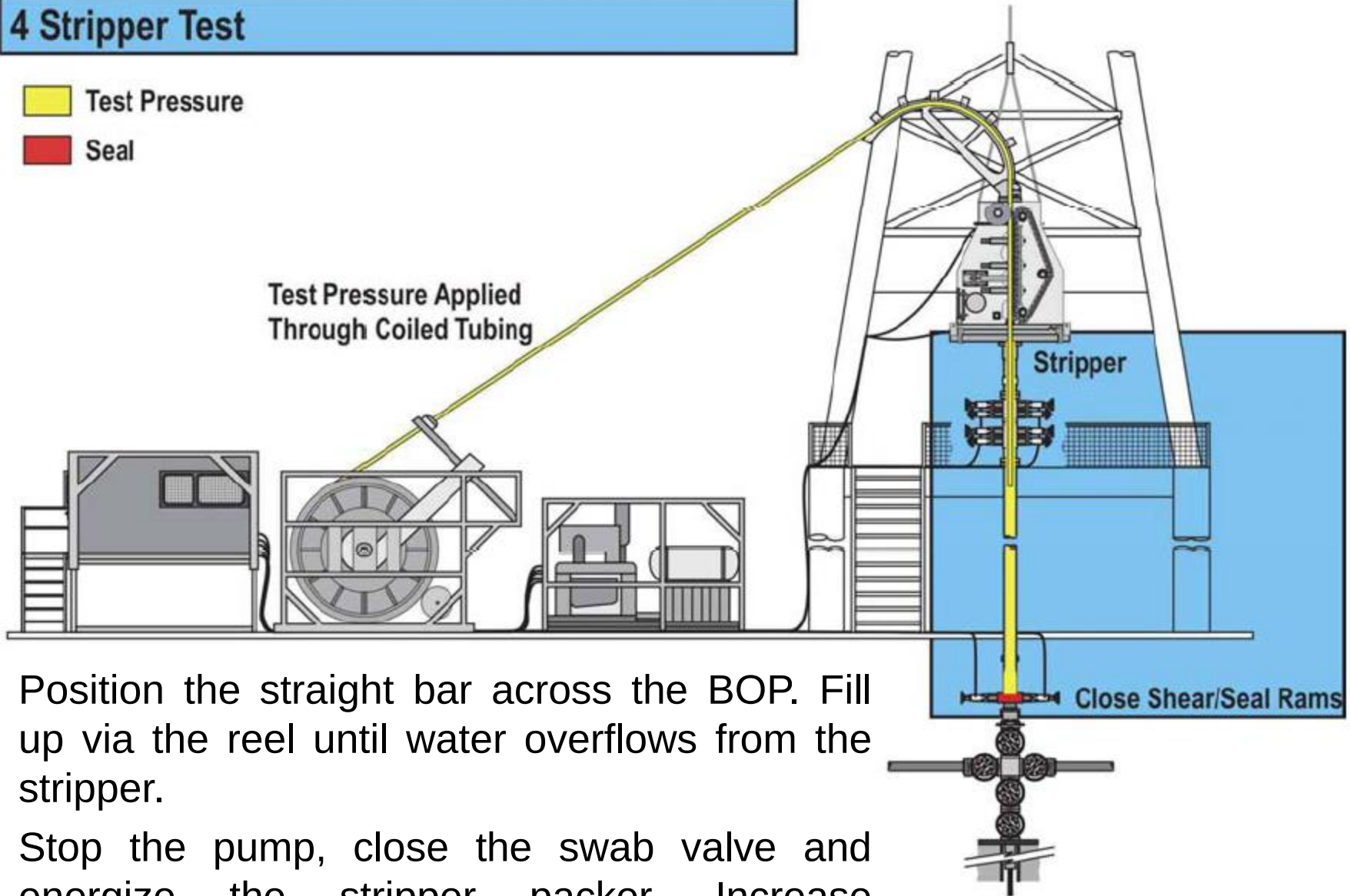
- Close the blind ram and test from below via the wing valve on the tree
- Increase the test pressure in 500 psi increments to maximum and hold stabilized pressure for the prescribed time. Record on a chart.

4 Stripper Test

Test Pressure

Seal

Test Pressure Applied
Through Coiled Tubing



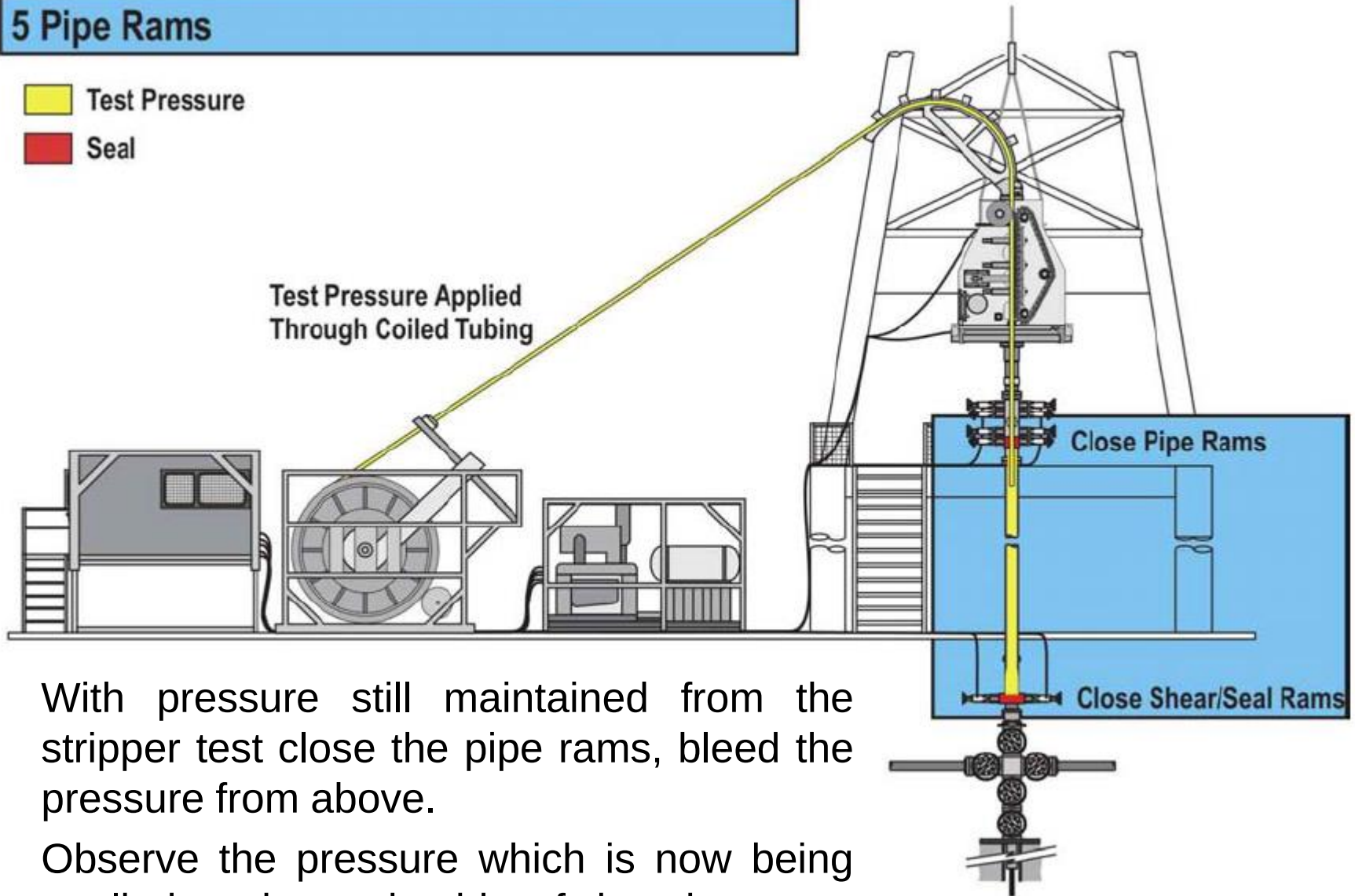
Position the straight bar across the BOP. Fill up via the reel until water overflows from the stripper.

Stop the pump, close the swab valve and energize the stripper packer. Increase pressure in 500 psi increments up to 5000 psi and hold for 15 minutes.

5 Pipe Rams

Test Pressure

Seal



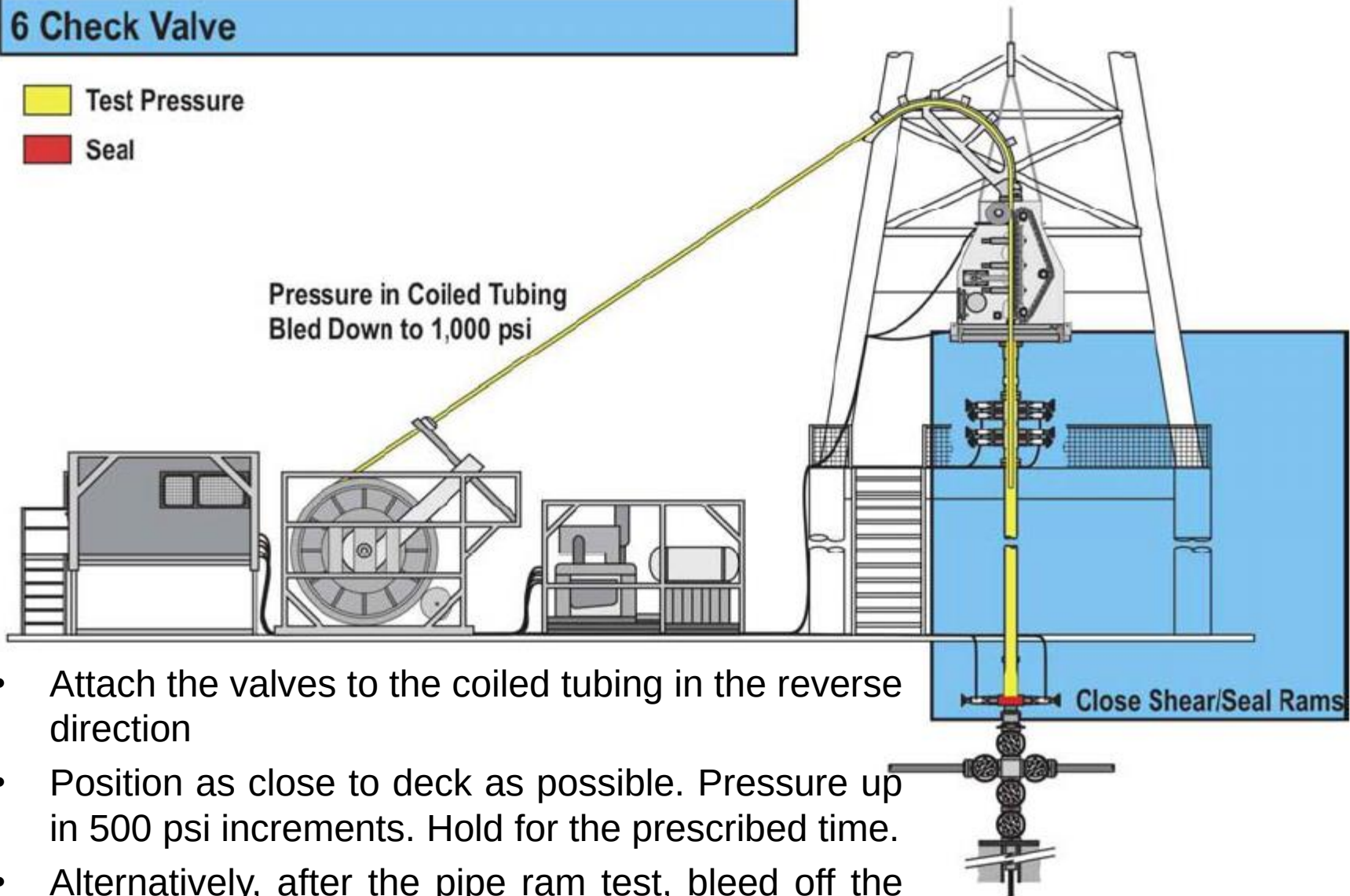
- With pressure still maintained from the stripper test close the pipe rams, bleed the pressure from above.
- Observe the pressure which is now being applied to the underside of the pipe rams for the prescribed time.

6 Check Valve

Test Pressure

Seal

Pressure in Coiled Tubing
Bled Down to 1,000 psi



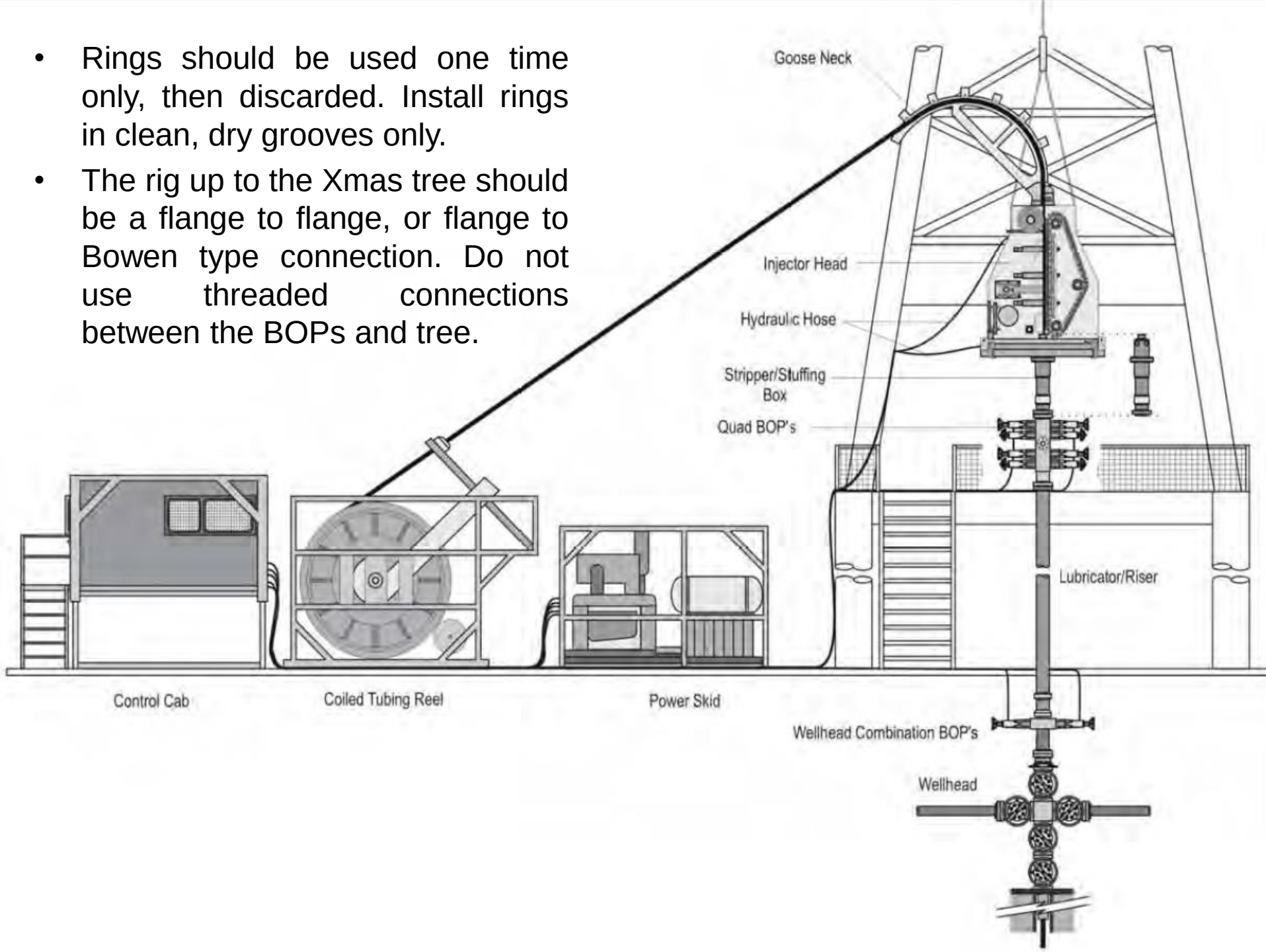
- Attach the valves to the coiled tubing in the reverse direction
- Position as close to deck as possible. Pressure up in 500 psi increments. Hold for the prescribed time.
- Alternatively, after the pipe ram test, bleed off the coiled tubing pressure to 1,000 psi and monitor the check valves are holding the pressure still inside the BOP body.

Coiled Tubing

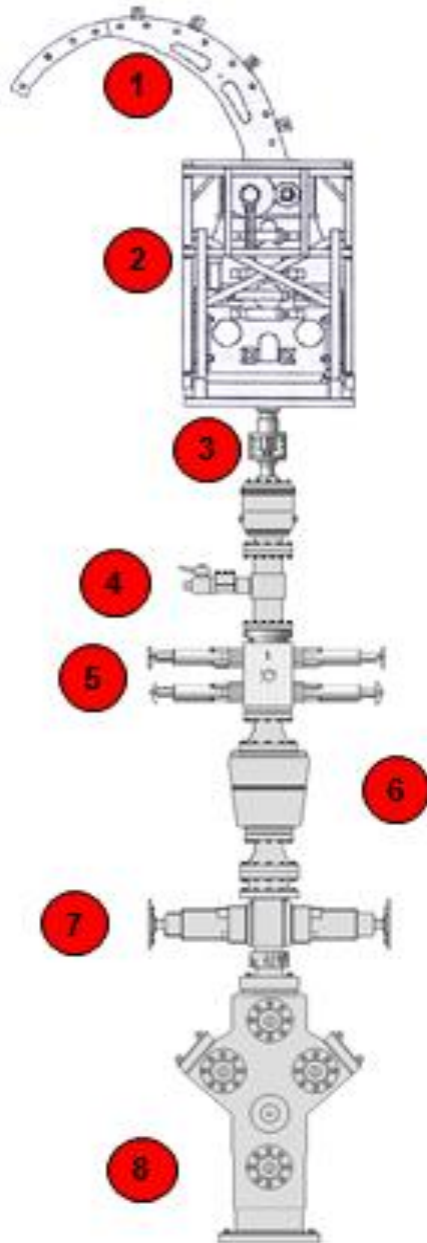
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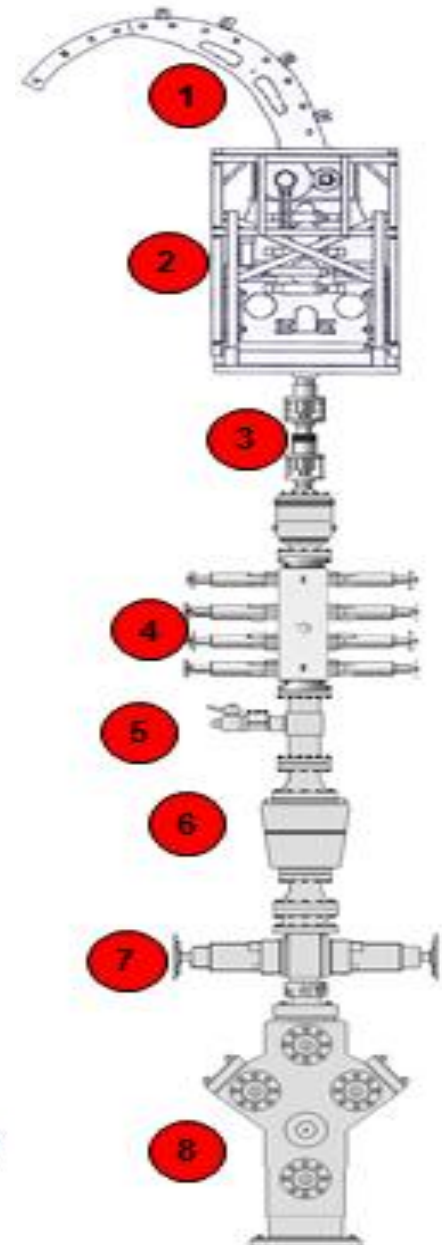
- Rings should be used one time only, then discarded. Install rings in clean, dry grooves only.
- The rig up to the Xmas tree should be a flange to flange, or flange to Bowen type connection. Do not use threaded connections between the BOP's and tree.



Low Pressure



High Pressure



Primary Barrier

Secondary Barrier

Tertiary Barrier

Blind Ram

Shear Ram

Slips Ram

Pipe Ram

Shear & Seal Ram

Coiled Tubing

Outlines

- I. Coiled Tubing Description
- II. Description of Fatigue
- III. Critical Control and Monitoring Equipment
- IV. Pressure Control Equipment
- V. BOP Test Procedures
- VI. General Rig Up Procedures
- VII. Contingencies**

- **Contingencies**

- **Holes In Tubing**

- ✓ Possible solutions depend on the severity of the hole, and what types of fluids are being pumped.
- ✓ Depending on the length, pressure, integrity of the tubing (i.e. if a pinhole exists), and whether any hazardous fluids are in use, the hole in the tubing should be run back into the well between the stripper and pipe ram and the well killed.
- ✓ If a circulating spool is provided below the stack, close the slip and pipe ram and then kill the well. Once killed, the pipe can be safely pulled from the well.
- ✓ If the fluid is not hazardous, if the hole is small, and if the hole in the tubing can be safely spooled onto the reel, this will effectively stop the leak. The tubing may then be pulled from the well if the supervisor deems that is it safe to continue.

• Contingencies

➤ Tubing Parts

- ✓ Should the coil **parts downhole**, the operator should be aware of a sudden change in pressure and/or a sudden loss of weight.
- ✓ By continuing to pump fluid down the pipe to prevent wellbore fluids from coming up the pipe, by estimating the length of pipe lost and by careful and regular checking when near the surface, the pipe can be safely pulled out of the hole.
- ✓ When near surface, SV can be gently closed (counting turns) every few ft. to check whether the pipe is still across the valve.
- ✓ If the pipe **parts on surface**, there are check valves that stop wellbore fluids from escaping. If also failed, close the slip and pipe rams, shear the pipe with the cutter rams, pull pipe from stack, and close the blind rams. The well must be killed
- ✓ If the tubing parts downhole, pull tubing in well above the blind ram, and close the blind ram. The well should be killed using the proper method (i.e. Bullhead or Volumetric).
- ✓ After the well has been killed, the tubing can be fished.

• Contingencies

➤ Leaks

- ✓ If a leak develops in a BOP, tool lubricator, crossover, spool, or any component above the tree, then attempts should be made to stop the leak. If the leak cannot be stopped, then the well should be killed.
- ✓ If the leak starts accelerating, or becomes too critical while attempting the kill, the decision may be made to cut CT& then the swab or master valves closed (count turns).
- ✓ If leaks develop in stripper or stuffing box, close pipe rams and replace packing element.
- ✓ If the leak develops in a lubricator or connection, close the pipe ram and try to stop the leak. If this does not stop the leak, the well should be killed and the leak repaired, or the tubing stripped out of the well, the crown or master valves closed and the leak repaired.

• Contingencies

➤ Stuck Coiled Tubing

- ✓ The pipe is said to be stuck when the pull required is greater than 80% of pipe yield strength.
- ✓ Some of the more common ways of sticking are:
 - Settled solids around the pipe during washing operations.
 - Obstruction in the well.
 - Increased friction drag. (Pumping friction reducers around CT outside).
 - Differential stuck or Mechanical stuck when it hangs on anything downhole.
- ✓ If the pipe is stuck and you can circulate, try pumping friction-reducing fluids, one of the pump cylinders should be crippled. If this doesn't work, spot a heavier fluid around the stuck point back to surface to increase the pipe's buoyancy. If the pipe remains stuck, work the pipe pulling minimum tensile.
- ✓ If the pipe is stuck and you cannot circulate, apply maximum pressure to tubing to try to hydraulically jack the tubing free or drop the ball and release the BHA.
- ✓ If the pipe cannot be freed by any of the above techniques, the free point must be determined and use a chemical cutter run of wireline.
- ✓ If the well is under pressure, the well should be killed prior to cutting the tubing.

- **Contingencies**

- **Runaway Tubing**

- ✓ This is the condition where the pipe is moving faster than the operator can control. It can occur when running in or pulling out.
- ✓ It is usually caused by the drive blocks losing their grip on the pipe. In high pressure wells the pipe may be ejected whilst in low pressure wells the pipe may run in.
- ✓ The operator should be able to stop the pipe from continuing to be ejected by increasing the stripper operating pressure to maximum and closing the slip rams.
- ✓ The operator should be able to stop the pipe from continuing to be injected by stopping injection and increasing the inside tension. Increasing the stripper operating pressure to maximum also creates more friction.
- ✓ Depending on company policy, applying reel brakes
- ✓ However, it should be realized that these options may cause sudden failure of the tubing.

• Contingencies

➤ Power Pack Failure

- ✓ The failure of a power pack during operations should not cause a well control problem.
- ✓ All functions on the unit are fail safe or there are manual locks and brakes that can be applied.
- ✓ The BOP do not require the power pack running to operate them because they use, as with any BOP system, stored pressure in an accumulator.
- ✓ In the event of a power pack failure:
 - Engage injector brake
 - Close pipe rams and manually lock
 - Close manual stems on pipe and slip rams as back-up
 - Apply the reel brake if it is not fail-safe applied
 - While maintaining circulation (if possible), repair or replace power pack
 - Equalize pressure across pipe rams and open pipe and slip rams
 - Release injector brake
 - Re-commence coiled tubing operations.

- **Contingencies**

- **Coiled Tubing Pinhole**

- ✓ If the pinhole is above the stripper, the leak will be very visible. There are check valves at the bottom of the pipe that should stop wellbore fluids from exiting the well via the pinhole.
- ✓ If the check valves are not holding the individual circumstances of the problem will determine the steps to be taken. This might well involve closing the BOPs, cutting the pipe, killing the well and fishing the pipe.

- **Check Valve Failure**

- ✓ If the check valves are no longer holding, by continuing to pump fluid down the pipe it is often possible to POOH to replace the check valves.
- ✓ This can depend on the work being performed and the pressures involved. It may prove necessary to kill the well before pulling out.